
Measure Theory

— *EYNTKA* Series —

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April 6, 2026

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Part I

Measure Theory

(I feel like I should split the book in two, with all the measure theory/integration build up, followed by the more function-analysis stuff afterwards)

1

Measures

As a start, we might want to define a reasonable definition of what it means to measure any subset of $\mathbb{R}^n$. Such a function would be of the form $\mu : P(\mathbb{R}^n) \rightarrow [0, \infty]$. If μ were to exist, we would want our common geometric intuitions to hold, that is:

1. If $E_1, E_2, \dots, E_n, \dots$ is a countable collection of pair-wise disjoint subsets, then

$$\mu\left(\bigcup_i E_i\right) = \sum_i \mu(E_i)$$

2. If E and F are sets such that there is a function that only translates, rotates, and reflects E to get to F (i.e. a rigid motion), then $\mu(E) = \mu(F)$
3. If C is a unit cube in $\mathbb{R}^n$, then $\mu(C) = 1$

However, these three axioms are inconsistent with one another! We have actually shown this by showing if μ is defined on $P(\mathbb{R}^n)$ (even when $n = 1$), then we can construct *Vitali sets* that will have to be both 0 and ∞ in measure – a contradiction.

Theorem 1.0.1: Vitali Sets Are Unmeasurable

Vitali sets are un-measurable

Proof:

Take $[0, 1]$. Partition this set as follows:

$$x \sim y \Leftrightarrow x - y \in \mathbb{Q}$$

In a sense, this is the set of cosets of $\mathbb{R}/\mathbb{Q}$ restricted to $[0, 1]$; we will rely on this intuition to label

the equivalence classes. Since $\mathbb{R}/\mathbb{Q}$ is the set of irrational numbers modulo rational numbers, let N_q be an equivalence class of contains x that is equivalent to the irrational number q . Notice that there are uncountably many such equivalence relations.

Now, pick one $p \in N_q$ for each equivalence class, and let N be the collection of all these p 's, so N contains one point from each equivalence class (notice that the axiom of choice must be used to create a non-empty N). We'll show that N is unmeasurable.

First, for each rational number $r \in \mathbb{Q} \cap [0, 1)$, define:

$$N + r = \{x + r \mid x \in N \cap [0, 1 - r)\} \cup \{x + r \mid x \in N \cap (1 - r, 1)\}$$

Then clearly, $N + r \subseteq [0, 1)$ for each $r \in \mathbb{Q} \cap [0, 1)$, and furthermore, for each $x \in [0, 1)$ there exists an r such that $x \in N + r$. To show this, let's say $y \in N$ is the element that was picked from the equivalence class of $[x]$. Then by our choice of y , $x \in N + r$ since $r = x - y$ if $x \geq y$, or $r = x - y + 1$ if $x < y$. Next, notice that the $N + r$ also forms a partition: if $(N + r) \cap (N + s) \neq \emptyset$ then $x - r$ (or $x - r + 1$) would equal $x - s$ (or $x - s + 1$), but they belong to distinct equivalence classes, and so $(N + r) \cap (N + s) = \emptyset$.

We have now reached the point where we can declare our contradiction. If μ satisfies the three wanted properties, by (1) and (2):

$$\mu(N) = \mu(N \cap [0, 1 - r)) + \mu(N \cap (1 - r, 1)) = \mu(N + r)$$

for any $r \in \mathbb{Q} \cap [0, 1)$. Since $\mathbb{Q} \cap [0, 1)$ is countable and $[0, 1)$ is the disjoint union of the $N + r$'s:

$$1 = \mu([0, 1)) = \sum_{r \in \mathbb{Q} \cap [0, 1)} \mu(N + r)$$

where $1 = \mu([0, 1))$ comes from (3). Now, we have two possibilities:

1. Either $\mu(N) = 0$, in which case each $\mu(N + r) = 0$ but then $0 = 1$ – a contradiction
2. Either $\mu(N) \neq 0$ (say k , so $\mu(N + r) = k$, but then $0 = \infty$ – a contradiction

In other words, no matter what $\mu(N)$ is, we will run into a contradiction!

As you can verify, this construction required that we take advantage of the countability condition (the 1st property). You might ask if weakening it to only finitely many will fix the issue, but sort of crazily it does not! In 1924, Banach and Tarski proved the following paradox now known as the *Banach Tarski Paradox*:

Let U and V be open subsets of $\mathbb{R}^n$, $n \geq 3$. Then there exists a $k \in \mathbb{N}$ and subsets $E_1, E_2, \dots, E_k, F_1, F_2, \dots, F_k$ such that

1. all E_j 's are disjoint and their union is U
2. all F_j 's are disjoint and their union is V
3. There exists a rigid motion from E_j to F_j for $j = 1, \dots, k$

This is kind of insane: this means that we can take a pea-size ball, and make into the size of the earth! This means that the notion of “geometry” is not actually a consistent concept in general set theory!

Fortunately for us, The sets E_j and F_j are very strange sets; most sets that are in fact measurable. To be more precise, it is consistent to say that the axiom of choice fails and that all sets are Lebesgue measurable (which will be the measure that will replace the Riemann integrability). For the purposes of this course, that means that any set we construct without the axiom of choice (and just doing normal ZF without any sneaky logical cheating) will be measurable. For more details, see

<https://math.stackexchange.com/questions/1847052/most-functions-are-measurable>

1.1 Sigma-Algebras and Properties

Since the axiom of choice is the source of the problem of unmeasurability in the previous problems, we start by defining a space on which we avoid unmeasurable sets, namely, we want to limit ourselves to *countable* choices. The definition should be pretty general, as it should accommodate measures in more general spaces. It should also be “arithmetically expandable”, as in we should be able to measure smaller parts of the space (perhaps in more convenient orientations) and add them all up. Thus, we define an *algebra*:

Definition 1.1.1: Algebra and σ -Algebra

Let X be an arbitrary set. Then let $\mathcal{A} \subseteq P(X)$ be a non-empty collection of subsets such that

1. for any finite set of sets $A_1, A_2, \dots, A_n \in \mathcal{A}$, $\bigcup_i A_i \in \mathcal{A}$
2. For any $A \in \mathcal{A}$, $A^c \in \mathcal{A}$.

If it is closed under countable union, we call it a σ -algebra. We will usually denote σ -algebra on X as $\mathcal{M}(X)$.

The σ -algebra will be our natural space in which we will show in the next section we can define a “measure function” without the previous contradiction.

I like to think of the finite closure as a more general way of saying that a space is “closed under a function”, the function here being $\cup$. In this way, it’s sort of the most general algebraic structure. Adding the fact that it’s closed under countable union brings us further way from the realm of algebra, and hence it’s a σ -algebra. The term sigma is very common to mean countable, for example, a space σ -compactness if it is the countable union of compact subspaces.

Definition 1.1.2: Measurable Space

Let X be a set and $\mathcal{M} \subseteq P(X)$ be a σ -algebra. Then $(X, \mathcal{M})$ is called a *measurable space* and a set in $\mathcal{M}$ is called a *measurable set*

Proposition 1.1.3: Properties of σ -algebra

Some immediate properties of a σ -algebra:

1. It is closed under countable intersection, since $\bigcap_i X_i = (\bigcup_i X)^c$, and by difference, since $X \setminus Y = X \cap Y^c$.
2. $\emptyset$ and $X \in \mathcal{A}$, since for any $E \in \mathcal{A}$, $E \cap E^c = \emptyset \in \mathcal{A}$ and $E \cup E^c = X \in \mathcal{A}$
3. Any countable collection of subsets $\{E_i\}_{i=1}^\infty \subseteq \mathcal{A}$ can be turned into a countable disjoint union, where

$$F_k = E_k \setminus \left(\bigcup_{i=1}^{k-1} E_i \right) = E_k \cap \left(\bigcup_{i=1}^{k-1} E_i \right)^c$$

and by what we've just shown, every $F_k \in \mathcal{A}$, and by construction $\bigcup_i E_i = \bigcup_i F_i$. We will use this technique often very soon (note that not all sets are nonempty, that proof requires more set-theory manipulation).

4. Given any family of σ -algebras on X , their intersection is also a σ -algebra. Because of this, given $Y \subseteq X$, there is always a unique smallest σ -algebra $\mathcal{M}(Y)$ that contains Y (there is at least always one, $P(Y)$, and so this concept is well-defined). $\mathcal{M}(Y)$ is called the σ -algebra generated by Y .

A lemma we'll point out right now to simplify future proofs is the following:

Lemma 1.1.4

If $Y \subseteq \mathcal{M}(X)$, then $\mathcal{M}(Y) \subseteq \mathcal{M}(X)$

Proof:

Since $\mathcal{M}(X)$ is a σ -algebra that contains Y , then it will contain all countable unions and compliment of Y , and hence it will contain $\mathcal{M}(Y)$ (i.e. proposition 1.1.3[4]).

Example 1.1.5: σ -Algebras

1. If X is any set, then $\{\emptyset, X\}$ and $P(X)$ are trivially σ -algebras. Once we define measurable function, these will be the initial and final object in measurable spaces.
2. If X is any set, then $P(X)$ is also a σ -algebra. Usually, we will use proposition 1.1.3[5] and other tools to construct a smaller σ -algebra to not work with $P(X)$ since it's too big for most cases.
3. If X is an uncountable set, define

$$\mathcal{A} = \{E \subseteq X \mid E \text{ is countable or } E^c \text{ is countable}\}$$

this set is called the σ -algebra of countable and co-countable sets. It is useful when we want to make sure that the countable union of sets is still countable (or it's compliment is) ^a

^aSneakily, we use the axiom of choice for countably many choices to prove this is true

Another common σ -algebra generated on a set X is related to a topology on X , that is, if $\mathcal{T}(X)$ is a topology, of X , we will define the following σ -algebra:

Definition 1.1.6: Borel σ -Algebra

Let X be a set and $\mathcal{T}(X)$ be a topology on X . Define the *Borel σ -algebra* to be

$$\mathcal{B}_X := \mathcal{M}(\{U \subseteq X \mid U \text{ is an open set contained in } X\})$$

The elements of $\mathcal{B}_X$ are called *Borel sets*

The set $\mathcal{B}_X$ will include the open sets, closed sets, countable union and intersection of closed sets, and the countable union of countable intersections of open/closed sets, et cetera.

Since the countable intersection of open sets and countable union of closed sets are not necessarily open or closed sets respectively¹, but are members in $\mathcal{B}_X$, we will give them names: A countable intersection of open sets is called a G_δ set (G for *Gebeit* in German), and a countable union of closed sets is called a F_σ set (F for *fermé* in french). The σ is for *Summe* (sum in German) since it's a union and the δ is for *Durchschnitt* (intersection in German) since it's an intersection. We can continue this pattern and define $G_{\delta\sigma}$ to be the countable union of countable intersection of open sets, $F_{\sigma\delta}$ to be the countable intersection of countable union, and so and so on and so forth for any length of interchanging σ 's and δ 's. In general, this "chain" of unions and intersections does not need to terminate, and a $F_{\sigma\delta}$ (as a simple example for a chain of length 2, $\mathbb{R} \setminus \mathbb{Q}$ is a $F_{\sigma\delta}$ set, but not a F_σ set; use Baire's Categories theorem). There will be cases in which it will terminate (in fact, the majority of measure's we'll work with will have this terminate after at just F_σ or G_δ "up to a zero set", see theorem 1.4.5, and example ref:HERE for a measure that does not terminate)

One of the most common Borel sets we will be working with is $\mathcal{B}_{\mathbb{R}}$ with the usual euclidean topology on $\mathbb{R}$. It is useful to have an of different generating sets for $\mathcal{B}_{\mathbb{R}}$:

Proposition 1.1.7: Generators For $\mathcal{B}_X$

Let $\mathbb{R}$ be the real numbers and $\mathcal{B}_{\mathbb{R}}$ be the Borel σ -aglebra on $\mathbb{R}$ given the standard euclidean topology $\mathcal{T}(\mathbb{R})$. Then the following generate $\mathcal{B}_{\mathbb{R}}$:

1. The set of open intervals
2. the set of closed intervals
3. the set of half-open intervals
4. the set of open rays
5. the set of closed rays

Proof:

These proofs are quite easy to prove, and should be done as an exercise. A fact that you might need to recall is that every open set in $\mathbb{R}$ can be written as a countable union of open intervals (hint: $\mathbb{R}$ has a countable basis).

¹it is possible that the countable intersection/union of open/closed sets are open/closed, and so the word "necessarily" is required

Products and σ -Algebras

Recall that the product between two sets is defined as $A \times B := \{(a, b) \mid a \in A, b \in B\}$, or more generally $\{X_\alpha\}_{\alpha \in A}$ for a collection of non-empty sets X_α ², with $X = \prod_{\alpha \in A} X_\alpha$. As we know from Category Theory, the product comes equipped with set functions π_α that satisfy the universal property of products (or more generally, this is a limit over the discrete category). Though we have yet to give a category where σ -algebras are the objects (we need to define what the morphisms are³), we can already construct our set $\prod_\alpha X_\alpha$ that will be the product of the X_α in the category of measurable spaces.

Definition 1.1.8: Product σ -Algebra

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of non-empty sets and $X = \prod_{\alpha \in A} X_\alpha$. Let

$$\bigotimes_{\alpha \in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in M_\alpha, \alpha \in A\})$$

to be the *product σ -Algebra* on $\{M_\alpha\}_{\alpha \in A}$, i.e., it is the σ -algebra generated by all of the pre-images of the sets in M_α .

It is not difficult to check that the product σ -algebra is indeed a σ -algebra. If $|A|$ is countable we usually write $\bigotimes_{i=1}^\infty M_i$, and if $|A|$ is finite, we can write $\bigotimes_{i=1}^n M_i$ or $M_1 \otimes M_2 \otimes \cdots \otimes M_n$. Notice that since we allow for countable unions, it is not so easy to represent $\bigotimes_{\alpha \in A} M_\alpha$ as the product of the elements of M_α , (just like for Groups in the category of **Grp**), and hence why we do not use the $\prod$ or $\bigoplus$ notation. On the other hand, the fact that only a countable union of elements is permitted will still limit the product in the number of non-trivial components, hence justifying the $\bigotimes$ notation instead of $\prod$. In fact, it is the same as $\bigoplus$, but instead of “all but finitely many”, we have “all but countably many”:

Proposition 1.1.9: Countable Product σ -algebra

Let $M_n = \mathcal{M}(\mathcal{E}_n)$ be a σ -algebra for $n \in \mathbb{N}$ where $\mathcal{E}_n \subseteq P(X_n)$. If $X_n \in \mathcal{E}_n$, then

$$\bigotimes_{n=1}^\infty M_n = \mathcal{M}\left(\left\{\prod_{i=1}^n E_i \mid E_i \in M_i\right\}\right)$$

Proof:

The $\supseteq$ direction is easy (Take $\bigcap_{i=1}^n \pi_i(E_i)$ for appropriate values). For $\subseteq$, Since $X_i \in M_i$, we have that every element is inside the right hand side. Make sure you see why $X_n \in \mathcal{E}_n$ makes a difference. By lemma 1.1.4, the result follows.

We can further simplify our representation of $\bigotimes_{\alpha \in S} M_\alpha$ given a generating set for each M_α (such a set always exists, namely M_α itself is always a generating set)

²If they were empty, then the product would be empty

³this is done in chapter 2

Proposition 1.1.10: Generating set for Product σ -Algebra

Let $\{M_\alpha\}_{\alpha \in A}$ be a non-empty collection of σ -algebras, and let $\mathcal{E}_\alpha \subseteq P(X_\alpha)$ be a generating set for M_α . If $\mathcal{F} = \cup_{\alpha \in A} \mathcal{E}_\alpha$, then:

$$\bigotimes_{\alpha \in A} M_\alpha = \mathcal{M}(\{\pi_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\}) = \mathcal{M}(\mathcal{F})$$

Furthermore, if $|A|$ is countable and $X_\alpha \in \mathcal{E}_\alpha$ for each $\alpha \in A$, then

$$\bigotimes_{\alpha \in A} M_\alpha = \mathcal{M}\left(\left\{\prod_{\alpha \in A} E_\alpha \mid E_\alpha \in \mathcal{E}_\alpha, \alpha \in A\right\}\right)$$

Proof:

Dealing with the first case, the $\supseteq$ is clear since $E_\alpha \in \mathcal{E}_\alpha$ implies $E_\alpha \in M_\alpha$, so $E_\alpha \in \bigotimes_{\alpha \in A} M_\alpha$. For the $\subseteq$ direction, If $E_\alpha \notin \mathcal{E}_\alpha$, then since $\mathcal{M}(\mathcal{E}_\alpha) = M_\alpha$, some countable unions and intersections of elements of $\mathcal{E}_\alpha$ equal E_α , proving the $\subseteq$ direction.

The countable case follows proposition 1.1.9

Borel Spaces and Products

A topological space of much study in Analysis is metric spaces. As a reminder, if $\{X_\alpha\}_{i=1}^n$ is a collection of metric spaces, then $X = \prod_{i=1}^n X_i$ has the product metric which is the generalization of the euclidean metric. As was seen in a topology class, the following metrics on a product space all define the same product topology (for notational simplicity, let's work with two metric spaces X and Y and consider $X \times Y$):

1. $d((x_1, x_2), (y_1, y_2)) = \max\{d_X(x_1, y_1), d_Y(x_2, y_2)\}$
2. $d((x_1, x_2), (y_1, y_2)) = \sqrt{d_X(x_1, y_1)^2 + d_Y(x_2, y_2)^2}$
3. $d((x_1, x_2), (y_1, y_2)) = d_X(x_1, y_1) + d_Y(x_2, y_2)$

We can ask about establishing Borel sets $\mathcal{B}_X$ on $X = \prod_{i=1}^n X_i$ with X having the product measure. It is tempting to say that this naturally splits into $\bigotimes \mathcal{B}_{X_i}$, however, this is not always the case! This is mainly due to the fact that not all topological spaces have a countable basis, and since we're allowing countable intersection of open sets, this will actually create more sets than the if we take the product of the Borel spaces:

Proposition 1.1.11: Borel sets on product Metric Space

Let $X_1, \dots, X_n$ be metric spaces, and $X = \prod_{i=1}^n X_i$ be the product metric space. Then

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$$

If the spaces X_i are separable (i.e. has a countable dense subset, and since it's a metric space it means there is a countable basis), then

$$\bigotimes_{i=1}^n \mathcal{B}_{X_i} = \mathcal{B}_X$$

Proof:

The proof of the $\subseteq$ direction rather simple. By proposition 1.1.10, $\bigotimes_{i=1}^n \mathcal{B}_{X_i}$ is generated by $\mathcal{E} = \{\pi_i^{-1}(U_j) \mid U_j \text{ is open in } X_j, 1 \leq j \leq n\}$. Since the sets of $\mathcal{E}$ are open in X , by lemma 1.1.4, $\bigotimes_{i=1}^n \mathcal{B}_{X_i} \subseteq \mathcal{B}_X$.

For the converse, since each X_j is separable, each have a countable dense subset. Since X_j is a metric space, we can form a countable basis with open balls with rational radii around each points in the countable dense subsets.

As a consequence, $\bigotimes_{i=1}^n \mathcal{B}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R}^n}$, since $\mathbb{R}$ has a countable dense subset.

Elementary Sets

This is a technical result needed for later (maybe introduce it as exercises?)

Definition 1.1.12: Elementary Family

Let X be a set and $\mathcal{E} \subseteq P(X)$. Then $\mathcal{E}$ is called an *elementary family* if

1. $\emptyset \in \mathcal{E}$
2. If $E, F \in \mathcal{E}$ then $E \cap F \in \mathcal{E}$
3. If $E \in \mathcal{E}$, then E^c is a finite disjoint union of elements of $\mathcal{E}$

Proposition 1.1.13: Elementary Families and Algebra

Let $\mathcal{E}$ be an elementary family. Then the collection of finite disjoint unions of element of $\mathcal{E}$ forms an algebra

Proof:

(TBD) Let $U, V \in \mathcal{E}$, and consider $V^c = \coprod_{j=1}^J F_j$ (for disjoint $F_j \in \mathcal{E}$). Then

$$U \setminus V = U \cap V^c = \coprod_{j=1}^J (U \cap F_j)$$

notice that each $U \cap F_j \in \mathcal{E}$. So

$$U \cup V = (U \setminus V) \cup V = \coprod_{j=1}^J (U \cap F_j) \coprod V$$

where $\coprod_{j=1}^J (U \cap F_j) \in \mathcal{A}$

Continuing inductively, **can check Folland for proof, or just do it yourself.**

(also, remember that separable means countable dense subset) Once the course starts

1. There is an interesting example with logical statements!!

Remark. If A is a set, and $\varphi(x)$ is a logical statement about x .
Then $\{x: \varphi(x) \text{ holds and } x \in A\}$
is a set.

Let $\mathcal{E} \subset \mathcal{P}(X)$, define:
 $\mathcal{E}_1 = \left\{ \bigcup_{i=1}^{\infty} \bigcap_{j=1}^{\infty} E_{i,j} \mid \begin{array}{l} \text{either } E_{i,j} \in \mathcal{E} \\ \text{or } E_{i,j}^c \in \mathcal{E} \end{array} \right\}$
 $\mathcal{E}_2 = \left\{ \bigcup_{i=1}^{\infty} \bigcap_{j=1}^{\infty} E_{i,j} \mid \begin{array}{l} \text{either } E_{i,j} \in \mathcal{E}_1 \\ \text{or } E_{i,j}^c \in \mathcal{E}_1 \end{array} \right\}$
 etc.
 $\bigcup \mathcal{E}_n$ is in general not a σ -alg. and
 there is no constructive way to
 understand $\mu(\mathcal{E})$.

Figure 1.1: fascinating!

1.2 Measure

We can finally define the function which will give us an idea of the “size” of sets. The key idea behind the idea of notion is that the “bigger” a set is, the “larger” the area. In other words, if μ is a measure on X , then (abusing notation a bit):

$$\mu(A \subseteq B) \Leftrightarrow \mu(A) \leq \mu(B) \quad (1.1)$$

This also means that we want the codomain of a measure to be *totally ordered*, and since we want limits to work, it also have to be *complete*. Furthermore, if we have a countable number of disjoint areas, it should possible to measure them all separately and or all together at once and get the same measure (a sort of “invariance of space”). In fact, if we just assume $\mu(\emptyset) = 0$ (i.e. “nothing” should have no area of measure), then using this “invariance of space” notion, we can deduce equation (1.1), as we’ll show in the next proposition, and hence we would want the codomain of a measure to be $[0, \infty]$. We thus define a measure on these intuitions:

Definition 1.2.1: Measure

Let X be a set along with a σ -algebra $\mathcal{M}$. A function $\mu : \mathcal{M} \rightarrow [0, \infty]$ is called a *measure* on $\mathcal{M}$ (or $(X, \mathcal{M})$ or even X if the measure is clear from context) if

1. $\mu(\emptyset) = 0$
2. **(Countably Additivity)** If $\{E_i\}_{i=1}^{\infty}$ is a sequence of disjoint sets, then $\mu(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mu(E_i)$

Given that $(X, \mathcal{M})$ is a measurable space, along with a μ we can say the space has a measure:

Definition 1.2.2: Measure Space

Let $(X, \mathcal{M})$ be a measurable space. Then if μ is a measure on X , then $(X, \mathcal{M}, \mu)$ is a *measure space*

If the 2nd condition of a measure was limited to a finite sequence (or more generally, all but countably many of the sets in the sequence or nonempty), then we would say that it respects finite additivity. If μ respects finite but not countable additivity, μ is called a *finite additive measure*.

Most measure's that we will work with have some finiteness condition associated with it:

Definition 1.2.3: Finiteness Condition on μ

Let $(X, \mathcal{M}, \mu)$ be a measure space.

1. If $\mu(X) < \infty$, then μ is called a *finite measure*
2. if there exists sets $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ such that $\bigcup_{i=1}^{\infty} E_i = X$ and $\mu(E_i) < \infty$ for all E_i , then μ is called a *σ -finite measure*. More generally, if $E = \bigcup_{i=1}^{\infty} E_i$ for some sequence of E_i , then E is called *σ -finite on μ* (some say that E is said to be *σ -finite on μ*)
3. if for all $E \in \mathcal{M}$ such that $\mu(E) = \infty$, there exists a $D \subseteq E$, $D \in \mathcal{M}$ such that $\mu(D) < \infty$, then μ is called a *semi-finite measure*

If $\mu(X) < \infty$, then this implies for all $S \in \mathcal{M}$, $\mu(S) < \infty$ (as can quickly be checked). Naturally, all finite measure's are σ -finite, and all σ -finite measure are semi-finite, but the converses are not true (as we'll explore in the following examples). Note too that most measure's we will be working over will be σ -finite (think of $\mathbb{R}$ with the intuitive notion of length of an interval).

Example 1.2.4: Measures

1. Let X be nonempty, $\mathcal{M} = P(X)$, and $f : X \rightarrow [0, \infty]$. Then we can define

$$\mu_f(E) = \sum_{e \in E} f(e)$$

where the sum can be infinite (the domain of f is the positive part of the extended real numbers). Notice that f is semifinite if and only if $f(x) < \infty$ for all $x \in X$, and μ is σ -finite if and only if μ is semi-finite and $\{x \mid f(x) > 0\}$ is countable.

If $f(x) = 1$ for all $x \in X$, then μ_f is called the *counting measure*. On the other hand, if for some $x_0 \in X$, $f(x_0) = 1$ and $f(x) = 0$ if $x \neq x_0$, then μ_f is called a *point mass* or *Dirac measure* at x_0 .

2. Let X be an uncountable set and $\mathcal{M}$ be the σ -algebra of countable or co-countable sets (recall example 1.1.5). Define

$$\mu(E) = \begin{cases} 0 & E \text{ is countable} \\ 1 & E \text{ is co-countable} \end{cases}$$

Then μ is a measure!

3. An example of a finite additive measure that is not a measure is the following: let X be some infinite set and $\mathcal{M} = P(X)$. Define μ to be

$$\mu(E) = \begin{cases} 0 & E \text{ is finite} \\ \infty & E \text{ is infinite} \end{cases}$$

Then μ is a finite additive measure, but not a measure

Proposition 1.2.5: Properties of Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then

1. **(Monotonicity)** Let $E, F \in \mathcal{M}$ and $E \subseteq F$. Then $\mu(E) \leq \mu(F)$
2. **(Subadditivity)** if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$, then $\mu(\cup_i E_i) \leq \sum_i \mu(E_i)$
3. **(Continuity from Below)** if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$, and $E_1 \subseteq E_2 \subseteq \dots$, then

$$\mu(\cup_i E_i) = \lim_{i \rightarrow \infty} \mu(E_i)$$

4. **(Continuity from Above)** if $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ and $E_1 \supseteq E_2 \supseteq \dots$ and $\mu(E_1) < \infty$. Then

$$\mu(\cap_i E_i) = \lim_{i \rightarrow \infty} \mu(E_i)$$

Proof:

here. Do them as exercises! I also like this solution by Hytham Farah for continuity from above:

$$\begin{aligned}
\mu\left(\bigcup_{n \geq 1} F_n\right) &= \mu\left(\bigcup_{n \geq 1} E_n\right) \\
&= \mu\left(\left(\bigcap_{n \geq 1} E_n^c\right)^c\right) \\
&= \mu(X) - \mu\left(\bigcap_{n \geq 1} E_n^c\right) \\
&= \mu(X) - \lim_{n \rightarrow \infty} \mu(E_n^c) \\
&= \mu(X) - \lim_{n \rightarrow \infty} (\mu(X) - \mu(E_n)) \\
&= \lim_{n \rightarrow \infty} \mu(E_n) \\
&= \sum_{i=1}^{\infty} \mu(F_i).
\end{aligned}$$

Notice that for (d), we can instead limit it to $\mu(E_i) < \infty$ for some i , and the proof remains the same. We still require that some i exists such that $\mu(E_i) < \infty$, for it's possible that all $\mu(E_i)$ is infinite, but $\mu(\cap_i E_i) < 0$. For example, take the measurable space $(\mathbb{N}, P(\mathbb{N}))$ with the counting measure. Then the sets

$$E_i = \{n \in \mathbb{N} \mid n \geq i\}$$

then every E_i is infinite but their intersection is empty and thus has measure 0 in the counting measure.

zero measure sets

Frequently, there are sets in our measure that have a “trivial size” in the current measure. This is akin to when we integrate, the border has no effect on the value of the integral. Such sets are said to be zero sets or null set:

Definition 1.2.6: Null Set

Let $(X, \mathcal{M}, \mu)$ be a measure space and $E \in \mathcal{M}$. Then if $\mu(E) = 0$, E is said to be a *null set* or *zero set*.

By countably subadditivity, the countable union of null sets is a null set, hence null sets are “invariant” under countable union. Due to this, we can work with measurable sets in $\mathcal{M}$ *up to null sets*. We shall often state in proofs that a statement is true *almost everywhere* (often abbreviated to a.e.) if it is true on the sets except for the null sets.

If we take $E \in \mathcal{M}$ such that $\mu(E) = 0$, then by monotonicity, if $F \subseteq E$ the $\mu(F) = 0$, *provided that* $F \in \mathcal{M}$. This is not always the case:

Example 1.2.7: Not Complete Space

Partition $[0, 1]$ into 10 (really any number) of subintervals, $[\frac{i}{n}, \frac{i+1}{n}]$ for $0 \leq i \leq 9$, and let each of their measures be 0. Take $\mathcal{M}$ to be the σ -algebra defined on this set. Then is clearly an uncountable number of sets that do not have a measure defined on them that are subset of zero-measure sets.

However, adding all of these F 's that are subsets of null sets does not break our measure. A measure whose domain includes all subsets of null sets is a *complete measure* (i.e. the σ -algebra is closed under subsets of null sets). Proving that this is still a measure is useful to eliminate future “trivial obstructions” to our proofs:

Theorem 1.2.8: Completing a Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} \mid \mu(N) = 0\}$ and

$$\overline{\mathcal{M}} = \{E \cup F \mid E \in \mathcal{M} \text{ and } F \subseteq N \text{ for some } N \in \mathcal{N}\}$$

Then $\overline{\mathcal{M}}$ is also a σ -algebra, and there is a *unique* extension $\overline{\mu}$ of μ that is a complete measure on $\overline{\mathcal{M}}$ that restricts to the measure of μ on $\mathcal{M}$

Proof:

First of all, since $\mathcal{M}$ and $\mathcal{N}$ are both closed under countable union, so $\overline{\mathcal{M}}$ is closed under countable union. To show it's closed under compliments, consider some $E \cup F \in \overline{\mathcal{M}}$ ($E \in \mathcal{M}$, $F \subseteq N \in \mathcal{N}$). Without loss of generality, assume $E \cap N = \emptyset$ (if not, take $F \setminus E$ and $N \setminus E$ and continue the proof, then this will have the same value as our original $E \cap N$). Then

$$E \cup F = (E \cup N) \cap (N^c \cup F)$$

So we get

$$(E \cup F)^c = (E \cup N)^c \cup (N \setminus F)$$

Since $E \cup N \in \mathcal{M}$, $(E \cup N)^c \in \mathcal{M}$. Since $N \setminus F \in \mathcal{N}$, Since it's closed under union, $(E \cup F)^c \in \overline{\mathcal{M}}$. Thus $\overline{\mathcal{M}}$ is a σ -algebra.

Since $\overline{\mathcal{M}}$ is a σ -algebra, we can try to define a measure, and indeed

$$\overline{\mu}(E \cup F) = \mu(E)$$

is a well-defined measure. If

$$E_1 \cup F_1 = E_2 \cup F_2$$

Then $E_1 \subseteq E_2 \cup F_2$, and so by monotonicity:

$$\mu(E_1) \leq \mu(E_2 \cup F_2) \leq (E_2 \cup N_2) = \mu(E_2) + \mu(N_2) = \mu(E_2)$$

Similarly, $\mu(E_1) \geq \mu(E_2)$. Thus $\overline{\mu}(E_1 \cup F_1) = \overline{\mu}(E_2 \cup F_2)$, and so $\overline{\mu}$ is well-defined. Since $\overline{\mu}$ has all subsets of zero-sets in its domain, it is a complete measure.

Furthermore, if $\overline{\nu}$ was another complete measure on $\overline{\mathcal{M}}$, then it is easy to see that $\overline{\nu} = \overline{\mu}$, (since if it were different, then some subset of a zero-set must have non-zero measure, a contradiction to monotonicity) showing that $\overline{\mu}$ is unique.

Definition 1.2.9: Completion of Measure

Let $(X, \mathcal{M}, \mu)$ be a measurable space. Then $\overline{\mathcal{M}}$ and $\overline{\mu}$ is called the *completion of $\mathcal{M}$ with respect to μ* and the *completion of μ* respectively

1.3 Outer Measure

We now move on to constructing a particular type of measure which draws inspiration from how we measured area's when we calculated Riemann Integrals. In the abstract, when we tried defining a Jordan measure (or content⁴), we approximated the inner area of a shape E in $\mathbb{R}^n$ by the limit of sums of rectangles contained in E , and the outer-area of E in $\mathbb{R}^n$ by the limit of the sums of rectangles containing E . The two numbers are called the inner and outer area of E . If the two matched, we would call the result the area of E . This intuition can be generalized to the countable case, except we can replace rectangles with some other “basic sets” that we might want that form an algebra. Furthermore, the intuition of having the “inner area” and “outer-area” is actually a bit restricting to work with in the general measure case, and so that condition will be replaced with a different measureability condition we will soon introduce.

For the sake of generality, we will start by definition a more general notion of the measure called the outer-measure, and then build-up on how from the outer-measure, we can get a measure that satisfies this generalization of Riemann integration:

Definition 1.3.1: Outer Measure

A function $\mu^* : P(X) \rightarrow [0, \infty]$ is called an *outer measure* if

1. $\mu^*(\emptyset) = 0$
2. if $A \subseteq B$, then $\mu^*(A) \leq \mu^*(B)$
3. $\mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (even if the sets are disjoint!)^a

^aWe will return to see when equality holds in definition 1.3.3

Note that we have not yet proven that an outer-measure is a measure; in particular notice that the domain is the powerset. What we will do is define a generalization of the Riemann integral that will be our “base-line” for the outer-measure:

⁴since “Jordan measures” only allow finite additivity, some don't like to use the word measure and opt for the word content. In this book, we would say *finite additive measure* as has already been stated earlier

Proposition 1.3.2: Outer Measure Construction

Let $\mathcal{E} \subseteq P(X)$ where $\emptyset, X \in \mathcal{E}$, and let $\rho : \mathcal{E} \rightarrow [0, 1]$ where $\rho(\emptyset) = 0$. For any $A \subseteq X$, define

$$\mu^*(A) = \inf \left\{ \sum_{j=1}^{\infty} \rho(E_j) \mid E_j \in \mathcal{E} \text{ and } A \subseteq \bigcup_{j=1}^{\infty} E_j \right\}$$

Then μ^* is an outer measure

Proof:

First, this measure is well-defined since for any $A \subseteq X$, If we take $E_i = X$, then $A \subseteq \cup_i E_i$. Next, we verify the 3 properties

1. Clearly, $\mu^*(\emptyset) = 0$ by taking $E_i = \emptyset$ for all i
2. If $A \subseteq B$, then let $\mathcal{A}$ and $\mathcal{B}$ sets that correspond to $\mu^*(A)$ and $\mu^*(B)$. Then since all of the elements of $\mathcal{A} \supseteq \mathcal{B}$, by the property of the infimum $\inf \mathcal{A} \leq \inf \mathcal{B}$, which transferring notations gives us $\mu^*(A) \leq \mu^*(B)$
3. Let $\{E_i\}_{i=1}^{\infty} \subseteq P(X)$ be a collection of (not necessarily disjoint) subsets. Let $\epsilon > 0$. Consider $\mu^*(E_k)$ for some $E_k \in \{E_i\}_{i=1}^{\infty}$. By the property of the infimum, there must exist some $\{F_j^k\}_{j=1}^{\infty}$ such that $E_k \subseteq \cup_j F_j^k$ where $\sum_{j=1}^{\infty} \rho(F_j^k) \leq \mu^*(E_k) + \epsilon 2^{-k}$.

Now, let $E = \cup_i E_i$. By construction, $E \subseteq \cup_{i,k=1}^{\infty} F_j^k$, and

$$\sum_{j,k} \rho(F_j^k) \leq \mu^*(A) + \epsilon$$

Since ϵ can be arbitrarily small, this completes the proof.

As mentioned before, an outer-measure is not necessarily a measure. What we will do to allow this to be a measure is single out sets that have the following property

Definition 1.3.3: μ^* -measureable

Let $E \subseteq X$ be a set and μ^* be an outer measure on X . Then E is called μ^* -measurable if

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

In other word, if E can naturally “split” a set S into a part that’s “inside” E or “outside” E . The set $S \subseteq X$ is called the *test set*.

By definition of the outer measure, since $S = (S \cap E) \cup (S \cap E^c)$:

$$\mu^*(S) \leq \mu^*(S \cap E) + \mu^*(S \cap E^c) \quad \forall S \subseteq X$$

Therefore, what is more important is to check the $\geq$ direction. Furthermore, if $\mu^*(A) = \infty$. Then $\geq$ holds trivially, so to show a set is μ^* measurable, it is equivalent to show that for all finite measure sets A , $\mu^*(S) \geq \mu^*(S \cap E) + \mu^*(S \cap E^c)$.

As mentioned in the definition, this gives us a sense of being able to measure the “inside” and “outside” of a set. If $E \subseteq A$, that is the μ^* -measurable set belong in a better set, then $\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \setminus E)$ which can be thought of as saying that we are measuring S from the “inside” of E and the “outside” of E , and getting the same result. In fact, if $\mu^*(X) < \infty$, then we can define $\mu_*(E) = \mu^*(X) - \mu(E^c)$, and have that E is μ^* -measurable if and only if $\mu_*(E) = \mu^*(E)$ aligning with our intuition for inside equalling outside (see exercise ref:HERE). We do not do this method since it requires some finiteness condition on μ . We can conversely define μ_* in terms of supremums and need that E contains the collection of $\cup_i A_i$, and that the elements of $\{A_i\}_{i=1}^\infty$ must be disjoint (see exercise ref:HERE).

The big thing that is in the way of saying that an outer measure is a measure is that its domain is $P(X)$, which can cause problems when X is uncountable. However, all sets that satisfy μ^* -measureability turn out form a σ -algebra and make μ^* a complete measure!

Theorem 1.3.4: Carathéodory's Theorem

Let μ^* be an outer measure on X , and let $\mathcal{M}$ be the collection of μ^* -measurable sets. Then $\mathcal{M}$ forms a σ -algebra, and the restriction of μ^* to $\mathcal{M}$ is a complete measure

Notice that in the process of defining this measure, we started with the powerset, and then simply choose all sets that restrict (which we will show forms a σ -algebra). In this way proving a function is an outer-measure is a little less finicky since we don't need to be so careful in defining over which σ -algebra we want to define it over: the σ -algebra comes with μ^*

Proof:

Let $\mathcal{M}$ be the collection of μ^* -measurable sets. Clearly, $\emptyset \in \mathcal{M}$ since for all $S \subseteq X$:

$$\mu^*(S \cap \emptyset) + \mu^*(S \cap X) = 0 + \mu^*(S)$$

implying μ^* -measureability. Next, $\mathcal{M}$ is closed under compliments, since if $E \in \mathcal{M}$, then

$$\mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \mu^*(S \cap E^c) + \mu^*(S \cap E)$$

which also shows that $X \in \mathcal{M}$. Finally, we need to show that $\mathcal{M}$ is closed under countable unions. We'll start by showing finite unions, which then means it suffices to show the union of two sets is closed (due to induction). Let $A, B \in \mathcal{M}$. As we remarked, it suffices to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$. Since A and B are μ^* measurable, we have

$$\begin{aligned} \mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) && E \text{ is the test set} \\ &= \mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) \\ &\quad + \mu^*(E \cap A^c \cap B) + \mu^*(E \cap A^c \cap B^c) && E \cap A \text{ and } E \cap A^c \text{ are the test sets} \end{aligned}$$

Then notice we can write

$$A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$$

Thus, using subadditivity, we can re-write 3 of the terms in the previous expression as:

$$\mu^*(E \cap A \cap B) + \mu^*(E \cap A \cap B^c) + \mu^*(E \cap A^c \cap B) \geq \mu^*(E \cap A \cup B)$$

Since $E \cap A^c \cap B^c = E \cap (A \cup B)^c$, we get:

$$\mu^*(E) \geq \mu^*(E \cap (A \cup B)) + \mu^*(E \cap (A \cup B)^c)$$

showing that $\mathcal{M}$ is closed under finite union. Since $\mathcal{M}$ is closed under complement and contains $\emptyset$, $\mathcal{M}$ is an algebra. Even better, If we let A and B be disjoint, and let $A \cup B$ be the test set, then since A is μ^* measurable:

$$\mu^*(A \cup B) = \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) = \mu^*(A) + \mu^*(B)$$

showing that $\mu^*|_{\mathcal{M}}$ is a finitely additive measure.

Next, we must show that $\mathcal{M}$ is a σ -algebra, which as a consequence will show that $\mu^*|_{\mathcal{M}}$ is a measure. Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$. To simplify our task, re-construct $\{E_i\}_{i=1}^{\infty}$ to be the set of disjoint sets $\{D_i\}_{i=1}^{\infty}$. Let $B_n = \bigcup_{i=1}^n D_i$ and $B = \bigcup_{i=1}^{\infty} D_i$. Our goal is to show that for any $E \subseteq X$ such that $\mu^*(E) < \infty$ that $\mu^*(E) \geq \mu^*(E \cap B) + \mu^*(E \cap B^c)$. We will start by considering $\mu^*(E \cap B_n)$:

$$\begin{aligned} \mu^*(E \cap B_n) &= \mu^*(E \cap B_n \cap D_n) + \mu^*(E \cap B_n \cap D_n^c) && E \cap B_n \text{ is the test set} \\ &= \mu^*(E \cap D_n) + \mu^*(E \cap B_{n-1}) \end{aligned}$$

Notice that on the right hand side we have diminished to the case of $E \cap B_{n-1}$ from $E \cap B_n$. Using induction, we can see that the expression simplifies to

$$\mu^*(E \cap B_n) = \sum_{i=1}^n \mu^*(E \cap D_i)$$

Continuing, since B_n is measurable

$$\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \cap B_n^c) \stackrel{!}{\geq} \mu^*(E \cap B_n) + \mu^*(E \cap B^c) = \sum_{i=1}^n \mu^*(E \cap D_i) + \mu^*(E \cap B_n^c)$$

where $\stackrel{!}{\geq}$ comes from monotonicity after switching B_n^c to B^c . Since this equality holds true for all n , we get:

$$\begin{aligned} \mu^*(E) &\geq \sum_{i=1}^{\infty} \mu^*(E \cap D_i) + \mu^*(E \cap B^c) \\ &\geq \mu^*\left(\bigcup_i E \cap D_i\right) + \mu^*(E \cap B^c) && \text{subadditivity} \\ &= \mu^*(E \cap B) + \mu^*(E \cap B^c) && \text{by construction of } E \cap B \\ &\geq \mu^*(E) && \text{subadditivity} \end{aligned}$$

Since we got $\mu^*(E) \geq \mu^*(E)$, we see that in fact all inequalities are equalities, getting us that

$$\mu^*(E) = \mu^*(E \cap B) + \mu^*(E \cap B^c)$$

showing us that $B \in \mathcal{M}$. Furthermore, replacing $E = B$ in the previous proof shows us that

$$\mu^*(B) = \sum_{i=1}^{\infty} \mu^*(D_i)$$

giving us that μ^* is a measure on $\mathcal{M}$.

Finally, we must show that $\mathcal{M}$ is complete, that is, if $E \in \mathcal{M}$ such that $\mu^*(E) = 0$, then all subsets of E are in $\mathcal{M}$. Notice that since μ^* is defined on all of $P(X)$, it is equivalent to show that if $\mu^*(A) = 0$, then $A \in \mathcal{M}$. Then for any test set E , by monotonicity:

$$\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c) = 0 + \mu^*(E \cap A^c) \leq \mu^*(E)$$

meaning equality holds, showing that $A \in \mathcal{M}$. Thus, $\mu^*|_{\mathcal{M}}$ is a complete measure, as we sought to show.

Carathéodory's Theorem is quite useful being able to state the existence of a measure given an outer-measure, but it's not good to construct a measure (for example, can you tell me a non-trivial element in $\mathcal{M}$ without directly appealing to the definition?). The following concept is used to more systematically construct the set $\mathcal{M}$ and the measure μ induced by μ^* :

Definition 1.3.5: Premeasure

Let $(X, \mathcal{A})$ be an algebra. Then μ_0 is called a *premeasure* if

1. $\mu_0(\emptyset) = 0$
2. **“Respects” countable union:** If $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ is a collection of disjoint such that $\cup_i A_i \in \mathcal{A}$, then $\mu_0(\cup_i A_i) = \sum_i \mu_0(A_i)$

Note that the premeasure is automatically finitely additive since we can let all but finitely many of the terms in (2) be $\emptyset$ (and hence are a stronger condition than finitely additive measure). However, it is not yet a measure since its domain is not necessarily closed under countable union. However, if it is closed for countable union for a particular collection, that collection must satisfy the “countable” condition. Interestingly, by the definition of an algebra, this implies that the countable union can be represented as a finite union and intersection of elements from $\mathcal{A}$.

As before, we can define semifinite and σ -finite in a similar way. Using μ_0 , we can define the following outer-measure:

$$\mu^* : P(X) \rightarrow [0, \infty] \quad \mu^*(E) = \inf \left\{ \sum_i^{\infty} \mu_0(A_i) \mid A_i \in \mathcal{A}, E \subseteq \cup_i A_i \right\}$$

Proposition 1.3.6: Premeasure To Measure

Let μ_0 be a premeasure on $\mathcal{A}$, and let μ^* be defined as above. Then

1. $\mu^*|_{\mathcal{A}} = \mu_0$
2. Every set in $\mathcal{A}$ is μ^* -measurable

Essentially, μ^* is an extension of μ_0 and the associated restriction of μ^* to a measure will contain $\mathcal{A}$

Proof:

1. Let's show $\leq$ and $\geq$ for any $E \in \mathcal{A}$. The $\leq$ direction is obvious since $\mu^*(E) \leq \mu_0(E)$ since $E \subseteq \bigcup_{i=1}^{\infty} A_i$ where $A_1 = E$ and $A_i = \emptyset$ for $i > 1$, and so it is part of the set over which

we take the infimum, and so the result *has* to be smaller.

For the $\geq$ direction, let's say $E \subseteq \bigcup_{i=1}^{\infty} A_i$ for any $\{A_i\}_{i=1}^{\infty} \in \mathcal{A}$. Let $B_n = E \cap (A_n \setminus \bigcup_{i=1}^{n-1} A_i)$. Then the collection $\{B_i\}_{i=1}^{\infty}$ is a disjoint collection of members of $\mathcal{A}$ whose union is E . Thus, by the property of premeasure

$$\mu_0(E) = \sum_1^{\infty} \mu_0(B_i) \stackrel{!}{\leq} \sum_1^{\infty} \mu_0(A_i)$$

where the $\stackrel{!}{\leq}$ inequality comes from the fact that $\mu_0(B_i) \leq \mu_0(A_i)$. Since this is true for any arbitrary collection $\{A_i\}_{i=1}^{\infty}$, it follows that

$$\mu_0(E) \leq \mu^*(E)$$

Thus establishing equality

2. We need to show that for any $E \subseteq X$ and $A \in \mathcal{A}$, $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$. As mentioned earlier, it suffice to prove $\geq$ where $\mu^*(E) < \infty$.

Let $\epsilon > 0$. By the property of the infimum, there exists a subset $\{B_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \bigcup_i B_i$ such that $\sum_i \mu_0(B_i) \leq \mu^*(E) + \epsilon$. By part (1), μ^* is [countably] additive on $\mathcal{A}$ (since it equals the pre-measure)

$$\begin{aligned} \mu^*(E) + \epsilon &\geq \sum_{i=1}^{\infty} \mu_0(B_i \cap A) + \sum_{i=1}^{\infty} \mu_0(B_i \cap A^c) && \text{additivity} \\ &\geq \mu^*(E \cap A) + \mu^*(E \cap A^c) && \text{monotonicity} \end{aligned}$$

and since ϵ was arbitrary, A must be μ^* -measurable, completing the proof

We can in fact be more precise with the measures that extend a premeasure:

Theorem 1.3.7: Premeasure Extensions

Let $\mathcal{A} \subseteq P(X)$ be an algebra, μ_0 a premeasure on $\mathcal{A}$, and $\mathcal{M} = \mathcal{M}(\mathcal{A})$ (i.e. $\mathcal{M}$ is the σ -algebra generated by $\mathcal{A}$). Let μ be the extension of μ_0 from proposition 1.3.6. Then if ν is another measure on $\mathcal{M}$ that extends μ_0 , then

$$\nu(E) \leq \mu(E) \quad \forall E \in \mathcal{M}$$

with equality when $\mu(E) < \infty$ (i.e., it possible that μ is infinite while ν is finite).

If μ_0 is σ -finite, then μ is the unique extension of μ_0 to a measure on $\mathcal{M}$

In other words, we can extend a pre-measure to a measure by knowing what it does on the algebra (which in most cases will be a collection of simpler sets, like intervals or rectangles), and then what happens on the rest of the measurable sets we use the infimum definition. If we define another extension of μ_0 , then that measure will be *finer* than the μ given by μ^* , i.e. μ is the *coarsest* extension. If μ_0 is σ -finite, then this will in fact be the only extension of μ_0

Proof:

Let's say ν is an extension of μ_0 so that $(X, \mathcal{M}(\mathcal{A}), \nu)$ is a measure space. To show that $\nu(E) \leq \mu(E)$, let's say $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$ where $E \subseteq \cup_i A_i$. Then since ν is a measure, it is sub-additive on non-disjoint sets, and so

$$\nu(E) \leq \sum_i \nu(A_i)$$

Since ν extends μ_0 , by proposition 1.3.6 $\nu|_{\mathcal{A}} = \mu_0$, so

$$\nu(E) \leq \sum_i \nu(A_i) = \sum_i \mu_0(A_i)$$

Since $\{A_i\}_{i=1}^{\infty}$ was an arbitrary choice, this works for any collection, and so

$$\nu(E) \leq \mu(E) \quad [= \mu^*(E)]$$

We show that $\nu(E) = \mu(E)$ if $\mu(E) < \infty$. Let $A = \cup_i A_i$ for some collection $\{A_i\}_{i=1}^{\infty} \subseteq \mathcal{A}$. Then

$$\nu(A) = \lim_{n \rightarrow \infty} \nu\left(\bigcup_{i=1}^n (A_i)\right) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{i=1}^n (A_i)\right) = \mu(A)$$

If μ is not σ -finite, then there can be more than one extension:

Example 1.3.8: 2 extensions of pre-measure

1. Let $\mathcal{A}$ be the collection of finite unions of the sets of the form $(a, b] \cap \mathbb{Q}$ with $-\infty \leq a \leq b \leq \infty$. Then $\mathcal{A}$ is an algebra on $\mathbb{Q}$

Proof:

Recall proposition 1.1.13 where we can construct an algebra using an elementary family. Thus, we will show that the collection $\mathcal{A}$ satisfies the 3 properties of an elementary family:

- (a) To get the empty-set in $\mathcal{A}$, we will understand the notion of “finite union” to also include “no union”. Otherwise, since $\mathbb{Q}$ is dense in $\mathbb{R}$, all intervals $(a, b]$ intersected with $\mathbb{Q}$ is nonempty (by MAT157 knowledge). Since $a < b$, the singleton interval is not admissible, and so we will have to assume this “no union” property
- (b) Let $E, F \in \mathcal{A}$. We want to show that $E \cap F \in \mathcal{A}$. By definition, $E = (a, b] \cap \mathbb{Q}$ and $F = (c, d] \cap \mathbb{Q}$. Then by some basic set-theory properties, we know that the intersection of two left-open/right-closed intervals is again a left-open/right-closed interval or $\emptyset$. If their intersection is $\emptyset$, then we’re done. If not, let $(e, f] = (a, b] \cap (c, d]$. Then $E \cap F = (e, f] \cap \mathbb{Q}$. Thus, $E \cap F \in \mathcal{A}$
- (c) Let $E \in \mathcal{A}$, so that $E = (a, b] \cap \mathbb{Q}$. Let $i \in \mathbb{N}$ ($0 \notin \mathbb{N}$) and:

$$E_i = (a - i, a - (i + 1)] \cap \mathbb{Q} \quad F_i = (b + (i - 1), b + i] \cap \mathbb{Q}$$

then each $E_i, F_i \in \mathcal{A}$, $E_i \cap E_j = F_i \cap F_j = \emptyset$ if $i \neq j$ and $E_i \cap F_j = \emptyset$ for all $i, j \in \mathbb{N}$. Let $E = \{E_i\}_{i=1}^{\infty}$ and $F = \{F_i\}_{i=1}^{\infty}$ and let $G = E \cup F$. Then

$$\bigcup_{g \in G} G = E^c$$

since G is a countable union of disjoint elements of $\mathcal{A}$, this completes the proof.

Thus, by proposition 1.7, $\mathcal{A}$ is an algebra.

2. The σ -algebra generated by $\mathcal{A}$ is $P(\mathbb{Q})$

Proof:

Let $x \in \mathbb{Q}$ Notice that

$$\{x\} = \bigcap_{k=1}^{\infty} \left(x - \frac{1}{k}, x \right] \cap \mathbb{Q}$$

hence, every singleton is inside the σ -algebra of $\mathbb{Q}$. Since every subset of $\mathbb{Q}$ is countable, every subset is the countable union of singletons. Since every singleton is inside the σ -algebra, we have that the σ -algebra is $P(\mathbb{Q})$.

3. Define μ_0 on $\mathcal{A}$ by $\mu_0(\emptyset) = 0$ and $\mu_0(A) = \infty$ for $A \neq \emptyset$. Then μ_0 is a premeasure on $\mathcal{A}$, and there is more than one measure on $P(\mathbb{Q})$ whose restriction to $\mathcal{A}$ is μ_0

Proof:

μ_0 is trivially a pre-measure. By definition, $\mu_0(\emptyset) = 0$. For any nonempty $A \in \mathcal{A}$, we have $\mu_0(A) = \infty$. When checking countable additivity, unless all the sets are empty, we get that both sides are infinite.

Thus, we can define μ to be the μ^* induced by μ_0 restricted to all measurable sets. By construction, all sets are measurable. If a set is non-empty, then it will always be covered by a nonempty set, which has measure ∞ . Thus, if $E \in P(X)$ for any nonempty test set $S \subseteq \mathbb{Q}$

$$\infty = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = \infty$$

and if $S = \emptyset$

$$0 = \mu^*(S) = \mu^*(S \cap E) + \mu^*(S \cap E^c) = 0$$

Thus, $\mathcal{M}_\mu = P(\mathbb{Q})$.

We can define another extension of μ_0 using the counting measure, in particular, let μ_C be the counting measure on $P(\mathbb{Q})$. Then since every $A \in \mathcal{A}$ has countable cardinality

$$\mu_C(A) = \infty$$

Thus, $\mu_C|_{\mathcal{A}} = \mu_0$. However

$$\mu_C(\{1\}) = 1 \neq \infty = \mu(\{1\})$$

Thus, we have defined two separate extensions of μ_0 .

1.4 Borel Measure on the Real Line

The full title of this section should really be “Borel Measures on the Real line defined through Distributions”, but that title seems too cumbersome. Nevertheless, this is exactly what we will be doing in this section. We will extensively concentrate on these types of measures, since they are, in a sense, the focus of “real analysis”. From a high-level perspective, the fact that we are studying “real analysis” justifies why we will take so much time understanding this measure. More concretely, these measures are pervasive in analysis in general: they are central to defining integration, to the study of Fourier series, to studying L^p spaces, to linking differentiation to integration, to generalizing the notation of integration of groups (in particular, it will make sense to integrate locally compact topological groups), and so much more. Another way of looking at it is that the euclidean topology is central to the study of many fields of mathematics and physics, and is the basis for our notion of manifolds (which are locally euclidean). Therefore, getting a sense of how to define notions of “size” in $\mathbb{R}$ (and then generalize to $\mathbb{R}^n$, $\mathbb{R}^\infty$, or manifolds) gives us a solid first step in gaining an intuition on measures.

One way to define a measure in $\mathbb{R}$ is by using our intuition of the fact that the length of (a, b) is $b - a$. This is a good intuition and will be the basis to the generalization of this idea we’ll do in order to be able to define a larger class of measures based on this idea which will also have more uses outside of $\mathbb{R}$. Here is how we will motivate the construction of these measures: let μ be a finite measure on Borel sets (a measure defined on Borel sets is called the *Borel Measure*). Define:

$$F(x) = \mu((-\infty, x])$$

F is sometimes called the *distribution function* of μ . Intuitively, it shows the “accumulated size” of the space. Clearly, F is an increasing function, and is right-continuous since

$$(\infty, x] = \bigcap_k (\infty, x + 1/k]$$

or any decreasing function, $x_n \searrow x$ (i.e. limits are preserved from the right). Furthermore, if we take $a < b$, then $(-\infty, b] = (-\infty, a] \cup (a, b]$, thus

$$\mu((a, b]) = F(b) - F(a)$$

Notice how similar this is to the usual result from Riemann integration! In the following, instead of defining F in terms of μ , we will define μ from an increasing right-continuous function F by showing how F defines a pre-measure and using Carathéodory. The special case of $F(x) = x$ will give us our usual notion of length in $\mathbb{R}$.

Throughout this section, we will use sets of the form $(a, b]$ where $a < b$, along with the edge cases of (a, ∞) and $\emptyset$. Let all intervals of this type be called *h-intervals* (*h* for half open). It is clear that the union or complement of two *h-intervals* are *h-intervals* (or two disjoint *h-intervals*) and that *h-intervals* are closed under finite operations, and so *h-intervals* form an algebra $\mathcal{A}$. By proposition 1.1.7, the σ -algebra by the set $\mathcal{A}$ is $\mathcal{B}_{\mathbb{R}}$. Using this, we'll first show that we can define a premeasure μ based off of an F :

Proposition 1.4.1: Distribution and Premeasure

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right continuous. If $(a_i, b_i]$ is a set of disjoint *h-intervals*. If we define μ_0 to be:

$$\mu_0 \left(\bigcup_i (a_i, b_i] \right) = \sum_{i=1}^{\infty} F(b_i) - F(a_i)$$

and $\mu_0(\emptyset) = 0$, then μ_0 is a premeasure on the algebra $\mathcal{A}$

Proof:

We will first show that μ_0 is well-defined, then we simply must show that $\mu(\bigcup_i (a_i, b_i]) = \sum_i \mu_0((a_i, b_i])$.

To show it's well-defined, we must show that for any finite representation of $(a, b]$, $\mu_0((a, b]) = F(b) - F(a)$. To that end, let $\{(a_i, b_i]\}_{i=1}^n$ be disjoint where $(a, b] = \bigcup_{i=1}^n (a_i, b_i]$. Since the co-domain is commutative, we can re-arrange the order that we are taking the union (i.e., re-index the set) so that

$$a < a_1 < b_1 = a_2 < b_2 = a_3 < \dots < b_n = b$$

Then the resulting μ_0 is simply an alternating series:

$$\sum_{j=1}^n F(b_j) - F(a_j) = F(b) - F(a)$$

showing that it is independent of representation of the intervals. Thus, if we have any disjoint collection $\{I_i\}_{i=1}^n$ and $\{J_j\}_{j=1}^n$ such that $\bigcup_i I_i = \bigcup_j J_j$, since we can always form alternative

series:

$$\sum_i \mu_0(I_i) = \sum_{i,j} \mu_0(I_i \cap J_j) = \sum_j \mu_0(J_j)$$

and so μ_0 is well-defined. Since it is well-defined, then we also get that μ_0 is finitely additive (essentially by definition), and since $\mu_0(\emptyset) = 0$, it is at least a finite measure.

Next, we need to show that it is a premeasure. Since $\mu_0(\emptyset) = 0$, it remains to check that if $\{A_i\}_{i=1}^\infty \subseteq \mathcal{A}$ where $\cup_i^\infty A_i \in \mathcal{A}$, then

$$\mu_0\left(\bigcup_{i=1}^\infty A_i\right) = \sum_{i=1}^\infty \mu_0(A_i)$$

Since $\cup_i^\infty I_i = I \in \mathcal{A}$, there exists (at least one) subsets $\{A_i\}_{i=1}^n \subseteq \mathcal{A}$ such that $\cup_i^\infty I_i = \cup_i^n A_i$. Therefore the elements from I can be partitioned so that the union of each partition represents a single h -interval. By finite additivity of μ_0 , it suffices to consider what happens at one of these partitions. So, let J be the union of one of the families so that $(a, b]J = \cup_{j=1}^\infty I_{i_j}$. We will show that $\mu_0(\cup_j^\infty J_j) = \sum_j^\infty \mu_0(J_j)$ by showing $\leq$ and $\geq$. For the $\geq$ direction, notice that

$$\mu_0(J) = \mu_0\left(\bigcup_j^n J\right) + \mu_0\left(J \setminus \bigcup_j^n J\right) \geq \mu_0\left(\bigcup_j^n J\right) = \sum_j^n \mu_0(J)$$

Thus, as $n \rightarrow \infty$ we have $\mu_0(J) \geq \sum_j^\infty \mu_0(J_j)$.

For the $\leq$ direction, we will take advantage of the right-continuity of F .

(Might not be the case TBD p.34 Folland) First, if either b is infinite, the result holds automatically. If a is infinite, then b would have to also be infinite, and so we have the zero-intervals. So, let a and b be finite.

Let $\epsilon > 0$. Then since F is right-continuous, there exists a δ such that $F(a + \delta) - F(a) < \epsilon$ or similarly $-F(a) < -F(a + \delta) + \epsilon$. Similarly, for the same epsilon, there exists a δ_j such that $F(b_j + \delta_j) - F(b_j) < \frac{\epsilon}{2^n}$. Notice that $(a_j, b_j] \subseteq (a_j, b_j + \delta_j)$, so the open intervals cover $[a, b]$, and since $[a, b]$ is compact, there exists a finite sub-cover! If $\{(a_j, b_j + \delta_j)\}_{i=1}^N$ is the finite subcover of $[a, b]$, if we discard all sets that are contained in a bigger set in this family, then we get that

$$b_j + \delta_j \in (a_{j+1}, \beta_{j+1} + \delta_{j+1}) \quad \text{for } j = 1, \dots, N-1$$

With this, we get the following sequence of inequalities:

$$\begin{aligned}
\mu_0(J) &= F(b) - F(a) \\
&\leq F(b) - F(a + \delta) + \epsilon \\
&\leq F(b_N + \delta_N) - F(a_1) + \epsilon && \text{extrema of cover} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(a_{j+1}) - F(a_j))\epsilon && \text{fancy 0} \\
&= F(b_N + \delta_N) - F(a_N) + \sum_{j=1}^{N-1} (F(b_{j+1} + \delta_j) - F(a_j))\epsilon && \text{expanding} \\
&< \sum_{j=1}^N [f(b_j - f(a_j) + \epsilon 2^{-n})] + \epsilon \\
&< \sum_{j=1}^{\infty} \mu_0(J_j) + 2\epsilon
\end{aligned}$$

and since ϵ was arbitrary, the result follows.

This result let's us prove that *any* increasing right-continuous function defines a measure!

Theorem 1.4.2: Distributions define Measure

If $F : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing right-continuous function, there is a unique Borel measure μ_F on $\mathbb{R}$ such that $\mu_F((a, b]) = F(b) - F(a)$ for all $a, b \in \mathbb{R}$. This function is unique up to the following condition: if G is another such function, then $F - G$ is constant.

Conversely, if μ is a Borel measure that is bounded on all finite Borel sets, we can define

$$F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((-x, 0]) & \text{if } x < 0 \end{cases}$$

Then F is an increasing right-continuous function

Proof:

First, by the previous proposition, F induces a premeasure on $\mathcal{A}$. If $F - G$ is constant, then the sums will cancel out in $F(b) - F(a)$, and so $\mu_F = \mu_G$. Similarly, if the two are equal, F and G can only differ up to a constant and that these measures are σ -finite (recall that $\mathbb{R} = \bigcup_{-\infty}^{\infty} (-i, i + 1]$).

For the converse, note that if μ is monotonic, then so is F . Next, above/bellow continuity of μ will imply right-continuity of F for $x \geq 0$ and $x < 0$. Therefore, by construction, $\mu = \mu_F$ on $\mathcal{A}$, and so by proposition 1.4.1, $\mu = \mu_F$ on $\mathcal{B}_{\mathbb{R}}$ by uniqueness in theorem 1.3.7.

Note that F need not be left-continuous. Something with the point-density measure that proves the counter-case.

We have been using intervals of the form $(a, b]$ and right-continuous function. The same theory can

be developed with $[a, b)$ intervals and left-continuous functions. If we restrict μ to being a finite Borel measure, then we get $\mu = \mu_F$ where $F(x) = \mu((-\infty, x])$ (i.e. it differs up to a constant of the μ_F when μ is a Borel measure). Next, by the construction of the measure from the pre-measure, the resulting measure is in fact a complete measure. Even better, taking completion of μ_F and then applying Carathéodory is the same as just taking completion of μ after applying Carathéodory to μ_F , and so taking the completion on μ_0 or μ_F does not make a difference! The complete measure is also usually denoted by μ_F and called the *Lebesgue-Stieljes measure* of F .

As a consequence of the Lebesgue-Stieljes measure μ being complete, if we have a measure on a Borel space (which we usually call a Borel measure), then then then the set of measurable sets $\mathcal{M}^*$ might be larger than the number of Borel sets (in particular in that it may contain some zero sets that are not Borel sets).

The Lebesgue-Stieljes measure has a couple of nice properties worth exploring. Let μ be a complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to F , and let $\mathcal{M}_\mu$ be the associated μ^* -measurable sets. Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf \left\{ \sum_1^\infty F(b_i) - F(a_i) \mid E \subseteq \cup_i^\infty (a_i, b_i] \right\} = \inf \left\{ \sum_1^\infty \mu((a_i, b_i]) \mid E \subseteq \cup_i^\infty (a_i, b_i] \right\}$$

Notice that the contribution of the right end points of every h intervals is clearly negligible and so we can replace it with open intervals

Lemma 1.4.3: h -Intervals to Open Intervals

Let μ be the complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to F . Then for any $E \in \mathcal{M}_\mu$:

$$\mu(E) = \inf \left\{ \sum_1^\infty \mu((a_i, b_i)) \mid E \subseteq \cup_i^\infty (a_i, b_i) \right\}$$

Proof:

This is the same epsilon trick you have seen before, right it down here as an exercise later

In fact, we can write down $\mu(E)$ in even simple terms

Theorem 1.4.4: Lebesgue-Stieljes in Terms of Open and Compact sets

Let μ be the complete Lebesgue-Stieljes measure on $\mathbb{R}$ associated to F . Then for any $E \in \mathcal{M}_\mu$:

$$\begin{aligned} \mu(E) &= \inf \{ \mu(U) \mid U \supseteq E \text{ and } U \text{ is open} \} \\ \mu(E) &= \sup \{ \mu(K) \mid K \subseteq E \text{ and } K \text{ is compact} \} \end{aligned}$$

Proof:

By lemma 1.4.3, for all $\epsilon > 0$, there exists open intervals (a_i, b_i) such that $E \subseteq \cup_i^\infty (a_i, b_i)$ and $\mu(E) \leq \sum_i^\infty \mu((a_i, b_i)) + \epsilon$. If we let $U = \cup_i (a_i, b_i)$, we have that $\mu(U) \leq \mu(E) + \epsilon$. Conversely, since $E \subseteq U$, $\mu(U) \geq \mu(E)$, and so the result for the first statements follows.

For the 2nd equivalence, let's first suppose E is bounded. If E is closed, then it is compact and

so E is contained in the set that we are taking the sup over, and so equality is immediate. If E was open, then we can choose an $\epsilon > 0$ such that $\overline{E} \setminus E \subseteq U$ and $\mu(U) \leq \mu(\overline{E} \setminus E) + \epsilon$. Set $K = \overline{E} \setminus U$. Then K is compact, and so

$$\begin{aligned}\mu(K) &= \mu(E) - \mu(E \cap U) \\ &= \mu(E) - [\mu(U) - \mu(U \setminus E)] \\ &\geq \mu(E) - [\mu(U) - \mu(\overline{E} \setminus E)] \\ &= \mu(E) - \mu(U) + \mu(\overline{E} \setminus E) \\ &\geq \mu(E) - \epsilon\end{aligned}$$

showing that any compact set arbitrarily approximates K .

If E is unbounded, let $E_i = E \cap (i, i + 1]$ so that $E = \cup_i E_i$. Then by what we've just shown, for all $\epsilon > 0$, there exists a $K_j \subseteq E_j$ such that $\mu(K_j) \leq \mu(E_j) - \frac{\epsilon}{2^n}$. Take $H_n = \cup_{-n}^n K_i$. Then H_n is compact, $H_n \subseteq E$, and

$$\mu(H_n) \geq \mu(\cup_{-n}^n E_i) - \epsilon$$

Since this equality holds when $n \rightarrow \infty$, the result follows

This gives us a strikingly easy representation of representing every measurable subset of $\mathbb{R}$!

Theorem 1.4.5: $E \subseteq \mathbb{R}$ Representation

Let $E \subseteq \mathbb{R}$. Then the following are equivalent

1. $E \in \mathcal{M}_\mu$
2. $E = V \setminus N_1$ where V is a G_δ set and $\mu(N_1) = 0$
3. $E = W \cup N_2$ where W is a H_σ set and $\mu(N_2) = 0$

Proof:

Clearly, (2) and (3) imply (1). For (1) implies (2) or (3), I would highly recommend you try and solve this in the $\mu(E) < \infty$ case to jog your memory.

For the $\mu(E) = \infty$, TBD (exercise 25 p.37 Folland)

This means that all borel sets, or more generally sets in $\mathcal{M}_\mu$ (also contain all unions of borel sets with measure zero sets). Note that this theorem and proof would be much harrier if we did not assume the measure was complete. ‘

I will leave this section with another way of interpreting finite measurable sets that Folland left as an exercise (we'll use the notation of $A \triangle B = (A \setminus B) \cup (B \setminus A)$)

Proposition 1.4.6: Diminishing Difference

Let $E \in \mathcal{M}_\mu$ and $\mu(E) < \infty$. Then for every $\epsilon > 0$, there exists a set A which is the finite union of open intervals and

$$\mu(E \triangle A) < \epsilon$$

1.4.1 Lebesgue Measure

We will study the case where the measure μ is defined through the distribution $F(x) = x$, i.e., when F represents our usual intuition of length! This is arguably the most important measure on $\mathbb{R}$, and most certainly the most used. For that reason, it has some special symbology: we will denote the measure by m instead of μ and the set of measurable sets by $\mathcal{L}$. If we restrict m to $\mathcal{B}_{\mathbb{R}}$, we will also write it as m .

The first properties worth establishing is to see that measurable sets in $\mathbb{R}$ inherit some of our intuitions how a “shape” should act under translation and dilation, that it’s invariant under translation (just like we assumed for the construction of Vitali sets!) and it scales linearly with dilation:

Proposition 1.4.7: Scaling and Dilation

Let $(X, \mathcal{L}, m)$ be a Lebesgue measure space. Define

$$E + s := \{e + s \mid e \in E\} \quad rE := \{re \mid e \in E\}$$

Then $E + s, rE \in \mathcal{L}$ for all $r, s \in \mathbb{R}$, and:

$$m(E + s) = m(E) \quad m(rE) = |r|m(E)$$

Proof:

We essentially notice that open intervals are invariant under translation and operate in the form described in the proposition, then apply theorem 1.3.7 (which is a really cool application of this theorem!!)

For any $E \in \mathcal{B}_{\mathbb{R}}$ define $m_s(E) = m(E + s)$ and $m^r(E) = m(rE)$. Then m_s and m^r clearly agree with m and $|r|m$ on finite unions of intervals, and so by theorem 1.3.7 they agree on all of $\mathcal{B}_{\mathbb{R}}$. As a consequence, if $m(E) = 0$, then $m(E + s) = m(rE) = 0$. Thus, for all sets in $\mathcal{L}$ (i.e. all unions of borel sets and Lebesgue zero-sets) must be preserved and linearly scaled (respectively) under translation and dilation:

$$m(E + s) = m(E) \quad m(rE) = |r|m(E)$$

as we sought to show

Since we are working over $\mathcal{M}(\mathcal{B}_{\mathbb{R}})$, the next interesting questions we may ask about m is how it interacts with the topological sets on $\mathbb{R}$. For starters, it is clear (or should be immediately proven) that any singleton has measure 0. Therefore, by countable additivity, every countable set has measure zero; in particular $m(\mathbb{Q}) = 0$. Notice that the rational are dense which is a “topologically large” concept, however it is measure-theoretically small. In fact, we can even have a collection of *open dense* sets, and still be measure-theoretically small. Take some enumeration $\{r_i\}_{i=1}^{\infty}$ of the rational numbers in $[0, 1]$ and let I be the indexing set for this enumeration $i \in I$. Let I_n be the interval of size $\frac{\epsilon}{2^n}$ which is centered at r_n . Then set $U = \bigcup_n I_n$. By construction, U must be dense and open in $(0, 1)$, however

$$\mu(U) \leq \sum_{i=1}^{\infty} \mu(I_n) = \sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon$$

and hence can be as measure-theoretically small as we want. From this we can also find the opposite,

where we have something topologically small but measure-theoretically big, by taking $K = [0, 1] - U$. By monotonicity

$$\mu(K) \geq 1 - \epsilon$$

and yet K is nowhere dense.

There is at least one intuition that can remain: a nonempty *open* set will have non-zero measure. For more on how open/closed sets interact with the measure, see the exercises.

Note that the converse that a non-zero measure set will have an open intervals, is not actually true! For that, we can explore the concept of *cantor sets*

Cantor Set

(was gonna look through: (commented so that this complies

Definition 1.4.8: Classical Cantor Set

Let $I = [0, 1]$. From this, we'll construct the cantor set iteratively

1. Start by removing the middle third $I_1 = I - (\frac{1}{3}, \frac{2}{3})$
2. Given I_i , remove the open middle third of each connected compoennt

Let $C = \cap_{n=0}^{\infty} I_n$ be the Cantor set. In particular, it is the *standard middle thirds Cantor set*. We'll usually refer to this set as "the" Cantor set.

Notice that C is measurable by continuity from above since $C_1 \subseteq C_2 \subseteq \dots \subseteq C_n \subseteq$.

Proposition 1.4.9: Properties Of Cantor Sets

Give C the standard subspace metric of $[0, 1]$. Then C is:

1. nonempty and uncountable
2. compact
3. nowhere dense ($\text{int}(\overline{A}) = \emptyset$ for all $A \subseteq C$)
4. $m(C) = 0$
5. totally disconnected

Proof:

Using some topological properties, since C is the intersection of compact spaces, it is compact.

Next, the cantor set is nonempty since $0 \in C_k$ for all k , and so $0 \in C$. To show C is uncountable, Notice that we can represent every element of the uncountable set $[0, 1]$ in it's base 3 expansion, namely for appropriate $a_k \in \{0, 1, 2\}$:

$$x = \sum_{k=1}^{\infty} a_k 3^{-k}$$

Then take:

$$f : [0, 1] \rightarrow C \quad x = \sum_{k=1}^{\infty} a_k 3^{-k} \mapsto$$

To show total disconnectedness, let's take the same I as before. I is closed in $\mathbb{R}$, and thus in C^n . Note that I^c is the union of finite closed sets, and thus is also closed. Thus, I is clopen in C^n . Thus, $C \cap I$ is also a clopen neighborhood of x in C . Since x was arbitrary, C is totally disconnected.

Finally uncountableness comes from C being compact and complete, which makes C uncountable (exercise).

This construction does not rely on the size of the middle-third; in fact, we can make it converge to any value we want. To see this, first notice that To do this, we first prove a lemma:

Lemma 1.4.10

Suppose $\{\alpha_i\}_{i=1}^{\infty} \subseteq (0, 1)$.

1. $\prod_1^{\infty} (1 - \alpha_i) > 0$ if and only if $\sum_1^{\infty} \alpha_i < \infty$ (Compare $\sum_1^{\infty} \log(1 - \alpha_i)$ to $\sum_1^{\infty} \alpha_i$)
2. Given $\beta \in (0, 1)$, there always exists a sequence $\{\alpha_i\}_{i=1}^{\infty}$ such that $\prod_1^{\infty} (1 - \alpha_i) = \beta$

Proof:

First, we see that we can convert $\prod_i^{\infty} (1 - \alpha_i)$ to $\sum_i^{\infty} \log(1 - \alpha_j)$ by considering

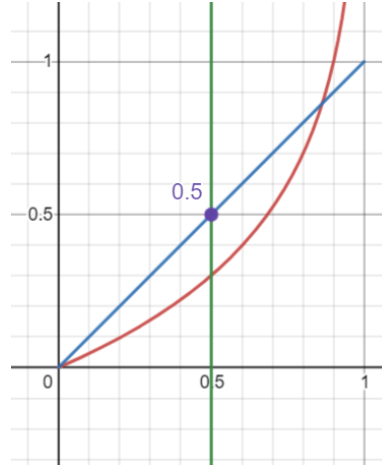
$$\begin{aligned} \log \left(\prod_i^{\infty} (1 - \alpha_i) \right) &= \log \left(\lim_{n \rightarrow \infty} \prod_i^n (1 - \alpha_i) \right) \\ &\stackrel{!}{=} \lim_{n \rightarrow \infty} \log \left(\prod_i^n (1 - \alpha_i) \right) \\ &= \lim_{n \rightarrow \infty} \sum_i^n \log(1 - \alpha_i) \\ &= \sum_{i=1}^{\infty} \log(1 - \alpha_i) \end{aligned}$$

where $\stackrel{!}{=}$ comes from continuity of $\log$ for $1 - \alpha_i > 0$. Notice that it's impossible that $1 - \alpha_i \leq 0$ for some α_i terms since $\alpha_i \in (0, 1)$, and so this proof works. Thus, we get the equivalent condition of

$$\sum_i^{\infty} \log(1 - \alpha_i) > -\infty$$

With this setup, we proceed with the proof

($\Leftarrow$) Let $\sum_i^{\infty} \alpha_i < \infty$. Then we know that $\alpha_i \rightarrow 0$. Thus, pick an N such that for all $i > N$, $\alpha_i < 0.5$. Then notice that $\log(1 - \alpha_i) < \alpha_i$. This is easier to see with a visual aid:



In particular, for all $0 < x < 1/2$, we have $|\log(1-x)| - x > 0$, showing that x is greater than $\log(1-x)$ for all $x \in (0, 1/2)$.

Then from here, this is a simple application of the Weiestrass M -test. Let $K = \sum_{i=1}^N \log(1 - \alpha_i)$ to eliminate the irregular part. Then since $|\log(1 - \alpha_i)| < \alpha_i$, for all $i > N$ and $\sum_{i=1}^{\infty} \alpha_i$ converges, by the Weiestrass M -test the $\sum_{i=N+1}^{\infty} \log(1 - \alpha_i)$ converges. Let's say it converges to L . Then

$$\sum_{i=1}^{\infty} \log(1 - \alpha_i) = K + L > -\infty$$

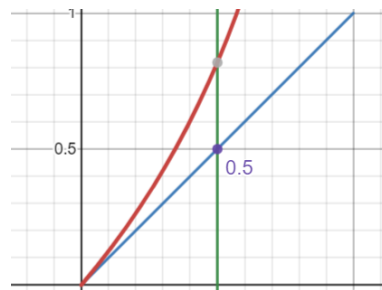
as we sought to show.

($\Rightarrow$) let $\prod(1 - \alpha_i) > 0$ so equivalently $\sum_{i=1}^{\infty} \log(1 - \alpha_i) > -\infty$ so $\sum_{i=1}^{\infty} \log(1 - \alpha_i) = K$ for appropriate $K \in \mathbb{R}$. Then since the series converges, we have that

$$eK = e \sum_{i=1}^{\infty} \log(1 - \alpha_i) = \sum_{i=1}^{\infty} e \log(1 - \alpha_i)$$

i.e., we can scale the terms by e . We will show why this scaling is important in a moment; before we do we must setup a little more.

Since the series converges, it must be that $e \log(1 - \alpha_i) \rightarrow 0$. So pick N such that for all $i > N$, $e \log(1 - \alpha_i) < 1/2$. Then notice that $\alpha_i < e \log(1 - \alpha_i)$. This is easier to see with a visual aid:



Or, to carry out some of the computations, if you raise $e \log(1 - \alpha_i) = \log(1 - \alpha_i)e$ to the power of e , you get

$$(e^{\log(1-\alpha_i)})^e = (1 - \alpha_i)^e \geq \alpha_i^e \quad \forall \alpha_i \in (0, 1/2]$$

and so the inequality holds. Then the proof continues as for the $\Leftarrow$ direction by setting up an appropriate constant term to bound the irregularities, then using the Weiestrass M test bound every α_i by $\log(1 - \alpha_i)$, and since $\sum_i^\infty \log(1 - \alpha_i)$ converges, we get that $\sum_{i=N}^\infty \alpha_i$ also converge, and since it will be the sum of a constant plus the convergent value, $\sum_{i=1}^\infty \alpha_i$ converges, as we sought to show.

Now, let β be given. Let $x_i = \prod_{i=1}^n (1 - \alpha_i)$. We'll construct a sequence where

$$x_1 = \beta + \frac{1 - \beta}{2} \quad x_2 = \beta + \frac{1 - \beta}{4} \quad \cdots \quad x_n = \beta + \frac{1 - \beta}{2^n} \cdots$$

to accomplish this, let

$$\alpha_i = 1 - \frac{\beta + \frac{1-\beta}{2^i}}{\beta + \frac{1-\beta}{2^{i-1}}} \quad (1.2)$$

This construction comes from solving the following equation:

$$\left(\beta + \frac{1 - \beta}{2}\right) y_2 = \beta + \frac{1 - \beta}{4} \Rightarrow y_2 = \frac{\beta + \frac{1-\beta}{4}}{\beta + \frac{1-\beta}{2}}$$

so $\alpha_2 = 1 - y_2$, and clearly $\alpha_2 \in (0, 1)$ since the denominator is bigger than the numerator in the above calculations. Generalizing this pattern, we get that α_i would be of the form presented in equation (1.2) and $\alpha_i \in (0, 1)$. Thus, we get that

$$\lim_{n \rightarrow \infty} \prod_i^n (1 - \alpha_i) = \lim_{n \rightarrow \infty} \beta + \frac{1 - \beta}{2^n} = \beta$$

as we sought to show

Now, let's say we are starting with $[0, 1]$ but on each iteration eliminates intervals that sum to α_i . Let K_i be the i th step in this iteration. Notice that:

$$m(K_{i+1}) = (1 - \alpha_i)m(K_i)$$

Then by what we've just shown, we can make $K = \cap_i^\infty K_i$ be any value up to and including 0 and 1.

Proposition 1.4.11: All Cantor Set Homeomorphism

Let C_n be a cantor set where $m(C_n) = n$. Then $C_n \cong C_0$

Proof:

section 9 of Pugh – will get back to it

You can also prove uncountability directly by making a string represent which side of the split interval you go to.

Next, you might've seen many uncountable sets which are dense in $\mathbb{R}$. Here we have an absolutely fascinating result: C is *nowhere dense* in $[0, 1]$.

Definition 1.4.12: Dense, Somewhere Dense, Nowhere Dense

If $S \subseteq M$ and $\overline{S} = M$, then S is dense in M . The set S is somewhere dense if there exists an open nonempty set $U \subseteq M$ such that $\overline{S \cap U} \supset U$. If S is not somewhere dense, then it is *nowhere dense*.

Proposition 1.4.13: C Nowhere Dense

The Cantor set contains no interval and is nowhere dense in $\mathbb{R}$.

Proof:

To prove it doesn't contain an interval, assume it does, then choose an n large enough, that is $(\frac{1}{3})^n < b - a$, so that it will not be in the interval.

For nowhere dense, assume it's somewhere then, then $\overline{C \cap U} \supset U \supset (a, b)$ – a contradiction.

We now present how Cantor sets are almost initial objects in compact metric space

Theorem 1.4.14: Cantor Surjection Theorem

Given a compact nonempty metric space M , there is a continuous surjection of C onto M .

Proof:

exercise

Finally, we present how all Cantor sets are homeomorphic.

Theorem 1.4.15: Moore-Kline Theorem

We say M is a Cantor space if it is compact, nonempty, perfect, and totally disconnected (like the Cantor set). Every Cantor space M is homeomorphic to the standard middle-thirds Cantor set C .

In other words, those properties *define* Cantor sets.

Proof:

Pugh. p.112

Corollary 1.4.16: Cantor And Fat Cantor Set

The fact Cantor set is homeomorphic to the standard Cantor set

Proof:

Using 1.4.15, the result is immediate.

This is interesting, since it shows that measure is *not* a topological property! This should make sense with $(0, 1)$ and $\mathbb{R}$ being homeomorphic with the same measures, but different, but this is an even more extreme example.

Corollary 1.4.17: Product Of Cantor Sets

A cantor set is homeomorphic to its own cartesian product; $C \equiv C \times C$.

Proof:

A fact that Folland added that I thought was cool is that since every subset of the cantor set is Lebesgue Measurable (since the cantor set has measure zero and the Lebesgue measure is complete), we have that

$$|\mathcal{L}| = |P(\mathbb{R})| > \aleph_1 \quad \text{but} \quad |\mathcal{B}_{\mathbb{R}}| = \aleph_1$$

so there are a *tone* of zero-measure sets, however, we can essentially ignore them because they are zero measure sets!!

For those who are Categorically inclined, Cantor sets are almost like the initial objects of the category of compact metric spaces. They are not initial because there does not exist a unique uniformly continuous function from the cantor set to any compact metric space, but there always will exist at least 1 such function.

Exercise 1.4.1

1. If $E \in \mathcal{L}$ and $m(E) > 0$, for any $\alpha < 1$, there is an open interval I such that $m(E \cap I) > \alpha m(I)$
2. If $E \in \mathcal{L}$, and $m(E) > 0$, the set $E - E = \{x - y \mid x, y \in E\}$ contains an interval centered at 0 (If I is as in the previous exercise with $\alpha > \frac{3}{4}$, then $E - E$ contains $(-\frac{1}{2}m(I), \frac{1}{2}m(I))$).
3. Let E be a Lebesgue measurable set (so $E \in \mathcal{L} = \overline{\mathcal{M}}(\mathcal{B}_{\mathbb{R}})$)
 1. If $E \subseteq N$ where N is a non-measurable set described in section 1.1, then $m(E) = 0$
 2. if $m(E) > 0$, then E contains a non-measurable set (it suffices to assume $E \subseteq [0, 1]$. In notation of section 1.1 $E = \bigcup_{r \in R} E \cap N_r$

2

Integration

In this chapter, we now work with functions between measure spaces that preserve measure structure! Measurable functions feel like the types of functions we “usually” work with, the same way measurable sets are the sets we “usually” work with. Being continuous can be a bit strong (ex. being a monotone function doesn’t imply continuous), but measurability rectifies that by in a sense expanding what are the possible “pre-images” we allow to take (ex. Monotone functions *are* always measurable, as we will demonstrate).

After defining measurable functions, we will focus on real and complex functions (i.e. functions with codomain $\mathbb{R}$ or $\mathbb{C}$). In the real case, I like to think of real functions as given every point in the domain X a value or a weight, in contrast to a measure which is a real function which simply assigns a constant value at each point (this will become more clear as we introduce the definitions). For the complex case, each point is given a weight and a *phase* or *state*¹. In terms of computation, we simply split the problem into the real and imaginary component. Then, we will define how to “measure” a real (resp. complex) measurable function. We then focus on measurable functions which have “finite measure”, and will call them (*Lebesgue*) *integrable functions*; integrable functions will allow us to have some regularity in how we can combine measurable functions without running into some problems with infinities (as we’ll soon show). One result of this is that we can think of the integral as being a “measure” for a function, and since integrable function will have “finite measure”, it will make it more meaningful to see which functions are “close” (in the appropriate sense of the word close) to a certain function.

A word must be said about Lebesgue integrability vs. Riemann integrability. The key difference is the allowed measurable sets. For Riemann integrability, all our measurable sets are *rectangles*: every partition of the area of integration can be seen as a sum of indicator functions over pair-wise disjoint

¹you can think of every input having a natural magnitude that periodically shifts. The phase $e^{i\theta}$ tracks this. So if $f(a) = 10$ and $f(b) = 10e^{-i\pi} = -10$, then a is at the beginning of its phase with magnitude 10, while b is half way through its phase which shifted it’s magnitude to -10 . Addition adds the interference (ex. $(2 + i) + (-2 - 2i) = -i$) while

rectangles. On the other hand, Lebesgue integration is over *measurable* sets. The strategy shall shift so that instead of partitioning the domain based off of its geometry, we shall take partition it using $f^{-1}([-k, k])$ (or really any measurable covering of $\mathbb{R}$, and appropriately modified for $\mathbb{C}$) where we shall show the pre-image of a measurable set is measurable if f is measurable. This shall solve many headache the Riemann integral imposes (ex. any rectangle of $\chi_{\mathbb{Q}}$ on $[0, 1]$ will have have rational and irrational points, making it unfeasible to measure).

After we have defined integrable functions, we will have to take a moment to compare different types of convergences, since at that stage we will have 4 different ways functions can converge. We will then move on to trying to bring the theory of the product of σ -algebras to measure and integrable functions more generally, allowing us to construct larger measures and integrable functions from smaller components. Importantly, we will show that we can find the a measure $\mu \times \nu$ by separating any problem to only measuring “slices” of the set with respect to μ , then “slices” of the set with respect to ν (this is Fubini’s Theorem).

2.1 Measurable Functions

Recall that if $f : X \rightarrow Y$ is a function, then $f^{-1} : P(Y) \rightarrow P(X)$ is a function where $f^{-1}(E) = \{x \in X \mid f(x) \in E\}$ which preserves unions, intersections, and compliments. Thus, if $\mathcal{N}$ is a σ -algebra on Y , $f^{-1}(\mathcal{N})$ is a σ -algebra on X . If X and Y are measure space, if $f^{-1}(\mathcal{N})$ maps into $\mathcal{M}$, we will call f *measurable*:

Definition 2.1.1: Measurable Function

Let $f : X \rightarrow Y$ be a function and $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. Then f is called a $(\mathcal{M}, \mathcal{N})$ -*measurable function* (or just measurable) if for all $E \in \mathcal{N}$

$$f^{-1}(E) \in \mathcal{M}$$

Just like in topology, this could be thought of as a surjection of the measurable space on X onto the measurable space on Y . It should be clear that the composition of two measurable functions is measurable (in particular, the composition of an $(\mathcal{M}, \mathcal{N})$ -measurable function with $(\mathcal{N}, \mathcal{O})$ -measurable function is $(\mathcal{M}, \mathcal{O})$ -measurable), and that the identity function $\text{id} : X \rightarrow X$ is clearly a measurable function, and so the collection of measurable functions and measurable spaces form a category, usually denoted **Meas**.

Just like lemma 1.1.4 was useful for simplifying proofs, we proof the following:

Lemma 2.1.2: Measurable Function and Generating Sets

Let $\mathcal{N}$ be generated by $\mathcal{E}$. Then $f : X \rightarrow Y$ is a $(\mathcal{M}, \mathcal{N})$ -measurable function if and only if for all $E \in \mathcal{E}$, $f^{-1}(E) \in \mathcal{M}$

Proof:

The $\Rightarrow$ direction is *a-fortiori* true by definition. The $\Leftarrow$ direction comes from f^{-1} preserving unions and compliments, so $\{E \subseteq Y \mid f^{-1}(E) \in \mathcal{M}\}$ is a σ -algebra that contains $\mathcal{E}$, and so contains $\mathcal{N}$.

From this, there is a very nice corollary when $\mathcal{M}$ is a collection of Borel sets:

Corollary 2.1.3: Continuous Function $\Rightarrow$ [Borel] Measurable Function

Let $f : X \rightarrow Y$ be a continuous function and Y have the borel σ -algebra. Then f is a measurable function.

Proof:

Since f is continuous, the pre-image of an open set (in particular, open intervals) is open, and hence by proposition 1.1.7 and lemma 2.1.2 it is measurable

Naturally, in analysis, the most important measurable functions will be of the form

$$f : \mathbb{R} \rightarrow \mathbb{R} \quad f : \mathbb{R} \rightarrow \mathbb{C}$$

This also means that the domain can either have the $\mathcal{B}_{\mathbb{R}}$ σ -algebra or $\mathcal{L}$ σ -algebra.

Example 2.1.4: Measurable Functions

Most functions you have worked with will in fact be measurable

1. As already shown, all continuous functions are measurable
2. All monotone functions are measurable: the pre-image of closed rays will either be open or closed rays, and so by proposition 1.1.7, they are measurable.
3. All indicator functions over measurable sets are measurable (hence, bump functions are essentially measurable)
4. All Riemann integrable functions are measurable (we will show this in section 2.3.1)
5. We will soon show that the point-wise limit of measurable functions is also measurable, hence almost all examples of functions that shall be studied are measurable.

It is in practice rare to work with non-measurable functions unless a contrived example is used, like a measurable function purposefully built to map to non-measurable sets (like in the case of the composition of two Lebesgue measurable functions)

2.1.5 Convention We will *always* assume the codomain will have $\mathcal{B}_{\mathbb{R}}$ (resp. $\mathcal{B}_{\mathbb{C}}$) σ -algebra. In fact, if we have a function $f : X \rightarrow \mathbb{R}$ (resp. $f : X \rightarrow \mathbb{C}$), then we will sometimes say f is $\mathcal{M}$ -measurable instead of $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$ -measurable (resp. $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$ -measurable). If the domain is also Borel, we will say that f is *Borel measurable* (and omit the borel measure if it's $\mathbb{R}$ or $\mathbb{C}$). Similarly, if the domain is $\mathcal{L}$, we will call f Lebesgue measurable.

Because of the convention of the co-domain being Borel, it is possible that the composition of two Lebesgue measurable functions is *not* Lebesgue measurable: if $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ are two Lebesgue measurable functions where $E \in \mathcal{B}_{\mathbb{R}}$, $g^{-1}(E) \in \mathcal{L}$ but $g^{-1}(E) \notin \mathcal{B}_{\mathbb{R}}$, it is possible that $f^{-1}(g^{-1}(E)) \notin \mathcal{L}$

Example 2.1.6: Composition Not Lebesgue Measurable

Let $f : [0, 1] \rightarrow [0, 1]$ be the cantor function from section 1.5, and let $g(x) = f(x) + x$.

1. g is a bijection from $[0, 1]$ to $[0, 2]$ and $h = g^{-1}$ is continuous from $[0, 2]$ to $[0, 1]$

Proof:

First, since f is increasing (though not injective) and x is strictly increasing, then $f + x$ is injective. Since f and x are both continuous, so is $g = f + x$. Hence the pre-image of a closed set must be closed. If we consider $g^{-1}([0, 2]) = D$, then using the fact that g is a monotonic functions, since $g(0) = 0$ and $g(1) = 2$, it must be that $D = [0, 1]$, showing that g is indeed surjective, and hence bijective.

To show that $g^{-1} : [0, 2] \rightarrow [0, 1]$, is continuous, bijective, and the codomain is Hausdorff, then g is homeomorphic (recall that these imply that g is an open map: if $U \subseteq [0, 1]$ then $S = [0, 1] - U$ is closed, and since g is continuous $g([0, 1] - U) = [0, 2] - g(U)$ is closed, so $g(U)^c$ is closed, so $g(U)$ is open), and hence g^{-1} is continuous.

2. If C is the cantor set, then $m(g(C)) = 1$

Proof:

We will take advantage of translation invariance of the Lebesgue measure. Let C be the cantor set on $[0, 1]$. Then C can be thought of as the interval $[0, 1]$ minus all of the open sets that are eliminated during construction:

$$C = [0, 1] - I_1 - I_{21} - I_{22} - \cdots - I_{ik} - \cdots$$

Let $I = \coprod_{i,k} I_{ik}$ collection of these sets so that $C = [0, 1] - I$. Since $m(C) = 0$, $m(I) = 1$. Consider now $g(I)$. Since g is injective and I is the disjoint union of sets, then there exists sets $\{J_j\}_{j=1}^{\infty}$ such that $J_j = g(I_{ik})$. We now examine the measure of the sets J_j .

By definition, I_{ik} does not contain any points of the cantor set, and so for appropriate $c_{ik} \in C$,

$$g_{I_{ik}}(x) = f(c_{ik}) + x$$

that is, g on the interval I_{ik} is of the form $x + f(c_{ik})$ for constant $f(c_{ik})$! Since the Lebesgue measure is translation invariant, we get:

$$m(g(I_{ik})) = m(g(I_{ik}) - f(c))$$

and so, since the I_{ik} intervals are all disjoint and g is injective:

$$g(\coprod I_{ik}) = \coprod g(I_{ik})$$

and so the measure of $g(I)$ is in fact exactly the measure of I , that is, $m(g(I)) = m(I) = 1$. Thus:

$$2 = g(m([0, 2])) = m(g(C) \coprod J) = m(C) + m(J) = m(g(C)) + 1$$

and so $m(g(C)) = 1$, as we sought to show.

3. by exercise 29, $g(C)$ contains a Lebesgue non-measurable set A . Let $B = g^{-1}(A)$. Then B is Lebesgue measurable, but not Borel

Proof:

Since $A \subseteq g(C)$, $B \subseteq g^{-1}(g(C)) = C$. Since $m(C)$ is zero, by completeness, $m(A) = 0$, and so A is Lebesgue measurable.

Next, to show that B is not Borel-measurable, For the sake of contradiction let's assume it was. Then since g^{-1} is continuous, it is Borel-measurable, meaning the pre-image (with respect to g^{-1}) of a measurable set is measurable. However, $g(B) = A$, which is not measurable, and hence B is not Borel measurable.

4. There exists a Lebesgue measurable function F and a continuous function G on $\mathbb{R}$ such that $F \circ G$ is not Lebesgue measurable

Proof:

Let F be the indicator function on B , I_B , and $G = g^{-1}$. Then F and G are clearly Lebesgue measurable, However, G^{-1} does not map all Lebesgue measurable sets to Lebesgue measurable sets (in particular, B maps to a Borel set), and so

$$(F \circ G)^{-1}(B) = (G^{-1} \circ F^{-1})(B) = (g(I_B^{-1}(B))) = g(B) = A$$

However, A is not Lebesgue measurable set, and so $F \circ G$ is not Lebesgue measurable.

Just like for Borel sets earlier, we have the following equivalence for measurable functions with codomain $\mathbb{R}$:

Proposition 2.1.7: Borel sets and Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $f : X \rightarrow \mathbb{R}$. Then the following are equivalent:

1. f is $\mathcal{M}$ -measurable
2. $f^{-1}((a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
3. $f^{-1}([a, \infty)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
4. $f^{-1}((-\infty, a)) \in \mathcal{M}$ for all $a \in \mathbb{R}$
5. $f^{-1}((-\infty, a]) \in \mathcal{M}$ for all $a \in \mathbb{R}$

Proof:

This follows from proposition 1.1.7 and lemma 2.1.2.

Similarly, we will soon be encountering co-domains which are product spaces (ex. $\mathbb{R}^n$ or $\mathbb{C}^n$). In particular, let's say we had some set X , and some family of measurable spaces $\{(Y_\alpha, \mathcal{M}_\alpha)\}_{\alpha \in A}$, and $f : X \rightarrow Y_\alpha$ is a map for each $\alpha \in A$. Then we can impose a unique smallest σ -algebra on X that makes each f_α a measurable function, in particular, $\mathcal{M}_X = \{f_\alpha^{-1}(E_\alpha) \mid E_\alpha \in \mathcal{M}_\alpha, \alpha \in A\}$ ². This σ -algebra on X is called the σ -algebra generated by $\{f_\alpha\}_{\alpha \in A}$.

²A similar strategy is used in the construction of *weak* topologies. If you're more categorical, this construction makes X satisfy the universal property of products in the category of **Meas**

Proposition 2.1.8: Product of Measurable Functions

Let $(X, \mathcal{M})$ be a measurable space and $(Y_\alpha, \mathcal{N}_\alpha)$ be a collection of measurable spaces, so $Y = \prod_{\alpha \in A} Y_\alpha$, $\mathcal{N} = \bigotimes_{\alpha \in A} \mathcal{N}_\alpha$, and $\pi_\alpha : Y \rightarrow Y_\alpha$ is the coordinate map. Then $f : X \rightarrow Y$ is $(\mathcal{M}, \mathcal{N})$ -measurable if and only if $f_\alpha = \pi_\alpha \circ f$ is $(\mathcal{M}, \mathcal{N}_\alpha)$ -measurable.

Proof:

Let's say f is $(\mathcal{M}, \mathcal{N})$ -measurable. Then since the composition of two measurable functions is measurable, so are each f_α is measurable (since π_α is measurable by definition of $\bigotimes$).

Conversely, if every f_α is $(\mathcal{M}, \mathcal{N}_\alpha)$ -measurable, then for each $E_\alpha \in \mathcal{N}_\alpha$, $f^{-1}(\pi_\alpha^{-1}(E_\alpha)) = f^{-1}(E_\alpha) \in \mathcal{M}$, and since these sets generate $\mathcal{N}$, we have that f is $(\mathcal{M}, \mathcal{N})$ -measurable by proposition 2.1.7

This proposition also let's us define what it means to take the “product” of two functions. As a consequence, we can give an easy criterion for when complex functions are measurable:

Corollary 2.1.9: Complex Function Measurability

Let $f : X \rightarrow \mathbb{C}$ be a complex function. Then f is $\mathcal{M}$ -measurable if and only if $\operatorname{Re} f$ and $\operatorname{Im} f$ are both measurable.

Proof:

This is simply taking advantage of the topology of $\mathbb{C}$, namely

$$\mathcal{B}_{\mathbb{C}} = \mathcal{B}_{\mathbb{R}^2} \stackrel{!}{=} \mathcal{B}_{\mathbb{R}} \otimes \mathcal{B}_{\mathbb{R}}$$

where the $\stackrel{!}{=}$ equality comes from the fact that $\mathbb{R}$ is separable (see proposition 1.1.11). And so proposition 2.1.8 with the fact that Re and Im can be considered projections completes the proof.

Sometimes, we want to work with the extended real line (or in general some compactification of a topological space). If $\overline{\mathbb{R}} = [-\infty, \infty]$, Then let

$$\mathcal{B}_{\overline{\mathbb{R}}} = \{E \subseteq \overline{\mathbb{R}} \mid E \cap \mathbb{R} \in \mathcal{B}_{\mathbb{R}}\}$$

Notice that is we put the metric ρ on $\overline{\mathbb{R}}$ to be $\rho(x, y) = |\tan^{-1}(x) - \tan^{-1}(y)|$, then $\mathcal{B}_{\overline{\mathbb{R}}}$ is indeed the usual definition of a Borel σ -algebra (exercise). It should be verified that $\mathcal{B}_{\overline{\mathbb{R}}}$ can be generated by open or closed rays, and so we will say $f : X \rightarrow \overline{\mathbb{R}}$ is $\mathcal{M}$ -measurable if it is $(\mathcal{M}, \mathcal{B}_{\overline{\mathbb{R}}})$ -measurable.

Next, if the codomain has some notion of $+$ and $\cdot$, we establish that the basic “ $\mathbb{C}$ -algebra operations” work on measurable functions:

Proposition 2.1.10: Arithmetics on Measurable Functions

Let $f : X \rightarrow \mathbb{C}$ and $g : X \rightarrow \mathbb{C}$ be $\mathcal{M}$ -measurable functions. Then $f + g$, fg , and zf for $z \in \mathbb{C}$ are $\mathcal{M}$ -measurable functions

Note that we have limited ourselves to $\mathbb{C}$. This still works with $\overline{\mathbb{R}}$ with the appropriate care of the $\infty - \infty$ and $0 \cdot \infty$ case (exercise 2 in book)

Proof:

The proof for zf being measurable is similar to that of $+$ and $\cdot$, so is left as an exercise.

Since f and g , are measurable, then their “cartesian product” is measurable by proposition 2.1.8

$$F : X \rightarrow \mathbb{C} \times \mathbb{C} \quad F(x) = (f(x), g(x))$$

In particular, F is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}} \otimes \mathcal{B}_{\mathbb{C}})$ -measurable. Next, consider $p : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$, $p(x, y) = x + y$ and $t : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$, $t(x, y) = xy$ (p for plus, t for times). Then since p and t are continuous, by corollary 2.1.3 p and t are $(\mathcal{B}_{\mathbb{C} \times \mathbb{C}}, \mathcal{B}_{\mathbb{C}})$ -measurable. Thus, $p \circ F$ and $t \circ F$ is $(\mathcal{M}, \mathcal{B}_{\mathbb{C}})$ -measurable, and by definition

$$f + g = p \circ F \quad fg = t \circ F$$

completing the proof

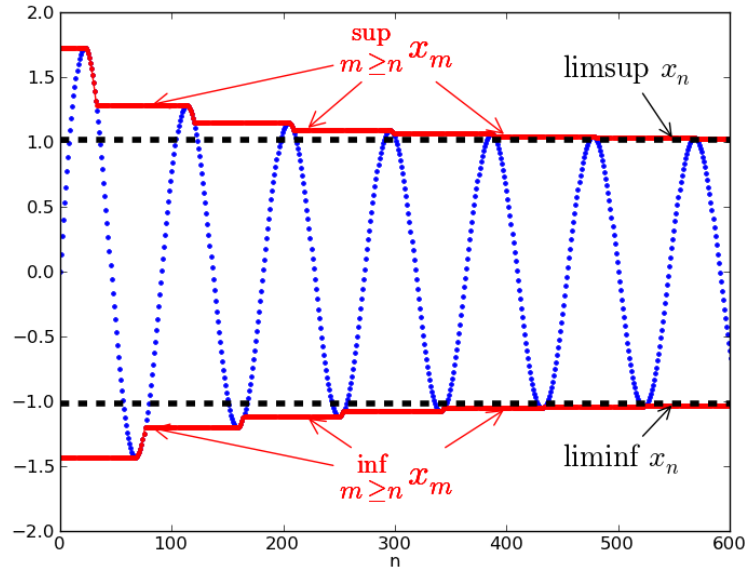
The next operations we’ll show are measurable are $\sup$, $\inf$, $\limsup$ and $\liminf$. This is perhaps one of the most important functions to be measurable since they are at the heart of many constructions in analysis. Recall that

$$\sup_i f_i(x) = \sup \{f_i(x) \mid i \in \mathbb{N}\}$$

and so this expression means we’re taking the set $f_i(x)$ for each i and taking the sup of that set. In other words, for every point x we’re taking the supremum of the set $\{f_1(x), f_2(x), \dots\}$. Symmetrically the same for $\inf$. For $\limsup$, recall that:

$$\limsup_i f_i(x) = \lim_{i \rightarrow \infty} \sup \{f_k(x) \mid k \geq i\}$$

that is, we are constantly shrinking the set over which we are doing the supremum, which in a sense means we’re trying to “hug” the value from above as close as possible in the limit. Symmetrically the same for $\liminf$. If you’re visual, I like to keep this image in mind: let’s say the blue line is the value of $f_i(x_0)$ for a specified x_0 . Then:



Comment here for future reference:

Proposition 2.1.11: Measurable functions and Limit Bounding

Let $f_i : X \rightarrow \overline{\mathbb{R}}$ be $\mathcal{M}$ -measurable functions for each i and consider $\{f_i\}_{i=1}^{\infty}$. Then

$$\begin{aligned} g_1(x) &= \sup_i f_i(x) & g_2(x) &= \inf_i f_i(x) \\ g_3(x) &= \limsup_{i \rightarrow \infty} f_i(x) & g_4(x) &= \liminf_{i \rightarrow \infty} f_i(x) \end{aligned}$$

The codomain $\overline{\mathbb{R}}$ was used so that the sup and inf is still considered defined if $\sup = \pm\infty$ or $\inf = \pm\infty$.

Proof:

To show $\sup_i f_i$ and $\inf_i f_i$ are measurable, since the codomain domain is $\overline{\mathbb{R}}$, which is equivalent measure-theoretically to $\mathbb{R}$, we will show the pre-image of open rays is the union of measurable sets:

$$\begin{aligned} x \in g_1^{-1}((a, \infty]) &\Leftrightarrow g_1(x) > a \\ &\Leftrightarrow f_i(x) > a && \text{for some } i \\ &\Leftrightarrow x \in f_i^{-1}((a, \infty]) && \text{for some } i \\ &\Leftrightarrow x \in \bigcup_{i=1}^{\infty} f_i^{-1}((a, \infty]) \end{aligned}$$

and similarly for g_2 , except considering $(-\infty, a]$, therefore:

$$g_1^{-1}((a, \infty]) = \bigcup_i f_i^{-1}((a, \infty]) \quad g_2^{-1}((-\infty, a]) = \bigcup_i f_i^{-1}((-\infty, a])$$

since each f_i is measurable, we have a union of measurable sets, so g_1 and g_2 are measurable by proposition 1.1.11. Next, let

$$h_k(x) = \sup_{i>k} f_i(x) = \sup \{f_k(x) \mid k > i\}$$

then by what we've just established, h_k is measurable for each k . Then by of $\limsup$:

$$g_3 = \inf_k h_k$$

and so g_3 is measurable since we just established the $\inf$ of measurable functions is measurable. Similarly for g_4 .

as an immediate corollary:

Corollary 2.1.12: Measurability of Max and Min

Let f, g be measurable functions. Then so is $\max\{f, g\}$ and $\min\{f, g\}$

Corollary 2.1.13: Measurable Preserved under Pointwise limit

Let $\{f_i\}$ be a sequence of measurable function. If $\lim_{i \rightarrow \infty} f_i(x)$ exists for every x , and we define $f : X \rightarrow \mathbb{R}$ to be

$$f(x) = \lim_{i \rightarrow \infty} f_i(x)$$

then f is measurable (i.e. point-wise convergence preserves measurability)

Recall that point-wise convergence *does not preserve continuity* (the steeple function's are the usual example). However, measurability is sufficiently general to be preserved under this weaker form of convergence! Naturally, if the sequence of functions $\{f_i\}$ are all $\mathcal{M}$ -measurable and uniformly convergent, they are measurable

Proof:

If f exists (i.e., every $f_n(x)$ converges pointwise to $f(x)$), then by proposition 2.1.11, $f = g_3 = g_4$, completing the proof.

2.1.14 Remark Note how important it is that *all* points converge! It is possible that a single point doesn't converge (ex. $\lim f_i(0)$ doesn't converge), and so the limit is *not* a measurable function, in fact it doesn't even need to be a function. Take for example the collection of functions $f_k : \mathbb{R} \rightarrow \mathbb{R}$ with the properties:

1. f_k is continuous (it can even be smooth)
2. $\int_{\mathbb{R}} f(x) dx = 1$ (In the Riemann sense), or really any constant $c \in \mathbb{R}_{>0}$
3. $\text{supp} f \subseteq \left[-\frac{1}{k}, \frac{1}{k}\right]$

In other words, the collection of f_k are all smooth bump functions with volume 1, but the area in which it has support is shrinking, and so the “bump” of each f_k must be getting higher and higher (to define such functions, consider starting with $f_1 = e^{-\frac{1}{1-x^2}}$ where $x \in (-1, 1)$ and 0 otherwise). Then $f_k \rightarrow 0$ at all but 1 point, namely $f_k(0)$ does *not* converge. As a consequence, this limit is *not* a function!

This example shows the importance of somehow “bounding” your area; we will get back to this observation in section 2.2. Some might think that this example is a bit cheeky, it still feels like the function is measurable if we just circumnavigate this problem by making the domain of the function $\mathbb{R} \cup \{\infty\}$, there is the trouble of defining a measure on the 1-point compactification of $\mathbb{R}$ while extending the measure of $\mathbb{R}$ ^a. If the reader wonders if there is a way of defining some generalization of a function that will accept the fact that the f_k converge to something that still somehow preserves the fact that $\int f = 1$, then we will show in chapter ?? that this converges to something called a *distribution*^b.

^aIn particular, $\mu(\{\infty\}) = \infty$, as can be shown using continuity from above, and so somehow we have “gained” an infinite amount of area. On the other-hand, if we just ignore this and consider that ∞ is just some point like any other point, then we have “lost” area

^bThe example named here is a distribution called the *delta distribution*, or sometimes *delta function* for reasons that is covered in chapter ??

Now that we proved that these operations preserving measurability, we will introduce some new notation to decompose real and complex functions which will soon become useful when defining integrable functions.

Definition 2.1.15: Real and Complex Decomposition

Let $f : X \rightarrow \mathbb{R}$ and $g : X \rightarrow \mathbb{C}$ be real and complex measurable functions respectively. Then

$$f^+ = \max\{f, 0\} \quad f^- = \max\{-f, 0\}$$

and

$$\text{sgn}(g(x)) = \begin{cases} \frac{g(x)}{|g(x)|} & g(x) \neq 0 \\ 0 & g(x) = 0 \end{cases} \quad |g(x)| = |z| = |x + iy| = \sqrt{x^2 + y^2}$$

which gives the decomposition of:

$$f = f^+ - f^-, \quad g = (\text{sgn} g)|g|$$

The functions f^+ and f^- are clearly well-defined and measurable. For the complex case, $|\cdot|$ is

continuous everywhere and $z \mapsto \operatorname{sgn} z$ is continuous everywhere except 0. It is measurable since if $U \subseteq \mathbb{C}$ is open, then $\operatorname{sgn}^{-1}(U)$ is either V for some open set V , or $V \cup \{0\}$, meaning sgn is Borel measurable. In some textbooks, $\arg$ is used instead of sgn .

2.1.1 Standard Representation

Recall that every measurable function is the point-wise limit of a bunch of easy to work with function, in particular, linear combination of characteristic functions! As a reminder, if $\chi_E : X \rightarrow \mathbb{R}$ (or $\chi_E : X \rightarrow \{0, 1\}$) where $E \subseteq X$, then:

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

Then χ_E is called the characteristic (or indicator) function. It is sometimes also denoted as 1_E or I_E .

Definition 2.1.16: Simple function

Let $\{E_i\} \subseteq P(\mathbb{C})$ be some finite collection of sets. Then the finite linear combination of characteristic functions on E_i multiplied by a complex constant:

$$f = \sum_{i=1}^n z_i \chi_{E_i}$$

is called a *simple function*

Example 2.1.17: Simple Functions

1. Show that $|f(X)| < \infty$ then f is in fact a simple function (hint: let $E_i = z_i$ for each $z_i \in f(X)$). In fact, each coefficient can also be distinct (let it be the respective z_i from the range)
2. Show that if f and g are simple functions, so are $f + g$ and fg

Using these, we can now show that any measurable function is simply the limit of simple functions!

Theorem 2.1.18: Simple Function Approximation

Let $(X, \mathcal{M})$ be a measurable space. Then

1. If $f : X \rightarrow [0, \infty]$ is measurable, there exists a sequence of functions $\{\varphi_i\}_{i=1}^{\infty}$ such that

$$0 \leq \varphi_1 \leq \varphi_2 \leq \cdots \leq f$$

where $\lim_{n \rightarrow \infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \rightarrow f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformly converges)

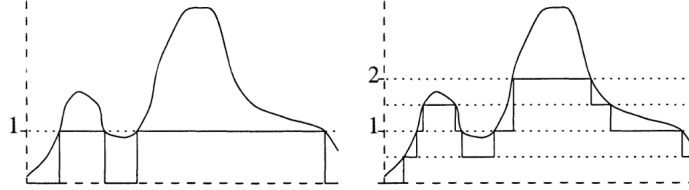
2. If $f : X \rightarrow \mathbb{C}$ is measurable, there exists a sequence of functions $\{\varphi_i\}_{i=1}^{\infty}$ such that

$$0 \leq |\varphi_1| \leq |\varphi_2| \leq \cdots \leq |f|$$

where $\lim_{n \rightarrow \infty} \varphi_i(x) = f(x)$ (i.e. $\varphi_n \rightarrow f$ pointwise). and $\varphi_n \rightrightarrows f$ on a bounded domain (i.e. uniformly converges)

Proof:

Essentially, you will get characteristic functions to converge to the value like so:



In particular, for $n \in \mathbb{N}_{\geq 0}$ and $0 \leq k \leq 2^{2n} - 1$, let

$$E_n^k = f^{-1} \left(\left(\frac{k}{2^n}, \frac{k+1}{2^n} \right) \right) \quad F_n = f^{-1}((2^n, \infty])$$

and

$$\varphi_n = \sum_{k=0}^{2^{2n}-1} \frac{k}{2^n} \chi_{E_n^k} 2^n \chi_{F_n}$$

then we get that $\varphi_n \leq \varphi_{n+1}$ for all n and $0 \leq f - \varphi_n \leq \frac{1}{2^n}$ where $f \leq 2^n$. The rest is left to check.

Generalizing to $\mathbb{C}$, decomposing to $f = g + ih = (g^+ + g^-) = i(h^+ + h^-)$, where the $+$ and $-$ functions are the positive and negative parts of f and g , then we can repeat the process for each of these and get $\psi_n^+, \psi_n^-, \zeta_n^+, \zeta_n^-$ and let $\varphi_n = (\psi_n^+ - \psi_n^-) + i(\zeta_n^+ - \zeta_n^-)$.

This shows that measurable functions are a very natural type of function to work with in analysis: they are all a pointwise-limit (and on a bounded domain a uniform limit) of simple functions. The particular sequence of simple functions will be called the *standard representation of f* . Using this

result, we can define a new definition of equality up to zero-measure sets:

Proposition 2.1.19: Measurability up to Zero Set

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be two measure with ν being a complete measure, and let $f : X \rightarrow Y$ be $(\mathcal{M}, \mathcal{N})$ -measurable. Then:

1. If $f = g$ up to a zero measure set with respect to μ (shorthand to μ -a.e.), then g is measurable
2. If f_n is measurable for each $n \in \mathbb{N}$ and $f_n \rightarrow f$ μ -a.e, then f is measurable

Proof:

1. Let $N = \{x \in X \mid f(x) \neq g(x)\}$. Let $B \in \mathcal{N}$ and consider $g^{-1}(B)$. Then by definition:

$$\begin{aligned} g^{-1}(B) &= \{x \in M \mid g(x) \in B\} \\ &= \{x \in \mathcal{M} \mid g(x) \in B, g(x) = f(x)\} \cup \{x \in \mathcal{M} \mid g(x) \in B, g(x) \neq f(x)\} \end{aligned}$$

the latter set is a subset of N , and so is measurable by the completeness of f . The former set is of the form:

$$\begin{aligned} \{x \in \mathcal{M} \mid g(x) \in B, g(x) = f(x)\} &= \{x \in \mathcal{M} \mid f(x) \in B, g(x) = f(x)\} \\ &= f^{-1}(B) \cap \{x \mid f(x) = g(x)\} \\ &= f^{-1}(B) \cap (X - \{x \mid f(x) \neq g(x)\}) \\ &= f^{-1}(B) \cap (X - N) \end{aligned}$$

and since f measurable the pre-image of a measurable set is too, and since N is measurable, $X - N$ is too, so we have that $g^{-1}(B)$ is measurable.

2. Let's say N is the set on which $f_n \not\rightarrow f$. Define

$$g_n = f_n \chi_{X-N} + f \chi_N$$

Then notice that $g_n = f_n$ μ -a.e., and so by part 1 are measurable, and by construction $g_n \rightarrow f$ everywhere, and so f is measurable!

However, if μ is not complete, it turns out to make little difference since we can complete the measure, produce the result, and restrict back to the original measure without changing the result:

Proposition 2.1.20: Extending To Complete Measures

Let $(X, \mathcal{M}, \mu)$ be a measure space and $(X, \overline{\mathcal{M}}, \overline{\mu})$ be its completion. if f is a $\overline{\mathcal{M}}$ -measurable function on X , there is a $\mathcal{M}$ -measurable function g such that $f = g$ $\overline{\mu}$ -a.e.

Proof:

If f is a simple function, then we see that the completion will at most remove finitely many points when we restrict every χ_E ($E \in \overline{\mathcal{M}}$) to g , and so $f = g$ $\bar{\mu}$ -a.e.

Now let f be a $\overline{\mathcal{M}}$ -measurable function, and by theorem 2.1.18 choose a sequence of $\overline{\mathcal{M}}$ -measurable functions $\{\varphi_i\}_{i=1}^\infty$ that converge pointwise to f . The idea is that each $\varphi_i = \psi_i$ $\bar{\mu}$ -a.e., that is, there exists a E_i where $\varphi_i(x) \neq \psi_i(x)$ for all $x \in E_i$ but $\bar{\mu}(E_i) = 0$, and the countable union of these will still be measure 0.

In particular Let $E = \bigcup_i^\infty E_i$. By definition of completion, there must exist some N such that $\mu(N) = 0$ and $E \subseteq N$. Let $g = \lim_{\chi_X - N} \psi_n$. Then g is measurable by corollary 2.1.13, and clearly $f = g$ on N^c , completing the proof.

2.2 Integration of non-negative Function

We are now in a position to define how to integrate functions (integrable functions coming soon)! Throughout this section, let $(X, \mathcal{M}, \mu)$ be a measure space

Definition 2.2.1: L^+

Let

$$L^+ = \{f : X \rightarrow [0, \infty] \mid f \text{ is a } \mathcal{M}\text{-measurable function}\}$$

Sometimes, L^+ is written as $L^+(X)$, $L^+(\mu)$, or $L^+(\mu, X)$ to emphasize which space of functions we are dealing with. If it is unambiguous, we use L^+ .

Definition 2.2.2: Integral of Simple Function

Let $f = \sum_1^n a_i \chi_{E_i}$ be a simple function. Then define the *integral of f with respect to μ* to be

$$\int f d\mu = \sum_i^n a_i \mu(E_i)$$

In other words, we simply took the measure of the domain of χ_{E_i} for each i . As usual, we take the convention that $0 \cdot \infty = 0$. We will also allow $\int f = \infty$ to be a valid result. If there is no confusion, we will write $\int f$ instead of $\int f d\mu$. It should also be emphasized that it is the measure of the *domain* and the coefficient from the *codomain* that we are taking.

If we want to integrate over just some $A \subseteq X$ where $A \in \mathcal{M}$, then $\varphi|_A$ is also a simple function (simply take $\chi_{E \cap A}$ for each characteristic function). We usually denote this by

$$\int_A f d\mu = \int f \chi_A d\mu$$

Because of this notation, we can write $\int f d\mu$ as

$$\int f d\mu = \int_X f d\mu$$

Proposition 2.2.3: properties of Integrals of Simple Functions

Let φ and ψ be simple functions in L^+ . Then

1. if $c \geq 0$, $\int c\varphi d\mu = c \int \varphi d\mu$
2. $\int (\varphi + \psi) d\mu = \int \varphi d\mu + \int \psi d\mu$
3. if $\varphi \leq \psi$ then $\int \varphi \leq \int \psi$
4. The map $A \mapsto \int_A 1 d\mu$ is a measure on $\mathcal{M}$ (more generally, $A \mapsto \int_A \varphi d\mu$ is a measure on $\mathcal{M}$ for a simple function φ)

Proof:

Do these as midterm exercises

With this definition, we can define integrable function more generally:

Definition 2.2.4: Integral of Measurable Function

Let $f \in L^+$. Then define

$$\int f d\mu := \sup \left\{ \int \varphi d\mu \mid \varphi \text{ is a simple function, } 0 \leq \varphi \leq f \right\}$$

It's important to see why the integral of a simple function will give the same result as the “simple integral” (i.e. integral over simple functions from earlier), in particular since the set over which we are containing the supremum will contain the simple function in question, and so the maximum will be achieved by itself (and proposition 2.2.3?). Furthermore, results 1 and 3 from proposition 2.2.3 holds for the definition of integration just given. The other results, in particular linearity, will be established soon. We can also see how the definition of the integral solidifies the idea that the integral is a sort of “weighted measure”: the integral of a simple function is just a sum of measure of sets weighted by some constant, and we take the supremum of all such sets. We will in fact take a closer look at defining measure in terms of integration of functions in chapter 3.

Given the supremum definition of the integral, it could be rather hard to find the integral of a function in L^+ . We essentially never check this, and treat the following theorem as the de-facto way of finding the integral:

Theorem 2.2.5: Monotone Convergence Theorem (MCT)

If $\{f_n\}_{n=1}^\infty \subseteq L^+$ is a sequence such that $f_i \leq f_{i+1}$ for all $i \in \mathbb{N}$ and $f = \lim_{n \rightarrow \infty} f_n$ ($= \sup_n f_n$), then

$$\int f = \lim_{n \rightarrow \infty} \int f_n$$

or, since $f = \lim_{n \rightarrow \infty} f_n$:

$$\int \lim_{n \rightarrow \infty} f = \lim_{n \rightarrow \infty} \int f_n$$

i.e. the limit commutes

It might be tempting to then say that $\int$ is “continuous” in the appropriate domain since it commutes with limits. However, the f_i must be *increasing* for the monotone convergence theorem to apply, and hence it is better to say that $\int$ is continuous from below. We will later show that the limit need not always commute (and hence it will be inappropriate to say that the integral is continuous on the space of measurable functions).

2.2.6 Remark any times in Analysis, when the limit can commute with something (ex. integral, another limit, functions), then the proof that it commutes is usually given a name (ex. sequential continuity, MCT, DCT, Fatou lemma, Moore-Osgood theorem, etc.) see:

[wikipedia: Interchanging Limits](#)

Proof:

We’ll show $\leq$ and $\geq$. For $\leq$, notice that $\{\int f_n\}$ is an increasing sequence, and so converges (possibly to ∞). Furthermore, $f_n \leq f$ for all n , and so by proposition 2.2.3 updated for general integral functions, we have:

$$\int f_n \leq \int f$$

and so

$$\lim_{n \rightarrow \infty} \int f_n \leq \int f$$

Conversely, we’ll show that if we “shrink” the left hand side by a factor of $\alpha \in (0, 1)$ so that we get $\geq$, then since this will be true for all α , we will get the result we want.

Fix some $\alpha \in (0, 1)$, and choose a simple function where $0 \leq \varphi \leq f$ (such a function exists by theorem 2.1.18). Let

$$E_n = \{x \in \mathbb{R} \mid f_n(x) \geq \alpha \varphi(x)\}$$

Then $\{E_n\}_{n=1}^\infty$ is an increasing sequence (via inclusion) of measurable sets whose union is X (since $\lim_{n \rightarrow \infty} f_n = f$ and the f_n ’s are increasing). Then

$$\int f_n \geq \int_{E_n} f_n \geq \alpha \int_{E_n} \varphi$$

Since the map $E_n \mapsto \int_{E_n} \varphi$ is measurable by proposition 2.2.3 and the E_n ’s are continuous from

above, we have

$$\lim_{n \rightarrow \infty} \int_{E_n} \varphi = \int \varphi$$

and so

$$\lim_{n \rightarrow \infty} \int f_n \geq \alpha \int \varphi$$

Since this is true for all $\alpha < 1$, it is true for $\alpha = 1$, and so $\int f_n \geq \int \varphi$. Taking the supremum over all φ , we get that

$$\lim_{n \rightarrow \infty} \int f_n \geq \int f$$

And since we showed the $\leq$ direction, we in fact have:

$$\lim_{n \rightarrow \infty} \int f_n = \int f$$

as we sought to show

Thus, to find the value of $\int f$, it suffices to find some sequence of simple functions $\{\varphi_n\}$ that converges to f , which always exists since every measurable function has a standard representation (theorem 2.1.18).

An immediate consequence of the Monotone Convergence Theorem is that linearity also applies to integrals:

Proposition 2.2.7: Linearity of Integrals

Let $\{f_i\}_{i=1}^N \subseteq L^+$ where $N \in \mathbb{N} \cup \{\infty\}$ and let $f = \sum_i^N f_i$. Then

$$\int f = \sum_i^N \int f_i$$

Proof:

Let's first consider the case where $n = 2$ so that we have f_1 and f_2 . Let $\{\varphi_i\}$ and $\{\psi_i\}$ be a sequence for the standard representation of f_1 and f_2 . Then clearly $\{\varphi_i + \psi_i\}$ is a sequence for a standard representation of $f_1 + f_2$. Then by the Monotone Convergence Theorem and linearity

of integrals with simple functions:

$$\begin{aligned}
 \int f_1 + f_2 &= \lim_{i \rightarrow \infty} \int \varphi_i + \psi_i \\
 &\stackrel{\text{MCT}}{=} \int \lim_{i \rightarrow \infty} \varphi_i + \psi_i \\
 &= \int \lim_{i \rightarrow \infty} \varphi_i + \lim_{i \rightarrow \infty} \int \psi_i \\
 &= \int \lim_{i \rightarrow \infty} \varphi_i + \int \lim_{i \rightarrow \infty} \psi_i \\
 &\stackrel{\text{MCT}}{=} \lim_{i \rightarrow \infty} \int \varphi_i + \int \lim_{i \rightarrow \infty} \psi_i \\
 &= \int f_1 + \int f_2
 \end{aligned}$$

By induction, it holds that

$$\int \sum_i^n f_i = \sum_i^n \int f_i$$

Letting $n \rightarrow \infty$, we can once again apply the Monotone Convergence Theorem and get

$$\begin{aligned}
 \int \sum_i^\infty f_i &= \int \lim_{n \rightarrow \infty} \sum_i^n f_i \\
 &\stackrel{\text{MCT}}{=} \lim_{n \rightarrow \infty} \int \sum_i^n f_i \\
 &= \lim_{n \rightarrow \infty} \sum_i^n \int f_i \\
 &= \sum_i^\infty \int f_i
 \end{aligned}$$

completing the proof

The next result is a generalization from the same result with Riemann integrals, except we replace the condition of “finitely many points” with “null-set amount of points”. It follows the trend of integration “forgetting” up to a zero-measure amount of points:

Proposition 2.2.8: Zero Integral

Let $f \in L^+$. Then $\int f = 0$ if and only if $f = 0$ a.e.

Proof:

Starting with simple functions, this is evident in both directions. If φ was our simple function and $\int \varphi = 0$, then $\sum_i^n a_i \mu(E_i) = 0$, so either each $a_i = 0$ or each $\mu(E_i) = 0$ which immediately implies φ is zero a.e. (not necessarily everywhere since it's possible that $E_i \neq \emptyset$ but $\mu(E_i) = 0$).

Conversely, if each $\mu(E_i) = 0$ a.e., then clearly $\int \varphi = 0$.

Moving onto to a measurable function f , the $\Leftarrow$ comes almost immediately. If $f = 0$ a.e., then for every simple function $\varphi \leq f$, then $\varphi = 0$ a.e., which we have established means $\int \varphi = 0$. Then by definition of $\int f$:

$$\int f = \sup_{\varphi \leq f} \int \varphi = 0$$

For the $\Rightarrow$ direction, assume that $\int f = 0$, and for the sake of contradiction let's say $f \neq 0$ a.e. The idea is that we can then construct a simple function φ where $\varphi \leq f$ whose integral is greater than zero, but $\int f = \sup_{\varphi \leq f} \int \varphi$. To that end take:

$$\bigcup_i E_n = \bigcup_i \left\{ x \mid f(x) > \frac{1}{n} \right\} = \{x \mid f(x) > 0\} = E$$

Since $f \neq 0$ a.e., it must be that $\mu(E) > 0$, and so by construction for some n , $\mu(E_n) > 0$. But then

$$f \geq n^{-1} \chi_{E_n} \quad \Leftrightarrow \quad \int f \geq n^{-1} \mu(E_n) > 0$$

so the supremum of values in f contain simple functions of nonzero measure, showing that $\int f > 0$, a contradiction to our original assumption.

This theorem also tells us that if $f = g$ a.e., then $\int f = \int g$, since $f - g = 0$ a.e., so $\int f - g = 0$, so $\int f = \int g$. Using this, we can upgrade the Monotone Convergence Theorem by needing the convergence to happen up to a measure zero set:

Corollary 2.2.9: MCT a.e.

Let $\{f_n\}_{n=1}^\infty \subseteq L^+$ and $f \in L^+$. Then if $f_1 \leq f_2 \leq \dots \leq f$ and $f_n \rightarrow f$ a.e., then

$$\int f = \lim_{n \rightarrow \infty} \int f_n$$

Proof:

Let $f_n(x) \uparrow f(x)$ for all $x \in E$ such that $\mu(E^c) = 0$. Then $f - f \chi_E = 0$ a.e. which implies $f_n - f_n \chi_E = 0$ a.e., so by the Monotone Convergence Theorem

$$\int f = \int f \chi_E \stackrel{\text{MCT}}{=} \lim_{n \rightarrow \infty} \int f_n \chi_E = \lim_{n \rightarrow \infty} \int f_n$$

as we sought to show.

We next address the fact that the sequence $\{f_i\}$ must be increasing (at least a.e.) for the MCT to work (recall we relied on the sequence being increasing in the construction of one of the directions of the inequality). If it were not, then it need not be the case that the limit passes through the integral!

Example 2.2.10: not increasing $\nrightarrow$ Limit commutes with $\int$

Let $X = \mathbb{R}$ and μ be the Lebesgue measure. Consider the two following examples:

$$\{n\chi_{(0,1/n)}\}_{n=1}^{\infty}$$

then it is clear that $n\chi_{(0,1/n)} \rightarrow 0$. However, notice that $\int n\chi_{(0,1/n)} = 1$ for each element in the sequence. Thus:

$$0 = \lim_{n \rightarrow \infty} \int n\chi_{(0,1/n)} \neq \int \lim_{n \rightarrow \infty} n\chi_{(0,1/n)} = 1$$

Even if the function is bounded, the problem remains: take $\chi_{(n,n+1)}$ which also converges pointwise to 0 but the sequence of the integrals of the functions is a sequence of constants 1 and so converges to 1 which is clearly not 0.

This is interesting if we interpret it from a sort of “physics” perspective: somehow, we have “lost our mass” in the process of taking our limit. There is in fact essentially 3 ways we can “lose mass”, the first two begin the counter-examples above, the last one being a function that oscillates faster and faster (this last one requires a function that is also defined in the negative, and so we will get back to this)

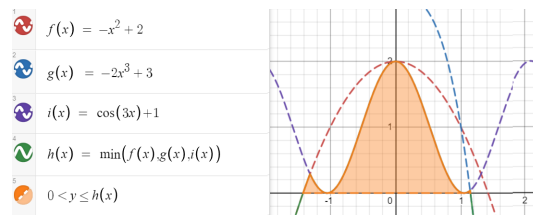
Notice that in both examples, there is some “escaping to infinity” going on. Perhaps if we have some overarching “bound” to the sequence, then we can pass the limit through, and that will indeed be the case. We address this when talking about the Dominated Convergence Theorem in the next section. However, not all hope is lost for the general case; there is weakening to only the $\liminf$ and the equality to $\leq$ which works in the general case:

Theorem 2.2.11: Fatou’s Lemma

If $\{f_n\} \subseteq L^+$ is *any* sequence, then

$$\int \liminf_n f_n \leq \liminf_n \int f_n$$

To visualize this, consider:



The integral of the orange region will be less than or equal to the $\liminf$ of those functions over that region. In this case, the integral would be equal, but in the case where the mass somehow “escapes”, we will have an inequality.

Proof:

This comes directly from properties of the $\liminf$. Let $k \geq 1$. Then

$$\inf_{i \geq k} f_i \leq f_j \quad \forall j \geq k$$

Then by theorem 2.2.3

$$\int \inf_{i \geq k} f_i \leq \int f_j \quad \forall j \geq k$$

and since this holds for all $j \geq k$, we get:

$$\int \inf_{i \geq k} f_i \leq \inf_{j \geq k} \int f_j$$

Since the $\liminf$ defines an increasing sequence of function, by letting $k \rightarrow \infty$ and applying the Monotone Convergence Theorem:

$$\int \liminf f_n = \lim_{k \rightarrow \infty} \int \inf_{n \geq k} f_n \leq \liminf \int f_n$$

as we sought to show

Like before, the result holds if we loosen the condition so that $f_n \rightarrow f$ a.e.

Corollary 2.2.12: Fatou's Lemma a.e.

let $\{f_n\} \subseteq L^+$ be any sequence and $f \in L^+$. If $f_n \rightarrow f$ a.e. Then

$$\int f \leq \liminf_n \int f_n$$

Proof:

Using proposition 2.2.9, make $f_n \rightarrow f$ everywhere arguing that it is equivalent to do so, and then the rest is Fatou's Lemma.

Finally, we show that if $\int f < \infty$, we can guarantee that f cannot “explode” too much, and that even better, the set over which the value of f is non-zero is σ -finite, i.e., it is not “pathological”

Proposition 2.2.13

Let $f \in L^+$. If $\int f < \infty$, then $\{x \mid f(x) = \infty\}$ is a null set and $\{x \mid f(x) > 0\}$ is σ -finite

Proof:

was left as an exercise to the reader in the book

2.3 Integration of Real and Complex Functions

We'll again let $(X, \mathcal{M}, \mu)$ be the fixed measure space for this section, except the codomain now will be either $\mathbb{R}$ or $\mathbb{C}$. For the case of $\mathbb{R}$, we can quickly extend our results from last section by splitting f into f^+ and f^- (which are both measurable if f is by corollary 2.1.12), and if at least one of f^+ or f^- is finite, then

$$\int f = \int f^+ - \int f^-$$

Notice that if they are both infinity, we have $\infty - \infty$ which is in general not well-defined. Thus, we will limit our attention to functions where $\int |f| < \infty$ since $|f| = f^+ + f^-$:

For the case of $\mathbb{C}$, using the triangle inequality we get that:

$$|f| = |\operatorname{Re} f + i \operatorname{Im} f| \leq |\operatorname{Re} f| + |\operatorname{Im} f| \leq 2|f|$$

and so both $\operatorname{Re} f$ and $\operatorname{Im} f$ must be finite to get a well-defined result. In that case, we define

$$\int f := \int \operatorname{Re} f + i \int \operatorname{Im} f$$

We give a name for complex functions (the real functions being a special case) that satisfy this finiteness condition:

Definition 2.3.1: Integrable Function

Let $f : X \rightarrow \mathbb{C}$ be measurable. We say f is *integrable* if

$$\int |f| < \infty$$

Let L^1 represent the set of all integrable functions.

Like with L^+ , we sometimes write $L^1(\mu)$, $L^1(X)$, or $L^1(X, \mu)$. If the codomain is $\mathbb{R}$, then we get that $\int |f^+| < \infty$ and $\int |f^-| < \infty$, as we claimed for our extension of the integral to $\mathbb{R}$. Furthermore, for any $E \in \mathcal{M}$, we will say that f is *integrable on E* if $\int_E |f| < \infty$.

The set of L^1 functions are well-behaved under addition and scalar multiplication in the sense that they form a vector-space:

Proposition 2.3.2: Vector Space on L^1

Let L^1 be the set of integrable functions. Then with $+$ and scalar multiplication, L^1 forms a vector-space (over $\mathbb{R}$ or $\mathbb{C}$ respective to the codomain). Furthermore, $\int$ is a linear functional on L^1 (i.e. $\int : L^1 \rightarrow \mathbb{R}$ is a linear functional)

Proof:

the proof that $cf + dg$ is integrable comes from the triangle inequality ($|cf + dg| \leq |c||f| + |d||g| < \infty$). The fact that $cf \in L^1$ for any $c \in \mathbb{R}$ and $f \in L^1$ follows similarly.

$\int$ being a functional, we need to show linearity and $\int cf = c \int f$. For linearity, assume $h = f + g$

is integrable. Then $h^+ - h^- = f^+ - f^- + g^+ - g^-$, or, re-arranging, we get:

$$h^+ + f^- + g^- = h^- + f^+ + g^+$$

These are all functions in L^+ , and so by linearity for of integrals for L^1 functions:

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+$$

re-arranging, we get:

$$\begin{aligned} &= \int f + g \\ &= \int h = \int h^+ - \int h^- = \int f^+ - \int f^- + \int g^+ - \int g^- \\ &= \int f + \int g \end{aligned}$$

showing linearity. Through a similar means, we can also show $\int cf = c \int f$, completing the proof.

2.3.3 Remark Since it is a vector-space, it has a basis and a dimension. What dimension is this vector-space? Since it is a real vector-space, it is isomorphic to $\mathbb{R}^N$ for appropriate cardinal N (N may be finite, countable, uncountable, etc). If $N \geq \aleph_0$, that means that it is rather difficult to define a measure on the space L^{1^a} . Trying to find ways to study “measure” notions infinite dimensional vector-spaces (ex. boundedness, σ -finiteness, etc.) is accomplished by defining various different concepts like *seminorms*, *norms*, or *inner products*, which we shall do in chapter ??.

^aFor an illustration of the oddities of infinite dimensional vector-spaces, consider the [hyper]-volume of a unit ball S^n as $n \rightarrow \infty$; you should find that it goes to 0!

Next, we would like to establish some properties of integrable functions. One result is a including a useful inequality including absolute values. Another is how the results still work as long as things happen “a.e.”. Finally, the set of all “interesting” points (points where $f(x) \neq 0$) are still σ -finite:

Proposition 2.3.4: Properties of Integrable Functions

Let $f, g \in L^1$. Then:

1. $|\int f| \leq \int |f|$
2. $\{x \mid f(x) \neq 0\}$ is σ -finite
3. $\int_E f = \int_E g$ for all $E \in \mathcal{M}$ if and only if $\int |f - g| = 0$ if and only if $f = g$ a.e.

Proof:

1. The result is essentially immediate in the $\mathbb{R}$ case. In the $\mathbb{C}$ case, let α where $|\alpha| = 1$ be such that it “rotates” the integral so that it’s in the real case, and then apply a similar strategy to that of the $\mathbb{R}$ case.
2. Notice that $\{x \mid f(x) \neq 0\} = \{x \mid f^+(x) > 0\} \cup \{x \mid f^-(x) < 0\}$, which we’ve shown in proposition 2.2.13 to be σ -finite
3. The fact that $\int |f - g| = 0$ if and only if $f = g$ a.e. comes from proposition 2.2.8, (since $|f - g| \in L^+$) so we will prove $\int_E f = \int_E g$ if and only if $\int |f - g| = 0$. Let’s say $\int |f - g| = 0$. Then to show that $\int_E f = \int_E g$, it is equivalent to show that

$$\left| \int_E f - \int_E g \right| = 0$$

by the epsilon principle (i.e. $a = b$ if and only if $|a - b| < \epsilon$ for all ϵ). Using proposition 2.3.4, we get that:

$$\left| \int_E f - \int_E g \right| \leq \int \chi_E |f - g| \leq \int |f - g| = 0$$

completing the $\Rightarrow$ direction.

For the other direction, let’s argue contra-positively and assume $\int |f - g| \neq 0$ a.e. Let

$$u = \operatorname{Re}(f - g) \quad v = \operatorname{Im}(f - g)$$

so that u^+, u^-, v^+, v^- are the corresponding functions. Since $f \neq g$ a.e., at least one of these must have a set with positive measure. Without loss of generality, let’s say $E = \{x \mid u^+(x) > 0\}$ and $\mu(E) > 0$. Then

$$\operatorname{Re} \left(\int_E f - g \right) = \operatorname{Re} \left(\int_E f - \int_E g \right) = \int_E u^+ + \int_E u^- = \mu(E) + 0 > 0$$

since $u^- = 0$ on E . Thus, the measure will be positive.

As a consequence of this, we usually think of integrable functions *up to a zero set*. If we wanted to, we can limit our scope to of integration to functions f that are measurable on E and E^c has zero-measure to functions where $f = 0$ on E^c . This also means that if we are working with an $\mathbb{R}$ -value function that is finite a.e., then it is equivalent to treat it as a $\mathbb{R}$ -value function.

Proposition 2.3.4 also allows us to re-define L^1 in terms of functions that are equivalent up to a measure zero set. That is, we form an equivalence relation on L^1 if and only if $f = g$ a.e., and label the collection of equivalence classes as L^1 . We abuse notation and write “ $f \in L^1$ ” to mean we are selecting a function that is integrable (or an a.e.-defined integrable function). This is because we will usually be working with functions instead of the equivalence classes, and so this is a common abuse of notation.

The reason to introduce this new definition of L^1 is so that we can define a new metric on L^1 :

Definition 2.3.5: L^1 metric

Let L^1 be the set of equivalence classes as defined above. Then

$$\rho(f, g) = \int |f - g|$$

is a metric on L^1

Symmetry and triangle inequality come immediately without even needing to use the fact that these are equivalence classes. Where the new construction of L^1 becomes important is to say $\int |f - g| = 0$ if and only if $f = g$. Since this is true a.e., then we need these two to be in an equivalence class. Since we defined a metric, we can define a notion of convergence, in particular, f_n converges to f in L^1 if and only if $\int |f - f_n| \rightarrow 0$, meaning $f_n \rightarrow f$ almost everywhere (however, L^1 convergence does not imply a.e. point-wise convergence, and a.e. point-wise convergence does not imply L^1 convergence, as we'll show in section 2.4).

We will soon explore more properties of this metric space, in particular, we would want this metric space to be complete so that the limit of a sequence of L^1 function's is L^1 and find some nice dense set of L^1 (so far, we only have the [point-wise] limit of measurable function is measurable, but no such result yet for integrable functions). To do so, we need more tools to analyze swapping limits and integrals. The next result we'll cover allows us to generalize the Monotone Convergence Theorem to the general case of $f_n \rightarrow f$ a.e., with the extra condition that each $|f_n|$ is bounded by a single $|g|$ with finite area (in this way, we are getting rid of the "escaping to infinity" cases in example 2.2.10)

Theorem 2.3.6: Dominated Convergence Theorem (DCT)

Let $\{f_n\}_{n=1}^{\infty} \subseteq L^1$ be a sequence of integrable functions such that

1. $f_n \rightarrow f$ a.e.
2. there exists a nonnegative $g \in L^1$ such that for all $n \in \mathbb{N}$, $|f_n| \leq g$ a.e.

Then

$$\int f = \lim_{n \rightarrow \infty} \int f_n$$

or equivalently:

$$\int \lim_{n \rightarrow \infty} f_n = \lim_{n \rightarrow \infty} \int f_n$$

Proof:

By proposition 2.1.19 and 2.1.20, f is measurable (up to a null set, in which case we can appropriately redefine f on that null-set). Since $|f_n| < g$ for all $n \in \mathbb{N}$, $f < g$, and so $f \in L^1$.

Since all L^1 functions are broken down to real parts (complex functions into 2 real parts), it suffices to show the result for real value f_n and f . By assumption, we have $g + f_n \geq 0$ and $g - f_n \geq 0$. Thus, by Fatou's Lemma:

$$\int g + \int f \leq \liminf \int (g + f_n) = \int g + \liminf \int f_n$$

$$\int g - \int f \leq \liminf \int (g - f_n) = \int g - \limsup \int f_n$$

Therefore, $\limsup \int f_n \leq \int f \leq \liminf \int f_n$, and so the two values are actually equal and so

$$\int f = \lim_{n \rightarrow \infty} \int f_n$$

as we sought to show

Note that functions can still converge even if the hypothesis of the DCT fails:

Example 2.3.7: Converges Without DCT

Construct a sequence of functions $\{f_n\}$ such that all the following hold:

1. $f_n : [0, 1] \rightarrow \mathbb{R}$ is continuous
2. $f_n \geq 0$
3. $\lim_{n \rightarrow \infty} f_n(x) = 0$ for all $x \in [0, 1]$
4. $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dm = 0$
5. $\sup_n f_n(x)$ is *not* integrable on $[0, 1]$ with respect to the Lebesgue measure

(that is, just because the hypothesis of the dominated convergence theorem fails does not mean the conclusion does not hold: it is not an if and only if)

Proof. The proof essentially is formalizing the following images:

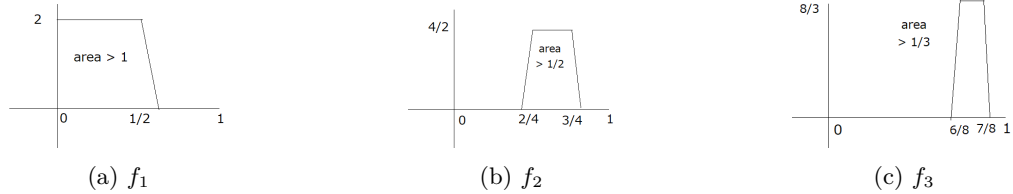


Figure 2.1: Visual intuition for question 3

We'll first define the indicator functions, then make these functions continuous. Let

$$g_1(x) = \begin{cases} 2^1/1 & x \in [0, \frac{1}{2}) \\ 0 & x \in [1 - \frac{1}{2}, 1] \end{cases} \quad g_2(x) = \begin{cases} 2^2/2 & x \in [\frac{1}{2}, \frac{3}{4}) \\ 0 & x \in [1 - \frac{1}{4}, 1] \end{cases} \quad g_3(x) = \begin{cases} 2^3/3 & x \in [\frac{3}{4}, \frac{7}{8}) \\ 0 & x \in [1 - \frac{7}{4}, 1] \end{cases}$$

and so on, defining:

$$g_n(x) = \begin{cases} 2^n/n & x \in [2^{n-1} - 2/2^{n-1}, 2^{n-1} - 1/2^{n-1}) \\ 0 & x \in [1 - 2^{n-1} - 1/2^{n-1}, 1] \end{cases}$$

Next, we can make each g_n continuous by simply adding a line to omit the jump:

$$f_1(x) = \begin{cases} 2^1/1 & x \in [0, 1/2) \\ -2^{10 \cdot 1} (x - (\frac{1}{2} - 1/2^{10 \cdot 1})) & x \in (1/2, 1/2 + 1/2^{10 \cdot 1}] \\ 0 & x \in [1 - (1/2 + 1/2^{10 \cdot 1}), 1] \end{cases}$$

and defining each f_n similarly. Then it is clear by our construction that $f_n \rightarrow 0$: since the trapezoids are moving away, they will never cross our values again once they pass them (this is the opposite of the proof for question 2 where they constantly loop back, since the proof is similar the details were omitted for this proof). Furthermore,

$$\int f_n = 1/n + 1/2^{10n}$$

Thus, $\int f_n \xrightarrow{n \rightarrow \infty} 0$, showing that the integral also converges to 0. However,

$$\int \sup_n f_n \geq \int \sup_n g_n = \sum_{k=1}^{\infty} 1/k = \infty$$

showing that the $\sup_n f_n$ is *not* integrable, as we sought to show. □

Therefore, the DCT is a sufficient, but not necessary condition for convergence. Sufficient and necessary conditions are a bit harder to come by, and we will not need to characterize limits commuting with integrals in these other ways, but for those who are curious:

<https://journalofinequalitiesandapplications.springeropen.com/articles/10.1186/s13660-020-02502-w>

This theorem will be key in finding the integral of many function constructed by limits, and will be used to prove completeness of L^1 (after defining Cauchy sequences in L^1 in section ref:HERE) A more immediate result is that the Dominated convergence theorem allows to define a form of countable linearity for L^1 functions:

Theorem 2.3.8: Linearity for Integrable Functions

Let $\{f_i\}_{i=1}^{\infty} \subseteq L^1$ be a sequence of functions, and $f = \sum_{i=1}^{\infty} f_i$. If $\sum_{i=1}^{\infty} |f_i| < \infty$, then $\sum_{i=1}^{\infty} f_i$ converges a.e. to a function in L^1 , and

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

Proof:

First, since $\sum_{i=1}^{\infty} |f_i| < \infty$, by proposition 2.2.7,

$$\int \sum_{i=1}^{\infty} |f_i| = \sum_{i=1}^{\infty} \int |f_i|$$

Thus, $g = |\sum_{i=1}^{\infty} f_i|$ is a function in L^1 . The function g is finite a.e. by proposition 2.3.4[2], so by the dominated convergence theorem, any partial sum of functions will commute, and so taking the limit, we get:

$$\int \sum_{i=1}^{\infty} f_i = \sum_{i=1}^{\infty} \int f_i$$

as we sought to show

The dominated convergence theorem also gives us a dense set in L^1 to work with, namely the simple functions:

Theorem 2.3.9: Simple Functions Dense in L^1

Let $f \in L^1$. Then for all $\epsilon > 0$, there exists an integrable simple function $\varphi = \sum a_i \chi_{E_i}$ such that

$$\int |f - \varphi| d\mu < \epsilon$$

Furthermore, if μ is a Lebesgue-Stieljes measure on $\mathbb{R}$, the sets E_i can be taken to be finite union of open intervals, and better, there is a continuous function g that vanishes outside a bounded interval (i.e. $g = 0$ outside the bounded interval) such

$$\int |f - g| d\mu < \epsilon$$

Proof:

p. 55 in the book. Also proven in fuller generality that simple functions are dense in L^p spaces, which includes L^1 .

For the purposes of Ordinary Differential Equations (ODE's), we want to establish is a quick result to relate differentiable functions and integral of functions:

Theorem 2.3.10: Relating Derivative and Integral

Let $(X, \mathcal{M}, \mu)$ be a measure space, and suppose $f : X \times [a, b] \rightarrow \mathbb{C}$ for $-\infty < a < b < \infty$ and that $f(\cdot, t) : X \rightarrow \mathbb{C}$ is integrable for every $t \in [a, b]$, so we can define $F(t) = \int_X f(x, t) d\mu$.

1. Suppose that there exists a $g \in L^1$ such that $|f(x, t)| \leq g(x)$ for all x, t . Then if $\lim_{t \rightarrow t_0} f(x, t) = f(x, t_0)$ for every x , $\lim_{t \rightarrow t_0} F(t) = F(t_0)$; that is, if $f(x, \cdot)$ is continuous at every x , so is F .
2. Suppose $\frac{\partial f}{\partial t}$ exists and there is a $g \in L^1(\mu)$ such that $\left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x)$ for all x, t . Then F is differentiable and

$$F'(x) = \int \frac{\partial f}{\partial t}(x, t) d\mu(x)$$

The way I like to think about is that there is a system X that at different times t acts/behaves differently. This difference in how the system acts at any given time t is captured by $f(x, t)$. The function F capture the “total value” of this action/behavior (note that f is non-negative too, so all behavior has a “non-negative” output).

Now, if the behavior varies continuously, then the total output varies continuously, and if the behavior is smooth enough so that it is differentiable (at any slice), then F is also differentiable, and it is in fact even easy to compute the value of F' .

Proof:

1. Since $|f(x, t)|$ is dominated by g , simply apply the Dominated Convergence Theorem:

$$\lim_{t \rightarrow t_0} F(t) = \lim_{t \rightarrow t_0} \int f(x, t) d\mu(x) = \int \lim_{t \rightarrow t_0} f(x, t) d\mu(x) = \int f(x, t_0) d\mu(x) = F(t_0)$$

where $\{t_n\}$ is any sequence converging to t (since any $f(x, t)$ satisfies the bound, no more restrictions are needed on the choice of sequence)

2. Since $\frac{\partial f}{\partial t}$ exists, the limit of the derivative exists, and so define

$$\frac{\partial f}{\partial t} = \lim_{t_n \rightarrow t} h_n(x) \quad h_n(x) = \frac{f(x, t_n) - f(x, t_0)}{t_n - t_0}$$

$\{t_n\}$ is any sequence converging to t . Since h_n difference and division of measurable functions, by proposition 2.1.10 it is measurable, and by the Dominated Convergence Theorem $\frac{\partial f}{\partial t}$ is measurable. Furthermore, by the Mean Value Theorem:

$$|h_n(x)| \leq \sup_{t \in [a, b]} \left| \frac{\partial f}{\partial t}(x, t) \right| \leq g(x)$$

and so by the Dominated Convergence Theorem again:

$$\begin{aligned} F'(t_0) &= \lim_{t_n \rightarrow t} \left(\frac{F(t_n) - F(t_0)}{t_n - t_0} \right) \\ &= \lim_{t_n \rightarrow t} \left(\int h_n(x) d\mu(x) \right) \\ &\stackrel{\text{D.C.T.}}{=} \int \lim_{t_n \rightarrow t} h_n(x) d\mu(x) \\ &= \int \frac{\partial f}{\partial t} d\mu(x) \end{aligned}$$

completing the proof

The final fact we shall present is a continuation of proposition 2.2.13 in the sense that we aim to further establish connections between some subsets of the domain and the resulting integral value in the co-domain, and see that if $\int$ is limited to some finite area of integration E ($\int_E < \infty$), does it actually respect continuity? In fact, it does! Even better, it is in a sense uniformly continuous with respect to the measure of the set that is being integrated over.

Proposition 2.3.11: $\int$ Uniform Continuity

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f \in L^1$. Then for all $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\mu(E) < \delta \Rightarrow \int_E |f| < \epsilon$$

Proof:

Without loss of generality, let $f \geq 0$. Let $\epsilon > 0$, and choose φ such that $\int f < \int \varphi + \epsilon/2$, i.e. $\int f - \int \varphi < \epsilon/2$. Next, since φ is a simple function, we have that $\sup \varphi = M < \infty$ (being the finite sum of constant terms). Now, set $\delta = \frac{\epsilon}{2M}$ so that $\mu(E) < \delta$.

$$\int_E f - \varphi \leq \int f - \varphi$$

then re-arranging we get:

$$\int_E f \leq \int f - \varphi + \int_E \varphi < \frac{\epsilon}{2} + \frac{M\epsilon}{2M} = \epsilon$$

as we sought to show.

As a consequence of this, if we have a space of integrable functions $\mathcal{F}$ with the sup norm (convergence in the sup norm is equivalent to convergence), then for any $\mu(E) < \infty$, $\int_E$ is in fact *continuous*, that is, if $f_n \rightrightarrows f$, then:

$$\int_E \lim f_n = \lim \int_E f_n$$

let's fix some E where $\int_E 1 = M < \infty$. Take $f_n \rightrightarrows f$ so that $|f(x) - f_n(x)| < \epsilon$ for all $x \in E$ when $n > N$. Then:

$$\begin{aligned} \left| \int_E f(x) dx - \int_E f_n(x) \right| &= \left| \int_E f(x) - f_n(x) \right| \\ &\leq \int_E |f(x) - f_n(x)| \\ &\leq \int_E \epsilon \\ &= \epsilon M \end{aligned}$$

It quickly follows from here that the limit does indeed commute.

2.3.1 Relating Lebesgue and Riemann Integral

Let's consider $f \in L^1(\mathbb{R})$ to be a Lebesgue integrable function on $\mathbb{R}$. As a preliminary to this book, Riemann integration of real-valued functions have been studied. So far, we have done nothing to show that the definitions of the integral will produce the same result. They surely must, since the Riemann integral correctly captures the area under surfaces. In this section, we show how the Lebesgue integral is in fact an extension of the Riemann integral, meaning if we restrict to the set

of all Riemann integrable functions (which will all be Lebesgue integral), then the Lebesgue and Riemann integral give the same result.

As a quick reminder of Riemann integration, choosing to follow Darboux' characterization, let $[a, b]$ be a closed interval (or more generally a compact subset of $\mathbb{R}^n$) Then for any finite partition $P = \{t_0, \dots, t_n\}$ where $a = t_0 < t_1 < \dots < t_n = b$, define:

$$U_P(f) = \sum_{i=1}^n M_i(t_i - t_{i-1}) \quad L_P(f) = \sum_{i=1}^n m_i(t_i - t_{i-1})$$

where M_i and m_i are the supremum and infimum of f on $[t_{i-1}, t_i]$. Then take:

$$\bar{I}_a^b(f) = \inf_P U_P(f) \quad \underline{I}_a^b = \sup_P L_P(f)$$

over all possible partitions P . If $\bar{I}_a^b(f) = \underline{I}_a^b$, then their common value is denoted as $\int_a^b f(x) dx$ and f is called Riemann integrable on $[a, b]$.

We'll now compare this to the Lebesgue integral:

Theorem 2.3.12: Lebesgue-Vitali Theorem

Let f be a bounded real-valued function on $[a, b]$.

1. If f is Riemann integrable, then f is Lebesgue measurable (and hence integrable on $[a, b]$ since f is bounded) and

$$\int_a^b f(x) dx = \int_{[a,b]} f dm$$

2. f is Riemann integrable if and only if $\{x \in [a, b] \mid f \text{ is discontinuous at } x\}$ has Lebesgue measure 0.

Note that $\chi_{\mathbb{Q}}$ on $[0, 1]$ is discontinuous everywhere, with discontinuities having Lebesgue measure 1, was shown to not be Riemann integrable (this was in fact the example of a non Riemann-integrable function presented in the preliminary chapter). However, it is certainly Lebesgue integrable!

Proof:

1. Let f be Riemann integrable. Let M_i and m_i are defined as the supremum and infimum as defined above, and for each partition P , let:

$$G_P = \sum_{i=1}^n M_i \chi_{(t_{i-1}, t_i]} \quad g_P = \sum_{i=1}^n m_i \chi_{(t_{i-1}, t_i]}$$

so $U_P(f) = \int G_P dm$ and $L_P(f) = \int g_P dm$. Now, there exists a sequence of partitions P_n such that $\max_i(t_i - t_{i-1})$ tends to zero and P_{n+1} is a refinement of P_n (meaning g_{P_n} increase in value while G_{P_n} decreases) such that $U_{P_n}(f)$ and $L_{P_n}(f)$ converge to $\int_a^b f(x) dx$. Let $G = \lim G_{P_n}$ and $g = \lim g_{P_n}$. Since $g \leq f \leq G$, by the dominated convergence theorem:

$$\int G dm = \int g dm = \int_a^b f(x) dx$$

Therefore, $\int G - g = 0$ and so $G = g$ a.e. by proposition 2.2.8, and so $G = f$ a.e.

Finally, since G is measurable (being the point-wise limit of measurable functions) and m is complete, f is measurable and

$$\int_{[a,b]} f dm = \int G dm = \int_a^b f(x) dx$$

2. This was exercise 23 in the textbook

The big advantage we have with this is that we can now use the computational techniques that were developed for Riemann integrals to compute particular Lebesgue integrals! In most cases in analysis, functions are locally Riemann integrable, and so we might what does the generality give us. The main thing it gives us is *completeness*. So though computing Lebesgue integrals might not be so easy (or if they are not an elementary Riemann integrable function), they are easier to work with in a theoretical framework, and help us define concepts like L^p space (which are important for Fourier Analysis).

(Note that in the infinite case, the Riemann and Lebesgue integral need not match. For example, $\sin(x)/x$ is Riemann integrable and produce a finite value, but it is not Lebesgue integral, producing an infinite value))

(A quick word on the Gamma function)

2.4 Modes of Convergence

So far, we have accumulated the following forms of convergence of functions:

1. uniform convergence $f_n \rightrightarrows f$
2. point-wise convergence $f_n \rightarrow f$
3. point-wise convergence a.e. $f_n \rightarrow f$ a.e.
4. L^1 convergence $f_n \rightarrow f$ if and only if $\int |f_n - f| \rightarrow 0$

The first 3 are weaker than then the next: uniform convergence $\implies$ point-wise convergence $\implies$ a.e. convergence, but the converse is false in each case. Furthermore, L^1 convergence implies none of those 3, and none of those 3 imply L^1 convergence. The examples to keep in mind to compare are:

1. $f_n = n^{-1} \chi_{[0,n]}$
2. $f_n = \chi_{[n,n+1]}$
3. $f_n = n \chi_{[0,1/n]}$
4. $f_1 = \chi_{[0,1]}$, $f_2 = \chi_{[0,1/2]}$, $f_3 = \chi_{[1/2,1]}$, $f_4 = \chi_{[0,1/4]}$, $f_5 = \chi_{[1/4,2/4]}$, and so on

We now introduce another way of thinking about convergence: instead of saying that the $\int f_n$ approaches that of f (i.e., evaluate the integral and look at the limiting behavior of the integral of the function), we will want there to exist an N such that $n \geq N$, the measure of the difference

between f_n and f are epsilon away from the function we are converging to f . This is not the same thing as L^1 convergence, since this (1), (3) and (4) all converge to 0 in this type of convergence. Before any further analysis let's define our terms:

Definition 2.4.1: Cauchy in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *cauchy in measure* if:

$$\forall \epsilon > 0, \exists N \text{ such that } \forall n, m \geq N, \mu(\{x \mid |f_m(x) - f_n(x)| \geq \epsilon\}) < \epsilon$$

Equivalently, we can write the definition as $\forall \epsilon > 0$,

$$\mu(\{x \mid |f_n(x) - f_m(x)| \geq \epsilon\}) \rightarrow 0 \text{ as } n, m \rightarrow \infty$$

Definition 2.4.2: Convergence in Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ a sequence of measurable functions. Then $\{f_n\}$ said to be *convergent in measure* or *converge in measure* if:

$$\forall \epsilon > 0, \lim_{n \rightarrow \infty} \mu(\{x \mid |f(x) - f_n(x)| \geq \epsilon\}) = 0$$

With these definition, notice that (2) is in fact *not* Cauchy in measure: the difference between the two functions.

Proposition 2.4.3: L^1 convergence Then Measure Convergence

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $f_n \rightarrow f$ in L^1 . Then $f_n \rightarrow f$ in measure

Proof:

Let ϵ be given, and define

$$E_{\epsilon, n} = \mu(\{x \mid |f_n(x) - f(x)| \geq \epsilon\})$$

Then:

$$\int |f - f_n| \geq \int_{E_{\epsilon, n}} |f - f_n| \geq \epsilon \mu(E_{\epsilon, n})$$

Thus:

$$\mu(E_{\epsilon, n}) \leq \epsilon^{-1} \int |f - f_n| \rightarrow 0$$

showing that it converges in measure.

Note that convergence in measure does not imply convergence in L^1 as we saw in example (1) and (3).

With this established, we can go onto establish why we care about convergence in measure: it gives us another way of finding point-wise convergent functions by establishing a condition weaker than L^1 convergence:

Theorem 2.4.4: Cauchy in Measure Then Pointwise a.e.

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}$ Cauchy in measure. Then there exists a f such that $f_n \rightarrow f$ in measure, and furthermore there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \rightarrow f$ a.e.

If $f_n \rightarrow g$ in measure, then $f = g$ a.e.

Proof:

We can choose a subsequence $\{g_j\} = \{f_{n_j}\}$ of $\{f_n\}$ such that if $E_j = \{x : |g_j(x) - g_{j+1}(x)| \geq 2^{-j}\}$, then $\mu(E_j) \leq 2^{-j}$. If $F_k = \bigcup_{j=k}^{\infty} E_j$, then $\mu(F_k) \leq \sum_{j=k}^{\infty} 2^{-j} = 2^{1-k}$, and if $x \notin F_k$, for $i \geq j \geq k$ we have

$$(2.31) \quad |g_j(x) - g_i(x)| \leq \sum_{l=j}^{i-1} |g_{l+1}(x) - g_l(x)| \leq \sum_{l=j}^{i-1} 2^{-l} \leq 2^{1-j}.$$

Thus $\{g_j\}$ is pointwise Cauchy on F_k^c . Let $F = \bigcap_{k=1}^{\infty} F_k = \limsup E_j$. Then $\mu(F) = 0$, and if we set $f(x) = \lim g_j(x)$ for $x \notin F$ and $f(x) = 0$ for $x \in F$, then f is measurable (see Exercises 3 and 5) and $g_j \rightarrow f$ a.e. Also, (2.31) shows that $|g_j(x) - f(x)| \leq 2^{1-j}$ for $x \notin F_k$ and $j \geq k$. Since $\mu(F_k) \rightarrow 0$ as $k \rightarrow \infty$, it follows that $g_j \rightarrow f$ in measure. But then $f_n \rightarrow f$ in measure, because

$$\{x : |f_n(x) - f(x)| \geq \epsilon\} \subset \{x : |f_n(x) - g_j(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |g_j(x) - f(x)| \geq \tfrac{1}{2}\epsilon\},$$

and the sets on the right both have small measure when n and j are large. Likewise, if $f_n \rightarrow g$ in measure,

$$\{x : |f(x) - g(x)| \geq \epsilon\} \subset \{x : |f(x) - f_n(x)| \geq \tfrac{1}{2}\epsilon\} \cup \{x : |f_n(x) - g(x)| \geq \tfrac{1}{2}\epsilon\}$$

for all n , hence $\mu(\{x : |f(x) - g(x)| \geq \epsilon\}) = 0$ for all ϵ . Letting ϵ tend to zero through some sequence of values, we conclude that $f = g$ a.e.

Corollary 2.4.5: L^1 convergence and Finding Subsequence

Let $f_n \rightarrow f$ in L^1 . Then there exists a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \rightarrow f$ a.e.

Proof:

By proposition 2.4.3, $f_n \rightarrow f$ in measure, and by theorem 2.4.4, there exists a subsequence that converges to f a.e.

As we saw, $f_n \rightarrow f$ a.e. does not imply $f_n \rightarrow f$ in measure, as we saw in example (2). This is due to

the “escape to infinity” that is happening, which is why we care about the dominated convergence theorem to help us stop such “sneaky infinity problem”. However, if μ was a finite measure, then $f_n \rightarrow f$ a.e. does imply $f_n \rightarrow f$ in measure. In fact, since the area is bounded, we can some even stronger convergent conditions:

Theorem 2.4.6: Ergoff’s Theorem

Let $(X, \mathcal{M}, \mu)$ be a measure space and suppose that $\mu(X) < \infty$. If $\{f_i\}$ are measurable complex-valued functions such that $f_n \rightarrow f$ a.e., then for every $\epsilon > 0$, there exists a $E \subseteq X$ such that $\mu(E) < \epsilon$ and $f_n \rightarrow f$ uniformly on $\mu(E^c)$

Proof:

Without loss of generality we may assume that $f_n \rightarrow f$ everywhere on X . For $k, n \in \mathbb{N}$ let

$$E_n(k) = \bigcup_{m=n}^{\infty} \{x : |f_m(x) - f(x)| \geq k^{-1}\}.$$

Then, for fixed k , $E_n(k)$ decreases as n increases, and $\bigcap_{n=1}^{\infty} E_n(k) = \emptyset$, so since $\mu(X) < \infty$ we conclude that $\mu(E_n(k)) \rightarrow 0$ as $n \rightarrow \infty$. Given $\epsilon > 0$ and $k \in \mathbb{N}$, choose n_k so large that $\mu(E_{n_k}(k)) < \epsilon 2^{-k}$ and let $E = \bigcup_{k=1}^{\infty} E_{n_k}(k)$. Then $\mu(E) < \epsilon$, and we have $|f_n(x) - f(x)| < k^{-1}$ for $n > n_k$ and $x \notin E$. Thus $f_n \rightarrow f$ uniformly on E^c .

The type of convergence in this theorem is sometimes called *almost uniform convergence*. Almost uniform convergence implies convergence a.e. and convergence in measure.

2.5 Product Measure

As we defined in definition 1.1.8, If $\mathcal{M}$ and $\mathcal{N}$ are σ -algebras, then we can define a new σ -algebra $\mathcal{M} \otimes \mathcal{N}$. If $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are measure spaces, we will show that we can define a new measure on $\mathcal{M} \otimes \mathcal{N}$ using μ and ν . We will use the tools we have done in constructing measure to construct this measure, and show that there is a natural measure $\mu \times \nu$ with respect to the outer-measure given our space is σ -finite (this is just the uniqueness clause of theorem 1.3.7).

First, we’ll give a name to sets in $\mathcal{M} \otimes \mathcal{N}$ that will be easier to measure (these will be our generating sets)

Definition 2.5.1: [Measurable] Rectangle

Let $A \in \mathcal{M}$ and $B \in \mathcal{N}$. Then $A \times B \in \mathcal{M} \otimes \mathcal{N}$ is called a *measurable rectangle* or just *rectangle* for short

Notice that if $A \times B$ and $C \times D$ are rectangles, they form an elementary family:

$$(A \times B) \cup (C \times D) = (A \cap C) \times (B \cap D) \quad (A \times B)^c = (X \times B^c) \cup (A^c \times B)$$

and so by proposition 1.1.13, the collection of finite disjoint unions of rectangles forms an algebra $\mathcal{A}$, and they generate $\mathcal{M} \otimes \mathcal{N}$.

Furthermore, not all sets in $\mathcal{M} \otimes \mathcal{N}$ are measurable rectangles: if we take $(\mathbb{R}, \mathcal{L}(\mathbb{R}), m)$ and consider $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$, then the set of points $\Delta = \{(x, x) \mid x \in \mathbb{R}\}$ is in $\mathcal{L}(\mathbb{R}) \otimes \mathcal{L}(\mathbb{R})$ (being the countable intersection of ever smaller squares on the line), however it is very far from being the product of two sets in $\mathcal{L}(\mathbb{R})$.

We'll now define how to integrate simple functions on $X \times Y$ with the associated measure spaces $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$: Let $A \times B$ be a rectangle that is the (finite or countable) disjoint union of rectangle $A_i \times B_i$. Then notice that:

$$\chi_{A \times B}(x, y) = \chi_A(x) \chi_B(y)$$

and so:

$$\chi_{A \times B}(x, y) = \sum_i \chi_{A_i \times B_i}(x, y) = \sum_i \chi_{A_i}(x) \chi_{B_i}(y)$$

If we integrate with respect to x , treating $\chi_{B_i}(y)$ as some constant, then by theorem 2.2.7

$$\int \chi_A(x) \chi_B(y) d\mu = \mu(A) \chi_B(y) d\mu = \mu(A) \chi_B(y) = \sum_i \mu(A_i) \chi_{B_i}(y)$$

It might be confusing which variable is associated to which measure, in which case it is common to write:

$$\int \chi_A(x) \chi_B(y) d\mu(x)$$

to emphasize the function with the x variable is the one being measured by μ . Integrating with respect to y yields:

$$\int \mu(A) \chi_B(y) d\nu = \mu(A) \nu(B) = \sum_i \mu(A_i) \nu(B_i)$$

Thus, we can define $\pi : \mathcal{A} \rightarrow \mathbb{R}$ where if $E \in \mathcal{A}$ is a finite disjoint union of rectangles $A_i \times B_i$, then:

$$\pi(E) = \sum_i \mu(A_i) \nu(B_i)$$

with the usual convention of $0 \times \infty = 0$ (i.e., if one of the dimensions is “0”, then we are making the value 0). Notice that π is independent of representation, since any finite disjoint union of rectangles have a common refinement. Thus π is a pre-measure, and so by theorem 1.3.7, π generates an outer measure on $X \times Y$ whose restriction to $\mathcal{M} \otimes \mathcal{N}$ is a measure where restricted to $\mathcal{A}$ is π (or conversely, whose measure extends π). Such a measure is given a name:

Definition 2.5.2: Product Measure

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces, and define π as in the previous construction. Then the induced outer-measure is called the *product measure* and is usually denoted $\mu \times \nu$.

Notice further that if μ and ν are σ -finite, so $\mu(X) = \sum_i^\infty \mu(A_i)$ and $\nu(Y) = \sum_i^\infty \nu(B_i)$, then $(\mu \times \nu)(X \times Y) = \sum_{i,j} \mu(A_i)\nu(B_j)$, and since all terms are finite, $\mu \times \nu$ is also σ -finite, in which case by the same theorem as invoked in defining $\mu \times \nu$, the product measure is the unique measure such that $\mu \times \nu(A \times B) = \mu(A)\nu(B)$.

Using induction, we can extend this construction to any finite product of measures, that is, for measure spaces $(X_i, \mathcal{M}_i, \mu_i)$, we can define $\mu_1 \times \cdots \times \mu_n$ on $\mathcal{M}_1 \otimes \mathcal{M}_2 \otimes \cdots \otimes \mathcal{M}_n$ such that

$$\mu_1 \times \mu_2 \times \cdots \times \mu_n(A_1 \times A_2 \times \cdots \times A_n) = \prod_i^n \mu_i(A_i)$$

and if every μ_i is σ -finite, then the result product measure is the unique extension from the premeasure on the rectangles of $\prod_i^n \mathcal{M}_i$.

Next, (should I prove associativity, it is exercise 45)

With what we've established so far, we can already measure sets using the outer-measure definition. However, this is very tedious, and ultimately comes down to some measure properties of the respective component measure. There is an easier way to use these component measure to get the resulting measure: notice in our construction of π that we integrated with respect to one variable first, and then the next. We will ostensibly generalize this trick so that we can measure spaces, or more specifically integrate functions on product spaces, by fixing all variables except one, integrating on it, and then moving the result out (since it's a constant) and integrate the next variable, until we integrate over all variables.

Since we can generalise our results by induction, let's take $n = 2$ and consider the measure space $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$. The following definition tries to capture the idea of fixing everything except one variable:

Definition 2.5.3: n -section

Let $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$ be a product measure space, and consider $E \subseteq X \times Y$. Then for any $x \in X$, $y \in Y$, define the x -section E_x and y -section E^y to be:

$$E_x = \{y \in Y \mid (x, y) \in E\} \quad E^y = \{x \in X \mid (x, y) \in E\}$$

If f is a function on $X \times Y$, we define the x -section f_x and y -section f^y to be:

$$f_x(y) = f(x, y) \quad f^y(x) = f(x, y)$$

The most simple example of the section of a function would be:

$$(\chi_E)_x = \chi_{E_x} \quad (\chi_E)^y = \chi_{E^y}$$

Proposition 2.5.4: Measure of Section

1. Let $E \in \mathcal{M} \otimes \mathcal{N}$. Then $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$ (i.e., all sets in a product σ -algebra can be broken down into x -sections and y -section)
2. If f is $\mathcal{M} \otimes \mathcal{N}$ -measurable, then f_x is $\mathcal{N}$ -measurable and f^y is $\mathcal{M}$ -measurable

Proof:

1. The proof of this fact comes down to taking advantage of the fact that the pre-measure of rectangles generates $\mathcal{M} \otimes \mathcal{N}$. Let $\mathcal{R}$ be the collection of all subsets of $E \subseteq X \times Y$ such that $E_x \in \mathcal{N}$ and $E^y \in \mathcal{M}$ for all $x \in X$ and $y \in Y$. This clearly contains the empty set, and the set must contain all rectangles, since if $A \neq \emptyset$ and $x \in A$, then $(A \times B)_x = B$ (if $x \notin A$, then $(A \times B)_x = \emptyset$). Now, since

$$\left(\bigcup_i^\infty E_i \right)_x = \bigcup_i^\infty (E_i)_x \quad (E^c)_x = (E_x)^c$$

by some basic set manipulation, we see that $\mathcal{R}$ is a σ -algebra. Since it contains the generating set of $\mathcal{M} \otimes \mathcal{N}$, we have that $\mathcal{R} \supseteq \mathcal{M} \otimes \mathcal{N}$. Thus, all the sets in $\mathcal{M} \otimes \mathcal{N}$ can be decompose in this way (note that we have not proved equality: we have not proved that if every slice is measurable then E is measurable)

2. By some set-theory manipulation, and part (1):

$$f_x^{-1}(B) = (f^{-1}(B))_x \quad f^y_-(B) = (f^{-1}(B))^y$$

and so f_x and f^y_- must be $\mathcal{N}$ and $\mathcal{M}$ measurable (respectively)

With this, we are almost ready to start proving we can integrate $A \times B \in \mathcal{M} \otimes \mathcal{N}$ with respect to 1 measure at a time. We first need to prove a technical result for the next theorem which gives (again) another way of constructing σ -algebra. This one is particularly useful, since it will let us prove certain sets are σ -algebras by using the MCT and DCT:

Definition 2.5.5: Monotone Class

Let X be a set and $P(X)$ the powerset of X . Then the subset $\mathcal{C} \subseteq P(X)$ is called a *monotone class* if

1. it is closed under countable unions of increasing sets, that is, if $E_1 \subseteq E_2 \subseteq \dots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\bigcup_i E_i \in \mathcal{C}$
2. it is closed under intersection unions of decreasing sets, that is, if $E_1 \supseteq E_2 \supseteq \dots$ where $\{E_i\} \subseteq \mathcal{C}$, then $\bigcap_i E_i \in \mathcal{C}$

Clearly, every σ -algebra is a monotone class (we've even used this property in a couple of proofs). Furthermore, the intersection of any family of monotone class is again a monotone class, and so we can define the unique smallest monotone class that contains the set $\mathcal{E} \subseteq P(X)$, where $\mathcal{E}$ is called the monotone class generated set, and the monotone class is called the monotone class generated by $\mathcal{E}$, and is denoted $\mathcal{C}(\mathcal{E})$

Lemma 2.5.6: Monotone Class Lemma

Let X be a set and $\mathcal{A} \subseteq \mathcal{M}$ be an algebra on some subsets of X . Then the monotone class $\mathcal{C}$ generated by $\mathcal{A}$ is equal to the σ -algebra $\mathcal{M}$ generated by $\mathcal{A}$, that is:

$$\mathcal{C}(\mathcal{A}) = \mathcal{M}(\mathcal{A})$$

Proof:

Since $\mathcal{M}$ is itself a monotone class, we automatically have $\mathcal{C} \subseteq \mathcal{M}$, so we'll prove $\mathcal{C} \supseteq \mathcal{M}$.

This is just a lot of set-theory manipulation, I'll leave it for another time

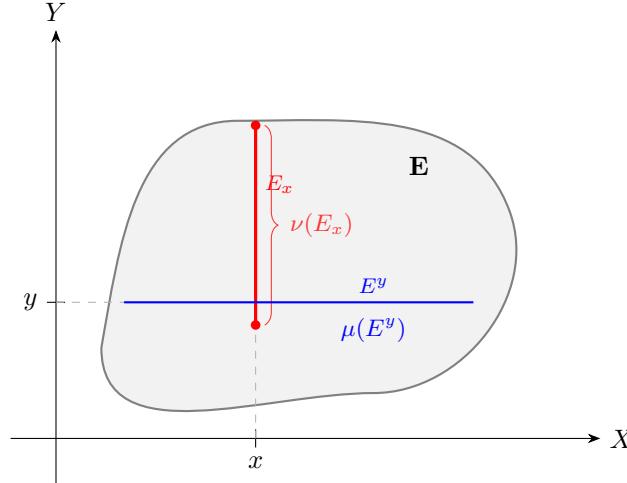
We now are able to start proving the main results:

Theorem 2.5.7: Fubini-Tonelli for Characteristic Functions

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces with μ, ν being σ -finite. Then if $E \in \mathcal{M} \otimes \mathcal{N}$, the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable on X and Y respectively, and

$$(\mu \times \nu)(E) = \int \nu(E_x) d\mu = \int \mu(E^y) d\nu$$

If you are visual, you can think of $\mu(E_x)$ as measuring the area under the slice (inside the appropriate measure so that it doesn't have measure 0), and see that are saying there exists a function that maps x to every slice of the graph:



In particular, it maps x the measure of that slice, so we get a new function which at every slice captures the effect of the X component. After that, we integrate the function to get the effect from the Y component.

Furthermore, it will turn out that we can measure the slices with respect to X , or with respect to Y , and get the same result, hence the second statement of this theorem! After proving the theorem, we'll go over why the assumption in the statement of the theorem are important.

Proof:

We'll first prove it for when μ and ν are finite, and then generalize for when they are σ -finite. The way we'll do it is by starting with a dense subset of $\mathcal{M} \otimes \mathcal{N}$ for which we know the conclusion holds, and then generalize the result using the MCT and DCT. In particular, we will be taking advantage that rectangles already satisfy the property of the theorem, and since rectangles are a generating set for $\mathcal{M} \otimes \mathcal{N}$, we will be able to extend this result any measurable set.

Let $\mathcal{C}$ be the subset of $\mathcal{M} \otimes \mathcal{N}$ for which the result of the theorem is true, that is, the two maps are measurable and we can integrate one way or another. Then if $A \in \mathcal{M}$ and $B \in \mathcal{N}$, we have that $A \times B \in \mathcal{C}$. To see this, first compute the values $\nu(E_x)$ and $\mu(E^y)$:

$$\nu(E_x) = \chi_A(x)\nu(B) \quad \mu(E^y) = \mu(A)\chi_B(y)$$

Then the functions is measurable since it's a slice (theorem 2.5.4), and:

$$\int \nu(E_x)d\mu = \int \chi_A(x)\nu(B)d\mu = \mu(A)\nu(B) = \int \mu(A)\chi_B(y)d\nu = \int \mu(E^y)d\nu$$

Showing the two integrals are equivalent, and since $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$, all equality hold, and so $E = A \times B \in \mathcal{C}$. By additivity of the integral, it follows that all finite disjoint unions of rectangles are also in $\mathcal{C}$. Thus, by lemma 2.5.6, it suffices to show that $\mathcal{C}$ is a monotone class, for then it is a σ -algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, meaning $\mathcal{C} \supseteq \mathcal{M} \otimes \mathcal{N}$, and so all measurable sets can be broken down in this way.

First, let $\{E_n\}_{n=1}^\infty \subseteq \mathcal{C}$ be an increasing subset and let $E = \bigcup_n E_n$. We want to show that $E \in \mathcal{C}$. Since $E_n \in \mathcal{C}$ for all $n \in \mathbb{N}$, the sequence of y -section functions $f_n, f_n(y) = \mu((E_n)^y)$ are measurable and increase pointwise to $f(y) = \mu(E^y)$. Since the limit of measurable functions is measurable (corollary 2.1.13), f is measurable. Thus, we can find the value of $\int \mu(E^y)d\nu$ by applying the Monotone Convergence Theorem:

$$\begin{aligned} \int \mu(E^y)d\nu &= \int \lim_{n \rightarrow \infty} \mu((E_n)^y)d\nu \\ &= \lim_{n \rightarrow \infty} \int \mu((E_n)^y)d\nu && \text{MCT} \\ &= \lim_{n \rightarrow \infty} (\mu \times \nu)(E_n) && E_n \in \mathcal{C} \\ &= (\mu \times \nu)(E) && \cup_i E_i = E \end{aligned}$$

Doing the same trick with respect to x , we can see that:

$$\int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

which combined together shows us that:

$$\int \mu(E^y)d\nu = \int \nu(E_x)d\mu = (\mu \times \nu)(E)$$

and so $E \in \mathcal{C}$.

Next, let $\{E_n\}_{n=1}^\infty \subseteq \mathcal{C}$ be a decreasing subset and let $\bigcap_n E_n = E$. Then since $E_1 \in \mathcal{C}$, the function $f_1, f_1(y) = \mu((E_1)^y)$ is measurable. In fact, it's in L^1 , since:

$$\mu((E_1)^y) \leq \mu(X) < \infty \quad \nu(Y) < \infty$$

and so the resulting integral must have finite measure. Furthermore, f_1 will dominate the sequence of functions f_n , and so we can apply the Dominated Convergence Theorem to do the same limit trick we have just done and conclude that $E \in \mathcal{C}$. Therefore, $\mathcal{C}$ is a monotone class, hence a σ -algebra containing the generating set of $\mathcal{M} \otimes \mathcal{N}$, and the result is true for all measurable sets.

Now, suppose that μ and ν are σ -finite. Then we can write $X \times Y$ to be the union of an increasing sequence of rectangles $\{X_i \times Y_i\}$ of finite measure. Thus, for any $E \in \mathcal{M} \otimes \mathcal{N}$, take $E \cap (X_i \times Y_i)$, then this set now has finite measure, and so what we have just proved applies giving us:

$$(\mu \times \nu(E \cap (X_i \times Y_i))) = \int \chi_{X_i}(x) \nu(E \cap Y_i) d\mu = \int \mu(E \cap X_i) \chi_{Y_i}(y) d\nu$$

and so, a final application of the Monotone Convergence Theorem gives us our desired result

The hypothesis that $\mu \times \nu$ is σ -finite is necessary:

Example 2.5.8: Necessity of σ -finite

To compute the iterated integrals, we start by finding the x and y section. Using the definition of the diagonal we get that: $(\chi_D)_x = \chi_{\{x\}}$ and $(\chi_D)_y = \chi_{\{y\}}$. Thus, we get:

$$\int \left[\int \chi_{\{y\}} d\mu(x) \right] d\nu(y) = \int 0 d\nu(y) = 0$$

and

$$\int \left[\int \chi_{\{x\}} d\nu(y) \right] d\mu(x) = \int 1 d\mu(x) = 1$$

showing that the iterated iterated iterated integrals are not equal. For the integral $\int \chi_D d(\mu \times \nu)$, since χ_D is a simple function, we have shown in class that the integral is the measure of D with respect to the product measure $\mu \times \nu$ (namely, since χ_D will be in the supremum set over which we are computing the integral), which is:

$$\int \chi_D d(\mu \times \nu) = \mu \times \nu(D) = \inf \left\{ \sum_i \mu(A_i) \nu(B_i) \mid A_i \times B_i \in \mathcal{A}, D \subseteq \bigcup_i A_i \times B_i \right\}$$

where $\mathcal{A}$ is the collection of all rectangles of $\mathcal{M} \otimes \mathcal{N}$. We'll show $\mu \times \nu(D) = \infty$.

Take any countable cover of D by rectangle $\mathcal{C}$ (so $\{A_i \times B_i\}_{i=1}^\infty = \mathcal{C}$ for appropriate $A_i \times B_i$). Without loss of generality, we can make these covers disjoint; we can refine the rectangles. If we can show that there must be a non-zero measure set A_i and an infinite measure set B_i such that $A_i \times B_i \in \mathcal{C}$ then we are done. For the sake of contradiction, let's say no such set exists in any countable cover $\mathcal{C}$. We'll show that $\mathcal{C}$ doesn't cover D .

Split $\mathcal{C}$ into $F \subseteq \mathcal{C}$ and $I \subseteq \mathcal{C}$ where F is the collection of all sets where A_i has *finite* non-zero measure (the sets have finite measure since if $A_i \subseteq X$, $\mu(A_i) \leq \mu(X) = 1$), and I is the set where B_i has *infinite* measure. By our contradiction assumption, these set's are disjoint, since the first component of the sets in I must have zero measure.

The set F can only cover countably many points of D : Since each B_i has finite measure, any set $A_i \times B_i \in F$ can cover at most finitely many points of D (given appropriate A_i), and so $\bigcup_{A_i \times B_i \in F} A_i \times B_i$ can at most cover countably many points of D (since the 2nd component of

D is covered by elements from B_i given appropriate A_i). Therefore, an uncountable number of points are not being covered by F .

The set I misses at least uncountably many points. Since each component A_i has zero-measure and the collection of x components of D (i.e. the set $[0, 1]$) has measure one, the union of the sets A_i cannot cover all of the x -components (given appropriate B_i components) or else $0 = \mu(\bigcup_i A_i) \geq \mu([0, 1]) = 1$, a contradiction. If the B_i component's don't line up correctly, then this set cover's even fewer points of D . Therefore, I can at most cover a measure zero amount of x -components points from D . Since any set two sets containing each other with a different measure (ex. $\alpha \subseteq \beta$, $0 = \mu(\alpha) \leq (\beta) = 1$) must have a difference with at least uncountable cardinality^a ($|B \setminus A| \geq \aleph_1$), the set I must miss at least uncountably many points.

However, the cover F can only cover at most countably many points. Therefore, the cover $F \cup I = \mathcal{C}$ cannot cover all of D , a contradiction to the assumption that $\mathcal{C}$ is a cover for D . Hence, there is at least one set $A_i \times B_i$ with $\mu(A_i) > 0$ and $\mu(B_i) = \infty$ for any cover of D , meaning all covers of D must have a set with infinite measure, and so $\infty \leq \mu \times \nu(D)$, and hence

$$\int \chi_D d(\mu \times \nu) = \infty$$

and so the two iterated integrals, as well as the integral of χ_D has unequal measures, as we sought to show.

^aIf two sets containing each other had differed by countably many point, then they would have the same measure

Notice that the functions within the integral of the previous theorem are in fact the values of characteristic functions! Hence by the linearity of the integral, we have the result for simple functions. Thus, with the statement proven for simple functions on $\mathcal{M} \otimes \mathcal{N}$, we move on to proving it for integrable functions in general:

Theorem 2.5.9: Fubini-Tonelli Theorem

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be σ -finite measures. Then:

1. (Tonelli) If $f \in L^+(X \times Y)$, then the functions $g(x) = \int f_x d\nu$ and $h(y) = \int f^y d\mu$ are in $L^+(X)$ and $L^+(Y)$ respectively, and

$$\int \left[\int f(x, y) d\nu(y) \right] d\mu(x) = \int f(x, y) d(\mu \times \nu) = \int \left[\int f(x, y) d\mu(x) \right] \nu(y)$$

2. (Fubini) If $f \in L^1(\mu \times \nu)$, then $f_x \in L^1(\nu)$ for a.e. $x \in X$ and $f^y \in L^1(\mu)$ for a.e. $y \in Y$. Furthermore, the a.e. defined function $g(x) = \int f_x d\nu$ and $h(y) = \int f^y d\mu$

The resulting integral is called the *iterated integral*

Proof:

1. This essentially comes down to using MCT and the standard representation of measurable functions. By theorem 2.5.7, we have the simple functions satisfy the theorem. To extend it to integrable functions, take $f \in L^1(X \times Y)$, and let $\{f_n\}$ be some simple representation for it, that is, a sequence of simple functions converging upwards to f . From $\{f_n\}$, we

have a corresponding $\{g_n\}$ and $\{h_n\}$ converging upwards to g and h : $g_n = \int (f_n)_x$ and $h_n = \int (f_n)_y$. By theorem 2.5.7, these functions are measurable, and so h and g are measurable. With this, we can put together all our information to form the following chain of equalities:

$$\begin{aligned}
 \int \left[\int f_x d\nu \right] d\mu &= \int g d\mu = \int \lim g_n d\mu \\
 &= \lim \int g_n d\mu \\
 &= \lim \int f_n d(\mu \times \nu) && \text{theorem 2.5.7} \\
 &= \int f d(\mu \times \nu) && \text{MCT} \\
 &= \lim \int f_n d(\mu \times \nu) \\
 &= \lim \int h_n d\nu && \text{theorem 2.5.7} \\
 \int \left[\int f_y d\mu \right] d\nu &= \int h d\nu = \int \lim h_n d\nu
 \end{aligned}$$

Thus establishing the equality of Tonelli's Theorem. Furthermore, since $\int f d(\mu \times \nu) < \infty$, then $g < \infty$ a.e. and $h < \infty$ a.e., so $\int f_x < \infty$ and $\int f_y < \infty$, and therefore $f_x \in L^1(\nu)$ and $f_y \in L^1(\mu)$, which some of the preliminary results for Fubini's Theorem.

2. Let $f \in L^1(\mu \times \nu)$, so $f^+ \in L^+(\mu \times \nu)$ and $f^- \in L^+(\mu \times \nu)$ (or Ref and $\text{sgn}f$). Thus, the result applies to each component function, and so by linearity applies to f

Often, the parenthesis are dropped, and we combine the integrals into a double integral:

$$\int \left[\int f(x, y) d\mu(x) \right] d\nu(y) = \int \int f(x, y) d\mu(x) d\nu(y) = \int \int f d\mu d\nu$$

Also, you might wonder if instead of assuming $f \in L^+(X \times Y)$ or $f \in L^1(X \times Y)$, is it possible to instead assume that each $f_x \in L^+(\nu)$, $f_y \in L^+(\mu)$ (L^1 respectively) and that the iterated integrals $\int \int f d\mu d\nu$ and $\int \int f d\nu d\mu$ existed, then $f \in L^+(X \times Y)$ (L^1 respectively), that is, is the converse true? The answer is actually no:

Example 2.5.10: Converse not True

Let $X = Y = \mathbb{N}$, $\mathcal{M} = \mathcal{N} = P(\mathbb{N})$, $\mu = \nu$ be the counting measures. Define:

$$f(m, n) = \begin{cases} 1 & m = n \\ -1 & m = n + 1 \\ 0 & \text{otherwise} \end{cases}$$

Then $\int |f| d(\mu \times \nu) = \infty$, and $\int \int f d\mu d\nu$, $\int \int f d\nu d\mu$ exist and are unequal.

This is not quite a converse because the two iterated integrals are unequal, and I don't know of an example where they are equal, but $\int f d(\mu \times \nu)$ is not equal to it (i.e. infinity).

One practical application of Fubini-Tonelli's Theorem is to solve integrals by integrating over the simpler variable first. Usually, one first uses Tonelli's theorem to evaluate the integral $\int f d(\mu \times \nu)$ as an iterated integral $\int \int f d\mu d\nu$ and show that the result is finite, at which point Fubini's Theorem can be invoked to get $\int \int f d\mu d\nu = \int \int f d\nu d\mu$.

To close off the discussion on product measures, notice that if μ and ν are complete measure, $\mu \times \nu$ is almost never a complete measure. Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be measure spaces, take $A \in \mathcal{M}$ is a non-empty set with $\mu(A) = 0$, and suppose $\mathcal{N} \neq P(Y)$ (For example, choose $(\mathbb{R}, \mathcal{L}, m)$ for both measure spaces). If $E \in P(Y) \setminus \mathcal{N}$, then $A \times E \notin \mathcal{M} \times \mathcal{N}$ since not all slices of $A \times E$ (in particular, the slice E) are in the respective measure spaces (proposition 2.5.4). However, $A \times E \subseteq A \times Y$, and $(\mu \times \nu)(A \times Y) = 0$ (since the measure is finite, any out-measure is equal to the usual outer-measure on these sets, which is $\mu(A)\nu(Y) = 0 \cdot n = 0$, since we take on the convention that $0 \cdot \infty = 0$). Thus, $\mu \times \nu$ is not complete. If we want, we can apply theorem 1.2.8 to complete the measure, however we can no longer directly apply Fubini-Tonelli's Theorem. (Put Why Here!). We thus have to slightly amend Fubini-Tonelli's theorem for the completion:

Theorem 2.5.11: Fubini-Tonelli's Theorem for Complete Measures

Let $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ be complete σ -finite measures and let $(X \times Y, \mathcal{L}, \lambda)$ be the completion of $(X \times Y, \mathcal{M} \otimes \mathcal{N}, \mu \times \nu)$. If f is $\mathcal{L}$ -measurable is either:

1. non-negative, that is $f \geq 0$
2. $f \in L^1(\lambda)$

Then f_x is $\mathcal{N}$ -measurable for a.e. x and f^y is $\mathcal{M}$ -measurable for a.e. y , and if (2) applies as well, then f_x and f^y are integrable for a.e. x and y . Furthermore, $x \mapsto \int f_x d\nu$ and $y \mapsto \int f^y d\mu$ are measurable (and in the case of (2), also integrable) and

$$\int f d\lambda = \int \int f(x, y) d\mu(x) d\nu(y) = \int \int f(x, y) d\nu(y) d\mu(x)$$

Proof:

We'll do the following:

1. If $E \in \mathcal{M} \otimes \mathcal{N}$ and $\mu \times \nu(E) = 0$, then $\nu(E_x) = \mu(E^y) = 0$ for a.e. x and y
2. If f is $\mathcal{L}$ -measurable and $f = 0$ λ -a.e., then f_x and f^y are integrable for a.e. x and y , and $\int f_x d\nu = \int f^y d\mu = 0$ for a.e. x and y (here, the completeness of μ and ν is needed)

lemma 1:

Proof. Note that $\chi_E \in L^+(X \times Y)$ and $(X, \mathcal{M}, \mu)$ $(Y, \mathcal{N}, \nu)$ are σ -finite, and so Tonelli's Theorem applies, meaning

$$\int \chi_E d(\mu \times \nu) = \int \left[\int (\chi_E)^y d\mu(x) \right] d\nu(y) = \int \left[\int (\chi_E)_x d\nu(y) \right] d\mu(x)$$

First, since χ_E is an indicator function, we see that $(\chi_E)_x = \chi_{E_x}$, since $\chi_{E_x}(y) = \chi_E(x, y)$.

Similarly, $(\chi_E)^y = \chi_{E^y}$. Therefore, we have:

$$\begin{aligned} 0 &= \int \chi_E d(\mu \times \nu) \\ &= \int \left[\int (\chi_E)^y d\mu(x) \right] d\nu(y) \\ &= \int \left[\int (\chi_E)_x d\nu(y) \right] d\mu(x) \end{aligned}$$

showing that the function $x \mapsto \int \chi_{E_x} d\nu$ and $y \mapsto \int \chi_{E^y} d\mu$ is 0 a.e., which implies $\nu(E_x) = 0$ and $\mu(E^y) = 0$ a.e. $\square$

lemma 2: Multiple parts of this proof take advantage of the fact that if μ is a measure and $\mu(A) = 0$, then $\int_A f d\mu = 0$ (since all simple functions φ approximating f with the property of $\varphi \leq f$ will have their integral be of the form $\sum_i^n c_i \chi_{A_i} = \sum_i^n c_i \cdot 0 = 0$ for $A_i \subseteq A$, and so the supremum of the integral of such simple functions must also be 0)

Proof. Let f be $\mathcal{L}$ -measurable and $f = 0$ λ -a.e. We want to show that f_x and f^y is integrable for a.e. x and y and $\int f_x d\nu = \int f^y d\mu = 0$ for a.e. x and y . What we'll do is take advantage of Fubini-Tonelli and theorem 2.12 (which states that there exists a $\mathcal{M} \otimes \mathcal{N}$ -measurable function g such that $f = g$ λ -a.e.)

First, since $f = g$ λ -a.e., $f - g = 0$ λ -a.e., meaning $(f - g)_x = f_x - g_x = 0$ for a.e. x (assuming completeness) and $f^y - g^y = 0$ for a.e. y (assuming completeness), and so $f_x = g_x$ and $f^y = g^y$ for a.e. x and y . This will be useful to us later.

Next, Consider $\int f d\lambda$. We want to take advantage of Fubini's theorem, which is true for g , and so we must re-write this expression in such a way that we get it equal to g . Let's say $S = \{(x, y) \in X \times Y \mid f(x, y) \neq g(x, y)\}$. We know that $\lambda(S) = 0$, and so:

$$0 = \int f d\lambda = \int_S f d\lambda + \int_{S^c} f d\lambda = \int_{S^c} f d\lambda \quad (2.1)$$

Since $S^c \in \mathcal{L}$, we have that $S^c = A \amalg N$ where $A \in \mathcal{M} \otimes \mathcal{N}$ and $N \subseteq B$ where $\mu \times \nu(B) = 0$. Thus, we get:

$$\int_{S^c} f d\lambda = \int_A f d\lambda + \int_N f d\lambda = \int_A f d\lambda$$

since $f = g$ on S^c , in particular $f = g$ on $A \in \mathcal{M} \otimes \mathcal{N}$, and λ is the extension of $\mu \times \nu$, we can re-write this as:

$$\int_A f d\lambda = \int_A g d(\mu \times \nu)$$

and so Fubini-Tonelli applies, giving us the iterated integrals are equal to 0 (carrying the 0 from equation (2.1)):

$$\int \left[\int_{A_x} g_x d\nu(y) \right] d\mu(x) = 0$$

and similarly for the y -section. Since $f = g$ on S^c , by what we've shown earlier, $f_x = g_x$ on A_x , (and similarly $f^y = g^y$ on A^y) and so:

$$\int_{A_x} f_x d\nu(y) = 0 \quad \text{for a.e. } x$$

and similarly for the y -sections, which brings us close to our goal (it is possible that $A_x \subsetneq X$ or $A^y \subsetneq Y$). We now have to add back in points which we eliminated since the integral was zero on S and N :

$$\int_S f d\lambda = 0 \quad \int_N f d\lambda = 0$$

Since they are measure zero sets in $\mathcal{L}$, there exists a $C, D \in \mathcal{M} \otimes \mathcal{N}$ such that $\mu \times \nu(C) = \mu \times \nu(D) = 0$ and $S \subseteq C, N \subseteq D$. Let $E = C \cup D$ so that $S \cup N \subseteq E$ and $\mu \times \nu(E) = 0$. Thus, we have:

$$\int_E f d\lambda = 0$$

since $\mu \times \nu(E) = 0$ and $E \in \mathcal{M} \otimes \mathcal{N}$, by the first lemma, $\mu(E^y) = \nu(E_x) = 0$ for a.e. x and y . But then, for a.e. x and y , $\int_{E_x} f_x d\nu(y) = \int_{E^y} f^y d\mu(x) = 0$. Since $X \times Y = S^c \cup S = A \cup E$ (by how we've broken the set down), then $X = A_x \cup E_x$ and $Y = A^y \cup E^y$, and so

$$0 \leq \int f_x d\nu(y) \leq \int_{A_x} f_x d\nu(y) + \int_{E_x} f_x d\nu(y) = 0$$

for a.e. x , implying $\int f_x d\nu(y) = 0$ for a.e. x (and similarly for y). Hence, they are integrable for a.e. x and y , completing the proof. $\square$

With this, we can prove Fubini-Tonelli's Theorem for complete measures. First, let's build-up some result and notation. Let $f \in L^+(\lambda)$ (resp. $f \in L^1(\lambda)$). Then by theorem 2.12, there exists a g such that $f = g$ λ -a.e., meaning that $f - g = 0$ λ -a.e. Thus, by the 2nd lemma, $\int (f - g)_x = 0$ and $\int (f - g)^y = 0$ are integrable for a.e. x and y . Therefore, $\int f_x - g_x = 0$ and $\int f^y - g^y = 0$ for a.e. x and y , and so $\int f_x = \int g_x$ and $\int f^y = \int g^y$ for a.e. x and y .

We can now start concluding the proof. Since μ and ν are complete and $f_x = g_x$ ($f^y = g^y$) for a.e. x and y , by proposition 2.11, f_x is $\mathcal{N}$ -measurable and f^y is $\mathcal{M}$ -measurable for a.e. x and y , completing the 1st assertion of the theorem. In the $f \in L^1(\lambda)$ case, the proof we've just done in the previous paragraph shows that each f_x and f^y are integrable for a.e. x and y .

Next, since $G_x : x \mapsto \int g_x d\nu$ and $G^y : y \mapsto \int g^y d\mu$ is measurable (resp. integrable), by theorem 2.11, $F_x : x \mapsto \int f_x d\nu$ and $F^y : y \mapsto \int f^y d\mu$ is measurable (resp. integrable). Finally, since $F_x = G_x$ for a.e. x and $F^y = G^y$ for a.e. y , their integrals match, and since g is $\mathcal{M} \otimes \mathcal{N}$ measurable, Tonelli's (resp. Fubini's) theorem applies, and so by one more application of theorem 2.11 :

$$\int f d\lambda = \int \int f(x, y) d\mu(x) d\nu(y) = \int \int f(x, y) d\nu(y) d\mu(x)$$

Completing the proof

2.6 The n-dimensional Lebesgue Integral

In this section, we focus on $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ with the completed product measure $m \times m \times \cdots \times m = m^n$ on $\mathcal{L} \otimes \mathcal{L} \otimes \cdots \otimes \mathcal{L} = \mathcal{L}^n$. The domain $\mathcal{L}^n$ of m^n is called the set of *Lebesgue measurable sets in $\mathbb{R}^n$* : $(\mathbb{R}^n, \mathcal{L}^n, m^n)$ (sometimes, we will restrict m^n to have domain $(\mathcal{B}_{\mathbb{R}})^n = \mathcal{B}_{\mathbb{R}^n}$). If it is unambiguous, we will usually write $(\mathbb{R}^n, \mathcal{L}^n, m)$, dropping the power on the m , and for the $n = 1$ case, we will often

write $\int f(x) dx := \int f dm$.

The following 3 theorems are simply generalizations of previous properties of the Lebesgue measure. In the following, let $E = \prod_{i=1}^n E_i$ where $E_i \subseteq \mathbb{R}$. We will call each E_i the *sides* of E

Theorem 2.6.1: Simplifying outer-measure

Suppose $E \in \mathcal{L}^n$

1. $m(E) = \inf \{m(U) \mid E \supseteq U, U \text{ is open}\} = \sup \{m(K) \mid K \subseteq E, K \text{ is compact}\}$
2. $E = A_1 \cup N_1 = A_2 \setminus N_2$ where $A_1 \in F_\sigma$, $A_2 \in G_\delta$, and $m(N_1) = m(N_2) = 0$
3. If $m(E) < \infty$, for all $\epsilon > 0$, there is a finite disjoint collection of rectangles $\{R_i\}_{i=1}^n$ whose sides are intervals such that $m(E \Delta \cup_i^n R_i) < \epsilon$

Proof:

p. 70 in foland

Theorem 2.6.2: Completeness of $L^1(m)$

Let $f \in L^1(m)$ and let $\epsilon > 0$. Then there exists a simple function $\varphi = \sum_{i=1}^n a_i \chi_{R_i}$ where each R_i is a product of intervals such that

$$\int |f - \varphi| < \epsilon$$

and there is a continuous function that vanishes outside a bounded set such that

$$\int |f - g| < \epsilon$$

Proof:

Just like in theorem 2.3.9, approximate f by a simple function, then apply proposition 2.6.1(3) to approximate φ appropriately. Finally, approximate φ by continuous functions by applying the natural generalization of the argument in theorem 2.3.9

Theorem 2.6.3: Translation Invariance in $\mathbb{R}^n$

Let $(\mathbb{R}^n, \mathcal{L}^n, m)$ be a measure space. Then m is translation invariant, that is, for any $a \in \mathbb{R}^n$, if we define $\tau_a : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\tau_a(x) = a + x$, then

1. If $E \in \mathcal{L}^n$, then $\tau_a(E) \in \mathcal{L}^n$ and $m(E) = m(\tau_a(E))$
2. If $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is Lebesgue measurable, then so is $f \circ \tau_a$. Furthermore, if either $f \geq 0$ or $f \in L^1(m)$, then $\int (f \circ \tau_a) dm = \int f dm$

Notice that we did not mention scaling. This is because scaling is not abstracted to linear transformations which requires more work to show how the scaling of a linear transformation affects the

value, and so we will prove that later in this section.

Proof:

Folland p.71

(here, some on cube approximation and Jordan content)

I'll for now simply state the effect of linear transformations and come back to this:

Theorem 2.6.4: Linear Transformation on Integrability

Suppose $T \in GL_n(\mathbb{R})$. Then:

1. If f is a Lebesgue measurable function on $\mathbb{R}^n$, so is $f \circ T$. If $f \geq 0$ or $f \in L^1(m)$, then:

$$\int f \, dm = |\det(T)| \int f \circ T \, dm$$

2. If $E \in \mathcal{L}^n$, then so is $T(E)$, and $m(T(E)) = |\det(T)|m(E)$

Proof:

here

There is also the statement on how diffeomorphisms affect the integral.

Signed Measure and Differentiation

In this chapter, we will take a close look at how to define measure's with respect to integrals. In particular, given the appropriate integrable function f , we will want to define a measure μ such that

$$\mu(A) = \int_A f$$

After defining some terms in section 3.1, we will focus in section 3.2 and 3.3 on how to find a function f given a measure μ so that $\mu(A) = \int_A f$. The last 2 sections then focus on the special case where the function f that we found can actually be the *derivative*. In the simple case where $X = \mathbb{R}$, then we can define:

$$F(x) = \int_{(-\infty, x]} f dx = \int_{-\infty}^x f dx$$

and ask ourselves what conditions does f or F requires so that $f = F'$ (or, $f = F'$ a.e.). This line of reasoning will ultimately lead to a generalisation of the Fundamental Theorem of Calculus, which will tell us when can we recover the function f given we know $\int_A f$ for all A .

3.1 Signed Measures

Definition 3.1.1: Signed Measure

Let $(X, \mathcal{M})$ be a measurable space. A *signed measure* on $(X, \mathcal{M})$ is a function: $\nu : \mathcal{M} \rightarrow [-\infty, \infty]$ such that:

1. $\nu(\emptyset) = 0$
2. ν has at most one value of $\pm\infty$
3. If $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{M}$ is a disjoint collection, then

$$\nu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \nu(E_i)$$

where if $\nu(\bigcup_{i=1}^{\infty} E_i) < \infty$, then $\sum_{i=1}^{\infty} \nu(E_i)$ converges absolutely.

Example 3.1.2: Signed Measures

Let $(X, \mathcal{M})$ be a measurable space.

1. Clearly, every measure on $\mathcal{M}$ is a signed measure. To distinguish between the two, we will often call measures *positive measures*.
2. Let μ_1 and μ_2 be two measure on $\mathcal{M}$ with at least one being finite. Then $\nu = \mu_1 - \mu_2$ is a signed measure.
3. If f is a $\mathcal{M}$ -measurable function, then if at least one of $\int f^+$ or $\int f^-$ is finite,

$$\nu(E) = \int_E f$$

is a signed measure. In this case, we call f the *extended μ -integrable function* (in contrast to $f \in L^1$ where both terms need to be finite)

The fact that it converges absolutely is to make sure that the order in which we sum the elements does not matter: If it did not converge absolutely, then it is possible to make the sum approach any value we want. Since $\mathcal{M}$ doesn't necessarily have an inherent order (in fact it almost never does), then having this dependence on the order of summing the elements is inconsistent with our notion of area (this is also why we only allow one of the direction to be infinite). It might seem that we must thus be careful on how we define our signed measure to make sure this condition holds, however we will soon show that every signed measure can be decomposed into 2 measures. If the measure of all sets with positive measure and negative measure are both infinite ($\pm\infty$), then we have contradicted the definition of a signed measure. As just mentioned, all signed measures can be decomposed into two positive measures. We will spend this section proving that fact. In the next section, we will show that under appropriate conditions and choices of signed measure ν and positive measure μ , we can find a measurable function f such that $\nu(E) = \int_E f d\mu$. This will essentially be the generalization of the derivative!

We start by building up to decomposing a signed measure into two positive measures:

Proposition 3.1.3: Continuity from Above/Below for Signed Measures

Let ν be a signed measure on $(X, \mathcal{M})$.

1. If $\{E_i\}$ is an increasing sequence in $\mathcal{M}$, then $\nu(\bigcup_i E_i) = \lim_{i \rightarrow \infty} \nu(E_i)$
2. If $\{E_i\}$ is a decreasing sequence in $\mathcal{M}$ where $\nu(E_1)$ is finite, then $\nu(\bigcap_i E_i) = \lim_{i \rightarrow \infty} \nu(E_i)$

Proof:

Very similar to proposition 1.2.5

For the next proposition, we introduce a quick new definition: a set $P \in \mathcal{M}$ is called *positive* (resp. *negative*) if for all $S \subseteq P$ such that $S \in \mathcal{M}$, then $\nu(S) \geq 0$ (resp. $\nu(S) \leq 0$)

Lemma 3.1.4

Any countable union of positive sets is a positive set

Proof:

Use proposition 3.1.3

Theorem 3.1.5: Hahn Decomposition Theorem

If ν is a signed measure on $(X, \mathcal{M})$, there exists a positive set P and a negative set N for ν $P \sqcup N = X$ and $P \cap N = \emptyset$. If P' and N' is another such pair, then $P \Delta P'$ and $N \Delta N'$ is a null set for ν

Proof:

Let ν be a signed measure, so only ∞ or $-\infty$ can be achieved. If ∞ is achieved, take $-\nu$ so that ∞ is not achieved. Now, let m be the supremum of $\nu(E)$ where E ranges over all positive sets. Thus, there is a sequence of positive sets $\{P_i\}$ such that $\nu(P_i) \rightarrow m$. Let $P = \bigcup_i P_i$. Then by the previous lemma and proposition, P is a positive set and $\nu(P) = m$. Since ν does not achieve ∞ , $m < \infty$. I claim that $N = X \setminus P$ is negative (thus completing the first assertion of the proof).

First, N cannot contain non-null positive sets. If $E \subseteq N$ was a positive set (i.e. $m(E) > 0$), then $E \cap P = \emptyset$ and $E \cup P$ is a positive set, so $m(E \cup P) = m(E) + m(P) > m$, which is a contradiction to m being the supremum.

Now, let's say there exists a $A \subseteq N$ such that $\nu(A) > 0$. We will reach a contradiction in using the following fact: since A cannot be positive, then there must exist a $B \subseteq A$ such that $\nu(B) < 0$. Then let $C = A \setminus B$ so that $\nu(C) > \nu(A)$ and $C \subseteq A$.

With this, we proceed in forming a contradiction: Define the following sequence $\{A_i\}$ and $\{n_i\}$ as follows:

1. Let n_1 be the smallest positive integer such that there exists a $A_1 \subseteq N$ where $\nu(A_1) > n_1^{-1}$ (this set exists since there is a $A \subseteq N$ where $\nu(A) > 0$)
2. Let n_k be the smallest positive integer such that there exists a $A_k \subseteq A_{k-1}$ such that $\nu(A_{k-1}) > \nu(A_k) + n_k^{-1}$ (since A_{k-1} is positive, we use the earlier remark fact to find such a set)

From this sequence of positive decreasing sets, $A_1 \supseteq A_2 \supseteq \dots$, let $A = \bigcap_i A_i$. Then by the fact that ν does not attain ∞ and the previous proposition:

$$\infty > \nu(A) = \lim \nu(A_i) > \sum_{i=1}^{\infty} n_i^{-1}$$

since the series converges, it must be that $n_i \rightarrow \infty$. Finally, once again take $B \subseteq A$ such that $\nu(A) > \nu(B) + n^{-1}$ for some positive integer n . Since $n_i \rightarrow \infty$, there exists an i sufficiently large so that $n < n_i$ and $B \subseteq A_i$, meaning we can replace n_i by n in the above construction. However, by construction, we picked the *minimum* integer every-time, and so this contradicts minimality! Therefore, it cannot be that there exists a set $A \subseteq N$ such that $\nu(A) > 0$, and so N is indeed negative.

If P' and N' where two other sets satisfying the decomposition condition, then we have that $P \setminus P' \subseteq P$ (since it's a subset) and $P \setminus P' \subseteq N'$ (since it is P minus all the positive set P'), so it is both positive and negative, which can only be possible if it is a null-set (similarly for $N \setminus N'$)

Using the Hahn decomposition, we can define a positive measure on P and N . Before proceeding to that theorem, we quickly give a name to the relation between the two measures that will be defined:

Definition 3.1.6: Mutually Singular Signed Measures

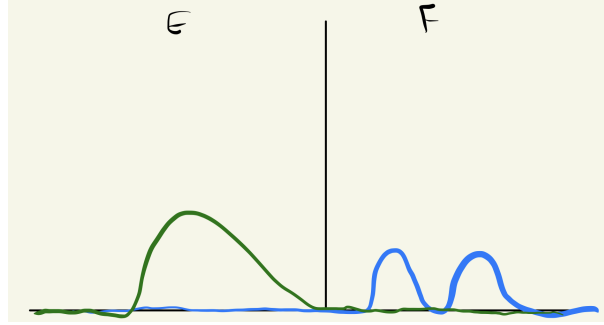
Let ν and μ be two signed measure on $(X, \mathcal{M})$. Then they are called *mutually singular* or μ is *singular with respect to ν* if there exists a $E, F \in \mathcal{M}$ such that $E \cup F = X$, $E \cap F = \emptyset$, and:

1. $\mu(E) = 0$
2. $\nu(F) = 0$

and we write mutually singular signed measures as:

$$\mu \perp \nu$$

In a sense, this means that these two sets are meaningful on disjoint sets, in that sense they are “perpendicular” to each other. I like to keep an image like this in mind



Notice that it is possible that μ have zero values on F , and ν have zero values on E . The key is that they *must* have zero values on their respective sets.

Mutually singular measures can be applied more generally than in the context of decomposing a signed measure. For example, if δ_0 is the dirac-delta measure at 0 (i.e. $\delta(\{0\}) = 1$ and zero everywhere else), then $\delta_0 \perp \lambda$.

Theorem 3.1.7: Jordan Decomposition Theorem

If ν is a signed measure, there exists unique positive measure ν^+ and ν^- such that $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$

Proof:

By theorem 3.1.5, let $X = P \cup N$ be the Hahn decomposition of X with respect to ν . Define $\nu^+(E) = \nu(E \cap P)$ and $\nu^-(E) = -\nu(E \cap N)$. Then by definition, $\nu = \nu^+ - \nu^-$ and $\nu^+ \perp \nu^-$.

If there was another decomposition $\nu = \mu^+ - \mu^-$ where $\mu^+ \perp \mu^-$, then we have $E, F \in \mathcal{M}$ such that $E \sqcup F = X$ and $E \cap F = \emptyset$, $\mu^+(F) = \mu^-(E) = 0$, so E, F is another Hahn decomposition for ν , and so $P \Delta E$ is null. Thus, for any $A \in \mathcal{M}$

$$\mu^+(A) = \mu^+(A \cap E) = \nu(A \cap E) = \nu(A \cap P) = \nu^+(A)$$

thus, $\mu^+ = \nu^+$, and similarly $\mu^- = \nu^-$, completing the proof.

Definition 3.1.8: Jordan Decomposition

Let ν be a signed measure on $(X, \mathcal{M})$. Then ν^+ and ν^- are called the *positive* and *negative variation* of ν , and $\nu = \nu^+ - \nu^-$ is called the *Jordan decomposition* of ν . The *total variation* of ν , denoted $|\nu|$ is defined as $|\nu| = \nu^+ + \nu^-$.

As an exercise, you should verify that $E \in \mathcal{M}$ is ν -null if and only if $|\nu|(E) = 0$, and $\nu \perp \mu$ if and only if $|\nu| \perp \mu$ if and only if $\nu^+ \perp \mu$ and $\nu^- \perp \mu$. With this definition, we will say that ν is finite (resp. σ -finite) if $|\nu|$ is finite (resp. σ -finite)

Finally, we can also get back to showing that any signed measure is of the form $\nu(E) = \int_E d\mu$ for appropriate μ , namely $\mu = |\nu|$ and $f = \chi_P - \chi_N$ for appropriate P and N from a Hahn decomposition. From this, we can define the integration with respect to a signed measure ν as follows:

Definition 3.1.9: Integration w.r.t. Signed Measures

Let ν be a signed measure, $\nu^+ - \nu^-$ be it's Hahn decomposition, and $f : X \rightarrow \mathbb{C}$ be a real function. Then we will say that f is *integrable with respect to the signed measure ν* if:

$$f \in L^1(\nu^+) \cap L^1(\nu^-) = L^1(\nu)$$

and define it's integral to be:

$$\int f d\nu = \int f d\nu^+ - \int f d\nu^- \quad (f \in L^1(\nu))$$

3.2 Lebesgue-Radon-Nikodym Theorem

In this section, we show under what circumstances we can decompose a signed measure μ into $\nu + \rho$ where $\nu(A) = \int_A f$ and ρ is the “left-over” (also known as the singular part). We first define the “opposite” notion of mutually singular signed measures. The motivation for the vocabulary will come in section ref:HERE

Definition 3.2.1: Absolutely Continuous w.r.t. μ

Let ν be a signed measure on $(X, \mathcal{M})$ and μ is a positive measure^a on $(X, \mathcal{M})$. We say that ν is *absolutely continuous with respect to μ* when if $\mu(E) = 0$ (given $E \in \mathcal{M}$) then $\nu(E) = 0$, and we write is as:

$$\nu \ll \mu$$

^athis can be extended to a signed measure by taking $\nu \ll \mu$ if and only if $\nu \ll |\mu|$, but we don't need this for this document

I like to remember the direction of the arrow by remembering $\mu(E) = 0 \Rightarrow \nu(E) = 0$, so the arrow is pointing towards ν , so we write the arrow's pointing towards ν : $\nu \ll \mu$. The ν is absolutely continuous with respect to μ , and so it's ν “w.r.t.” μ : ν is on the left, the μ on the right. If μ is any measure and 0 is the zero measure, it is always the case that $\mu \ll \mu$ and $0 \ll \mu$. The most important example of an absolutely continuous measures is given by measures defined via integration:

Definition 3.2.2: Measures Defined By Densities

Let μ be a measure on x and let $f \in L^1(X)$. Then:

$$\nu(E) = \int_E f d\mu$$

is a measure with respect to f .

Clearly $\nu \ll \mu$, since if $\mu(E) = 0$, then $\int_E f d\mu = 0 = \nu(E)$.

It should be verified that $\nu \ll \mu$ if and only if $|\nu| \ll \mu$ if and only if $\nu^+ \ll \mu$ and $\nu^- \ll \mu$. Notice that absolute continuity is sort of the opposite of modular singularity in the sense that if $\nu \perp \mu$ and $\nu \ll \mu$, then it must be that $\nu = 0$. Note that $\mu \perp \nu$ does not imply $\mu(E) = 0 \Rightarrow \nu(E^c) = 0$: take

ν to be the Lebesgue signed measure on $\mathbb{R}$, so that the positive and negative sets are $[0, \infty)$ and $(-\infty, 0)$ and experiment.

The following theorem should give you a taste on why the term “continuity” comes around when ν is finite. In particular, it should give you a feeling of uniformly continuous. Later on, we’ll show that absolute continuity is stronger than uniform continuity, and so this feeling should make sense:

Theorem 3.2.3: Epsilon-Delta Absolute Continuity

Let ν be a finite signed measure on $(X, \mathcal{M})$ and μ a positive measure on $(X, \mathcal{M})$. Then $\nu \ll \mu$ if and only if for every $\epsilon > 0$, there exists a $\delta > 0$ such that $\mu(E) < \delta \Rightarrow |\nu(E)| < \epsilon$.

Proof:

Since $\nu \ll \mu$ if and only if $|\nu| \ll \mu$, and moreover $|\nu(E)| \leq |\nu|(E)$, it suffices to show it in the case where $\nu = |\nu|$ is positive.

($\Leftarrow$) Let the ϵ - δ definition hold so that for any $\epsilon > 0$, there exists a $\delta > 0$ such that for all $E \in \mathcal{M}$ that satisfy $\mu(E) < \delta$, then $\nu(E) < \epsilon$. If we pick all E such that $\mu(E) = 0$, then for any ϵ arbitrarily small, any δ will work, and so $\nu(E) < \epsilon$ for all ϵ , and so it must be that $\nu(E) = 0$

($\Rightarrow$) assume the contra-positive; that the ϵ - δ condition is not satisfied, meaning there exists a $\epsilon > 0$ such that for all $\delta > 0$, there is a E_δ such that $\mu(E_\delta) < \delta \Rightarrow \nu(E_\delta) \geq \epsilon$. In particular, for every $n \in \mathbb{N}$, $\mu(E_n) < 2^{-n} \Rightarrow \nu(E_n) \geq \epsilon$. let $F_k = \bigcup_{n=k}^{\infty} E_n$ and $F = \bigcap_k F_k$. Then

$$\mu(F_k) < \sum_{n=k}^{\infty} \frac{1}{2^n} = 2^{1-k}$$

and since $F \subseteq F_k$ for all F_k , it must be that $\mu(F) = 0$. However, by assumption $\nu(F_k) \geq \epsilon$ for all k . Since ν is finite, $\nu(F_1)$ is finite, so continuity from above implies

$$\nu(F) = \lim \nu(F_k) \geq \epsilon$$

and so $\nu \not\ll \mu$

An easy way to find a ν that is absolutely continuous with respect to μ is if we have a measure μ and a extended μ -integrable function f and define $\nu(E) = \int_E f d\mu$. If $f \in L^1$, then ν is finite, in which case we have the following corollary:

Corollary 3.2.4: Absolute Continuity And L^1

If $f \in L^1(\mu)$, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\forall E \in \mathcal{M}, \mu(E) < \delta \Rightarrow \left| \int_E f \right| < \epsilon$$

Proof:

apply the previous theorem on Ref and $\text{Im}f$

The definition of ν with respect to $\int_E f d\mu$ is common enough that we define the following notation for it:

Definition 3.2.5: Density Notation

$$d\nu = f d\mu$$

This should give you the feeling of change of variables, and indeed the two concepts will be related. I like to remember this notation by thinking that if we add an integral with respect to E , $\int_E$ to both sides, and imagine there is the constant function in-front of the $d\nu$ on the left hand side, we get back our previous notation of $\nu(E) = \int_E f d\mu$.

We are now in a position to fully classify signed measures given positive measures. The way we will generalize this is to start by remember that an extended μ integrable function f has one part that allow's for an infinite value. Thus, we can think of the domain of f as being split between a signed measure and a positive measure. To avoid pathologies, we will work over σ -finite measures (as usual). We first prove some technical lemma before continuing to the main result:

Lemma 3.2.6

Let μ and ν be finite measures on $(X, \mathcal{M})$. Then either $\mu \perp \nu$, or there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\nu \geq \epsilon\mu$ on E (that is, ν and μ have domain $M_E = \{E \cap F \mid F \in \mathcal{M}\}$, and E is a positive set for $\nu - \epsilon\mu$)

Proof:

Let $X = P_n \amalg N_n$ be the Hahn decomposition for $\nu - n^{-1}\mu$, and let $P = \bigcup_n P_n$ and $N = \bigcap_n N_n = P^c$. Then N is a negative set on $\nu - n^{-1}\mu$ for all n a.e., and

$$0 \leq \nu(N) \leq n^{-1}\mu(N)$$

for all n , so $\nu(N) = 0$. If $\mu(P) = 0$, then $\nu \perp \mu$. If $\mu(P) > 0$, then $\mu(P_n) > 0$ for some n , and P_n is a positive set for $\nu - n^{-1}\mu$

Lemma 3.2.7: Sum and Absolute Continuity

Suppose $\{\nu_i\}$ is a sequence of positive measures. If $\nu_i \perp \mu$ for all i , then $\sum_i^\infty \nu_i \perp \mu$ and if $\nu_i \ll \mu$ for all i then $\sum_i^\infty \nu_i \ll \mu$

Proof:

Let $\nu_i \perp \mu$ for all i , so that $X = A_i \amalg B_i$ where $\nu_i(A_i) = 0$ and $\mu(B_i) = 0$ for all i . Since $\mu(B_i) = 0$ for all i , then $\mu(\bigcup_i B_i) = 0$. Similarly, $\sum_i^n \nu_i(\bigcap_i A_i) = 0$ since $\bigcap_i A_i \subseteq A_i$ for each i . By some basic set theory

$$X = \left(\bigcap_i A_i \right) \amalg \left(\bigcup_i B_i \right) \quad \left(\bigcap_i A_i \right) \cap \left(\bigcup_i B_i \right) = \emptyset$$

since each x is in either A_i or B_i and $A_i \amalg B_i = X$ for each i . Therefore, we have:

$$\sum_i \nu_i \perp \mu$$

For the second statement, let's say $\nu_i \ll \mu$ for all i . This implies that:

$$\begin{aligned} \mu(E) = 0 &\Rightarrow \nu_1(E) = 0 \\ &\Rightarrow \nu_2(E) = 0 \\ &\vdots \\ &\Rightarrow \nu_n(E) = 0 \\ &\vdots \end{aligned}$$

meaning if $\mu(E) = 0$, then each $\nu_i(E) = 0$. But then $\sum_i \nu_i(E) = 0$, and so

$$\sum_i \nu_i \ll \mu$$

as we sought to show.

Theorem 3.2.8: Lebesgue-Radon-Nikodym Theorem

Let ν be a signed σ -finite measure on $(X, \mathcal{M})$ and μ be a positive σ -finite measure on $(X, \mathcal{M})$. Then, there exists unique σ -finite measure λ and ρ on $(X, \mathcal{M})$ such that

$$\lambda \perp \mu, \quad \rho \ll \mu \quad \text{and} \quad \nu = \lambda + \rho$$

Furthermore, there is an extended μ -integrable function f such that $d\rho = f d\mu$, and any two such functions are equal μ -a.e. and so we get:

$$\nu(A) = \int_A f d\mu + \lambda(A)$$

In other words, λ and ρ “split” the measure μ based on where μ is zero or not, and together they can be combined to make ν . Notice that there is no prior relation between the signed measure ν and the positive measure μ . This means that we can replace μ with μ' , and find $\lambda' \perp \mu'$ and $\rho' \ll \mu'$, and still get $\nu = \lambda' + \rho'$, and so this decomposition of a signed measure with respect to a measure μ heavily depends on μ . This decomposition might make more sense if we consider the special case where $\nu \ll \mu$, in which case we are left with $\nu = \rho$. The last statement of the theorem tells us that:

$$d\rho = f d\mu \quad \text{which is shorthand for} \quad \rho(E) = \int_E f d\mu$$

Notice that this looks like the change of variables we were talking about earlier. Thus, in this case, this theorem is ultimately about a “change of variables” into the μ measure!

Finally, notice that this “change of variables” shows that the measure ν is sort of the “integral” of the measure of f , and so in a sense f is thus the “derivative” of ν since you “get back” ν by integrating. In other words, this should start giving you Fundamental Theorem of Calculus where $d/dx(\int f) = f$. However, be wary that the function f is not [always] obtained by taking the limit of some function F , i.e. does not always exist a function F such that $f(a) = F'(a) = \lim_{x \rightarrow a} \frac{F(x+a) - F(a)}{x-a}$. However, in the next section, we will see that in some special cases we *can* define a function F (which we obtain through ν) where $F' = f$. Furthermore, this does not imply that we can compute real measures by defining a distribution F like in the case of Lebesgue measures (which is the other result of the FTC). Thus, the function f can be thought of as a sort of generalization of the notion of derivative.

Proof:

We will break down this proof into 3 cases: when ν and μ are finite positive measures, when ν and μ are σ -finite positive measures, and finally when ν is a signed measure.

For the first case, let:

$$\mathcal{F} = \left\{ f : X \rightarrow [0, \infty] \mid \int_E f d\mu \leq \nu(E) = \int_E 1 d\nu \text{ for all } E \in \mathcal{M} \right\}$$

Remember that we can define a measure through the integral, so what we are asking for is if we can find all possible functions which can define us a measure that will be smaller than ν . We will end up taking the largest such function f , and labelling $d\rho = f d\mu$, and $d\lambda = d\nu - d\rho$. We can already see that $\rho \ll \mu$. What's left to show is that such an f exists, that $\lambda \perp \mu$ and that the result is unique.

Notice also how the construction can make you think of change of variables. Let's say for a moment that $\int_E f d\mu = \int_E 1 d\nu$. Then it is as though f is correcting the value so that the values are equal. This is actually what we are looking for, so we look for all f such that $\int_E f d\mu \leq \int_E 1 d\nu$ and take the supremum.

First, $\mathcal{F}$ is nonempty since $0 \in \mathcal{F}$. Thus, let $a = \sup \left\{ \int f d\mu \mid f \in \mathcal{F} \right\}$ (notice that $a \leq \nu(X) < \infty$ since ν is finite), and choose a sequence $\{f_n\} \subseteq \mathcal{F}$ such that $\int f_n d\mu \rightarrow a$ and let $f = \sup_n f_n$. We would like to show that $a = \int f$. To that end, notice that if $f, g \in \mathcal{F}$, then $\max(f, g) \in \mathcal{F}$: if $A = \{x \mid f(x) > g(x)\}$, for any $E \in \mathcal{M}$:

$$\int_E \max(f, g) d\mu = \int_{E \cap A} f d\mu + \int_{E \setminus A} g d\mu \leq \nu(E \cap A) + \nu(E \setminus A) = \nu(E)$$

Let $g_n = \max(f_1, f_2, \dots, f_n)$. Then $g_n \in \mathcal{F}$, g_n increases pointwise to f , and $\int g_n d\mu \geq \int f_n d\mu$ by monotonicity. Thus, since a is the supremum, it must be that $\lim \int g_n d\mu = a$, and so by the MCT

$$a = \lim \int g_n d\mu = \int \lim g_n d\mu = \int f$$

and so $f \in \mathcal{F}$ (i.e. the supremum is “achieved”) and $\int f d\mu = a$. In particular, $f < \infty$ a.e., so we may take f to be real-valued everywhere. Let $d\rho = f d\mu$ (we will soon show this $d\rho$ satisfies the conditions of the theorem).

We now define the measure λ which is singular with respect to μ . Let $d\lambda = d\nu - f d\mu$ (i.e. $\lambda(E) = \nu(E) - \int_E f d\mu$). Then $d\lambda$ is positive, since $f \in \mathcal{F}$, and is singular with respect to

μ . If it weren't, then by lemma 3.2.6, there exists a $\epsilon > 0$ and $E \in \mathcal{M}$ such that $\mu(E) > 0$ and $\lambda \geq \epsilon\mu$ on E . Then $\epsilon\chi_E d\mu \leq d\lambda = d\nu = f d\mu$, or rearranging $(f + \epsilon\chi_E)d\mu \leq d\nu$. Since $f \in \mathcal{F}$ and the $\epsilon\chi_E$ only effects the value on E , we just showed that $f + \epsilon\chi_E \in \mathcal{F}$. However, $\int f + \epsilon\chi_E d\mu = a + \epsilon\mu(E) > a$, contradicting the maximality of a . Hence it must be that $\lambda \perp \mu$. Therefore, we have that $d\nu = f d\mu + d\lambda$ where $\lambda \perp \mu$. Recalling we set $d\rho = f d\mu$, we get that $d\nu = d\rho + d\lambda$, and by definition of ρ we have $\rho \ll \mu$ (see definition 3.2.2). Converting into our more usual notation, we see that $\nu(E) = \rho(E) + \lambda(E)$ for all E , and so $\nu = \rho + \lambda$ with $\lambda \perp \mu$ and $\rho \ll \mu$. It remains to show that this decomposition is unique. To that end, let's say $d\nu = d\lambda' + f' d\mu$. Then $d\lambda - d\lambda' = (f - f')d\mu$. Then (show that $d\lambda - d\lambda' \perp \mu$) and $(f' - f)d\mu \ll d\mu$, and since both sides must be equal, this forces that $d\lambda - d\lambda' = (f' - f)d\mu = 0$. Thus, $\lambda = \lambda'$, and by proposition 2.3.4, $f = f'$ μ -a.e. Thus, if ν and μ are of finite measure, the theorem is done.

Now let ν and μ be σ -finite measures so that X is a countable union of μ -finite sets and a countable union of ν -finite sets. By refining the two by taking intersections, we get a new collection $\{A_i\} \subseteq \mathcal{M}$ $\mu(A_i)$ and $\nu(A_i)$ is finite for all i and $X = \bigcup_{i=1}^{\infty} A_i$. We can define

$$\mu_i(E) = \mu(E \cap A_i) \quad \nu_i(E) = \nu(E \cap A_i)$$

Then by what we've just shown, we have the decomposition:

$$d\nu_i = d\lambda_i - f_i d\mu$$

where $\lambda_i \perp \mu_i$. Now $\mu_i(A_i^c) = \nu_i(A_i^c) = 0$, so $\int_{A_i^c} f d\mu = 0$ and

$$\lambda_i(A_i^c) = \nu_i(A_i^c) - \int_{A_i^c} f d\mu = 0$$

for our construction, let f_i be 0 on A_i^c . Now, let $\lambda = \sum_{i=1}^{\infty} \lambda_i$ and $f = \sum_{i=1}^{\infty} f_i$. Then by lemma 3.2.7:

$$d\nu = d\lambda - f d\mu$$

and $d\lambda$ and $f d\mu$ are σ -finite, as desired.

Finally, taking the case of ν being a signed measure, using the Jordan decomposition, apply the preceding argument to ν^+ and ν^- , completing the proof

Since the decomposition exists, it is given a name:

Definition 3.2.9: Lebesgue Decomposition

Let ν be a signed σ -finite measure and μ a positive σ -finite measure. Then the decomposition

$$\nu = \lambda + \rho \quad \lambda \perp \mu \quad \rho \ll \mu$$

is called the *Lebesgue decomposition of ν with respect to μ* .

if $\nu \ll \mu$, then the Lebesgue-Radon-Nikodym theorem tell's us that $d\nu = f d\mu$. If the extra condition that $\nu \ll \mu$ is added, the theorem is often called *Radon-Nikodym's Theorem*, and f is given a special name:

Definition 3.2.10: Radon-Nikodym Derivative

Let $\nu \ll \mu$ and f be the function obtain from the Lebesgue Decomposition such that $d\nu = f d\mu$. Then f is called the *Radon-Nikodym derivative of ν with respect to μ* . We often denote f by:

$$\frac{d\nu}{d\mu} := f$$

As a consequence of this notation, we get the following representation of $\nu(E) = \rho(E)$ that looks like a fraction cancellation:

$$d\nu = \frac{d\nu}{d\mu} d\mu$$

Strictly speaking, $\frac{d\nu}{d\mu}$ should really be the class of function's that are equal to f μ -a.e., however just like with L^1 , we often abuse notation and pick some function in this class when we use this notation). Most of the time, we will be working with the case of $\nu \ll \mu$. In Distribution theory, we will be studying the case where $\nu \not\ll \mu$.

Writing f in this form is very fitting, as in f respects the same properties as the derivative! Once we reach the special case of the measures being Lebesgue measures and the functions being differentiable functions, then f will in fact be the derivative of f .

Without yet doing concrete computations of f , if we take the Radon-Nikodym of $\nu_1 + \nu_2$ where $\nu_1 + \nu_2 \ll \mu$ then we have

$$\frac{d(\nu_1 + \nu_2)}{d\mu} = \frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu}$$

to see this, let $d(\nu_1 + \nu_2)/d\mu = f$. Since $\nu_1 + \nu_2 \ll \mu$, $\nu_1 \ll \mu$ and $\nu_2 \ll \mu$. Then taking $d\nu_1 = f_1 d\mu$ and $d\nu_2 = f_2 d\mu$, then:

$$\begin{aligned} &= \int \frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} d\mu \\ &= \int \frac{d\nu_1}{d\mu} d\mu + \int \frac{d\nu_2}{d\mu} d\mu \\ &= \int 1 d\nu_1 + \int 1 d\nu_2 \\ &= \nu_1(E) + \nu_2(E) \\ &= (\nu_1 + \nu_2)(E) \\ &= \int 1 d(\nu_1 + \nu_2) \\ &= \int_E \frac{d(\nu_1 + \nu_2)}{d\mu} d\mu \end{aligned}$$

and since the two integrals are equal over every measurable set, we get:

$$\frac{d\nu_1}{d\mu} + \frac{d\nu_2}{d\mu} = \frac{d(\nu_1 + \nu_2)}{d\mu}$$

which is the same property of differentiable functions. In particular, notice that similarly we have the following chain-rule property:

Proposition 3.2.11: Chain Rule For Measures

Let ν be a signed σ -finite measure on $(X, \mathcal{M})$, μ and λ positive σ -finite measures on $(X, \mathcal{M})$ such that $\nu \ll \mu$ and $\mu \ll \lambda$. Then:

1. If $g \in L^1(\nu)$, then $g(d\nu/d\mu) \in L^1(\mu)$ and

$$\int g d\nu = \int g(d\nu/d\mu) d\mu$$

2. Absolute continuity, $\ll$, is transitive: $\nu \ll \lambda$ and

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda}$$

Proof:

1. This proof uses (what now should be) the standard measure-theory way to generalize results. First, we can consider ν^+ and ν^- separately, so without loss of generality let's say $\nu \geq 0$. If we let $g = \chi_E$, then:

$$\int g \frac{d\nu}{d\mu} d\mu = \int \int \chi_E f d\mu = \int_E f d\mu$$

which is true by definition of f . Thus, it is true for any simple functions since the integral and the construction is linear, and so it is also true for non-negative function's by the MCT, and so it is true for functions in $L^1(\nu)$ by linearity again.

2. If we take ν to be μ and μ to be λ and set $g = \chi_E \frac{d\nu}{d\mu}$, we get:

$$\nu(E) = \int_E \frac{d\nu}{d\mu} d\mu = \int_E \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda} d\lambda$$

for all $E \in \mathcal{M}$, including X , and so by proposition 2.3.4

$$\frac{d\nu}{d\lambda} = \frac{d\nu}{d\mu} \frac{d\mu}{d\lambda} \quad \lambda\text{-a.e.}$$

as we sought to show

As a consequence, if $\mu \ll \lambda$ and $\lambda \ll \mu$, then $(d\lambda/d\mu)(d\mu/d\lambda) = 1$ λ -a.e. and μ -a.e. Note that all of this is for the special case of $\nu \ll \mu$. If this is not the case (say $\nu \not\ll \mu$) we get a different notion of “ $d\nu/d\mu$ ” (the example of $\delta_0 \perp \lambda$ has a “Radon-Nikodym derivative” of the dirac delta function).

Note that σ -finiteness is necessary:

Example 3.2.12: Without σ -Finiteness

Let $X = [0, 1]$, $\mathcal{M} = \mathcal{B}_{[0,1]}$, m the Lebesgue measure and μ the counting measure on $\mathcal{M}$

1. $m \ll \mu$ but $dm \neq f d\mu$ for any f

Proof. First, if $\mu(E) = 0$, then $|E| = 0$, which we have shown that this implies $m(E) = 0$, and so $m \ll \mu$.

However, there does not exist an f such that $m(E) = \int_E f d\mu$. To see this, consider any $p \in [0, 1]$. Then

$$0 = m(\{p\}) = \int_{\{p\}} f d\mu = \int_{[0,1]} \chi_{\{p\}} f d\mu \stackrel{!}{=} f(p)\mu(\{p\}) = f(p)$$

where $\stackrel{!}{=}$ comes from the fact that $\chi_{\{p\}}f$ is in fact a simple function (being of the form $\sum_i c_i \chi_{E_i}$ with $i = 1$, $E_i = \{p\}$ and $c_i = f(p)$), and so matches the definition of the integral for simple functions (which we have shown in class is the value of the usual integral). Hence, we have that $f(p) = 0$ for all $p \in [0, 1]$. However, this implies that $f \equiv 0$. This leads to a contradiction, since:

$$1 = m([0, 1]) = \int_{[0,1]} f d\mu = \int_{[0,1]} 0 d\mu = 0$$

but $1 \neq 0$ □

Next, show that μ has no Lebesgue decomposition with respect to m

Proof. For the sake of contradiction, let $\mu = \lambda + \rho$ where $\lambda \perp m$ and $\rho \ll m$. Then for any $c \in [0, 1]$

$$\mu(\{c\}) = \lambda(\{c\}) = 1$$

since $m(\{c\}) = 0$ and $\rho \ll m$, $\rho(\{c\}) = 0$. But then, λ is never null on non-zero sets by monotonicity, implying that m must *always* be null. However, this is not true since m is the Lebesgue measure – a contradiction. Thus, μ does not have such a decomposition with respect to m . □

Before we move on to our last result of the section, I want to take a moment to talk about generalizations of the Radon-Nikodym theorem. In the theorem, we assumed that the measures must be σ -finite. However, in contrast to previous example where without σ -finiteness the theorem fails badly, we seem to have these assumption because the generalization requires more build-up. In particular, to fix the lack of control of σ -finiteness, we would have to define the notion of being *truly continuous*. If μ is σ -finite, being absolutely continuous and truly continuous are equivalent. The statement of the Radon-Nikodym can be reformulated in terms of truly continuous measures, and I found a stack-exchange post on this:

<https://mathoverflow.net/questions/258936/radon-nikodym-theorem-for-non-sigma-finite-measures>

Finally, we conclude with this final proposition we'll use later on:

Proposition 3.2.13

Let $\mu_1, \dots, \mu_n$ be measures on $(X, \mathcal{M})$. Then there exists a measure μ (namely $\sum_i \mu_i$) such that $\mu_i \ll \mu$

Proof:
exercise

3.3 Complex Measure

We know show how we can generalize the results when the codomain is $\mathbb{C}$. The reason to define complex measure is generalize the notion of the Radon-Nikodym derivative to the complex case so that we may define complex results. Most of the work we have done before translates one-to-one for the complex measures, with the exception of defining the absolute variation which we will spend most of this section covering.

Definition 3.3.1: Complex Measure

Let $(X, \mathcal{M})$ be a measure space. A function $\nu : \mathcal{M} \rightarrow \mathbb{C}$ is called a *complex measure* if

1. $\nu(\emptyset) = 0$
2. If $\{E_i\} \subseteq \mathcal{M}$ is a pair-wise disjoint sequence, then if $\sum_i^\infty \nu(E_i)$ converges absolutely, then

$$\nu\left(\bigcup_i^\infty E_i\right) = \sum_i^\infty \nu(E_i)$$

Just like for signed measures, the absolute convergence is so that the order in which we measure the sets does not matter.

Notice that the sequence $X, \emptyset, \emptyset, \dots$ is a valid disjoint sequence, and so $\sum_i \nu(X_i) = \nu(X)$ must converge absolutely, and so ν must be a finite measure. Thus, a measure is a complex measure if it is finite. Another way of saying it is that if μ is a positive measure and $f \in L^1(\mu)$ (recall f is a complex function), then $f d\mu$ is also a complex measure. As is usually done with complex valued function, we define the signed measures $\nu_r(E) = \pi_r(\nu(E))$ and $\nu_i(E) = \frac{\pi_i(\nu(E))}{i}$ to take on the real and complex values of ν , so ν_i and ν_r are *finite* signed measures and $\nu = \nu_r + i\nu_i$. As a consequence, the range of ν is a bounded subset of $\mathbb{C}$.

Many of the previous concepts we've seen naturally generalize to the complex case:

1. We define $L^1(\nu)$ to be $L^1(\nu_r) \cap L^1(\nu_i)$
2. For $f \in L^1(\nu)$, set $\int f d\nu := \int f d\nu_r + i \int f d\nu_i$
3. If ν and μ are complex measures, then we that $\mu \perp \nu$ if $\nu_a \perp \nu_b$ for all $a, b \in \{r, i\}$.
4. If λ is a positive measure, we say $\nu \ll \lambda$ if $\nu_r \ll \lambda$ and $\nu_i \ll \lambda$
5. The Lebesgue-Radon-Nikodym Theroem generlizes to complex measures nicely; you would apply theorem 3.2.8 to the real and imaginary part seperately. To make sure there is no ambiguity in understanding the generalization:

Let ν be a complex measure and μ a σ -finite measure on $(X, \mathcal{M})$. Then there exists a complex measure λ and a $f \in L^1(\mu)$ such that

$$d\nu = d\lambda + f d\mu \quad \lambda \perp \mu$$

Furthermore, if there existed another λ' such that $\lambda' \perp \mu$ and $d\nu = d\lambda' + f'd\mu$, then $\lambda = \lambda' \mu$ -a.e. and $f = f' \mu$ -a.e.

If $\nu \ll \mu$, then we denote f a $\frac{d\nu}{d\mu}$

We next want to generalize the notion of *total variation* to the complex case. In the signed measure case, this was an easy task since $\mathbb{R}$ has an inherent order. In the complex case, this is not so straight forward: Given that a surface $S \subseteq \mathbb{C}$ is the image of some complex measure $\nu(E) = S$, what does it mean to capture the “total-variation” on that surface? There is more than one sensible answer to this question, namely since there is no “natural order” to $\mathbb{C}$, and so it would be nice to have succinct way to capture it's *total variation*. There is in fact a nice and concise way of defining this which also generalizes the previous way of looking at total variation that captures the idea in 1 formula, and so we will build-up to it.

First, let ν be a complex measure and let $\mu = |\nu_r| + |\nu_i|$. Next, notice that $\nu \ll \mu$, since if $\mu(E) = 0$, then it must be that both $|\nu_r|$ and $|\nu_i|$ are zero, and so $\nu_r(E) = \nu_i(E) = 0$, and so $\nu(E) = 0$. Therefore, the Lebesgue decomposition of ν with respect to μ gives us a f such that $d\nu = f d\mu$. As an example of such an f , let's say that ν was simply a finite signed measure (which is a special case of a complex measure), so $\mu = |\nu|$. Then $f = (\chi_P - \chi_N)$ and:

$$d\nu = (\chi_P - \chi_N)d\mu = (\chi_P - \chi_N)d|\nu|$$

Which is the Jordan decomposition theorem we have shown earlier! Now, notice that $|\chi_P - \chi_N| = 1$ so that we can write:

$$d|\nu| = |\chi_P - \chi_N|d\mu \quad \text{which is shorthand for} \quad |\nu|(E) = \int_E |\chi_P - \chi_N|d\mu$$

If we re-write the last equation as $d|\nu| = |f|d\mu$, then we have that the measure $|\nu|$ is defined by the absolute value of the measure defined by the derivative with respect to $|\nu|$. If we come back to the case of ν being a general complex measure, then we would get $\mu = |\nu_r| + |\nu_i|$, giving us a less obvious, but certainly existent decomposition $d|\nu| = |f|d\mu$ (i.e. $|\nu|(E) = \int_E |f|d\mu$). Thus, it seems that this property somehow “captures” the idea of total variation. We chose a particular μ for both decomposition, however it turns out that this property we have stated is independent of choice of μ up to an appropriate zero-measure set, and so we will define total-variation with respect to the following property:

Definition 3.3.2: Total Variation And Complex Differentiation

Let ν be a complex measure. Then $|\nu|$ is defined such that when $d\nu = f d\mu$ where μ is a positive measure, then $d|\nu| = |f|d\mu$

What we have shown with the μ we saw earlier is that $d\nu$ is always decomposeable into $d\nu = f d\mu$. We must now show that $|\nu|$ is in-fact independent of choice of μ (i.e. any choice will get the same result up zero an appropriate zero-measure set) given μ has the property of the definition. Let's say $d\nu = f_1 d\mu_1 = f_2 d\mu_2$ satisfy the total-variation property. Let $\rho = \mu_1 + \mu_2$. By proposition 3.2.11, we have that:

$$f_1 \frac{d\mu_1}{d\rho} d\rho = d\nu = f_2 \frac{d\mu_2}{d\rho} d\rho$$

Thus $f_1 \frac{d\mu_1}{d\rho} = f_2 \frac{d\mu_2}{d\rho}$ ρ -a.e. Since $\frac{d\mu_1}{d\rho}$ and $\frac{d\mu_2}{d\rho}$ are both non-negative, we have:

$$|f_1| \frac{d\mu_1}{d\rho} d\rho = \left| f_1 \frac{d\mu_1}{d\rho} d\rho \right| = \left| f_2 \frac{d\mu_2}{d\rho} d\rho \right| = |f_2| \frac{d\mu_2}{d\rho} d\rho \quad \rho\text{-a.e.}$$

Thus:

$$|f_1|d\mu_1 = |f_1|\frac{d\mu_1}{d\rho}d\rho = |f_2|\frac{d\mu_2}{d\rho}d\rho = |f_2|d\mu_2$$

and so, ν is independent of choice of μ . We next give some useful properties of total variation:

Proposition 3.3.3: Properties Of Total Variation

Let $(X, \mathcal{M}, \nu)$ be a complex measure space.

1. $|\nu(E)| \leq |\nu|(E)$ for all $E \in \mathcal{M}$
2. $\nu \ll |\nu|$ and $\frac{d\nu}{d|\nu|}$ has absolute value 1 $|\nu|$ -a.e.
3. $L^1(\nu) = L^1(|\nu|)$ and if $f \in L^1(\nu)$, then $|\int f d\nu| \leq \int |f| d|\nu|$

Proof:

Pick f such that $d\nu = fd\mu$ (as we've shown exists when defining $|\nu|$). Then:

$$|\nu(E)| = \left| \int_E fd\mu \right| \leq \int_E |f|d\mu = |\nu|(E)$$

showing the first claim, and also showing that $\nu \ll |\nu|$. For the next claim, let $g = \frac{d\nu}{d|\nu|}$ so

$$fd\mu = d\nu = gd|\nu|$$

so $g|f| = f$ μ -a.e. and so $|\nu|$ -a.e. Since $|f| > 0$ $|\nu|$ -a.e. as well, it must be that $g = 1$ $|\nu|$ -a.e.

Next, (left as an exercise by Folland)

The next property we introduce shows that it interacts well with the triangle inequality

Proposition 3.3.4: Triangle Inequality Of Total Variation

Let ν_1 and ν_2 be complex measures on $(X, \mathcal{M})$. Then

$$|\nu_1 + \nu_2| \leq |\nu_1| + |\nu_2|$$

Proof:

By proposition 3.2.13, we can write $\nu_1 = f_1d\mu$ and $\nu_2 = f_2d\mu$ for the same μ . Then:

$$d|\nu_1 + \nu_2| = |f_1 + f_2|d\mu \leq |f_1|d\mu + |f_2|d\mu = d|\nu_1| + d|\nu_2|$$

as we sought to show

Finally, though this characterization of total variation is a good definition, it can be hard to work with. The following gives an equivalent definition of total variation that is easier computationally:

Proposition 3.3.5: Equivalence Of Total Variation

Let ν be a complex measure on $(X, \mathcal{M})$. If $E \in \mathcal{M}$, define

1. $\mu_1(E) = \sup \left\{ \sum_{i=1}^n |\nu(E_i)| \mid n \in \mathbb{N}, E_1, \dots, E_n \text{ disjoint}, E = \bigsqcup_{i=1}^n E_i \right\}$
2. $\mu_2(E) = \sup \left\{ \sum_{i=1}^\infty |\nu(E_i)| \mid E_1, \dots \text{ disjoint}, E = \bigsqcup_{i=1}^\infty E_i \right\}$
3. $\mu_3(E) = \sup \left\{ \left| \int_E f d\nu \right| \mid |f| \leq 1 \right\}$

Then $\mu_1 = \mu_2 = \mu_3 = |\nu|$

Proof:

Throughout this proof, let:

$$S_1^E = \left\{ \sum_{i=1}^n |\nu(E_i)| \mid n \in \mathbb{N}, E_1, \dots, E_n \text{ disjoint}, E = \bigsqcup_{i=1}^n E_i \right\}$$

and

$$S_2^E = \left\{ \sum_{i=1}^\infty |\nu(E_i)| \mid E_1, \dots \text{ disjoint}, E = \bigsqcup_{i=1}^\infty E_i \right\}$$

($\mu_1 \leq \mu_2$) Notice that $S_1^E \subseteq S_2^E$ (since for any finite disjoint sequence, we can extend it to an infinite disjoint sequence by letting $E_i = \emptyset$ for the rest of the terms), we get $\sup S_1^E \leq \sup S_2^E$, and since this is true for all E , $\mu_1 \leq \mu_2$

($\mu_2 \leq \mu_3$) In the process we will use the fact that there exists an f such that:

$$d\nu = f d|\nu|$$

By proposition 3.3.3, $\left| \frac{d\nu}{d|\nu|} \right| = |f| = 1$ $|\nu|$ -a.e, and so $|f| \leq 1$ $|\nu|$ -a.e. For the $|\nu|$ -zero-measure points that are not less than or equal to 1, we can simply choose them to be equal to 0. This also implies that $|\bar{f}| \leq 1$. With this information we get that:

$$\bar{f} d\nu = \bar{f} f d|\nu| = |f|^2 d|\nu| = 1 d|\nu| = d|\nu| \quad |\nu| \text{-a.e.}$$

and so:

$$\begin{aligned} \sum_i |\nu(E_i)| &\leq \sum_i |\nu|(E_i) \\ &= \int_E d|\nu| \\ &= \int_E \bar{f} d\nu \\ &\leq \left| \int_E \bar{f} d\nu \right| \end{aligned}$$

and since $|f| \leq 1$, this is an element of $\left\{ \left| \int_E f d\nu \right| \mid |f| \leq 1 \right\}$, and since this was true for arbitrary E , we get: $\mu_2 \leq \mu_3$

($\mu_3 = |\nu|$) We'll first show $\mu_3 \leq |\nu|$. By the definition of the supremum, for all $\epsilon > 0$, there exists an f where $|f| \leq 1$ such that

$$\begin{aligned}\mu_3 &\leq \left| \int_E f d\nu \right| + \epsilon \\ &\leq \int_E |f| d|\nu| + \epsilon && \text{prop. 3.13[c]} \\ &\leq \int_E d|\nu| + \epsilon && |f| \leq 1 \\ &\leq |\nu|(E) + \epsilon\end{aligned}$$

and as $\epsilon \rightarrow 0$, we have that $\mu_3 \leq |\nu|(E)$. Going the other way, we can again define $\frac{d\nu}{d|\nu|}$, which by proposition 3.13 its absolute value is 1 $|\nu|$ -a.e. Let $\bar{f} = \frac{d\nu}{d|\nu|}$ so that $1 = |f|^2 = f\bar{f}$ $|\nu|$ -a.e. If f ever has value greater than 1, we can simply redefine them to be 0 since those value have measure 0. Then:

$$\begin{aligned}|\nu|(E) &= \int_E d|\nu| \\ &= \int_E 1 d|\nu| \\ &= \int_E |f|^2 d|\nu| \\ &= \int_E |f| d|\nu| && \text{since } |f| = 1 \text{ } |\nu| \text{-a.e.} \\ &= \int_E f \bar{f} d|\nu| \\ &= \int_E f \frac{d\nu}{d|\nu|} d|\nu| \\ &= \int_E f d\nu && \text{prop. 3.9} \\ &\leq \left| \int_E f d\nu \right|\end{aligned}$$

and since $|f| \leq 1$,

$$\left| \int_E f d\nu \right| \in \left\{ \left| \int_E f d\nu \right| \mid |f| \leq 1 \right\}$$

and since this is true for arbitrary E :

$$|\nu| \leq \mu_3$$

and so, combining the two inequalities, we get $\mu_3 = |\nu|$

($\mu_3 \leq \mu_1$) We will do as the hint suggests and use simple functions to approximate f , in particular we will use the standard representation from theorem 2.10, let's label the increasing sequence $\{\varphi_k\}_{k=1}^\infty$. To take advantage of these functions, first

break up ν into positive measures, $\nu_r^+, \nu_i^+, \nu_r^-, \nu_i^-$ so that:

$$\left| \int_E f d\nu \right| = \left| \int_E f d\nu_r^+ - \int_E f d\nu_r^- + i \left(\int_E f d\nu_i^+ - \int_E f d\nu_i^- \right) \right|$$

Then by the definition of integral, for all $\epsilon/4 > 0$, there must exist an $n \in \mathbb{N}$ such that

$$\begin{aligned} &= \left| \int_E f d\nu_r^+ - \int_E f d\nu_r^- + i \left(\int_E f d\nu_i^+ - \int_E f d\nu_i^- \right) \right| \\ &\leq \left| \int_E \varphi_n d\nu_r^+ + \frac{\epsilon}{4} - \int_E \varphi_n d\nu_r^- - \frac{\epsilon}{4} + i \left(\int_E \varphi_n d\nu_i^+ + \frac{\epsilon}{4} - \int_E \varphi_n d\nu_i^- - \frac{\epsilon}{4} \right) \right| \\ &\leq \left| \int_E \varphi_n d\nu_r^+ - \int_E \varphi_n d\nu_r^- + i \left(\int_E \varphi_n d\nu_i^+ - \int_E \varphi_n d\nu_i^- \right) \right| + \epsilon \\ &= \left| \int_E \varphi_n d\nu \right| + \epsilon \end{aligned}$$

Then by the definition of a simple function, $\varphi_n = \sum_{i=1}^n c_i \chi_{F_i}$ for appropriate F_i and c_i . Note that $c_i \leq 1$ since $|\varphi_n| \leq |f| \leq 1$. Also, note that $\chi_E \chi_{F_i} = \chi_{E \cap F_i}$. Then by definition of an integral:

$$\left| \int_E \varphi_n d\nu \right| + \epsilon = \left| \sum_{i=1}^n c_i \nu(F_i \cap E) \right| + \epsilon \leq \sum_{i=1}^n |c_i| |\nu(F_i \cap E)| + \epsilon$$

This is almost the result we need. The problem is that the collection $\{F_i \cap E\}$ is not guaranteed to cover E since there might be portions that are yet to be “precise” enough to bring it into the approximation. Since we only need to get bigger, we can solve this by adding $\bigcap_{i=1}^n F_i^c \cap E$, which is disjoint of every $F_i \cap E$, and so:

$$\begin{aligned} \sum_{i=1}^n |c_i| |\nu(F_i \cap E)| + \epsilon &\leq \sum_{i=1}^n |c_i| |\nu(F_i \cap E)| + \left| \nu \left(\bigcap_{i=1}^n F_i^c \cap E \right) \right| + \epsilon \\ &\leq \sum_{i=1}^n |\nu(F_i \cap E)| + \left| \nu \left(\bigcap_{i=1}^n F_i^c \cap E \right) \right| + \epsilon \end{aligned}$$

and so, combining all the inequalities, we get:

$$\left| \int_E f d\nu \right| \leq \sum_{i=1}^n |\nu(F_i \cap E)| + \left| \nu \left(\bigcap_{i=1}^n F_i^c \cap E \right) \right| + \epsilon$$

and so, by the definition of the supremum:

$$\left| \int_E f d\nu \right| \leq \sup S_1^E$$

and since this is true for arbitrary $\left| \int_E f d\nu \right|$:

$$\mu_3 \leq \mu_1$$

and so, we’ve established $\mu_1 \leq \mu_2 \leq \mu_3 [= |\nu|] \leq \mu_1$, and so they are all equal, as we sought to show

3.4 Differentiation of Euclidean Space

We now analyze the special case where $\mu = m$ is the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$. Since we are working over the Borel sets of $\mathbb{R}^n$, we have a metric available to us, and so we can define the pointwise derivative of ν with respect to m at x :

Definition 3.4.1: Pointwise Derivative

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, m)$ be our measure space, ν a signed or complex measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$, and $x \in \mathbb{R}^n$. Then if $B_r(x)$ is a ball of radius r around x , then if the following limit exists:

$$\lim_{r \rightarrow 0} \frac{\nu(B_r(x))}{m(B_r(x))} = F(x)$$

then we say that $F(x)$ is the *pointwise derivative of ν with respect to m at x*

We can replace balls of radius x with sets that shrink suitably quickly in some “regular” way. This will be advantageous to us later, since it will turn out we can make these “nicely shrinking” sets to be $(x, x+h)$ (or the appropriate generalisation in the $\mathbb{R}^n$ case) which will give us

$$\lim_{h \rightarrow 0} \frac{\nu(x+h) - \nu(x)}{h}$$

notice that if we define ν in terms of a distribution as we’ve done for the outer measure in section 1.3, say G is our distribution, then our numerator becomes $G(x+h) - G(x)$, so in this case the definition *exactly* matches the definition of the derivative in one dimension:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \stackrel{\text{F.t.C.}}{=} \frac{1}{h} \int_x^{x+h} f'(x) dx$$

We will get back to this important case after some more examination of the general case.

If $\nu \ll m$ so that $d\nu = f dm$, then $\nu(B_r(x))/m(B_r(x)) = \frac{1}{m(B_r(x))} \int_{B_r(x)} f d\mu$ is the average value of f on $B_r(x)$ (recall the 1-dimensional intuition of $\frac{1}{b-a} \int_a^b f(x) dx$). Since $r \rightarrow 0$, we can hope that $F = f$ m -a.e. (since the averaging is getting more precise with the smaller range), telling us that we can recover a density F with information about f . This will indeed be the case if $\nu(B_r(x))$ is finite for all r, x ; this will be what we explore in this section. Essentially, from the point of view of f , what we’re showing is that the derivative of the integral of f is f , i.e., the Fundamental Theorem of Calculus (FTC). For the rest of this section, the term integrable and a.e. is with respect to the Lebesgue measure m .

We start with an important technical lemma:

Lemma 3.4.2: build-up lemma I

Let $\mathcal{C}$ be a collection of open balls in $\mathbb{R}^n$ and let $U = \bigcup_{B \in \mathcal{C}} B$. Then for any $c \in \mathbb{R}$ such that $c < m(U)$, there exists a finite disjoint subset of $\mathcal{C}$, $B_1, \dots, B_k$, such that

$$\sum_{i=1}^k m(B_i) > \frac{c}{3^n}$$

In other words, for any value c less than $m(U)$, we can show that we only need to pick a finite number of balls from $\mathcal{C}$ and still keep the inequality, up to a correction factor of 3^{-n} where n is the dimension of our space, $\mathbb{R}^n$, and 3 comes from making sure the radius of the balls will be big enough to cover what we need.

Proof:

We essentially sneakily get compactness by using theorem 2.6.1[1]. Using this theorem, we know that there exists a compact set $K \subseteq U$ such that $c < \mu(K)$, so there exists a finite subcover of balls from $\mathcal{C}$, say $A = \{A_1, \dots, A_m\}$, that cover K . Since balls have finite radius and there are finitely many balls, let B_1 be the ball with largest radius from A . Then let B_2 be the largest ball from A that are disjoint from B_1 . Continuing until we have exhausted all possibilities, we get a collection of balls $B = \{B_1, \dots, B_k\}$. If there is an A_i that is not in B , then there must be a B_j such that $A_i \cap B_j \neq \emptyset$, and if j is the smallest integer with this property, then by our construction A_i can have radius that is at most that of B_j (or else we would have picked A_i instead of B_j in our construction). Thus, we can take a ball B_j^* whose radius is 3 times that of B_j and have the same center as B_j (i.e., they are concentric) so that $A_i \subseteq B_j^*$. Then by this construction, $K \subseteq \bigcup_j^k B_j^*$, and so

$$c < \mu(K) \leq \sum_j^k m(B_j^*) = 3^n \sum_j^k m(B_j)$$

and so $\frac{c}{3^n} \leq \sum_j^k m(B_j)$, as we sought to show

With this lemma ready to be referenced, we define another important concept that we will use

Definition 3.4.3: Locally Integrable

Let $f : \mathbb{R}^n \rightarrow \mathbb{C}$ be a function. Then we say that f is *locally integrable* (with respect to the Lebesgue measure) if:

$$\forall K \in \mathcal{B}_{\mathbb{R}^n}, m(K) < \infty, \int_K |f(x)| dx < \infty$$

and is denoted L_{loc}^1

Definition 3.4.4: Average

Let $f \in L_{\text{loc}}^1$. Then we define the average $A_r f(x)$ on $B_r(x)$ to be

$$A_r f(x) = \frac{1}{m(B_r(x))} \int_{B_r(x)} f(y) dy$$

This is simply the observation we made in the previous paragraph but made into a definition. We next prove that varying r or x by a bit will only change $A_r f(x)$ by a bit, i.e., $A_r f(x)$ is continuous:

Lemma 3.4.5: Build-Up Lemma II

Let $f \in L^1_{\text{loc}}$. Then $A_r f(x)$ is continuous on both r ($r > 0$) and x ($x \in \mathbb{R}^n$)

Notice that f certainly need not be continuous, while $1/m(B_r(x))$ is clearly continuous. It is the $\int$ operation that is forcing this result to be continuous.

Proof:

First, taking advantage that we're working over the Lebesgue measure in $\mathbb{R}^n$, so that if $c = B_1(0)$, then $cr^n = B_r(x)$ and $m(\partial B_r(x)) = 0$ (where $\partial B_r(x) = \{y \mid \|x - y\| = r\}$). This will come back soon.

To actually prove continuity, we'll show that $A_{\lim_{r \rightarrow r_0}} f(x) = \lim_{r \rightarrow r_0} A_r f(x)$ and $A_r f(\lim_{x \rightarrow x_0}) = \lim_{x \rightarrow x_0} A_r f(x)$. To that end, take a sequence $r \rightarrow r_0$ and $x \rightarrow x_0$, and notice that the $\chi_{B_r(x)} \rightarrow \chi_{B_{r_0}(x_0)}$ pointwise on $\mathbb{R}^n \setminus \partial B_{r_0}(x_0)$ (imagine $x = x_0$ is fixed and the radius goes to r_0 by oscillating around r_0 so that the pointwise limit doesn't exist for the border). Thus, $\chi_{B_r(x)} \rightarrow \chi_{B_{r_0}(x_0)}$ a.e., and so we have that $\chi_{B_{r_0+1}(x_0)}$ dominates the $\chi_{B_r(x)}$ if $r < r_0 + 1/2$ and $|x - x_0| < 1/2$. Thus, by the Dominated Converting Theorem:

$$\begin{aligned} A_{\lim_{r \rightarrow r_0}} f(\lim_{x \rightarrow x_0} x) &= \int_{B_{\lim_{r \rightarrow r_0}}(\lim_{x \rightarrow x_0} x)} f(y) dy \\ &= \int_{B_{r_0}(x_0)} \lim_{r, x} f(y) dy \\ &= \lim_{r, x} \int_{B_r(x)} f(y) dy \end{aligned} \quad \text{DCT}$$

and so $\int_{B_r(x)} f(y) dy$ is continuous. Since $B_r(x) = cr^n$ and r^n is a continuous function, and the product of two continuous function is continuous, we have that

$$A_r f(x) = \frac{1}{cr^n} \int_{B_r(x)} f(y) dy$$

is also continuous, as we sought to show

Since $A_r f$ is continuous, it is measurable. Next, we can define the largest possible average given a point x :

Definition 3.4.6: Hardy-Littlewood Maximal Function

Let $f \in L^1_{\text{loc}}$. Then the *Hardy-Littlewood Maximal function* is defined as

$$Hf(x) := \sup_{r>0} A_r |f|(x) = \sup_{r>0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y)| dy$$

Note that Hf is measurable since $(Hf)^{-1} = \bigcup_{r>0} (A - r|f|)^{-1}((a, \infty))$ is open for any $a \in \mathbb{R}$ by the previous lemma. What the next theorem shows us is that if we fix some number $\alpha \in \mathbb{R}_{>0}$, and try

to take the measure of all points where $Hf(x) > \alpha$ (i.e., we are measuring the area of the largest possible average over an area of points with that larger average), then this value is in fact bounded by $\int |f(x)|$ provided an appropriate constant C/α :

Theorem 3.4.7: The Maximal Theorem

Let $(\mathbb{R}^n, \mathcal{L}, m)$ be a measure space. There exists a constant C such that for all $f \in L^1$ and all $\alpha > 0$

$$m(\{x \mid Hf(x) > \alpha\}) \leq \frac{C}{\alpha} \int |f(x)| dx$$

Proof:

Let $E_\alpha = \{x \mid Hf(x) > \alpha\}$. By supremum property of $Hf(x)$, there exists a $r_x > 0$ such that

$$\frac{1}{m(B_{r_x}(x))} \int_{B_{r_x}(x)} |f(y)| dy = A_{r_x} f(x) > \alpha \quad \Leftrightarrow \quad m(B_{r_x}(x)) < \frac{1}{\alpha} \int_{B_{r_x}(x)} |f(y)| dy \quad (3.1)$$

Clearly, the balls $B_{r_x}(x)$ cover E_α . So for any $c < m(E_\alpha)$, by lemma 3.4.2, there exists a finite collection $r_{x_1}, \dots, r_{x_k} \in E_\alpha$ such that

$$\sum_{i=1}^n m(B_{r_{x_j}}(x_j)) > \frac{c}{3^n}$$

and so, combining the inequalities we have and moving values around, we get:

$$c < 3^n \sum_{i=1}^n m(B_{r_{x_j}}(x_j)) \stackrel{\text{eq. 3.1}}{\leq} \frac{3^n}{\alpha} \int_{B_{r_{x_j}}} |f(y)| dy \leq \frac{3^n}{\alpha} \int_{\mathbb{R}^n} |f(y)| dy$$

Since this is true for all c , letting $c \rightarrow m(E_\alpha)$, we get the desired result, as we sought to show.

With all this build-up, we are ready to show the result we stated at the beginning, that $A_r f(x) \rightarrow f(x)$ a.e.

Before stating the theorem, it is useful to define the notion of limit superior for real-valued functions:

Definition 3.4.8: Limit Superior

Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$. Then:

$$\limsup_{r \rightarrow R} \varphi(r) := \lim_{\epsilon \rightarrow 0} \sup_{0 < |r-R| < \epsilon} \varphi(r) = \inf_{\epsilon > 0} \sup_{0 < |r-R| < \epsilon} \varphi(r)$$

As an exercise, show that:

$$\lim_{r \rightarrow R} \varphi(r) = c \quad \text{iff} \quad \limsup_{r \rightarrow R} |\varphi(r) - c| = 0$$

Theorem 3.4.9: Almost Lebesgue Differentiation Theorem

If $f \in L^1_{\text{loc}}$. Then

$$\lim_{r \rightarrow 0} A_r f(x) = f(x)$$

for a.e. $x \in \mathbb{R}^n$. Phrased another way:

$$\lim_{r \rightarrow 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} f(x) - f(y) dy = 0$$

for a.e. $x \in \mathbb{R}^n$.

Proof:

First, let's show we can limit our attention to function in L^1 . In particular, If $A_r f(x) \rightarrow f(x)$ for a.e. x , for any $N \in \mathbb{N}$, we can bound what x 's we choose by $|x| \leq N$. Since for any $r \leq 1$, the values $A_r f(x)$ depend only on $f(y)$ for $|y| \leq N+1$, we can replace f with $f\chi_{B_{N+1}(0)}$, so $f \in L^1$

By theorem 2.6.2, for every $\epsilon > 0$, there exists a continuous integrable function g such that $\int |g(y) - f(y)| dy < \epsilon$. Since g is continuous, for every $x \in \mathbb{R}^n$, for all $\delta > 0$, there exists an $r > 0$ such that $|y - x| < r$ implies $|g(y) - g(x)| < \delta$. From this, we get the result to work for g :

$$\begin{aligned} |A_r g(x) - g(x)| &= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y) dy - g(x) \right| \\ &= \left| \frac{1}{m(B_r(x))} \int_{B_r(x)} g(y) dy - \frac{m(B_r(x))}{m(B_r(x))} g(x) \right| \\ &= \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} g(y) - g(x) dy \right| \\ &< \frac{1}{m(B_r(x))} \left| \int_{B_r(x)} \delta dy \right| \\ &= \frac{1}{m(B_r(x))} |m(B_r(x))\delta| \\ &= \delta \end{aligned}$$

and so, since δ can be anything, we have that $A_r g(x) \rightarrow g(x)$ for a.e. x .

Now we use this result to prove it for f . We will use the Maximal theorem to bound the result. Let:

$$\begin{aligned} \limsup_{r \rightarrow 0} |A_r f(x) - f(x)| &= \limsup_{r \rightarrow 0} |A_r f(x) - A_r g(x) + A_r g(x) - g(x) + g(x) - f(x)| \\ &= \limsup_{r \rightarrow 0} |A_r(f - g)(x) + (A_r g - g)(x) + (g - f)(x)| \\ &\leq H(f - g)(x) + 0 + |f - g|(x) \end{aligned}$$

Or, in terms of sets, if

$$E_\alpha = \left\{ x \mid \limsup_{r \rightarrow 0} |A_r f(x) - f(x)| > \alpha \right\} \quad F_\alpha = \{x \mid |f - g| > \alpha\}$$

then

$$E_\alpha \subseteq F_{\alpha/2} \cup \{x \mid H(f - g)(x) > \alpha/2\}$$

Furthermore, we have that

$$(\alpha/2)m(F_{\alpha/2}) \leq \int_{F_{\alpha/2}} |f(x) - g(x)| dx < \epsilon$$

Thus, by the maximal theroem:

$$m(E_\alpha) \leq \frac{2\epsilon}{\alpha} + \frac{2C\epsilon}{\alpha}$$

and since ϵ is arbitrary, $m(E_\alpha) = 0$ for all $\alpha > 0$. However, $\lim_{r \rightarrow 0} A_r f(x)$ for all $x \notin \bigcup_{i=1}^{\infty} E_{1/n}$, finishing the proof.

In the theorem, we proved it for $f(x)$; it can in fact be shown that it can be proven for $|f|$. To show this, we define the following concept for reference:

Definition 3.4.10: Lebesgue Set

Let f be a measurable function. Then the *Lebesgue set* is :

$$L_f := \left\{ x \mid \lim_{r \rightarrow 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(x) - f(y)| dy = 0 \right\}$$

The following theorem shows that the points on which the Lebesgue set fails on is measure 0, and so the result is true for L_f as well

Theorem 3.4.11: Upgrade To Lebesgue Set

Let $f \in L^1_{\text{loc}}$. Then $m((L_f)^c) = 0$

Proof:

This proof will take advantage of the zero measure set in theorem 3.4.9, and that we can form a dense set of $\mathbb{C}$ which will correspond the zero-measure set from this previous theorem, meaning we can make a limit argument that will work for the proof.

First, for each $c \in \mathbb{C}$, we can apply theorem 3.4.9 to $g_c = |f(x) - c|$ to get that:

$$\lim_{r \rightarrow 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - c| dy = |f(x) - c|$$

up to some Lebesgue zero measure set, say N_c . Now, Let D be any countable dense subset of $\mathbb{C}$ and let $N = \bigcup_{c \in D} N_c$ (so $m(N) = 0$). If $x \notin N$, then for any $\epsilon > 0$, we can choose some $c \in D$ such that $|f(x) - c| < \epsilon$. Thus:

$$|f(y) - f(x)| < |f(y) - c| + \epsilon$$

Thus, we have that:

$$\limsup_{r \rightarrow 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| dy \leq |f(x) - c| + \epsilon \leq 2\epsilon$$

and since ϵ can be arbitrary small, we have that $\limsup_{r \rightarrow 0} \frac{1}{m(B_r(x))} \int_{B_r(x)} |f(y) - f(x)| dy = 0$, and so if $x \notin N$, the theorem applies. But then $0 = m(N) = m(L_f^c)$, as we sought to show.

We have essentially proven the main results of this section; the last thing we want to do is generalize the types of sets that shrink to x . So far, we have been using balls, however that is not necessary. Thus, we will generalize the types of sets and state the most general version of this “averaging” theorem

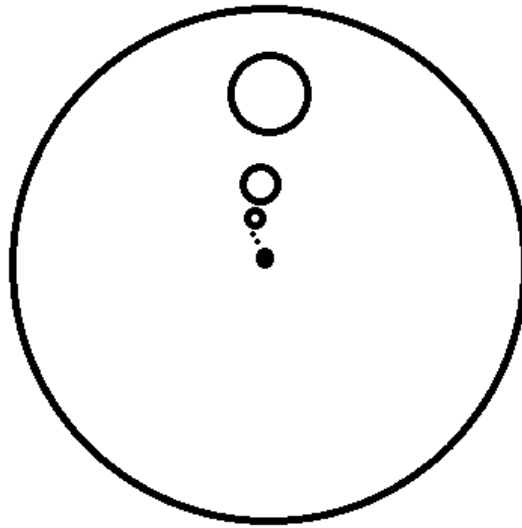
Definition 3.4.12: Sets That Shrink Nicely

A family of set $\{E_r\}_{r>0}$ of Borel subsets of $\mathbb{R}^n$ is said to *shrink nicely* if :

1. $E_r \subseteq B_r(x)$ for each r (so $m(E_r) \leq m(B_r(x))$)
2. There is a constant $\alpha > 0$ that is independent of r such that $m(E_r) > \alpha m(B_r(x))$

Example 3.4.13: Nicely Shrinking Sets

1. Let $U \subseteq B_1(0)$ be any borel subset. such that $m(U) > 0$, and consider $E_r = \{x + ry \mid y \in U\}$. You can imagine that U is some circle of $B_1(0)$ and see that, if $x = 0$:



Then we see that E_r shrinks to to x (even if $x \notin B_1(0)$)

2. in $\mathbb{R}$, the set $E_r = (x, x + r]$ also shrink nicely.

With this, we can finally state the result we’ve building up to in full generality:

Theorem 3.4.14: The Lebesgue Differentiation Theorem

Let $f \in L^1_{\text{loc}}$. Then for every x in the Lebesgue set, that is almost every x :

$$\lim_{r \rightarrow 0} \frac{1}{m(E_r)} \int_{E_r} |f(y) - f(x)| dy = 0$$

and

$$\lim_{r \rightarrow 0} \frac{1}{m(E_r)} \int_{E_r} f(y) dy = f(x)$$

for every family of nicely shrinking sets $\{E_r\}_{r>0}$ that shrink nicely to x

Written in terms of measure's this tells us that if $d\nu = f d\mu$, then:

$$\lim_{r \rightarrow 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

Proof:

To prove the first equality, we simply use the definition of nicely shrinking sets:

$$\begin{aligned} \frac{1}{m(E_r)} \int_{E_r} |f(y) - f(x)| dy &\leq \frac{1}{m(E_r)} \int_{B_r(x)} |f(y) - f(x)| dy \\ &\leq \frac{1}{\alpha m(B_r(x))} \int_{E_r} |f(y) - f(x)| dy \end{aligned}$$

so we can use theorem 3.4.11.

For the second equality, instead of putting absolute value, can start without them, in which case we would do the same think and then use theorem 3.4.9, and move the $f(x)$ to the other side.

3.4.1 Regular Measures

We will apply these results to a special class of measures which, as is common with such definitions, we take some aspects that we like from the Lebesgue measure on $\mathbb{R}^n$, and want to study measure's more generally (in this case, borel measures on $\mathbb{R}^n$) that have the same property:

Definition 3.4.15: Regular Measure

Let $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, \nu)$ be a borel measure space. Then ν is called *regular* if:

1. $\nu(K) < \infty$ for all compact subset $K \subseteq \mathbb{R}^n$
2. $\nu(E) = \inf \{\nu(U) \mid U \text{ open}, E \subseteq U\}$ for every $E \in \mathcal{B}_{\mathbb{R}^n}$

It is in fact provable that the first condition implies the second. We have shown this for $\mathbb{R}$ (i.e. $n = 1$) in theorem 1.4.2 and 1.4.4. In the study of Radon measures, this proof can be done for arbitrary n (Folland chapter 7.2).

Notice that a regular measure is automatically σ -finite, since $\mathbb{R}^n$ is σ -compact. As usual, a signed or complex measure ν is regular if $|\nu|$ is regular.

If $f \in L^+(\mathbb{R}^n)$, then the measure $f dm$ is regular if and only if $f \in L^1_{\text{loc}}$. This means that we can generalize the Lebesgue differentiation theorem to regular measures in general instead of measures defined on densities which are locally integrable. Let's first check that that assertion is indeed the case. First, it should be clear that $f \in L^1_{\text{loc}}$ is equivalent to $\nu(K) < \infty$ for all compact sets. To show 2, we'll first show it works for bounded borel sets E . By theorem 2.6.1, for any $\delta > 0$, there exists an open set $U \supseteq E$ such that $\mu(U) < \mu(E) + \delta$, thus $\mu(U \setminus E) < \delta$. By theorem 3.2.3, for all $\epsilon > 0$, there exists a $U \supseteq E$ such that $\int_{U \setminus E} f dm < \epsilon$, so $\int_U f dm < \int_E f dm + \epsilon$, thus $\nu(U) < \nu(E) + \epsilon$. This completes the requirement of the infimum. For the unbounded case, take $E = \cup_i E_i$ where each E_i is finite, finding it in each case while choosing $\int_{U_i \setminus E_i} f dm < \epsilon 2^{-i}$.

With this equivalence, it becomes worthwhile exploring Lebesgue Differentiation Theorem in the case of regular measures:

Theorem 3.4.16: Lebesgue Differentiation With Regular Measure

Let ν be a regular signed or complex Borel measure on $\mathbb{R}^n$, and let $d\nu = d\lambda + f dm$ be its Lebesgue decomposition. Then for m -almost every $x \in \mathbb{R}^n$:

$$\lim_{r \rightarrow 0} \frac{\nu(E_r)}{m(E_r)} = f(x)$$

for every nicely shrinking family $\{E_r\}_{r>0}$ that nicely shrinks to x

Proof:

We have essentially already done this proof, except now we have to consider we are adding $d\lambda$, $d\nu = d\lambda + f dm$, and so we would like to show that $\frac{\lambda(E_r)}{m(E_r)} \rightarrow 0$.

First, notice that $d|\nu| = d|\lambda| + |f| dm$ (or $|f| dm$?). Thus, regularity in ν implies regularity of λ and $f dm$ (this should be proven, though it is a bit tedious^a). Since $f \in L^1_{\text{loc}}$, it is enough to show that if $\lambda \perp m$, then for all x , $\lambda(E_r)/m(E_r) \rightarrow 0$ m -a.e as $r \rightarrow 0$. Without loss of generality, we can let $E_r = B_r(x)$ and assume λ is positive since:

$$\left| \frac{\lambda(E_r)}{m(E_r)} \right| \leq \frac{|\lambda|(E_r)}{m(E_r)} \leq \frac{|\lambda|(B_r(x))}{m(E_r)} \leq \frac{|\lambda|(B_r(x))}{\alpha m(B_r(x))}$$

for some $\alpha > 0$. Now, let's say $\lambda \geq 0$ (if $\lambda = 0$, then we'd be done), and take $A \in \mathcal{B}_{\mathbb{R}^n}$ such that $\lambda(A) = m(A^c) = 0$ (considering that $\lambda \perp m$). We will do the usual trick when it comes to showing that something has measure zero by assuming it doesn't, havign some sets define define in terms of a "1/k sequence", so to speak, and show we reach a contradiction. Consider:

$$F_k = \left\{ x \in A \mid \limsup_{r \rightarrow 0} \frac{\lambda(B_r(x))}{m(B_r(x))} > \frac{1}{k} \right\}$$

We will show that $m(F_k) = 0$ for all k , meaning the set of points for which $\lambda(E_r)/m(E_r) \rightarrow 0$ is 0 m -a.e. First, since λ is regular, for all $\epsilon > 0$, there exists a $U_\epsilon \supseteq A$ such that $\lambda(U_\epsilon) < \epsilon$. Since U_ϵ is open, for each point $x \in U_\epsilon$, there exists a ball such that $B(x) \subseteq U_\epsilon$. By definition of F_k , $\lambda(B(x)) > \frac{m(B(x))}{k}$. or $k\lambda(B(x)) > m(B(x))$. Next, take $V_\delta = \cup_x B(x)$. Then by construction

$F_k \subseteq V_\epsilon$. We'll show that $m(V_\epsilon) \leq 3^n k \epsilon$ for all ϵ , and so $m(F_k) = 0$ for all k . To do this, notice that by assumption there is a constant c such that $c < m(V_\epsilon)$. Thus, by lemma 3.4.2, there exist disjoint balls around points $x_1, \dots, x_n$ such that

$$c < 3^n \sum_{i=1}^n m(B_{x_i}) \leq 3^n k \sum_{i=1}^n \lambda(B_{x_i}) \leq 3^n k \lambda(V_\epsilon) \leq 3^n k \lambda(U_\epsilon) \leq 3^n k \epsilon$$

showing that $m(F_k) \leq m(V_\epsilon) < 3^n k \epsilon$, and since ϵ is arbitrary, we have that $m(F_k) = 0$ for all k , as we sought to show.

^aIt was homework 8, q26

3.5 Functions of Bounded Variation

We will now work towards the Fundamental Theorem of Calculus in its generalized form. This generalization is not in the direction of higher dimensions, as is the case with Stokes Theorem which is a generalization into higher dimensions, but requires much nicer conditions like working over smooth manifolds and differential forms. Rather, we want to stay in $\mathbb{R}$ (or $\mathbb{C}$) and see if we can weaken the conditions (i.e. Weaker than differentiable) and see when we can still apply FTC. Essentially, we want to see if we can define a *density* from probability using the closest notion of a derivative of a function.

In the case of $\mathbb{R}$, we have already explored in section 1.4 that Borel measures defined on $\mathbb{R}$ are defined through an associated distribution F , which is a right-continuous increasing function. In particular, we defined μ_F based on a increasing right-continuous function F (i.e. a distribution) to be $\mu([a, b]) = F(b) - F(a)$. We will use the tools we build up in the last section to finally prove the Fundamental theorem of Calculus in full generality. We will then explore when we can define distributions when we have a complex function, and substitute the notion of increasing function with a function of *bounded variation*.

The first thing we will do is show how increasing functions are a.e. differentiable (which to me is sort of crazy)!!

Proposition 3.5.1: Increasing Then a.e. Differentiable

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing function. Then the set of points where F is discontinuous is countable.

Furthermore, if $G(x) = \lim_{y \rightarrow x+} F(y) = F(x+)$, then F and G are differentiable a.e. and $F' = G'$ a.e.

Proof:

First, if x is continuous at x , then $F(x+) = F(x-)$, so the interval $(F(x-), F(x+))$ is empty. If it is not empty, then since F is increasing, each $(F(x-), F(x+))$ must be disjoint. So, for any $N \in \mathbb{N}$, for all $|x| \leq N$, we have that $(F(x-), F(x+)) \subseteq (F(-N), F(N))$. Now, here is the trick of this proof: notice that

$$\sum_{|x| \leq N} [F(x+) - F(x-)] \leq F(N) - F(-N) < \infty$$

since the $F(x-)$ is equal to $F(x_+)$ at appropriate locations, and so the sum cancels out (note that this sum is absolutely continuous since $F(x+) \geq F(x-)$). To see more clearly what's going on, if we label all the points in $|x| < N$ through an indexing set A , and let $x_\alpha = (F(x_\alpha-) - F(x_\alpha-))$, then:

$$\sum_{\alpha < A} x_\alpha < \infty$$

From here, we use the fact that this the sum of uncountably many positive x_α , and use the fact that the sum of uncountably many positive elements must be infinite if more than countably elements are non-zero, but since the sum *must* be finite $\{x \in (-N, N) \mid f(x+) \neq F(x-)\}$ is countable. Since this is true for all N , the set of discontinuities of F must be countable, proving the first assertion.

For the second assertion, first note that G is increasing and right continuous and $G = F$ at all points excepts where F is discontinuous. To show G is differentiable a.e, by proposition 1.4.2

$$G(x+h) - G(x) = \begin{cases} \mu_G((x, x+h]) & \text{if } h > 0 \\ \mu_G((x+h, x]) & \text{if } h < 0 \end{cases}$$

Notice that $\{(x-r, x]\}$ and $\{(x, x+r]\}$ shrink nicely to x as $r = |h| \rightarrow 0$. Furthermore μ_G is regular by theorem 1.4.2. Thus, by theorem 3.4.16, we have that:

$$G'(x) = \lim_{r \rightarrow 0} \frac{G(x+h) - G(x)}{h} = \lim_{r \rightarrow 0} \frac{\mu_G(E_r)}{m(E_r)}$$

exists for a.e. x . The final part of this proof is to show that if $H = G - F$, then H' exists and is equal to zero a.e.

Let $\{x_i\}$ be an enumeration of the points where $H \neq 0$. Since $H = G - F$, $H(x_i) > 0$. Let's say $\delta_i : \mathbb{R} \rightarrow \mathbb{R}$ is a function where $\delta_i(x_i) = 1$ and zero everywhere else. Then for any $N \in \mathbb{N}$, since H is increasing we have $\sum_{\{i \mid |x_i| < N\}} H(x_i) < \infty$. Thus, we can define a new measure:

$$\mu = \sum_i H(x_i) \delta_i$$

By what we've shown, μ is finite on all compact sets by the previous lines reasoning, and so by the theorem's we've been using μ is regular. Furthermore, $\mu \perp m$ since $m(E) = \mu(E^c) = 0$ where $E = \{x_i\}_{i=1}^\infty$. Thus:

$$\left| \frac{H(x+h) - H(x)}{h} \right| \leq \frac{H(x+h) + H(x)}{|h|} \stackrel{!}{\leq} \frac{\mu(x-2|h|, x+2|h|)}{|h|}$$

where $\stackrel{!}{\leq}$ comes from trying to engulf $H(x+h)$ and $H(x)$ into an interval.

Since all measure are regular, by theorem 3.4.16 this limit exists and $H' = 0$ a.e., completing the proof.

With this, we now have the tools to try and explore how to think about “increasing function” of the form $F : \mathbb{R} \rightarrow \mathbb{C}$. Since $\mathbb{C}$ has no inherent order, we must be more precise on how we define “increasing” and how do we make sure it doesn't explode (just like how when defining the borel measures through a distribution)

To understand the motivation behind the following definitions, imagine the following scenario: let F be the function representing the movement of a particle as it travels accross in time with respect to t (the graph being the position of the particle relative to, say, where it started). The *total variation* from time a to time b of the particle would be the total distance the particle has travelled in that time frame. if F were differentiable, this would simply be:

$$\int_a^b |F'(x)| dx$$

However, what if F is not differentiable? In that case, we will take the following approach: partition $[a, b]$ into $[a, t_1], [t_1, t_2], \dots, [t_{n-1}, b]$. Then, linearly approximate F by a straight line from $(t_{i-1}, F(t_{i-1}))$ to $(t_i, F(t_i))$. Then, take the limit over all partitions (in particular, as partitions get finer). If you are worried that such a limit is not well-defined, remember that polynomials can arbitrarily well approximate continuous functions on $\mathbb{R}$, and in most cases F is right-continuous. This heuristic might be more rigorous, but it at least make me feel comfortable with doing this idea

We will do a similar idea for the complex case, except we will take the norm (which we'll also call absolute value) of the distances:

Definition 3.5.2: Total Variation

Let $f : \mathbb{R} \rightarrow \mathbb{C}$ and $x \in \mathbb{R}$. Then define the *total variation* of F to be:

$$T_F(x) = \sup \left\{ \sum_{i=1}^n |F(x_i) - F(x_{i-1})| \mid n \in \mathbb{N}, -\infty < x_0 < x_1 < \dots < x_n = x \right\}$$

It should be clear by construction that T_F is an increasing function. Since we put absolute values inside the sum, then as we refine the partition, the sum will get bigger. Therefore, If we have $a < b$ and consider $T_F(b)$, it is harmless to assume a is one of the subdivision points (since we can refine and get a partition with a being part of it). It follows that we have:

$$T_F(b) - T_F(a) = \sup \left\{ \sum_{i=1}^n |F(x_i) - F(x_{i-1})| \mid n \in \mathbb{N}, a = x_0 < x_1 < \dots < x_n = x \right\}$$

Using this fact, we can define the following:

Definition 3.5.3: Bounded Variation

Let T_F be the total variation of the function F . Then if:

$$\lim_{t \rightarrow \infty} T_F(t) = T_F(\infty) < \infty$$

Then F is said to be of *bounded variation* (on $\mathbb{R}$). We denote the set of of all functions of bounded variation as BV . If we are concerned with all function of bounded variation on an interval $[a, b]$, we will denote this as $BV([a, b])$.

Note that if $F \in BV$, then $F \in BV([a, b])$, since it's total variation on $[a, b]$ is $T_F(b) - T_F(a)$ which is less than the total variation of T_F . If $F \in BV([a, b])$, then we can extend F to be in BV by letting $F(x) = F(a)$ for all $x \leq a$ and $F(b) = F(x)$ for all $b \leq x$. Further note that we rarely actually compute $T_F(b)$, but we use the fact that a function is in BV to get many theoretical results:

Example 3.5.4: Bounded Variation

1. If $F : \mathbb{R} \rightarrow \mathbb{R}$ is bounded and increasing, then $F \in BV$ ($T_f(x) = F(x) - F(-\infty)$)
2. If $F, G \in BV$ and $a, b \in \mathbb{C}$ then $aF + bG \in BV$
3. If F is differentiable on $\mathbb{R}$ and F' is bounded, then $F \in BV([a, b])$ for $-\infty < a < b < \infty$ (by the mean value theorem)
4. If $F(x) = \sin(x)$, then $F \in BV([a, b])$ for all $-\infty < a < b < \infty$, but $F \notin BV$
5. if $F = x \sin\left(\frac{1}{x}\right)$ for $x \neq 0$ and $F(0) = 0$, then $F \notin BV([a, b])$ for $a \leq 0 < b$ and $a < 0 \leq b$

Before we prove stuff about BV functions, let's first prove a lemma:

Lemma 3.5.5: Sum Is Increasing

Let $F \in BV$ be a real valued function. Then $T_F + F$ and $T_F - F$ are increasing functions.

Proof:

Take $x < y$ and $\epsilon > 0$. Since T_F is a supremum, take appropriate $x_0 < \dots < x_n = x$ such that

$$\sum_i^n |F(x_i) - F(x_{i-1})| \geq T_F(x) - \epsilon$$

Next, notice that we can make this into an element of the supremum set of $T_f(y)$ (also known as an approximating sum) by doing

$$\sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)|$$

With this sum, take $F(y) = [F(y) - F(x)] + F(x)$. With some manipulation, we get:

$$T_f(y) \pm F(y) \geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x)$$

Since F is a real valued function, we can simply take cases to notice that:

+	$F(x) - F(y)$	+	$F(x) - F(y)$	$2 F(y) - F(x) $
+	$F(x) - F(y)$	-	$F(x) - F(y)$	0
-	$F(y) - F(x)$	+	$F(x) - F(y)$	0
-	$F(y) - F(x)$	-	$F(x) - F(y)$	$2 F(y) - F(x) $

$$\begin{aligned} T_f(y) \pm F(y) &\geq \sum_i^n |F(x_i) - F(x_{i-1})| + |f(y) - f(x)| \pm [F(y) - F(x)] \pm F(x) \\ &\geq T_f(x) - \epsilon \pm F(x) \end{aligned}$$

letting $\epsilon \rightarrow 0$, we see that:

$$T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$$

and since x and y was arbitrary, we have establish that $T_F + F$ and $T_F - F$ are both increasing functions, we as sought to show

Theorem 3.5.6: Properties Of BV

1. $T_F \in BV$ is an increasing function
2. $F \in BV$ if and only if $\operatorname{Re}F \in BV$ and $\operatorname{Im}F \in BV$
3. If $F : \mathbb{R} \rightarrow \mathbb{R}$ then $F \in BV$ if and only if F is the difference of two bounded increasing functions; for $F \in BV$ these functions may be taken to be $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$
4. If $F \in BV$, then $F(x+) = \lim_{y \rightarrow x-} F(y)$ and $F(x-) = \lim_{y \rightarrow x+} F(y)$ exist for all $x \in \mathbb{R}$, as do $F(\pm\infty) = \lim_{y \rightarrow \pm\infty} F(y)$
5. If $F \in BV$, the set of points at which F is discontinuous is countable
6. If $F \in BV$ and $G(x) = F(x+)$, then F' and G' are exist and are equal a.e.

Proof:

1. This essentially comes from the definition
2. This is essentially immediate, and so is left as exam review
3. Look at example 3.5.4 to see how the $\Rightarrow$ direction goes. For the $\Leftarrow$ direction, by lemma 3.5.5, the equation $F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$ express F as the difference of two increasing functions. To show boundedness, we showed that $T_F(y) \pm F(y) \geq T_F(x) \pm F(x)$ for $y > x$. Re-arranging and taking advantage of the fact that $F \in BV$, we get:

$$|F(y) - F(x)| \leq T_F(y) - T_F(x) \leq T_F(\infty) - T_F(-\infty) < \infty$$

showing that F (and hence T_F) is bounded

4,5,6 follow from 1,2,3 along with proposition 3.5.1.

Note that (6) is particularly powerful since we now have that a function simply needs to be of bounded variation to have a derivative a.e. (not necessarily increasing anymore). The converse is false: being differentiable a.e. (even everywhere) does not imply BV (think of $\sin(1/x)$ on $(0, 1]$).

The equation in part 2 are important enough to be given a name:

Definition 3.5.7: Jordan Decomposition

Let $F \in BV$ be a real-valued function. Then the *jordan decomposition* of F is:

$$F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$$

where $\frac{1}{2}(T_F + F)$ and $\frac{1}{2}(T_F - F)$ are called the *positive variation of F* and *negative variation of F* respectively.

Since $x^+ = \max(x, 0) = \frac{1}{2}(|x| + x)$ and $x^- = \max(-x, 0) = \frac{1}{2}(|x| - x)$ for $x \in \mathbb{R}$, we get that:

$$\frac{1}{2}(T_F \pm F)(x) = \sum \left\{ \sum_i^n [F(x_i)F(x_{i-1})]^\pm \mid x_0 < \cdots < x_n = x \right\} \pm \frac{1}{2}F(-\infty)$$

We are almost at a stage to connect borel measures to BV functions. There is one more peice of the puzzle that we need:

Definition 3.5.8: Normalized Bounded Variation (NBV)

Let $F \in BV$ be a function of bounded variation. Then if F is right-continuous and $F(-\infty) = 0$, then F is said to be a function of *normalized bounded variation*. We will denote the collection of such functions as:

$$NBV := \{F \in BV \mid F \text{ is right continuous and } F(-\infty) = 0\}$$

Example 3.5.9: NBV

Let $F \in BV$. Define $G = F(x+) - F(-\infty)$. Then $G \in NBV$, and $F' = G'$ a.e.

First, to show that $G \in BV$, by theorem 3.5.6[2,3], if F is real and $F = F_1 - F_2$ where F_1, F_2 are increasing, then $G(x) = F_1(x+) - [F_2(x+) - F(-\infty)]$, which is the difference of two increasing functions. That $G(-\infty) = 0$ comes by definition. Hence $G \in NBV$.

In fact, every $F \in BV$ almost defines an NBV function through T_F :

Lemma 3.5.10: BV, then T_F almost NBV

If $F \in BV$, then $T_F(-\infty) = 0$. If F is also right continuous, then T_F is also right-continuous (making $T_F \in NBV$)

Proof:

Let $\epsilon > 0$ so that for any x and $x_0 < \cdots < x$ we ahve:

$$\sum F(x_{i+1}) - F(x_i) = T_f(x) - T_f(x_0) > T_F(x) - \epsilon$$

So $T_f(x_0) < \epsilon$ or more generally for any $y \leq x_0$ $T_f(y) < \epsilon$. Since epislon is arbitray, as we let $y \rightarrow 0$ we get $T_f(-\infty) = 0$

For the next assertion, suppose F is right-continuous. I'll finish this proof later:

Now suppose that F is right continuous. Given $x \in \mathbb{R}$ and $\epsilon > 0$, let $\alpha = T_F(x+) - T_F(x)$, and choose $\delta > 0$ so that $|F(x+h) - F(x)| < \epsilon$ and $T_F(x+h) - T_F(x) < \epsilon$ whenever $0 < h < \delta$. For any such h , by (3.24) there exist $x_0 < \dots < x_n = x+h$ such that

$$\sum_{j=1}^n |F(x_j) - F(x_{j-1})| \geq \frac{3}{4}[T_F(x+h) - T_F(x)] \geq \frac{3}{4}\alpha,$$

and hence

$$\sum_{j=2}^n |F(x_j) - F(x_{j-1})| \geq \frac{3}{4}\alpha - |F(x_1) - F(x_0)| \geq \frac{3}{4}\alpha - \epsilon.$$

Likewise, there exist $x = t_0 < \dots < t_m = x_1$ such that $\sum_{j=1}^m |F(t_j) - F(t_{j-1})| \geq \frac{3}{4}\alpha$, and hence

$$\begin{aligned} \alpha + \epsilon &> T_F(x+h) - T_F(x) \\ &\geq \sum_{j=1}^m |F(t_j) - F(t_{j-1})| + \sum_{j=2}^n |F(x_j) - F(x_{j-1})| \\ &\geq \frac{3}{2}\alpha - \epsilon. \end{aligned}$$

Thus $\alpha < 4\epsilon$, and since ϵ is arbitrary, $\alpha = 0$. ■

We finally get to establish the connection between complex measures and functions in BV :

Theorem 3.5.11: Complex Measures And BV

If μ is a complex Borel measure on $\mathbb{R}$ (i.e. a finite signed measure) and $F(x) = \mu((-\infty, x])$, then $F \in NBV$. Conversely, if $F \in NBV$, then there is a unique complex Borel measure μ_F such that $F(x) = \mu_F((-\infty, x])$

Proof:

Let μ a complex measure. Then we can decompose it into $\mu = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$ where $\mu_i^\pm$ are positive finite measures. If $F_i^\pm(x) = \mu_i^\pm((-\infty, x])$, then $F_i^\pm$ is increasing, right continuous, $F_i^\pm(-\infty) = 0$, and $F_i^\pm(\infty) = \mu_i^\pm(\mathbb{R}) < \infty$. Thus, by theorem 3.5.6[2,3], the function

$$F = F_1 + -F_1^- + i(F_2^+ - F_2^-) \in NBV \quad (3.2)$$

Conversely, by theorem 3.5.6 and lemma 3.5.10, any $F \in NBV$ can be written in the form of equation (3.2) with the $F_i^\pm$ increasing and in NBV . By Theorem 1.4.2, each $F_i^\pm$ gives a measure $\mu_i^\pm$, so $F(x) = \mu_F((-\infty, x])$ where $\mu_F = \mu_1^+ - \mu_1^- + i(\mu_2^+ - \mu_2^-)$.

Finally, to show $|\mu_F| = \mu_{T_F}$ (this was left as an outlined exercise 28)

Corollary 3.5.12: Total Variation Of μ_F

Let $F \in NBV$, μ_F be the associated complex measure on $\mathbb{R}$, and $G(x) = |\mu_F|((-\infty, x])$. Then:

$$|\mu_F| = \mu_{T_F}$$

Proof:

We break down the proof in the following steps:

1. From the definition of T_F , $T_F \leq G$

Proof. We need to show $T_F(x) \leq G(x)$ In the previous homework, we showed that:

$$|\mu_F|((-\infty, x]) = \sup \left\{ \sum_i |\mu_F(E_i)| \mid \prod_i E_i = (-\infty, x] \right\}$$

In particular, we can choose are disjoint sets to be h -intervals and get:

$$\begin{aligned} |\mu_F|((-\infty, x]) &= \sup \left\{ \sum_i |\mu_F((x_i, x_{i+1}])| \mid \prod_i (x_i, x_{i+1}] = (-\infty, x] \right\} \\ &= \sup \left\{ \sum_i |F(x_{i+1}) - F(x_i)| \mid \prod_i (x_i, x_{i+1}] = (-\infty, x] \right\} \end{aligned}$$

and by definition of T_F

$$T_F(x) = \sup \left\{ \sum_i^N |F(x_{i+1}) - F(x_i)| \mid N \in \mathbb{N}, x_0 < \cdots < x_N = x \right\}$$

Notice that the set for $|\mu_F|$ contains that of $T_F(x)$ since we can set the rest of the infinite value for T_F to 0, and so $T_F \leq G = |\mu_F|$ $\square$

2. $|\mu_F(E)| \leq \mu_{T_F}(E)$ when E is an interval, and hence E is a borel set.

Proof. Without loss of generality, we will work with intervals of the form $(a, b]$. We only require these type of intervals for the σ -algebra that will be generated in part (c).

First, since $F \in NBV$, then it is right-continuous, and so by lemma 3.28 T_F is right-continuous, and so by lemma 3.28 again $T_F \in NBV$. Thus, by theorem 3.29, there is an associated complex borel measure such that $T_F(x) = \mu_{T_F}((-\infty, x])$. So $\mu_{T_F}((a, b]) = T_F(b) - T_F(a)$. With that, we proceed with the proof.

First, the emptyset (the trivial interval) is immediate since $|\mu_F(\emptyset)| = 0 = \mu_{T_F}(\emptyset)$. Next, to show the inequality is true for bounded intervals, notice that:

$$|\mu_F((a, b])| = |\mu_F((-\infty, b]) + \mu_F((-\infty, a])| = |F(b) - F(a)|$$

Next, recall that:

$$\mu_{T_F}((a, b]) = T_F(b) - T_F(a) = \sup \left\{ \sum_i^n |F(x_{i+1}) - F(x_i)| \mid n \in \mathbb{N} \ a = x_0 < \cdots < x_n = b \right\}$$

So, $|F(b) - F(a)|$ is inside the supremum set, and so:

$$= |F(b) - F(a)| \leq T_F(b) - T_F(a) = \mu_{T_F}((a, b])$$

Next, for rays, notice we can split a ray into the countable union of h -intervals:

$$|\mu_F((-\infty, b])| = \left| \sum_{i=0}^{\infty} \mu_F((b-i-1, i-k]) \right| \leq \sum_{i=0}^{\infty} |\mu_F((b-i-1, i-k])|$$

where we can now apply the results for bounded intervals to get that:

$$\leq \sum_{i=0}^{\infty} |\mu_F((b-i-1, i-k])| \leq \sum_{i=0}^{\infty} \mu_{T_F}((a, b]) = \mu_{T_F}((-\infty, b])$$

Thus $|\mu_F((-\infty, b])| \leq \mu_{T_F}((-\infty, b])$, and similarly for rays of the form (a, ∞) .

To show it's true for all Borel sets, first recall that h -intervals form an algebra (their disjoint union forms an elementary family, and hence by proposition 1.7). Next, let $\mathcal{C} = \{E \in \mathcal{B}_{\mathbb{R}} \mid |\mu_F(E)| \leq \mu_{T_F}(E)\}$. This set clearly contains all h -intervals by what we've just shown. We'll show that this set is in fact a monotone class, and hence by the monotone class lemma contains all Borel sets.

This is simply a matter of using the appropriate definitions. Let $\{E_i\}_{i=1}^{\infty} \subseteq \mathcal{C}$ be an increasing sequence. Then:

$$\begin{aligned} \left| \mu_F \left(\bigcup_i^{\infty} E_i \right) \right| &= \left| \lim_{i \rightarrow \infty} \mu_F \left(\bigcup_i^{\infty} E_i \right) \right| && \text{definition} \\ &= \lim_{i \rightarrow \infty} \left| \mu_F \left(\bigcup_i^{\infty} E_i \right) \right| && \text{limit property} \\ &\leq \lim_{i \rightarrow \infty} \mu_{T_F}(E_i) && \text{all } E_i \in \mathcal{C} \\ &= \mu_{T_F} \left(\bigcup_i^{\infty} E_i \right) && \text{definition} \end{aligned}$$

and hence $\mathcal{C}$ is closed under increasing sequences. A similar argument works for decreasing sequences (in particular since μ_{T_F} is finite, it will hold true for any decreasing sequence). Hence, by the monotone class lemma, $\mathcal{C}$ contains the Borel set, showing that $|\mu_F(E)| \leq \mu_{T_F}(E)$ for all $E \in \mathcal{B}_{\mathbb{R}}$, completing the proof. $\square$

3. $|\mu_F| \leq \mu_{T_F}$, and hence $G \leq T_F$ (use exercise 21.)

Proof. Recall that

$$|\mu_F|(E) = \sup \left\{ \sum_i^\infty |\mu_F(E_i)| \mid E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E \right\}$$

where I modified what's inside the parenthesis to make it more explicit that the sets are inside $\mathcal{B}_{\mathbb{R}}$. Using what we have just proven:

$$\begin{aligned} |\mu_F|(E) &= \sup \left\{ \sum_i^\infty |\mu_F(E_i)| \mid E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E \right\} \\ &\leq \sup \left\{ \sum_i^\infty \mu_{T_F}(E_i) \mid E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E \right\} \\ &\leq \sup \left\{ \mu_{T_F}(\coprod_i E_i) \mid E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E \right\} && \text{disjoint} \\ &\leq \sup \left\{ \sum_i^\infty \mu_{T_F}(E) \mid E_i \in \mathcal{B}_{\mathbb{R}}, \coprod_i E_i = E \right\} \\ &= \mu_{T_F}(E) \end{aligned}$$

Thus, $|\mu_F|(E) \leq \mu_{T_F}(E)$. If we take $E = (-\infty, x]$, then we get $|\mu_F|(E) = G(E) \leq \mu_{T_F}(E)F$, completing the proof.

In particular, since we've shown the other direction, we have that $G = \mu_{T_F}$. $\square$

With the connection established for between NBV functions and complex measures, it is natural to ask which function in NBV correspond to measure μ such that $\mu \perp m$ or $\mu \ll m$? The following proposition answers that:

Proposition 3.5.13: NBV And Radon-Nikodym with respect to m

If $F \in NBV$, then $F' \in L^1(m)$. Moreover:

1. $\mu_F \perp m$ if and only if $F' = 0$ a.e.
2. $\mu_F \ll m$ if and only if $F(x) = \int_{-\infty}^x F'(t)dt$

Proof:

First, note that μ_F is regular by theorem 1.4.4. Next, observe that:

$$F'(x) = \lim_{r \rightarrow 0} \frac{\mu_F(E_r)}{m(E_r)}$$

where $E_r = (x, x+r]$ or $(x-r, x]$ by applying theorem 3.4.16

Since we have now characterized μ_F in terms of F , we can characterize $\mu_F \ll m$ in terms of F :

Definition 3.5.14: Absolute Continuity w.r.t. F

A function $F : \mathbb{R} \rightarrow \mathbb{C}$ is said to be *absolutely continuous* if

$$\forall \epsilon > 0, \exists \delta > 0 \text{ such that}$$

for any set of disjoint intervals $(a_1, b_1), \dots, (a_N, b_N)$,

$$\sum_1^N (b_i - a_i) < \delta \Rightarrow \sum_i^N |F(b_i) - F(a_i)| < \epsilon$$

More generally, F is absolutely continuous on $[a, b]$ if all the open intervals lie in $[a, b]$.

Notice that if F is absolutely continuous, then F is automatically uniform (pick $N = 1$). If F is everywhere differentiable and F' is bounded, then F is absolutely continuous, since $|F(b_i) - F(a_i)| \leq (\max |F'|)(b_i - a_i)$ by the Mean Value Theorem. More generally, on a compact set, we have that:

$$\text{absolutely continuous} \subseteq \text{uniformly continuous} \subseteq \text{continuous}$$

and on a compact interval:

$$\begin{aligned} &\text{continuously differentiable} \\ &\subseteq \\ &\text{Lipchitz continuous} \\ &\subseteq \\ &\text{absolutely continuous} \\ &\subseteq \\ &\text{Bounded Variation} \\ &\subseteq \\ &\text{differentiable a.e.} \end{aligned}$$

where Lipchitz continuous means for all $x, y \in \mathbb{R}$, there exists an M such that $|f(x) - f(y)| \leq M|x - y|$. After going over the FTC for Lebesgue integrals, you can prove as an exercise that Lipchitz continuous implies absolutely continuous. Usually, Lipchitz continuity is used as a simple way of finding absolutely continuous functions.

To make sure this new notion links with the absolute continuity of the measure:

Proposition 3.5.15: Linking Absolute Continuity Of F And Measures

Let $F \in NBV$. Then F is absolutely continuous if and only if $\mu_F \ll m$

Proof:

- ($\Leftarrow$) Let $\mu_F \ll m$. Then since μ_F is finite, we can apply theorem 3.2.3 to the sets $E = \bigcup_i^N (a_i, b_i]$, and get that F is absolutely continuous
- ($\Rightarrow$) Let E be a borel set such that $m(E) = 0$. Let $\epsilon > 0$ so that we can choose a δ that

satisfies the definition of absolute continuity of F . By theorem 1.4.4, we can find open sets $U \supseteq U_1 \cdots \supseteq E$ such that $m(U_1) < \delta$ (which also implies $m(U_i) < \delta$ by monotonicity) and $\mu_F(U_i) \rightarrow \mu_F(E)$. Since each U_i is the countable union of disjoint intervals $(a_{i,j}, b_{i,j})$, for all $N \in \mathbb{N}$:

$$\sum_k^N |\mu_F((a_{i,j}, b_{i,j}))| \leq \sum_k^N |F(b_{i,j}) - F(a_{i,j})| < \epsilon$$

by absolute continuity. Letting $N \rightarrow \infty$, we get that $|\mu_F(U_i)| < \epsilon$, and so $|\mu_F(E)| < \epsilon$. Since this is true for all epsilon, it must be that $|\mu_F(E)| = 0$, showing that $\mu_F \ll m$, as we sought to show

With this, we can relate L^1 functions to NBV functions:

Corollary 3.5.16: Relating L^1 And NBV Functions

Let $f \in L^1$. Then the function $F(x) = \int_{-\infty}^x f(t)dt$ is in NBV , is absolutely continuous, and $f = F'$ a.e. Conversely, if $F \in NBV$, is absolutely continuous, then $F' \in L^1(m)$ and $F(x) = \int_{-\infty}^x F'(t)dt$

Proof:

This follows from proposition 3.5.13 and 3.5.15.

If the function is bounded, then we get a slightly stronger result:

Lemma 3.5.17

If F is absolutely continuous on $[a, b]$, then $F \in BV([a, b])$

Proof:

We will find an upper bound for the total variation of F , that is, we'll find an N such that $T_f(b) < N$. Choose $\epsilon = 1$ and let δ be the appropriate choice to satisfy absolute continuity: for any finite set of disjoint intervals (a_i, b_i) :

$$\sum (b_i - a_i) < \delta \Rightarrow \sum |F(b_i) - F(a_i)| < \epsilon$$

Choose N to be the greatest integer that is smaller than $\delta^{-1}(b-a)+1$. Then, for any subdivision $a = x_0 < x_1 < \cdots < x_n = b$, we can collect the subintervals (x_i, x_{i+1}) into at most N groups such that the sum of each group is less than δ (adding subdivision's if necessary). Then the sum $\sum |F(x_{i+1}) - F(x_i)|$ over each group is 1, and so the total variation of F on $[a, b]$ is at most N

Is the converse true: If $F \in BV$, then does it imply that F is absolutely continuous? The answer is no. Even better, F can even be NBV , uniformly continuous, and increasing (so differentiable a.e.), but still not absolutely continuous:

Example 3.5.18: NBV, Continuous, But Not Absolutely Continuous

Let F be the cantor function. Note that F is increasing, $F(-\infty) = 0$, and $T_F(\infty) = 1/2$ (since it's continuous, it is integrable, and integrating give $1/2$). Thus μ_F is defined. However, it is not absolutely continuous with respect to m . However, F is not absolutely continuous (you should prove this).

Theorem 3.5.19: Fundamental Theorem Of Calculus For Lebesgue Integrals

Let $-\infty < a < b < \infty$ and $f : [a, b] \rightarrow \mathbb{C}$. Then the following are equivalent:

- (a) F is absolutely continuous
- (b) $F(x) - F(a) = \int_a^x f(t)dt$ for some $f \in L^1([a, b], m)$
- (c) F is differentiable a.e. on $[a, b]$, $F' \in L^1([a, b], m)$ and $F(x) - F(a) = \int_a^x F'(t)dt$

Proof:

First, to show $(a) \Rightarrow (c)$, we can subtract some constant from F so that $F(a) = 0$. Doing the usual extension of setting $F(x) = 0$ for $x < a$ and $F(x) = F(b)$ for $x > b$, then $F \in NBV$. Thus, by lemma 3.5.17 using corollary 3.5.16. Then, that $(c) \Rightarrow (b)$ comes immediately since (c) is a stronger condition than (b) . For $(b) \Rightarrow (a)$, if we set $f(t) = 0$ for $t \notin [a, b]$ then $f \in L^1(m)$ and so by corollary 3.5.16 we get the final result.

If $F \in NBV$, then we usually write an integral of a function g with respect to the associated μ_F as $\int g dF$ or $\int g(x) dF(x)$. These types of integrals are called *Lebesgue-Stieltjes integrals*. I'm not sure why they were introduced, but this theorem was presented for generalizing integration by parts to these types of integrals

Theorem 3.5.20: Integration By Parts For Lebesgue-Stieltjes Integrals

Let $F, G \in NBV$ where at least one of them is continuous. Then for $-\infty < a < b < \infty$

$$\int_{(a,b]} F dG + \int_{(a,b]} G dF = F(b)G(b) - F(a)G(a)$$

Proof:

F and G are linear combinations of increasing functions in NBV (by Theorem 3.27(a,b) in Folland), so a simple calculation shows that it suffices to assume F and G increasing. Suppose for the sake of definiteness that G is continuous, and let $\Omega = \{(x, y) : a < x \leq y \leq b\}$. We use Fubini's theorem to compute $\mu_F \times \mu_G(\Omega)$ in two ways:

$$\begin{aligned} \mu_F \times \mu_G(\Omega) &= \int_{(a,b]} \int_{(a,y]} dF(x) dG(y) = \int_{(a,b]} [F(y) - F(a)] dG(y) \\ &= \int_{(a,b]} F dG - F(a)[G(b) - G(a)], \end{aligned}$$

and since $G(x) = G(x-)$,

$$\begin{aligned}\mu_F \times \mu_G(\Omega) &= \int_{(a,b]} \int_{[x,b]} dG(y) dF(x) = \int_{(a,b]} [G(b) - G(x)] dF(x) \\ &= G(b)[F(b) - F(a)] - \int_{(a,b]} G dF.\end{aligned}$$

Subtracting these two equations, we get the desired result.

Finally, we end this chapter with a quick discussion about different terminology when it comes down to decomposing measures.

Definition 3.5.21: Discrete And Continuous Measures

Let μ be a complex Borel measure on $\mathbb{R}^n$

1. μ is called *discrete* if there is a countable subset $\{x_i\}_i \subseteq \mathbb{R}^n$ such that $\sum_i c_i < \infty$ and $\mu = \sum c_i \delta_{x_i}$ where $\delta_{x_i}(x_i) = 1$ and is zero everywhere else (i.e. the pointmass at x).
2. μ is called *continuous* if for all $x \in \mathbb{R}^n$, $\mu(\{x\}) = 0$

It is easy to see that any measure can be broken down into $\mu = \mu_d + \mu_c$ where μ_d is discrete and μ_c is continuous. The main thing is to insure that μ_d is over a countable set: let $E = \{x \mid \mu(\{x\}) \neq 0\}$. Taking any countable subset $F \subseteq E$, then $\sum_{x \in F} \mu(\{x\})$ converges absolutely to $\mu(F)$ so $\{x \in E \mid \mu(\{x\}) > k^{-1}\}$ is finite, and so it must be that E is countable, so $\mu_d(A) = \mu(A \cap E)$ is discrete and $\mu_c(A) = \mu(A \setminus E)$ is continuous.

If μ is discrete, then $\mu \perp m$, and if μ is continuous then $\mu \ll m$. Thus, we can further break down the decomposition into:

$$\mu = \mu_d + \mu_{ac} + \mu_{sc}$$

where μ_d is the discrete part, μ_{ac} is the absolutely continuous part, and μ_{sc} is the *singularly continuous* part. So far, we have mainly worked with measures with zero singularly continuous parts. There do exist measures with nonzero singularly continuous parts (Folland says the surface measure of the sphere for $n > 1$ is an example). An example for $n = 1$ is rarer, but they do exist. By proposition 3.5.13, we know that we are looking for a corresponding $F \in NBV$ such that $F' = 0$ a.e. In fact, the cantor function is such an example (when extended in the way we keep doing to make sure functions are in NBV). In fact, using the cantor function, we can construct an increasing function F that is continuous, $F' = 0$ a.e., and that is *strictly increasing*:

Example 3.5.22: Interesting Singularly Continuous Function

1. If $\{F_i\}_{i=1}^\infty$ is a sequence on nonnegative increasing functions on $[a, b]$ such that $F(x) = \sum_{i=1}^\infty F_i(x) < \infty$ for all $x \in [a, b]$, then $F'(x) = \sum_{i=1}^\infty F'_i(x)$ for a.e. $x \in [a, b]$ (It suffices to assume $F_i \in NBV$. Consider the measure μ_{F_i})

Proof:

First, notice that since the F_i are nonnegative, by letting $F(x) = 0$ for $x < a$ and $F(b) = F(y)$ for $b < y$, we have that $F_i(-\infty) = 0$. Furthermore, the function $G(x) = F(x+)$ and $G_i(x) = F_i(x+)$ is equal a.e. with the same derivative wherever it is defined (which it is a.e.), and so if G and G_i has a particular derivative, so do F and F_i . Thus, Without loss of generality, we'll assume $F, F_i \in NBV$.

Since they are in NBV , consider the induced complex borel measure μ_F and μ_{F_i} given by theorem 3.29. Since these measures are finite, they are σ -finite, and so admit a Radon-Nykodym derivative:

$$d\mu_F = f dm + d\lambda \quad d\mu_{F_i} = f_i dm + d\lambda_i$$

where $f dm \ll m$, $f_i dm \ll m$ and $d\lambda \perp m$, $d\lambda_i \perp m$. We will show that $f dm = \sum f_i dm$. First, we need to show that $\mu_f = \sum \mu_{F_i}$, and then use the previous homework exercise (question 8) to conclude the equality.

By definition of F , $\mu_F((-\infty, b]) = \sum_i^\infty \mu_{F_i}((-\infty, b])$. Since the two measures agree on a generating set (closed rays form a generated set), they agree everywhere, meaning $\mu_F = \sum_i^\infty \mu_{F_i}$. Thus, we have that (using absolute convergence to re-arrange the sum)

$$f dm + d\lambda = d\mu_F = \sum_{i=1}^\infty d\mu_{F_i} = \sum f_i dm + \sum \lambda_i$$

Since $(\sum d\lambda_i) \perp m$ and $(\sum f_i dm) \ll m$, by question 8:

$$d\lambda = \sum d\lambda_i \quad f dm = \sum f_i dm$$

Finally, we can invoke theorem 3.22 to connect the information about f and f_i to derivative information. By letting $E_r = (x, x+r]$ (since the F and F_i 's are always positive), we get:

$$\begin{aligned} F'(x) &= \lim_{r \rightarrow 0} \frac{F(x+r) - F(x)}{r} \\ &= \lim_{r \rightarrow 0} \frac{\mu_F(E_r)}{m(E_r)} \\ &= f(x) && \text{thm. 3.22} \\ &= \sum f_i(x) && m\text{-a.e.} \\ &= \sum \left(\lim_{r \rightarrow 0} \frac{\mu_{F_i}(E_r)}{m(E_r)} \right) && \text{thm 3.22} \\ &= \sum \left(\lim_{r \rightarrow 0} \frac{F_i(x+r) - F_i(x)}{r} \right) \\ &= \sum F'_i(x) \end{aligned}$$

as we sought to show.

2. Let F denote the cantor function on $[0, 1]$ (see 1.5), and set $F(x) = 0$ for $x < 0$ and $F(x) = 1$ for $x > 1$. Let $\{[a_n, b_n]\}$ be an enumeration of the closed subintervals of $[0, 1]$ with rational endpoints, and let $F_n(x) = F\left(\frac{x-a_n}{b_n-a_n}\right)$. Then $G = \sum_1^\infty 2^{-n} F_n$ is continuous and strictly

increasing on $[0, 1]$ and $G' = 0$ a.e. (use exercise 39)

Proof:

We break this proof down in many steps.

First, F is differentiable a.e. Notice that F is increasing and bounded, and so by theorem 3.27(e) F is a.e. differentiable. Notice that we can extend F as $F(x) = F(0)$ $x \leq 0$ and $F(1) = F(y)$ $1 \leq y$. In this way, $F \in NBV$.

Next, $F' = 0$ a.e. To see this, notice that by definition of F , the function is constant on C^c where C is the Cantor set. Since the Cantor set is closed, C^c is open. On any open interval of C^c , F is constant, and so $F' = 0$ on C^c .

As a consequence, every F_n has zero derivative almost everywhere. At any point where F is differentiable:

$$F'_n = F' \left(\frac{x - a_n}{b_n - a_n} \right) \left(\frac{x - a_n}{b_n - a_n} \right)' = 0 \left(\frac{x - a_n}{b_n - a_n} \right)' = 0$$

Showing that F'_n is a.e. 0.

Using this result, we can apply exercise 39 to get $G' = \sum_n 2^{-n} F'_n$. Since each $2^{-n} F_n$ is a non-negative increasing function has derivative 0 a.e.:

$$G' = \sum_{n=1}^{\infty} 2^{-n} F'_n = 0 \quad \text{a.e.}$$

Which gives us our first result.

Next, let's prove G is continuous. To see this, let $G_n = \sum_{k=1}^n 2^{-k} F_k$. Notice that $2^{-k} F_k$ is continuous being the product of two continuous functions (F_k is continuous being the composition of two continuous functions). By definition of F_k , the largest value is 1, and so $|\sum_1^n 2^{-k} F_n| \leq \sum_1^n 2^{-k} = 1 - 2^{-n}$. Then we can combine these two pieces of information to show that G_n uniformly converges to G . In particular, for all $\epsilon > 0$, choose $n > \frac{1}{2^\epsilon}$ so that $|G - G_n| < \epsilon$ for all $x \in [0, 1]$. Thus, G is continuous.

Finally, to show G is strictly increasing, choose $x, y \in [a, b]$ such that $x < y$. Since x is strictly less than y , there exists an ϵ such that $x \notin [y - \epsilon, y]$. Since the rationals are dense, we can choose $[a_n, b_n]$ from our enumeration such that $x < a_n < b_n < y$ so that $x, y \notin [a_n, b_n]$. As a consequence, notice that $G(x)$ must be at least 2^{-n} away from $G(y)$, since:

$$\sum_n 2^{-n} F_n(x) < \sum_n 2^{-n} F_n(y)$$

and so G is strictly increasing.

I also want to give a quick word on convexity. I'd like to write more about convex functions, like their definition and some properties, but for now I'll just write down in bullet form what I found interesting:

1. Convex functions on compact intervals are absolutely continuous
2. As a consequence, FTC applies. In fact convex functions can be characterized like so : f is convex if and only if f' is strictly increasing. If f is twice differentiable, this leads to the

common notion of $f''(x) \geq 0$

3. In a sense, linear functions are a special type of convex function. Convex functions are closed under addition and scalar multiplication. Thus, linear functions being both convex and concave mean they are closed under subtraction, which is why they form a vector-space, but convex functions do not. The inequalities of linear functions are also all “trivial”, precisely because of linearity, while those of convex functions are more complex.

Hausdorff Dimension and Kekeya Sets

The Lebesgue measure m assigns a notion of “volume” to measurable subsets of $\mathbb{R}^n$, and throughout the preceding chapters this has been the principal tool for quantifying the size of sets. But m is a blunt instrument for the many sets it assigns measure zero: a single point, a line segment in $\mathbb{R}^2$, and the classical Cantor set C all satisfy $m = 0$, yet they differ in every geometric sense. A single point is 0-dimensional, a line segment is 1-dimensional, and the Cantor set—uncountable, perfect, totally disconnected (proposition 1.4.9)—seems to occupy some intermediate regime. To make this distinction precise requires a family of measures that probes sets at every “dimensional scale,” and the critical scale at which the measure transitions from infinity to zero defines a notion of dimension that applies to arbitrary subsets of $\mathbb{R}^n$.

This chapter develops this notion—the *Hausdorff dimension*—and then applies it to a geometric problem whose simplicity belies its difficulty: if a subset of $\mathbb{R}^n$ contains a unit line segment in every direction, how small can it be? Such sets, called *Besicovitch sets* or *Kekeya sets*, can have Lebesgue measure zero. The Kekeya conjecture asserts that despite having zero volume, they must still have full Hausdorff dimension n . The conjecture is resolved for $n = 2$ (Davies, 1971) and $n = 3$ (Wang–Zahl, 2025), but remains open in all higher dimensions. The tools developed in this chapter—Hausdorff measure, Minkowski dimension, and the geometry of directional sets—form the foundation for this problem and its connections to harmonic analysis.

4.1 Hausdorff Measure and Hausdorff Dimension

The construction of Hausdorff measure follows the same outer-measure paradigm used to build the Lebesgue measure in proposition 1.3.2: cover the set by countably many pieces, assign a cost to

each piece, and take the infimum over all such covers. For Lebesgue measure the cost of a piece is its volume; for Hausdorff measure the cost is the s -th power of its diameter, where $s \geq 0$ is a free parameter. By varying s , one obtains a one-parameter family of measures: when s is large, coverings are cheap and the measure collapses to zero; when s is small, no covering is efficient and the measure blows up. The transition point between these two regimes is the Hausdorff dimension.

Definition 4.1.1: Hausdorff Measure

Let $E \subseteq \mathbb{R}^n$ and $s \geq 0$. For each $\delta > 0$, define the δ -approximate s -dimensional Hausdorff measure of E by

$$\mathcal{H}_\delta^s(E) = \inf \left\{ \sum_{j=1}^{\infty} (\text{diam } U_j)^s \mid E \subseteq \bigcup_{j=1}^{\infty} U_j, \text{ diam } U_j < \delta \right\}$$

where the infimum is taken over all countable covers of E by sets $U_j \subseteq \mathbb{R}^n$ with $\text{diam } U_j < \delta$. The convention $0^0 = 1$ is adopted.

As δ decreases, fewer covers are admissible, so $\mathcal{H}_\delta^s(E)$ is non-decreasing in $1/\delta$. Define the s -dimensional Hausdorff measure of E by

$$\mathcal{H}^s(E) = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^s(E) = \sup_{\delta > 0} \mathcal{H}_\delta^s(E) \in [0, \infty]$$

The parameter δ forces the cover to use “fine” pieces: one cannot cheat by covering E with a single large set and paying a cost proportional only to the diameter of that set. As $\delta \rightarrow 0$, only the local geometry of E contributes. The parameter s controls the “dimensional weight” assigned to each covering set: a set of diameter r is penalized as r^s , so higher s makes small-diameter covers cheaper. This interplay between δ (resolution) and s (dimensional weight) is what allows $\mathcal{H}^s$ to detect dimension.

Proposition 4.1.2: $\mathcal{H}^s$ Is a Metric Outer Measure

For each $s \geq 0$, the function $\mathcal{H}^s : P(\mathbb{R}^n) \rightarrow [0, \infty]$ is an outer measure. Furthermore, $\mathcal{H}^s$ is a metric outer measure: if $A, B \subseteq \mathbb{R}^n$ satisfy $\text{dist}(A, B) > 0$, then

$$\mathcal{H}^s(A \cup B) = \mathcal{H}^s(A) + \mathcal{H}^s(B)$$

As a consequence, by theorem 1.3.4, all Borel sets in $\mathbb{R}^n$ are $\mathcal{H}^s$ -measurable.

Proof:

Step 1: $\mathcal{H}^s$ is an outer measure. The argument follows the same template as proposition 1.3.2. First, $\mathcal{H}^s(\emptyset) = 0$: the empty cover (no sets) has cost 0. Monotonicity is immediate: if $A \subseteq B$, every cover of B is a cover of A , so $\mathcal{H}_\delta^s(A) \leq \mathcal{H}_\delta^s(B)$ for each δ , and the inequality persists in the limit.

For countable subadditivity, let $\{E_i\}_{i=1}^\infty$ be a countable collection and fix $\delta > 0$ and $\epsilon > 0$. For

each i , choose a cover $\{U_j^i\}_{j=1}^\infty$ of E_i with $\text{diam } U_j^i < \delta$ and

$$\sum_{j=1}^\infty (\text{diam } U_j^i)^s \leq \mathcal{H}_\delta^s(E_i) + \frac{\epsilon}{2^i}$$

Then $\{U_j^i\}_{i,j}$ covers $\bigcup_i E_i$, so

$$\mathcal{H}_\delta^s\left(\bigcup_i E_i\right) \leq \sum_{i=1}^\infty \sum_{j=1}^\infty (\text{diam } U_j^i)^s \leq \sum_{i=1}^\infty \mathcal{H}_\delta^s(E_i) + \epsilon$$

Since ϵ was arbitrary, $\mathcal{H}_\delta^s(\bigcup_i E_i) \leq \sum_i \mathcal{H}_\delta^s(E_i)$. Taking $\delta \rightarrow 0$, countable subadditivity of $\mathcal{H}^s$ follows.

Step 2: $\mathcal{H}^s$ is a metric outer measure. Let $d = \text{dist}(A, B) > 0$ and choose any $\delta < d$. If $\{U_j\}$ is a cover of $A \cup B$ with $\text{diam } U_j < \delta < d$, then each U_j satisfies $U_j \cap A = \emptyset$ or $U_j \cap B = \emptyset$: if some $x \in U_j \cap A$ and $y \in U_j \cap B$ existed, then $|x - y| \leq \text{diam } U_j < d = \text{dist}(A, B)$, a contradiction. Partitioning the cover into $\mathcal{A} = \{U_j \mid U_j \cap A \neq \emptyset\}$ and $\mathcal{B} = \{U_j \mid U_j \cap B \neq \emptyset\}$ gives

$$\sum_j (\text{diam } U_j)^s = \sum_{U_j \in \mathcal{A}} (\text{diam } U_j)^s + \sum_{U_j \in \mathcal{B}} (\text{diam } U_j)^s \geq \mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B)$$

Taking the infimum over all such covers: $\mathcal{H}_\delta^s(A \cup B) \geq \mathcal{H}_\delta^s(A) + \mathcal{H}_\delta^s(B)$. The reverse inequality holds by subadditivity, so equality holds for each $\delta < d$, and hence in the limit $\delta \rightarrow 0$.

Step 3: Borel measurability. The fact that every metric outer measure on $\mathbb{R}^n$ makes all Borel sets measurable (in the sense of definition 1.3.3) follows from a standard argument: the metric property implies that every open set G satisfies the Carathéodory condition $\mathcal{H}^s(S) = \mathcal{H}^s(S \cap G) + \mathcal{H}^s(S \cap G^c)$ for all test sets S , and since open sets generate the Borel σ -algebra $\mathcal{B}_{\mathbb{R}^n}$, the claim follows from theorem 1.3.4. The verification that open sets satisfy the Carathéodory condition uses a telescoping decomposition of $S \cap G$ into annular shells $A_k = \left\{x \in S \cap G \mid \frac{1}{k+1} \leq d(x, G^c) < \frac{1}{k}\right\}$ with pairwise positive separation, and is left as an exercise.

This gives a second major application of theorem 1.3.4: the Carathéodory machinery, applied to the Hausdorff outer measure, produces a Borel measure for each dimensional parameter s . The resulting measure space $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n}, \mathcal{H}^s)$ is the arena in which Hausdorff dimension is defined. Note that the σ -algebra of $\mathcal{H}^s$ -measurable sets may be strictly larger than $\mathcal{B}_{\mathbb{R}^n}$ (just as the Lebesgue σ -algebra $\mathcal{L}$ is strictly larger than $\mathcal{B}_{\mathbb{R}}$), but for the purposes of dimension theory the Borel sets suffice.

A point of contrast with the Lebesgue outer measure from proposition 1.3.2 is worth emphasizing. The Lebesgue outer measure is constructed from a premeasure on intervals (or rectangles), and its restriction to Borel sets is the unique extension of that premeasure (theorem 1.3.7). The Hausdorff outer measure has no such premeasure origin—it is defined directly on all subsets of $\mathbb{R}^n$ using coverings by *arbitrary* sets, not just intervals. The cost functional $(\text{diam})^s$ plays the role that the volume functional played for the Lebesgue construction. This generality is what allows $\mathcal{H}^s$ to detect fractal behavior: by optimizing over all possible coverings (not just grids), the measure can adapt to the local geometry of irregular sets.

Before defining dimension, it is useful to ground the new construction in the familiar. The case $s = n$ recovers the Lebesgue measure up to a normalizing constant:

Proposition 4.1.3: Hausdorff Measure and Lebesgue Measure

Let m denote the Lebesgue measure on $\mathbb{R}^n$. Then for every Lebesgue measurable set $E \subseteq \mathbb{R}^n$:

$$\mathcal{H}^n(E) = \frac{2^n}{\omega_n} m(E)$$

where $\omega_n = \frac{\pi^{n/2}}{\Gamma(n/2+1)}$ is the volume of the unit ball in $\mathbb{R}^n$. The constant $c_n = 2^n/\omega_n$ arises because $\mathcal{H}^n$ measures sets using $(\text{diam})^n$ while m uses side-lengths (or ball volumes).

Proof Sketch:

For the inequality $\mathcal{H}^n(E) \leq c_n m(E)$: fix $\delta > 0$. By the definition of Lebesgue measure (or by theorem 2.6.1), E can be covered by countably many cubes $\{Q_j\}$ of side length $\delta/\sqrt{n}$, each having diameter $\text{diam}(Q_j) = \delta/\sqrt{n} \cdot \sqrt{n} = \delta$. The s -cost of each cube is $(\text{diam } Q_j)^n = \delta^n$, while its Lebesgue measure is $(\delta/\sqrt{n})^n$. Since $\sum_j m(Q_j) \geq m(E)$, the number of cubes needed is at least $m(E)/(\delta/\sqrt{n})^n = m(E)n^{n/2}/\delta^n$. Then

$$\mathcal{H}_\delta^n(E) \leq \sum_j \delta^n = N \cdot \delta^n$$

where N is the number of cubes. By choosing the cover to have $\sum_j m(Q_j) \leq m(E) + \epsilon$, the bound becomes $\mathcal{H}_\delta^n(E) \leq (m(E) + \epsilon)n^{n/2}$. Taking $\delta \rightarrow 0$ and $\epsilon \rightarrow 0$: $\mathcal{H}^n(E) \leq n^{n/2}m(E)$. (The sharper constant $c_n = 2^n/\omega_n$ requires a more careful argument using ball covers instead of cube covers.)

The reverse inequality $\mathcal{H}^n(E) \geq c_n m(E)$ uses the *isodiametric inequality*: among all sets of diameter d in $\mathbb{R}^n$, the ball of diameter d has the largest volume, namely $\omega_n(d/2)^n$. Hence for any cover $\{U_j\}$ of E with $\text{diam } U_j < \delta$:

$$m(E) \leq \sum_j m(U_j) \leq \sum_j \omega_n \left(\frac{\text{diam } U_j}{2} \right)^n = \frac{\omega_n}{2^n} \sum_j (\text{diam } U_j)^n$$

Taking the infimum over all such covers: $m(E) \leq \frac{\omega_n}{2^n} \mathcal{H}_\delta^n(E)$. Taking $\delta \rightarrow 0$: $m(E) \leq \frac{\omega_n}{2^n} \mathcal{H}^n(E)$, which rearranges to $\mathcal{H}^n(E) \geq \frac{2^n}{\omega_n} m(E) = c_n m(E)$.

4.1.4 Remark: The Isodiametric Inequality The isodiametric inequality states that among all subsets of $\mathbb{R}^n$ with a given diameter, the ball achieves the maximal volume. A proof using Steiner symmetrization can be found in Evans–Gariepy *Measure Theory and Fine Properties of Functions*. The inequality is logically independent of the rest of this chapter and is used only in the proof of proposition 4.1.3.

The normalization means $\mathcal{H}^n$ and m carry the same information on $\mathbb{R}^n$. Some authors define $\mathcal{H}^n$ with the normalizing constant built in; the convention used here (no constant in the definition, constant appearing in the comparison) is more common in fractal geometry.

Example 4.1.5: Hausdorff Measure at Integer Dimensions

1. $\mathcal{H}^0$ is the counting measure. If $|E| = k$ is finite, then $\mathcal{H}^0(E) = k$: cover each point by a set of diameter $< \delta$, and each contributes $(\text{diam})^0 = 1$. If E is infinite, $\mathcal{H}^0(E) = \infty$. This is consistent with the convention $0^0 = 1$.
2. If $\gamma : [a, b] \rightarrow \mathbb{R}^2$ is a C^1 curve with $|\gamma'(t)| > 0$, then $\mathcal{H}^1(\gamma([a, b]))$ equals the arc length $\int_a^b |\gamma'(t)| dt$. The Hausdorff 1-measure thus recovers the classical notion of length for smooth curves. More generally, $\mathcal{H}^k$ restricted to a smooth k -dimensional submanifold of $\mathbb{R}^n$ agrees (up to a normalizing constant) with the k -dimensional volume form induced by the ambient metric.
3. For non-integer values of s , the measure $\mathcal{H}^s$ does not correspond to any classical geometric quantity. The set E at which $\mathcal{H}^s(E)$ transitions from infinity to zero has “effective dimension” s —it is “too large” to be measured by any $\mathcal{H}^t$ with $t < s$ (giving infinity) and “too small” for $\mathcal{H}^t$ with $t > s$ (giving zero). The next proposition makes this transition precise.

The central observation about $\mathcal{H}^s$ is that for a given set E , the value $\mathcal{H}^s(E)$ jumps from ∞ to 0 at a single critical exponent. This jump is a consequence of the power-law relationship between the cost functions at different exponents:

Proposition 4.1.6: Critical Exponent

Let $E \subseteq \mathbb{R}^n$. If $\mathcal{H}^s(E) < \infty$ for some $s \geq 0$, then $\mathcal{H}^t(E) = 0$ for all $t > s$. Equivalently, if $\mathcal{H}^t(E) > 0$ for some t , then $\mathcal{H}^s(E) = \infty$ for all $s < t$.

Proof:

Fix $\delta > 0$ and let $\{U_j\}$ be a cover of E with $\text{diam } U_j < \delta$. For $t > s$:

$$\sum_j (\text{diam } U_j)^t = \sum_j (\text{diam } U_j)^{t-s} (\text{diam } U_j)^s \leq \delta^{t-s} \sum_j (\text{diam } U_j)^s$$

Taking the infimum: $\mathcal{H}_\delta^t(E) \leq \delta^{t-s} \mathcal{H}_\delta^s(E)$. If $\mathcal{H}^s(E) < \infty$, then $\mathcal{H}_\delta^s(E) \leq \mathcal{H}^s(E) + 1$ for small δ , so $\mathcal{H}_\delta^t(E) \leq \delta^{t-s}(\mathcal{H}^s(E) + 1) \rightarrow 0$ as $\delta \rightarrow 0$. Thus $\mathcal{H}^t(E) = 0$.

This jump behavior defines a unique threshold:

Definition 4.1.7: Hausdorff Dimension

Let $E \subseteq \mathbb{R}^n$. The *Hausdorff dimension* of E is

$$\dim_H(E) = \inf \{s \geq 0 \mid \mathcal{H}^s(E) = 0\} = \sup \{s \geq 0 \mid \mathcal{H}^s(E) = \infty\}$$

where the two expressions are equal by proposition 4.1.6. At the critical value $s_0 = \dim_H(E)$, the measure $\mathcal{H}^{s_0}(E)$ can be 0, ∞ , or any value in between.

By proposition 4.1.3, $\dim_H(E) = n$ whenever $m(E) > 0$, and $\dim_H(E) \leq n$ for all $E \subseteq \mathbb{R}^n$. The utility of $\dim_H$ is precisely for sets where $m(E) = 0$: it provides a finer stratification of null sets.

A countable set has dimension 0; a smooth curve in $\mathbb{R}^2$ has dimension 1; a “thick dust” like the Cantor set has some intermediate fractional dimension. The value $\mathcal{H}^{s_0}(E)$ at the critical exponent $s_0 = \dim_H(E)$ itself is often difficult to compute and can be 0, positive and finite, or ∞ . For many problems in geometric measure theory (including the Keakeya conjecture), only the dimension matters, and the precise value of $\mathcal{H}^{s_0}(E)$ is secondary.

Proposition 4.1.8: Properties of Hausdorff Dimension

Let $E, F \subseteq \mathbb{R}^n$ and let $\{E_i\}_{i=1}^\infty \subseteq \mathbb{R}^n$ be a countable collection.

1. **(Monotonicity)** If $E \subseteq F$, then $\dim_H(E) \leq \dim_H(F)$.
2. **(Countable stability)** $\dim_H(\bigcup_{i=1}^\infty E_i) = \sup_i \dim_H(E_i)$.
3. **(Countable sets)** If E is countable, then $\dim_H(E) = 0$.
4. **(Lipschitz maps)** If $f : E \rightarrow \mathbb{R}^m$ satisfies $|f(x) - f(y)| \leq L|x - y|$ for all $x, y \in E$, then $\dim_H(f(E)) \leq \dim_H(E)$. In particular, $\dim_H$ is invariant under bi-Lipschitz maps.

Proof:

1. If $E \subseteq F$, then $\mathcal{H}^s(E) \leq \mathcal{H}^s(F)$ for all s , so $\mathcal{H}^s(F) = 0$ implies $\mathcal{H}^s(E) = 0$, giving $\dim_H(E) \leq \dim_H(F)$.
2. Let $d = \sup_i \dim_H(E_i)$. By monotonicity, $\dim_H(\bigcup_i E_i) \geq d$. For the reverse, if $s > d$, then $\mathcal{H}^s(E_i) = 0$ for all i , so by countable subadditivity $\mathcal{H}^s(\bigcup_i E_i) \leq \sum_i \mathcal{H}^s(E_i) = 0$, hence $\dim_H(\bigcup_i E_i) \leq s$. Since $s > d$ was arbitrary, $\dim_H(\bigcup_i E_i) \leq d$.
3. A single point $\{x\}$ has $\dim_H(\{x\}) = 0$ since $\mathcal{H}^s(\{x\}) = 0$ for all $s > 0$ (cover by a ball of diameter δ : $\text{cost} = \delta^s \rightarrow 0$). By countable stability, any countable set has dimension 0.
4. If $\{U_j\}$ covers E with $\text{diam } U_j < \delta$, then $\{f(U_j \cap E)\}$ covers $f(E)$ with $\text{diam } f(U_j \cap E) \leq L \text{diam } U_j < L\delta$. Thus

$$\mathcal{H}_{L\delta}^s(f(E)) \leq \sum_j (L \text{diam } U_j)^s = L^s \sum_j (\text{diam } U_j)^s$$

Taking the infimum and the limit $\delta \rightarrow 0$: $\mathcal{H}^s(f(E)) \leq L^s \mathcal{H}^s(E)$. If $\mathcal{H}^s(E) = 0$ then $\mathcal{H}^s(f(E)) = 0$, giving $\dim_H(f(E)) \leq \dim_H(E)$. If f is bi-Lipschitz, apply the same argument to f^{-1} to get the reverse inequality.

Property (4) implies that $\dim_H$ is a bi-Lipschitz invariant but *not* a topological invariant. The fat Cantor set C_α of measure $\alpha > 0$ from lemma 1.4.10 satisfies $\dim_H(C_\alpha) = 1$ (since $m(C_\alpha) > 0$ forces $\mathcal{H}^1(C_\alpha) > 0$), yet C_α is homeomorphic to the standard Cantor set C by corollary 1.4.16. Since the homeomorphism need not be Lipschitz, the dimensions are free to differ. This distinction matters: the Keakeya conjecture (which asserts that certain sets have full Hausdorff dimension) would be trivially false if dimension were a topological invariant, since any compact set is homeomorphic to a set of dimension 0 in a sufficiently large ambient space.

Property (2)—countable stability—is the feature that most sharply distinguishes Hausdorff dimension from the Minkowski dimension studied in the next section. It ensures that the dimension of a countable

union is controlled by the dimensions of the individual pieces, a property that is indispensable for measure-theoretic arguments but that the Minkowski dimension fails to satisfy.

The upper bound on dimension is typically obtained by exhibiting an explicit covering and computing its cost, as in the standard representation of the Cantor set by stage- k intervals. Lower bounds are harder: one must show that *no* covering can be too cheap. The most effective general technique is to “distribute mass” on the set and use the following duality:

Lemma 4.1.9: Mass Distribution Principle

Let $E \subseteq \mathbb{R}^n$ and suppose there exists a Borel measure μ with $\mu(E) > 0$ and a constant $C > 0$ such that

$$\mu(B(x, r)) \leq Cr^s \quad \text{for all } x \in \mathbb{R}^n \text{ and all } r > 0$$

Then $\dim_H(E) \geq s$.

Proof:

Let $\{U_j\}$ be any cover of E with $\text{diam } U_j < \delta$. Each U_j is contained in a ball of radius $\text{diam } U_j$, so $\mu(U_j) \leq C(\text{diam } U_j)^s$. Then

$$0 < \mu(E) \leq \mu\left(\bigcup_j U_j\right) \leq \sum_j \mu(U_j) \leq C \sum_j (\text{diam } U_j)^s$$

Hence $\sum_j (\text{diam } U_j)^s \geq \mu(E)/C > 0$ for every admissible cover, giving $\mathcal{H}_\delta^s(E) \geq \mu(E)/C > 0$ and thus $\mathcal{H}^s(E) > 0$, so $\dim_H(E) \geq s$.

The lemma reduces dimension lower bounds to the construction of a measure with controlled local growth. The idea is a duality: to show $\dim_H(E) \geq s$, one produces a measure supported on E that “sees” the set as s -dimensional. The condition $\mu(B(x, r)) \leq Cr^s$ says that μ distributes its mass at rate r^s per ball—matching the growth rate of an s -dimensional object. If such a measure exists, no covering can be too efficient, and the Hausdorff measure at exponent s must be positive.

For self-similar sets, the natural “equidistributed” measure typically satisfies the required ball condition. For the Cantor set, this is the Cantor measure, which assigns equal mass to each interval at every stage of the construction.

Example 4.1.10: Hausdorff Dimension of the Middle-Thirds Cantor Set

Let C denote the standard middle-thirds Cantor set (definition 1.4.8). Then $\dim_H(C) = \frac{\log 2}{\log 3} \approx 0.631$.

Upper bound. At stage k of the construction, C is covered by 2^k closed intervals, each of length 3^{-k} . These intervals have diameter 3^{-k} , so for $s = \frac{\log 2}{\log 3}$:

$$\mathcal{H}_{3^{-k}}^s(C) \leq 2^k \cdot (3^{-k})^s = 2^k \cdot 3^{-k \log 2 / \log 3} = 2^k \cdot 2^{-k} = 1$$

The computation uses $3^{-\log 2 / \log 3} = 2^{-1}$, which follows from exponentiating $-\frac{\log 2}{\log 3} \cdot \log 3 = -\log 2$. Thus $\mathcal{H}^s(C) \leq 1 < \infty$, giving $\dim_H(C) \leq s$. Note that for any $s' < s$, the same covering gives $\mathcal{H}_{3^{-k}}^{s'}(C) \leq 2^k \cdot 3^{-ks'} = (2 \cdot 3^{-s'})^k$, and since $s' < s$ implies $2 \cdot 3^{-s'} > 2 \cdot 3^{-s} = 1$, this diverges as $k \rightarrow \infty$. The covering at exponent s is thus “perfectly balanced”—the cost neither diverges nor

converges to zero.

Lower bound. Define the Cantor measure μ on C as follows: at stage k , assign mass 2^{-k} to each of the 2^k intervals I_k^j . Concretely, if E is a union of stage- k intervals, then $\mu(E) = (\text{number of stage-}k \text{ intervals in } E) \cdot 2^{-k}$. This definition is consistent across stages (each stage- k interval contains exactly two stage- $(k+1)$ intervals, and $2 \cdot 2^{-(k+1)} = 2^{-k}$), so it defines a premeasure on the algebra of finite unions of stage- k intervals, and extends to a Borel measure μ on C with $\mu(C) = 1$ by theorem 1.3.7. The claim is that $\mu(B(x, r)) \leq Cr^s$ for a constant C and $s = \frac{\log 2}{\log 3}$.

For any $x \in C$ and $r > 0$, choose k so that $3^{-(k+1)} \leq r < 3^{-k}$. The ball $B(x, r)$ intersects at most 2 stage- k intervals (since consecutive stage- k intervals are separated by a gap of length $\geq 3^{-k}/2$). Therefore

$$\mu(B(x, r)) \leq 2 \cdot 2^{-k} = 2^{-(k-1)}$$

Since $r \geq 3^{-(k+1)}$, raising to the power s :

$$r^s \geq 3^{-(k+1)s} = 3^{-s} \cdot 3^{-ks} = 3^{-s} \cdot 2^{-k}$$

Combining: $\mu(B(x, r)) \leq 2^{-(k-1)} = 2 \cdot 2^{-k} \leq 2 \cdot 3^s \cdot r^s$. By the mass distribution principle, $\dim_H(C) \geq s = \frac{\log 2}{\log 3}$.

The same technique applies to any self-similar Cantor set where a fixed fraction is removed at each step. The formula $\dim_H = \log(\text{pieces})/\log(\text{scaling factor})$ is a general phenomenon for self-similar sets satisfying an “open set condition,” though the proof of this general statement requires additional machinery beyond the mass distribution principle.

Example 4.1.11: Further Dimension Computations

1. **Fat Cantor sets.** The fat Cantor set C_α with $m(C_\alpha) = \alpha > 0$ satisfies $\dim_H(C_\alpha) = 1$. The lower bound follows from $\mathcal{H}^1(C_\alpha) \geq c_1 m(C_\alpha) > 0$ by proposition 4.1.3, and $\dim_H(C_\alpha) \leq 1$ since $C_\alpha \subseteq \mathbb{R}$. Despite being homeomorphic to the standard Cantor set by corollary 1.4.16, the two sets have different Hausdorff dimensions ($\log 2/\log 3 < 1$). Hausdorff dimension is thus a metric—not topological—invariant.
2. **The rationals and irrationals.** $\dim_H(\mathbb{Q}) = 0$ since $\mathbb{Q}$ is countable. By proposition 4.1.8(2), $1 = \dim_H(\mathbb{R}) = \max\{\dim_H(\mathbb{Q}), \dim_H(\mathbb{R} \setminus \mathbb{Q})\}$, so $\dim_H(\mathbb{R} \setminus \mathbb{Q}) = 1$.
3. **Smooth submanifolds.** If $M \subseteq \mathbb{R}^n$ is a smooth k -dimensional submanifold (for instance, a smooth curve in $\mathbb{R}^2$ or a smooth surface in $\mathbb{R}^3$), then $\dim_H(M) = k$. The upper bound follows by covering M with coordinate patches, each of which is bi-Lipschitz equivalent to an open subset of $\mathbb{R}^k$. The lower bound follows from proposition 4.1.3 applied within each coordinate chart: the k -dimensional Lebesgue measure of the chart is positive, so $\mathcal{H}^k(M) > 0$. This confirms that Hausdorff dimension extends the classical notion of dimension for smooth objects.

The computations above illustrate a general principle: for “tame” sets (smooth manifolds, self-similar fractals satisfying the open set condition), Hausdorff dimension is computable and agrees with geometric intuition. The subtlety arises for sets constructed through limiting processes without self-similar structure—such as the Besicovitch sets studied in the next section, where even establishing a lower bound on $\dim_H$ requires fundamentally new ideas.

Exercise 4.1.1

1. Let C_r be the Cantor set obtained by removing a central interval of length $r \in (0, 1)$ at each stage (the middle-thirds Cantor set corresponds to $r = 1/3$). Show that $\dim_H(C_r) = \frac{\log 2}{\log(2/(1-r))}$.
2. Let $f : [0, 1] \rightarrow \mathbb{R}$ be α -Hölder continuous, that is, $|f(x) - f(y)| \leq C|x - y|^\alpha$ for some $C > 0$ and $\alpha \in (0, 1]$. Show that $\dim_H(\text{graph}(f)) \leq 2 - \alpha$, where $\text{graph}(f) = \{(x, f(x)) \mid x \in [0, 1]\} \subseteq \mathbb{R}^2$. (*Hint*: cover $[0, 1]$ by N intervals of length $1/N$; each piece of the graph is contained in a rectangle of width $1/N$ and height C/N^α .)
3. Show that every metric outer measure on $\mathbb{R}^n$ makes all Borel sets measurable. (*Hint*: it suffices to show that every open set G satisfies the Carathéodory condition. For a test set S , write $S \cap G = \bigcup_{k=1}^\infty A_k$ where $A_k = \{x \in S \cap G \mid \frac{1}{k+1} \leq d(x, G^c) < \frac{1}{k}\}$, and use the metric property on A_k 's with indices differing by ≥ 2 .)
4. Show that $\dim_H(rE) = \dim_H(E)$ for all $r \neq 0$ and $\dim_H(E + a) = \dim_H(E)$ for all $a \in \mathbb{R}^n$ (compare with proposition 1.4.7).
5. Construct an uncountable set $E \subseteq \mathbb{R}$ with $\dim_H(E) = 0$. (*Hint*: consider a Cantor-type construction where the pieces shrink faster than geometrically.)

4.2 Minkowski Dimension

The Hausdorff dimension has excellent theoretical properties—countable stability, Lipschitz invariance, and a clean relationship to Hausdorff measure—but computing it can be demanding, since the definition requires optimizing over *all* countable covers. In practice, a coarser notion that counts δ -balls rather than arbitrary covers is often more tractable. This leads to the *Minkowski dimension* (also called the box-counting dimension), which uses only one type of covering set (balls or cubes of a fixed radius) and extracts the dimension from the growth rate of the covering number.

Definition 4.2.1: Minkowski Dimension

Let $E \subseteq \mathbb{R}^n$ be a bounded set. For each $\delta > 0$, let $N(E, \delta)$ denote the smallest number of sets of diameter at most δ needed to cover E . Define the *upper Minkowski dimension* and *lower Minkowski dimension* of E by

$$\overline{\dim}_M(E) = \limsup_{\delta \rightarrow 0} \frac{\log N(E, \delta)}{-\log \delta} \quad \underline{\dim}_M(E) = \liminf_{\delta \rightarrow 0} \frac{\log N(E, \delta)}{-\log \delta}$$

If the two coincide, their common value is the *Minkowski dimension* $\dim_M(E)$.

The ratio $\log N(E, \delta)/(-\log \delta)$ measures the exponent in the scaling $N(E, \delta) \sim \delta^{-d}$: if covering E at resolution δ requires $\sim \delta^{-d}$ balls, then E “looks d -dimensional” at that scale. The $\limsup$ and $\liminf$ capture the worst-case and best-case scaling behavior as $\delta \rightarrow 0$.

There is an equivalent volume-based formulation that is sometimes more convenient. Let $E_\delta = \{x \in \mathbb{R}^n \mid d(x, E) < \delta\}$ denote the open δ -neighborhood of E . The volume $m(E_\delta)$ grows as E is

“thickened” by δ , and the rate of growth encodes the dimension:

$$\overline{\dim}_M(E) = n - \liminf_{\delta \rightarrow 0} \frac{\log m(E_\delta)}{\log \delta}$$

To see the connection: E_δ is approximately the union of the $N(E, \delta)$ balls of radius δ in an optimal cover, so $m(E_\delta) \approx N(E, \delta) \cdot \omega_n \delta^n$. Taking logarithms and dividing by $\log \delta$ recovers the covering-number formulation. This perspective is particularly useful for relating Minkowski dimension to Lebesgue measure estimates, which is the context in which it appears in the Keakeya conjecture.

The covering number $N(E, \delta)$ admits several equivalent formulations, and these equivalences are useful in different computational contexts:

Proposition 4.2.2: Equivalent Formulations

Let $E \subseteq \mathbb{R}^n$ be bounded. The following quantities all have the same lim sup and lim inf as $\delta \rightarrow 0$ when divided by $-\log \delta$:

1. $N(E, \delta)$: smallest number of sets of diameter $\leq \delta$ covering E .
2. $N'(E, \delta)$: smallest number of closed balls of radius δ covering E .
3. $M(E, \delta)$: largest number of disjoint balls of radius δ centered at points of E .
4. $N_{\text{cube}}(E, \delta)$: number of cubes in a δ -grid of $\mathbb{R}^n$ that intersect E .

Proof:

The equivalences follow from volume comparison. By definition, $N(E, \delta) \leq N'(E, \delta)$ since balls of radius δ have diameter 2δ , but a cover by sets of diameter $\leq \delta$ can be replaced by enclosing balls, giving $N'(E, \delta) \leq N(E, \delta)$. For $M(E, \delta)$ and $N'(E, \delta)$: any ball of radius δ covers at most one center of a disjoint packing, so $M(E, \delta) \leq N'(E, \delta)$. Conversely, the balls of radius 2δ centered at packing points cover E , so $N'(E, 2\delta) \leq M(E, \delta)$.

For N_{cube} : any cube of side δ has diameter $\delta\sqrt{n}$, and any ball of radius δ is contained in at most 3^n cubes of side δ , giving $N_{\text{cube}}(E, \delta) \leq 3^n N'(E, \delta)$ and $N'(E, \delta\sqrt{n}) \leq N_{\text{cube}}(E, \delta)$.

In each case, passing from δ to $c\delta$ for a fixed constant c shifts $\log N$ by at most $\log(c^n)$, which is $O(1)$ and hence negligible compared to $-\log \delta$ as $\delta \rightarrow 0$. Therefore all formulations yield the same lim sup and lim inf.

The relationship between Minkowski and Hausdorff dimensions is governed by a one-sided inequality:

Proposition 4.2.3: Hausdorff Versus Minkowski

For any bounded $E \subseteq \mathbb{R}^n$:

$$\dim_H(E) \leq \underline{\dim}_M(E) \leq \overline{\dim}_M(E)$$

Proof:

Fix $s > \overline{\dim}_M(E)$ and $\delta > 0$. By definition of the upper Minkowski dimension, for all sufficiently small δ there holds $N(E, \delta) \leq \delta^{-s}$. Cover E by $N(E, \delta)$ sets of diameter $\leq \delta$:

$$\mathcal{H}_\delta^s(E) \leq N(E, \delta) \cdot \delta^s \leq \delta^{-s} \cdot \delta^s = 1$$

Hence $\mathcal{H}^s(E) = \lim_{\delta \rightarrow 0} \mathcal{H}_\delta^s(E) \leq 1 < \infty$, so $\dim_H(E) \leq s$. Since $s > \overline{\dim}_M(E)$ was arbitrary, $\dim_H(E) \leq \overline{\dim}_M(E)$. The inequality $\underline{\dim}_M(E) \leq \overline{\dim}_M(E)$ is immediate from $\liminf \leq \limsup$.

The inequality $\dim_H \leq \underline{\dim}_M$ can be strict: the Hausdorff dimension optimizes over all covers (including covers with variable-diameter sets), while the Minkowski dimension uses only covers at a single scale δ . This constraint prevents Minkowski dimension from “adapting” to sets with non-uniform local structure. The proof of proposition 4.2.3 makes the source of the inequality transparent: the uniform cover at scale δ used to bound $\mathcal{H}_\delta^s(E)$ is only one of the many possible covers over which $\mathcal{H}_\delta^s$ takes an infimum. The Hausdorff measure can find cheaper covers, which is why $\dim_H$ can be strictly smaller.

Despite this deficiency, Minkowski dimension has a significant practical advantage: the covering number $N(E, \delta)$ can often be computed or estimated directly from the geometry of E , without the need to optimize over all covers. For this reason, many results in geometric measure theory (including the Keakeya conjecture) are first formulated in terms of Minkowski dimension, and the stronger Hausdorff statement is established separately. The volume-based formulation of Minkowski dimension also connects it directly to the “ δ -tube” estimates that appear in the discretized Keakeya problem.

Example 4.2.4: The Set $\{1/n\}$: Separation of Dimensions

Let $E = \{1/n : n \geq 1\} \subseteq \mathbb{R}$. Since E is countable, $\dim_H(E) = 0$ by proposition 4.1.8(3). The Minkowski dimension is positive: the clustering of points near the origin inflates the box count.

To compute $N(E, \delta)$, partition E into a “sparse” part and a “dense” part. Fix $\delta > 0$ small and let $n_0 = \lfloor 1/\sqrt{\delta} \rfloor$.

Sparse part. For $n \leq n_0$, consecutive points $1/n$ and $1/(n+1)$ have spacing $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)} \geq \frac{1}{n_0(n_0+1)} \geq \frac{\delta}{2}$ for δ small. Each such point requires its own δ -interval, contributing $n_0 \sim \delta^{-1/2}$ intervals.

Dense part. For $n > n_0$, all points lie in $[0, 1/n_0] \subseteq [0, 2\sqrt{\delta}]$. This interval is covered by at most $\lceil 2\sqrt{\delta}/\delta \rceil = \lceil 2/\sqrt{\delta} \rceil \sim \delta^{-1/2}$ intervals of length δ .

Combining both parts: $N(E, \delta) \sim \delta^{-1/2}$. Therefore

$$\dim_M(E) = \lim_{\delta \rightarrow 0} \frac{\log(\delta^{-1/2})}{-\log \delta} = \frac{1}{2}$$

The gap $\dim_H(E) = 0 < 1/2 = \dim_M(E)$ arises because the points cluster near 0, inflating the box count at small scales. The Hausdorff dimension ignores this clustering by using a cover with variable-diameter sets (large balls for the sparse region, small balls near 0). The Minkowski dimension is constrained to a single scale δ and pays a penalty for the dense cluster.

For well-behaved sets such as self-similar fractals, the two dimensions coincide:

Example 4.2.5: Minkowski Dimension of the Cantor Set

The middle-thirds Cantor set C satisfies $\dim_M(C) = \log 2 / \log 3$.

At stage k , C is covered by 2^k intervals of length 3^{-k} , so $N(C, 3^{-k}) \leq 2^k$. Conversely, each of the 2^k stage- k intervals contains a point of C at distance $\geq 3^{-k}/2$ from the others, so $N(C, 3^{-k}) \geq 2^k$. Thus $N(C, 3^{-k}) = 2^k$ for each k , and

$$\dim_M(C) = \lim_{k \rightarrow \infty} \frac{\log 2^k}{\log 3^k} = \frac{\log 2}{\log 3}$$

This confirms $\dim_M(C) = \dim_H(C)$.

Products of sets interact predictably with Minkowski dimension, which is useful for constructing higher-dimensional examples:

Proposition 4.2.6: Minkowski Dimension of Products

Let $A \subseteq \mathbb{R}^m$ and $B \subseteq \mathbb{R}^n$ be bounded. Then

$$\overline{\dim}_M(A \times B) \leq \overline{\dim}_M(A) + \overline{\dim}_M(B)$$

If $\dim_M(A)$ and $\dim_M(B)$ both exist (upper equals lower), then $\dim_M(A \times B) = \dim_M(A) + \dim_M(B)$.

Proof:

Any cover of A by N_A sets of diameter $\leq \delta$ and cover of B by N_B sets of diameter $\leq \delta$ induces a cover of $A \times B$ by $N_A \cdot N_B$ sets of diameter $\leq 2\delta$ (using the product metric). Therefore $N(A \times B, 2\delta) \leq N(A, \delta) \cdot N(B, \delta)$, giving

$$\frac{\log N(A \times B, 2\delta)}{-\log(2\delta)} \leq \frac{\log N(A, \delta) + \log N(B, \delta)}{-\log(2\delta)}$$

Taking $\limsup$ and using $-\log(2\delta) \sim -\log \delta$ establishes the upper Minkowski bound. The lower bound $\underline{\dim}_M(A \times B) \geq \underline{\dim}_M(A) + \underline{\dim}_M(B)$ follows from a packing argument (a maximal δ -separated set in $A \times B$ projects to δ -separated sets in A and B), and when the limits exist, the two bounds combine to give equality.

The product formula confirms the intuition that taking a Cartesian product “adds dimensions”: the product of a d_1 -dimensional set and a d_2 -dimensional set is $(d_1 + d_2)$ -dimensional. For Hausdorff dimension, the analogous inequality $\dim_H(A \times B) \geq \dim_H(A) + \dim_H(B)$ holds, but equality can fail without additional regularity assumptions (though equality does hold when one of the factors has equal upper and lower Minkowski dimensions).

4.2.7 Remark: Countable Stability Fails for Minkowski Dimension Unlike Hausdorff dimension, Minkowski dimension is *not* countably stable. The set $E = \{1/n\}$ is a countable union of points, each of Minkowski dimension 0, yet $\dim_M(E) = 1/2$. This failure is the principal theoretical deficiency of Minkowski dimension. For compact sets arising from geometric constructions—including the Besicovitch sets studied in the next section—the two dimensions often coincide, and the Keakeya conjecture asserts that both equal n .

The tools developed in this section and the previous one—Hausdorff dimension for sharp theoretical bounds, Minkowski dimension for computationally tractable estimates, and the inequality $\dim_H \leq \dim_M$ relating the two—form the dimensional framework for studying the Keakeya conjecture. The next section constructs the sets to which this framework will be applied: the Besicovitch sets, which contain a line segment in every direction yet have zero Lebesgue measure.

Exercise 4.2.1

1. Let $E_\alpha = \{n^{-\alpha} : n \geq 1\}$ for $\alpha > 0$. Show that $\dim_M(E_\alpha) = \frac{1}{1+\alpha}$.
2. Show that $\dim_M(\mathbb{Q} \cap [0, 1]) = 1$ while $\dim_H(\mathbb{Q} \cap [0, 1]) = 0$. This gives a maximal gap between the two dimensions.
3. Verify proposition 4.2.6 explicitly for $A = B = C$ (the middle-thirds Cantor set). Compute $\dim_M(C \times C)$ and confirm it equals $2 \log 2 / \log 3$.
4. Show that if $E \subseteq \mathbb{R}^n$ is bounded and $\dim_M(E)$ exists, then $\dim_M(\overline{E}) = \dim_M(E)$.
5. Show that for any bounded $E \subseteq \mathbb{R}^n$ and $\lambda > 0$, $\overline{\dim}_M(\lambda E) = \overline{\dim}_M(E)$ where $\lambda E = \{\lambda x \mid x \in E\}$. Contrast this with the behavior of Lebesgue measure under dilation (proposition 1.4.7).

4.3 Besicovitch Sets

In 1917, Sōichi Keakeya posed the following question: what is the smallest area of a region in the plane within which a unit needle can be continuously rotated through 180 degrees? If the needle is rotated about its center, the region is a circle of area $\pi/4$. A more efficient motion traces a *deltoid* (a three-cusped hypocycloid) of area $\pi/8$. In 1928, Pál proved that among *convex* regions, the equilateral triangle of unit height has the smallest area. One might expect that the answer for general (non-convex) regions is some small but positive number.

Two distinct questions emerge from Keakeya’s original problem. The first, called the *Keakeya needle problem*, asks for the infimum of areas of regions in which a needle can be *continuously* rotated. Besicovitch showed in 1928 that this infimum is zero: the needle can be rotated continuously through all directions within a set of arbitrarily small area, using a “shifting” motion that translates the needle to avoid sweeping large regions. The second question, which is the focus of this section, drops the continuity requirement entirely and asks: what is the smallest measure of a set that *contains* a unit segment in every direction (without any requirement that the segments be related by a continuous motion)? Besicovitch answered this in 1919: such sets can have Lebesgue measure *zero*. This result predates his solution to the needle problem and is the more fundamental of the two—the set problem concerns the static geometry of directional containment, stripped of all topological or kinematic

constraints.

The existence of measure-zero sets containing a segment in every direction is one of the most counterintuitive facts in geometric measure theory. Such a set carries full directional information—it “sees” every direction in S^{n-1} —yet is negligible from the perspective of Lebesgue measure. The connection to the Cantor set constructions in proposition 1.4.9 is not a coincidence: both phenomena exploit the gap between geometric richness and measure-theoretic size.

Definition 4.3.1: Besicovitch Set

A set $K \subseteq \mathbb{R}^n$ is called a *Besicovitch set* (or *Keakeya set*) if for every direction $e \in S^{n-1}$, there exists a point $a \in \mathbb{R}^n$ such that the unit segment $\{a + te \mid 0 \leq t \leq 1\}$ is contained in K .

A Besicovitch set need not be measurable *a priori*, but all known constructions produce Borel (in fact, compact) sets. Any closed disk of radius $1/2$ or larger is a trivial Besicovitch set—every unit segment can be translated to fit inside it—and has positive measure. The content of Besicovitch’s theorem is that the measure can be driven to zero while retaining all directions. The existence question thus splits into two parts: first, construct sets of arbitrarily small measure containing all directions; then, pass to a set of measure exactly zero.

The first part uses the *Perron tree* construction (named after Oskar Perron, who gave an early version of this argument in 1928). The key geometric mechanism is that thin triangles pointing in nearly parallel directions can be translated to overlap heavily: the overlap region is large near the apex (where the triangles are narrow and close together) and small near the base (where they spread apart). This produces a net reduction in area while preserving all directional content.

Lemma 4.3.2: Perron Tree

Let $T \subseteq \mathbb{R}^2$ be a triangle with base of length b on a horizontal line and apex at height h above the base. For each point P on the base, the segment from P to the apex defines a direction $\theta(P) \in (0, \pi)$, and as P ranges over the base, $\theta(P)$ sweeps an interval $[\theta_{\min}, \theta_{\max}]$. Then for any $\epsilon > 0$, there exists a compact set $\mathcal{P} \subseteq \mathbb{R}^2$ with $m(\mathcal{P}) < \epsilon$ containing, for each $\theta \in [\theta_{\min}, \theta_{\max}]$, a line segment of length h in direction θ .

Proof:

The construction proceeds by iterated splitting and sliding.

Step 1: One splitting. Divide the base of T into 2 equal segments, producing two sub-triangles T_L and T_R sharing the same apex. Each sub-triangle contains segments in half of the directions: T_L handles $[\theta_{\min}, \theta_{\text{mid}}]$ and T_R handles $[\theta_{\text{mid}}, \theta_{\max}]$. Translate T_R horizontally so that its apex coincides with that of T_L . In the resulting figure $\mathcal{P}_1 = T_L \cup T'_R$, all original directions are present, but the two sub-triangles overlap near the apex, reducing the total area.

To compute the overlap, observe that each sub-triangle narrows linearly from base to apex. At height y above the base ($0 \leq y \leq h$), each sub-triangle occupies a horizontal interval of width $\frac{b}{2}(1 - y/h)$: the factor $(1 - y/h)$ reflects the linear taper from base width $b/2$ to apex width 0. Before the translation, T_L and T_R sit side-by-side with no overlap. After translating T_R so that the two apexes coincide, the horizontal offset between the sub-triangles at height y is $\frac{b}{2}(1 - y/h)$ (the same as the width, since the apex shift equals the base half-width). The overlapping region

at height y therefore has width

$$2 \cdot \frac{b}{2}(1 - y/h) - \frac{b}{2} = \frac{b}{2} \left(1 - 2\frac{y}{h} + \frac{y}{h}\right) = \frac{b}{2} \cdot \frac{y}{h}$$

(valid for y where the sub-triangles genuinely overlap). Integrating the area of the overlap:

$$m(T_L \cap T'_R) = \int_0^h \frac{b}{2} \cdot \frac{y}{h} dy = \frac{bh}{4}$$

Since $m(T_L) + m(T_R) = m(T) = bh/2$, the area of the union is $m(\mathcal{P}_1) = bh/2 - bh/4 = bh/4 = m(T)/2$. The overlap saved exactly half the original area.

Step 2: Iteration. Apply the same splitting-and-sliding procedure to each of the two sub-triangles in $\mathcal{P}_1$, obtaining 4 sub-sub-triangles with total area $m(T)/4$. After k iterations, the construction produces 2^k thin triangles with total area $\leq m(T)/2^k$. Choosing k large enough that $m(T)/2^k < \epsilon$ completes the construction.

Step 3: Directional preservation. At each stage, every sub-triangle retains segments in its assigned sub-interval of directions, and the translation preserves these segments (it shifts their starting points but does not rotate them). The union $\mathcal{P}_k$ therefore contains a segment of length $h(1 - 2^{-k}) \geq h/2$ in each direction $\theta \in [\theta_{\min}, \theta_{\max}]$. A final rescaling by a factor of $h/(h \cdot (1 - 2^{-k}))$ adjusts the segment lengths to exactly h , at the cost of increasing the area by a bounded factor. Since k can be chosen arbitrarily large, the total area can still be made $< \epsilon$.

The area estimate $m(\mathcal{P}_k) = m(T)/2^k$ computed above is for a specific splitting scheme. In fact, with a more aggressive sliding strategy (translating the sub-triangles further before each iteration), the area can be reduced even faster. The precise optimum is not needed for the existence result; what matters is that the area can be made arbitrarily small.

The construction should be visualized as follows. At stage 0, the set is a single triangle—a solid wedge containing segments in all directions in a fixed angular interval. At stage 1, the triangle is split into two thinner wedges that are slid together so their tips overlap, producing a figure shaped like an elongated “V” or a narrow tree with two branches. At stage 2, each branch is split and slid again, producing four branches. The process continues: at stage k , the set consists of 2^k very thin triangular “branches” that overlap extensively near their tips. The total area shrinks exponentially with k , yet every original direction is still represented by some branch. The visual resemblance to a tree with an increasingly dense canopy and decreasing total leaf area explains the name “Perron tree.”

Example 4.3.3: Perron Tree: First Two Stages

Start with an isosceles triangle T with base $[-1, 1] \times \{0\}$ and apex $(0, 1)$. This triangle has area 1 and contains segments from the base to the apex sweeping directions in the interval $[\pi/4, 3\pi/4]$ (approximately, for this aspect ratio). A segment from the base point $(-1, 0)$ to the apex $(0, 1)$ points in direction $\arctan(1/1) = \pi/4$, and a segment from $(1, 0)$ to $(0, 1)$ points in direction $\pi - \pi/4 = 3\pi/4$. As the base point sweeps from -1 to 1 , the direction sweeps from $\pi/4$ to $3\pi/4$.

Stage 1. Split the base at 0. The left sub-triangle T_L has vertices $(-1, 0)$, $(0, 0)$, $(0, 1)$; the right sub-triangle T_R has vertices $(0, 0)$, $(1, 0)$, $(0, 1)$. Translate T_R left by 1 unit so its apex coincides with T_L 's. The translated T'_R has vertices $(-1, 0)$, $(0, 0)$, $(-1, 1)$. The union $T_L \cup T'_R$ is a parallelogram-like figure of area $1/2$: the overlap occurs in the upper portion where both triangles taper to width zero. T_L retains segments in directions $[\pi/4, \pi/2]$ and T'_R retains directions in

$[\pi/2, 3\pi/4]$, so all original directions are present.

Stage 2. Split each of T_L and T'_R at their base midpoints, obtaining 4 thin triangles. Slide each pair together as before. The resulting figure has area $1/4$ and still contains segments in all directions in $[\pi/4, 3\pi/4]$. The figure now consists of 4 thin triangles that share significant overlap in their upper portions, creating a branching pattern.

After k stages, the area is 2^{-k} , the set consists of 2^k thin triangles of base width 2^{1-k} , and the figure visually resembles a fractal “tree”—hence the name Perron tree. The tree structure arises because at each stage, the splitting doubles the number of branches while the sliding compresses them together.

With the Perron tree in hand, the passage to a measure-zero Besicovitch set proceeds in two steps: first cover all directions, then iterate the compression to reduce the area to zero.

Theorem 4.3.4: Existence of Measure-Zero Besicovitch Sets

There exists a compact Besicovitch set $K \subseteq \mathbb{R}^2$ with $m(K) = 0$.

Proof:

Step 1: Covering all directions. The Perron tree from lemma 4.3.2 produces a set containing segments in all directions within a single angular interval. Since $[0, \pi)$ is compact, finitely many such intervals cover it: choose N triangles $T^{(1)}, \dots, T^{(N)}$, suitably rotated, so that their angular intervals cover $[0, \pi)$. For any $\epsilon > 0$, applying lemma 4.3.2 to each $T^{(j)}$ produces compact sets $\mathcal{P}^{(j)}$ with $m(\mathcal{P}^{(j)}) < \epsilon/N$. Their union $K_\epsilon = \bigcup_{j=1}^N \mathcal{P}^{(j)}$ is compact, satisfies $m(K_\epsilon) < \epsilon$, and contains a segment (of some fixed length $\ell > 0$, independent of ϵ) in every direction. By rescaling, a segment of length ℓ in every direction can be promoted to a *unit* segment in every direction, at the cost of multiplying the area by $(1/\ell)^2$; since ϵ was arbitrary, the area of the rescaled set can still be made $< \epsilon$.

Step 2: Passage to measure zero. The sets K_ϵ from Step 1 are not nested (different values of ϵ may produce geometrically unrelated configurations), so taking $K = \bigcap_n K_{1/n}$ does not yield a set of measure zero—the intersection may even be empty. Instead, the construction must be refined to produce a nested sequence.

Choose a compact set K_0 containing unit segments in all directions (for instance, a sufficiently large closed disk). The strategy is to apply the Perron tree compression *within* the existing set K_0 , producing a subset $K_1 \subseteq K_0$ of smaller measure that still contains all directions, and then iterate.

Concretely, partition the directions $[0, \pi)$ into N sub-intervals $I_1, \dots, I_N$ and identify the corresponding “sectors” of K_0 : for each I_j , the set of points in K_0 lying on unit segments with directions in I_j forms a roughly triangular region Σ_j . Within each sector Σ_j , apply the splitting-and-sliding procedure of lemma 4.3.2, translating sub-triangles *inward* (toward the interior of Σ_j). The resulting set $K_1 = \bigcup_{j=1}^N \mathcal{P}_j$ consists of thin sub-triangles that are contained in K_0 (the inward sliding ensures this), and satisfies $m(K_1) \leq m(K_0)/2$. Each direction in $[0, \pi)$ is still represented, since the Perron tree within each sector preserves all directions in I_j .

Iterating this procedure produces a nested sequence $K_0 \supseteq K_1 \supseteq K_2 \supseteq \dots$ with $m(K_k) \leq m(K_0)/2^k$.

Define $K = \bigcap_{k=0}^{\infty} K_k$. Since $m(K_0) < \infty$ and the sequence is nested, continuity from above

(proposition 1.2.5) gives $m(K) = \lim_{k \rightarrow \infty} m(K_k) = 0$.

Step 3: K contains all directions. For each direction $\theta \in [0, \pi)$ and each k , the set of starting points a such that $\{a + te_\theta : 0 \leq t \leq 1\} \subseteq K_k$ is a nonempty compact set $S_k(\theta) \subseteq K_k$ (nonempty by construction, compact as a closed subset of K_k). The sequence $S_0(\theta) \supseteq S_1(\theta) \supseteq \dots$ is nested and nonempty, so $\bigcap_k S_k(\theta) \neq \emptyset$ by the finite intersection property. Any point $a \in \bigcap_k S_k(\theta)$ satisfies $\{a + te_\theta : 0 \leq t \leq 1\} \subseteq K_k$ for all k , hence $\{a + te_\theta : 0 \leq t \leq 1\} \subseteq K$. Therefore K is a Besicovitch set.

The passage from “small area” to “zero area” in Step 2 is the most delicate part of the construction. The argument that the Perron tree compression can be made to produce nested sets requires geometric care—in particular, the sub-triangles must be slid toward the interior of the parent triangle, not away from it. A complete treatment of this nesting, with explicit coordinates at each stage, can be found in Falconer’s *The Geometry of Fractal Sets* or Mattila’s *Geometry of Sets and Measures in Euclidean Spaces*.

The comparison to the Vitali set construction (theorem 1.0.1) is instructive. Vitali sets are *unmeasurable*—they cannot be assigned any consistent measure. Besicovitch sets are measurable (in fact, compact and Borel) with measure *exactly zero*. Both constructions reveal fundamental limitations: Vitali sets show that not all subsets of $\mathbb{R}$ can be measured; Besicovitch sets show that measure zero does not preclude rich geometric structure. However, Vitali sets require the axiom of choice, while Besicovitch sets are constructed explicitly (within ZF). This constructivity makes the Besicovitch set a more “concrete” pathology—it is not an artifact of set-theoretic axioms but a genuine feature of Euclidean geometry.

The existence of measure-zero Besicovitch sets shows that the Lebesgue measure is too coarse to capture the directional complexity of subsets of $\mathbb{R}^n$. A finer invariant is needed, and Hausdorff dimension from section 4.1 provides exactly this: while $m(K) = 0$ for all Besicovitch sets K , the Hausdorff dimension $\dim_H(K)$ can still be positive and potentially as large as n . The question of how large $\dim_H(K)$ must be is the Keakeya conjecture.

The construction generalizes to higher dimensions via a product argument:

Proposition 4.3.5: Besicovitch Sets in $\mathbb{R}^n$

For every $n \geq 2$, there exists a compact Besicovitch set $K \subseteq \mathbb{R}^n$ with $m(K) = 0$.

Proof:

Let $K_2 \subseteq \mathbb{R}^2$ be a compact Besicovitch set with $m_2(K_2) = 0$ (where m_2 denotes the 2-dimensional Lebesgue measure). The product $K_2 \times [0, 1]^{n-2} \subseteq \mathbb{R}^n$ has n -dimensional measure $m_n(K_2 \times [0, 1]^{n-2}) = m_2(K_2) \cdot 1 = 0$ by theorem 2.5.9. However, this product set only contains segments whose directions lie in the 2-plane $\mathbb{R}^2 \times \{0\}^{n-2}$: a unit segment in $K_2 \times [0, 1]^{n-2}$ has the form $(a + te_1, a_3, \dots, a_n)$ where $e_1 \in S^1 \subseteq \mathbb{R}^2$ and $a_3, \dots, a_n$ are fixed. Any direction $e \in S^{n-1}$ with a nonzero component outside the first two coordinates is missing.

To capture all directions in S^{n-1} , take a finite collection of orthogonal transformations $R_1, \dots, R_M \in O(n)$ such that

$$\bigcup_{j=1}^M R_j(\mathbb{R}^2 \times \{0\}^{n-2})$$

covers a neighborhood of every direction in S^{n-1} . Such a finite collection exists by a compactness

argument: for each direction $e \in S^{n-1}$, there is a rotation R mapping e into $\mathbb{R}^2 \times \{0\}^{n-2}$ (and in fact into a neighborhood of directions), and the resulting open cover of S^{n-1} has a finite subcover. For each R_j , the set $R_j(K_2 \times [0, 1]^{n-2})$ is compact and has $m_n = 0$ (orthogonal transformations preserve Lebesgue measure by theorem 2.6.4 since $|\det R_j| = 1$). The union

$$K = \bigcup_{j=1}^M R_j(K_2 \times [0, 1]^{n-2})$$

is a finite union of measure-zero compact sets, hence compact and measure zero. For any direction $e \in S^{n-1}$, some R_j maps e into $\mathbb{R}^2 \times \{0\}^{n-2}$, and the corresponding rotated copy $R_j(K_2 \times [0, 1]^{n-2})$ contains a unit segment in direction e . Therefore K is a Besicovitch set.

This product-and-rotate construction is crude—it gives a Besicovitch set but reveals nothing about its dimension. To understand why, note that $K_2 \times [0, 1]^{n-2}$ has Hausdorff dimension at most $\dim_H(K_2) + (n - 2) \leq 2 + (n - 2) = n$ by proposition 4.2.6 (applied to Hausdorff dimension, where the analogous inequality holds). Rotating and taking a finite union does not change the dimension, so the construction gives $\dim_H(K) \leq n$ —which is already known from $K \subseteq \mathbb{R}^n$. The construction provides no *lower* bound on $\dim_H(K)$ beyond the trivial $\dim_H(K) \geq 1$ (from the presence of line segments).

The dimension of a Besicovitch set is the central open question. By proposition 4.1.8(1), every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) \leq n$ (since $K \subseteq \mathbb{R}^n$). The question is whether $\dim_H(K) = n$ always holds—can the dimension be strictly less than n ? In $\mathbb{R}^2$, the answer is no:

Theorem 4.3.6: Davies' Theorem

Every Besicovitch set in $\mathbb{R}^2$ has Hausdorff dimension 2.

Davies' proof (1971) uses a projection argument that is specific to $\mathbb{R}^2$. The key idea is as follows. If $K \subseteq \mathbb{R}^2$ is a Besicovitch set, consider the projection $\pi_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}$ onto the line perpendicular to direction θ . Since K contains a unit segment in direction θ , the projection $\pi_\theta(K)$ contains an interval of length $\sin(\alpha)$ for some angle α depending on θ . By a result of Marstrand (1954) relating the Hausdorff dimension of a set to the dimensions of its projections, the fact that $\pi_\theta(K)$ has positive length for *every* direction θ forces $\dim_H(K) \geq 2$. (In precise terms: if $\dim_H(K) < 2$, then $\pi_\theta(K)$ would have zero length for *almost every* θ by Marstrand's projection theorem, contradicting the Besicovitch property.)

This argument fails in $\mathbb{R}^n$ for $n \geq 3$ because the relationship between a set and its projections becomes weaker in higher dimensions. A set of Hausdorff dimension $d < n$ in $\mathbb{R}^n$ projects to a set of positive measure in *some* directions but not necessarily in enough directions to force $d = n$. The combinatorial structure of how lines in $\mathbb{R}^n$ interact with lower-dimensional projections is far richer and less constrained than in $\mathbb{R}^2$, and exploiting the Besicovitch property (a segment in *every* direction) requires fundamentally different techniques. The natural generalization of Davies' theorem—that every Besicovitch set in $\mathbb{R}^n$ has Hausdorff dimension n —is the Keakeya conjecture, which forms the subject of the remaining sections.

To summarize the situation: Besicovitch sets of measure zero exist in every $\mathbb{R}^n$ with $n \geq 2$, so the Lebesgue measure cannot distinguish between a “large” Besicovitch set (like a disk) and a “small” one (measure zero). The Hausdorff dimension provides the finer distinction. In $\mathbb{R}^2$, Davies' theorem settles the matter: $\dim_H(K) = 2$ for every Besicovitch set. In $\mathbb{R}^3$, the Keakeya conjecture was open

for over 50 years until Wang and Zahl resolved it in 2025. In $\mathbb{R}^n$ for $n \geq 4$, it remains one of the central open problems in geometric measure theory.

Exercise 4.3.1

1. Verify that at stage k of the Perron tree construction in lemma 4.3.2, the total area of the 2^k sub-triangles (before accounting for overlaps) is $m(T)$, and that the overlap estimate $m(\mathcal{P}_k) \leq m(T)/2^k$ holds by induction.
2. Show that if $K \subseteq \mathbb{R}^n$ is a Besicovitch set and $T \in GL_n(\mathbb{R})$ is an invertible linear map, then $T(K)$ is also a Besicovitch set. Does this remain true if T is injective but not surjective?
3. Show that a Besicovitch set $K \subseteq \mathbb{R}^n$ cannot be contained in any affine hyperplane. Conclude that $\dim_H(K) \geq 1$ for all $n \geq 2$. (This lower bound is far from optimal.)
4. Explain why the product construction $K_2 \times [0, 1]^{n-2}$ does not directly contain unit segments in all directions of S^{n-1} (identify the missing directions), and verify that the finite rotation argument in proposition 4.3.5 resolves this.
5. (Nikodym set) A *Nikodym set* $N \subseteq \mathbb{R}^2$ is a set such that for every point $x \in \mathbb{R}^2$, there exists a line ℓ_x through x with $\ell_x \setminus \{x\} \subseteq N$. Show that Nikodym sets of measure zero exist. (*Hint*: the construction is closely related to the Besicovitch construction.)
6. Show that the Besicovitch set K constructed in theorem 4.3.4 is nowhere dense (definition 1.4.12). Conclude that K is topologically “small” (meager, being a compact set with empty interior) yet geometrically “rich” (containing a unit segment in every direction). Compare this with the discussion of the Cantor set in proposition 1.4.13.

4.4 The Kakeya Conjecture and Maximal Functions

The preceding section established that Besicovitch sets—compact sets of Lebesgue measure zero containing a unit segment in every direction—exist in every $\mathbb{R}^n$ with $n \geq 2$. Davies’ theorem (theorem 4.3.6) showed that in $\mathbb{R}^2$, such sets must have full Hausdorff dimension despite having zero measure. The central open problem is whether this remains true in all dimensions:

Conjecture 4.4.1: Kakeya Set Conjecture

Every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) = n$. Equivalently (since $\dim_H \leq \dim_M$ by proposition 4.2.3), every Besicovitch set satisfies $\dim_M(K) = n$.

The equivalence stated above is one-directional: the Hausdorff statement is strictly stronger than the Minkowski statement (since $\dim_H \leq \dim_M$, full Hausdorff dimension implies full Minkowski dimension, but not conversely). Both versions are commonly referred to as “the Kakeya conjecture,” and both are open for $n \geq 4$. In $n = 2$, both are settled by theorem 4.3.6. In $n = 3$, both were resolved by Wang and Zahl in 2025, as discussed at the end of this chapter.

The conjecture admits a natural discretization in terms of “ δ -tubes” that makes it more amenable to combinatorial and analytic methods. The passage from continuous sets to discrete collections of tubes is a standard technique in geometric measure theory: it replaces the measure-zero set K with a family of small-but-positive-volume objects, converting a dimension problem into a volume problem.

Definition 4.4.2: δ -Tube

Let $e \in S^{n-1}$ be a direction, $a \in \mathbb{R}^n$ a center point, and $\delta > 0$. The δ -tube centered at a in direction e is the set

$$T_e^\delta(a) = \{x \in \mathbb{R}^n \mid |(x-a) - ((x-a) \cdot e)e| \leq \delta, \quad 0 \leq (x-a) \cdot e \leq 1\}$$

This is a cylinder of length 1, radius δ , with its long axis in direction e and centered at the point $a + \frac{1}{2}e$. Its volume is $|T_e^\delta(a)| = \omega_{n-1}\delta^{n-1}$, where ω_{n-1} is the volume of the unit ball in $\mathbb{R}^{n-1}$.

A δ -tube should be thought of as a “thickened” line segment: as $\delta \rightarrow 0$, it shrinks to the unit segment $\{a + te : 0 \leq t \leq 1\}$. A Besicovitch set K containing a segment in every direction can be “thickened” by replacing each segment with a δ -tube, and the Kakeya conjecture can be rephrased in terms of the volume of the resulting union of tubes.

A family of directions is called δ -separated if any two directions $e, e' \in S^{n-1}$ satisfy $|e - e'| \geq \delta$. A maximal δ -separated subset of S^{n-1} has cardinality $\sim \delta^{-(n-1)}$ (since S^{n-1} has $(n-1)$ -dimensional surface area $n\omega_n$ and each δ -cap has area $\sim \delta^{n-1}$). The discretized Kakeya conjecture asserts that any such family of tubes has nearly maximal union volume:

Proposition 4.4.3: Discretized Kakeya Conjecture

The Kakeya set conjecture (?? 4.4.1) is equivalent to the following statement: for every $\epsilon > 0$, there exists a constant $C_\epsilon > 0$ such that if $\mathbb{T}$ is a collection of δ -tubes $\{T_{e_j}^\delta(a_j)\}_{j=1}^N$ with δ -separated directions and $N \geq c\delta^{-(n-1)}$, then

$$\left| \bigcup_{j=1}^N T_{e_j}^\delta(a_j) \right| \geq C_\epsilon \delta^\epsilon$$

The bound $|\bigcup T_j| \geq C_\epsilon \delta^\epsilon$ is a weak lower bound: it says the volume cannot decay faster than any power of δ . For comparison, the total volume of all N tubes (if disjoint) would be $N \cdot \omega_{n-1}\delta^{n-1} \sim 1$, while the trivial lower bound from a single tube is $\omega_{n-1}\delta^{n-1}$. The conjecture sits between these extremes, asserting that the union cannot be much smaller than a fixed positive constant (independent of δ).

Proof:

Step 1: Kakeya conjecture implies the discretized statement. Assume the Kakeya set conjecture holds, and let $\mathbb{T} = \{T_{e_j}^\delta(a_j)\}_{j=1}^N$ be a collection of δ -tubes with δ -separated directions and $N \geq c\delta^{-(n-1)}$. Define $K = \bigcap_{k=1}^\infty \bigcup_{j=1}^N \overline{T_{e_j}^{1/k}(a_j)}$, which is a compact set containing a unit segment in each direction e_j . By extending to a full Besicovitch set (adding segments in the finitely many missing directions), the Kakeya conjecture gives $\dim_M(K) = n$.

The δ -neighborhood $K_\delta = \{x \mid d(x, K) < \delta\}$ contains all the tubes $T_{e_j}^\delta(a_j)$. By the volume formulation of Minkowski dimension (section 4.2), $\dim_M(K) = n$ implies that for any $\epsilon > 0$ and all sufficiently small δ :

$$|K_\delta| \geq c_\epsilon \delta^{n-n+\epsilon} = c_\epsilon \delta^\epsilon$$

More precisely, the definition $\dim_M(K) = n$ means $\lim_{\delta \rightarrow 0} \frac{\log |K_\delta|}{\log \delta} = n - n = 0$ (using the n -dimensional volume formulation), so $|K_\delta|$ decays slower than any power of δ . Since $|\bigcup T_j| \leq |K_\delta|$, the discretized bound follows.

Step 2: Discretized statement implies the Kakeya conjecture. Conversely, let K be a Besicovitch set and fix $\epsilon > 0$. For each $\delta > 0$, choose a maximal δ -separated set of directions $\{e_j\}_{j=1}^N$ on S^{n-1} (so $N \sim \delta^{-(n-1)}$). For each e_j , the set K contains a unit segment in direction e_j , so there exist center points a_j such that the δ -tube $T_{e_j}^\delta(a_j)$ is contained in the δ -neighborhood K_δ . In particular, $|\bigcup_j T_{e_j}^\delta(a_j)| \leq |K_\delta|$.

By the discretized statement, $|K_\delta| \geq |\bigcup T_j| \geq C_\epsilon \delta^\epsilon$. Taking logarithms and dividing by $\log \delta$ (which is negative for $\delta < 1$):

$$\frac{\log |K_\delta|}{\log \delta} \leq \frac{\log C_\epsilon + \epsilon \log \delta}{\log \delta} = \epsilon + \frac{\log C_\epsilon}{\log \delta} \rightarrow \epsilon \quad \text{as } \delta \rightarrow 0$$

By the volume formulation of Minkowski dimension, $n - \dim_M(K) = \lim_{\delta \rightarrow 0} \frac{\log |K_\delta|}{\log \delta} \leq \epsilon$, giving $\dim_M(K) \geq n - \epsilon$. Since $\epsilon > 0$ was arbitrary, $\dim_M(K) = n$.

The passage from $\dim_M(K) = n$ to $\dim_H(K) = n$ requires an additional argument: the discretized statement must be applied at multiple scales simultaneously (not just a single δ) to produce efficient covers for the Hausdorff measure. The details of this “multi-scale” refinement are standard and can be found in Wolff’s lecture notes or Mattila’s textbook.

The discretized formulation transforms the Kakeya conjecture from a question about abstract Hausdorff dimension into a concrete volume estimate for unions of tubes. This reformulation is the form in which the conjecture is typically attacked, since tubes are rigid geometric objects whose intersections can be analyzed explicitly. The ϵ -loss in the bound $|\bigcup T_j| \geq C_\epsilon \delta^\epsilon$ may seem artificial, but it is essential: a bound of the form $|\bigcup T_j| \geq c$ (with no δ -dependence) is equivalent to the stronger statement that K has positive Lebesgue measure, which is false for Besicovitch sets. The δ^ϵ factor allows the volume to decay, but only sub-polynomially.

The intersection structure of tubes is governed by a basic geometric estimate: two tubes at angle α intersect in a region of controlled volume. This estimate is the foundation for all results on the Kakeya problem:

Lemma 4.4.4: Tube Intersection Estimate

Let $T_e^\delta(a)$ and $T_{e'}^\delta(a')$ be two δ -tubes in $\mathbb{R}^n$ with $|e - e'| = \alpha \geq \delta$. Then

$$|T_e^\delta(a) \cap T_{e'}^\delta(a')| \lesssim \frac{\delta^{n-1}}{\alpha}$$

where the implicit constant depends only on n . In particular, when $\alpha \sim 1$ (nearly perpendicular tubes), the intersection has volume $\sim \delta^{n-1}$, while when $\alpha \sim \delta$ (nearly parallel tubes), the intersection can be as large as $\sim \delta^{n-2}$.

Proof:

The intersection of two cylinders of radius δ and length 1 whose axes meet at angle α is contained in a “slab” of thickness $\delta/\sin \alpha$ in the direction bisecting the two axes, and of cross-section $\sim \delta^{n-2}$ in the remaining directions. Since $\sin \alpha \geq c\alpha$ for $\alpha \in [0, \pi/2]$, the volume of the intersection is bounded by

$$|T_e^\delta(a) \cap T_{e'}^\delta(a')| \lesssim \delta^{n-2} \cdot \frac{\delta}{\alpha} = \frac{\delta^{n-1}}{\alpha}$$

completing the estimate.

The tube intersection estimate controls the degree to which two tubes can overlap as a function of their angular separation. When $\alpha \sim 1$, the tubes cross nearly perpendicularly, and their intersection is a small “dot” of volume $\sim \delta^{n-1}$. When $\alpha \sim \delta$, the tubes are nearly parallel and can overlap along their entire length, giving an intersection of volume $\sim \delta^{n-2}$ —much larger. The worst case for the Kakeya problem is when many tubes are nearly parallel, since their overlaps waste volume. The Kakeya conjecture asserts that no arrangement of tubes can exploit this overlap efficiently enough to make the total union volume small.

The total overlap of a collection of N tubes with δ -separated directions can be quantified. Denoting $F = \sum_{j=1}^N \chi_{T_j}$, the L^2 norm squared of F counts pairwise intersections:

$$\|F\|_{L^2}^2 = \sum_{i,j} |T_i \cap T_j|$$

The diagonal terms contribute $N \cdot |T_j| \sim N\delta^{n-1}$. The off-diagonal terms are controlled by the tube intersection estimate and depend on the distribution of angles between pairs of tubes. Bounding this L^2 norm is the key to Córdoba’s argument and the starting point for all higher-dimensional approaches.

With the tube intersection estimate in hand, the Kakeya problem can be approached through a maximal function that encodes the “worst-case” averaging behavior of a function along tubes. The connection to the Hardy–Littlewood maximal function (definition 3.4.6) is direct: the Hardy–Littlewood function takes suprema over balls centered at a point, while the Kakeya maximal function takes suprema over tubes in a given direction. The directional constraint is what makes the Kakeya problem harder: the Hardy–Littlewood maximal theorem (theorem 3.4.7) gives L^p bounds for $p > 1$ with constants independent of the ball radius, while the Kakeya maximal function incurs a loss of $\delta^{-\epsilon}$ (and the conjecture asserts this loss is the worst possible).

Definition 4.4.5: Kakeya Maximal Function

Let $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $\delta > 0$. The *Kakeya maximal function* of f is the function $f_\delta^* : S^{n-1} \rightarrow [0, \infty]$ defined by

$$f_\delta^*(e) = \sup_{a \in \mathbb{R}^n} \frac{1}{|T_e^\delta(a)|} \int_{T_e^\delta(a)} |f(x)| dx$$

where the supremum is over all positions $a \in \mathbb{R}^n$ of the δ -tube in direction e .

For each direction e , $f_\delta^*(e)$ captures the maximum average value of $|f|$ over any tube pointing in direction e . If $f = \chi_K$ for a Besicovitch set K , then $f_\delta^*(e) \geq c > 0$ for every direction (since K contains a segment, hence a substantial fraction of a tube, in every direction), and bounding f_δ^* from above constrains how “concentrated” the set K can be. The Kakeya maximal conjecture asserts a

quantitative L^p bound:

Conjecture 4.4.6: Kakeya Maximal Conjecture

For every $\epsilon > 0$, there exists $C_\epsilon > 0$ such that for all $\delta > 0$ and $f \in L^n(\mathbb{R}^n)$:

$$\|f_\delta^*\|_{L^n(S^{n-1})} \leq C_\epsilon \delta^{-\epsilon} \|f\|_{L^n(\mathbb{R}^n)}$$

The exponent $p = n$ is critical: for $p < n$, the bound fails (as can be shown using a bush-like configuration), and for $p > n$, the bound follows from simpler covering arguments. Testing the maximal conjecture on $f = \chi_K$ recovers the set conjecture: if $\|f_\delta^*\|_{L^n} \leq C_\epsilon \delta^{-\epsilon}$ and $f_\delta^*(e) \geq c$ for every e , then the surface area of S^{n-1} gives $c^n \cdot n\omega_n \leq C_\epsilon^n \delta^{-n\epsilon} \|f\|_{L^n}^n$, forcing $\|f\|_{L^n}^n \geq c' \delta^{n\epsilon}$, hence $|K_\delta| \geq c' \delta^{n\epsilon}$ for any ϵ , establishing $\dim_M(K) = n$. The maximal conjecture is thus strictly stronger than the set conjecture—it provides quantitative control, not just a dimension bound.

Example 4.4.7: The Maximal Function for Simple Configurations

Consider $n = 2$ and $f = \chi_R$ where R is a rectangle of width δ and length 1 aligned with the x -axis. Then $f_\delta^*(e_1) = 1$ (the tube in direction $e_1 = (1, 0)$ can be chosen to coincide with R), while $f_\delta^*(e) \lesssim \delta/|e - e_1|$ for $|e - e_1| \gg \delta$ (a tube at angle α to R intersects R in a set of area $\sim \delta^2/\alpha$, and dividing by the tube area δ gives the average $\sim \delta/\alpha$). Integrating over S^1 : $\|f_\delta^*\|_{L^2}^2 \lesssim 1 + \delta^2 \sum_{k=1}^{1/\delta} 1/(k\delta)^2 \cdot \delta \sim \log(1/\delta)$. This shows the $\log(1/\delta)$ factor in Córdoba's estimate is achieved (up to constants) by a single rectangle, confirming the bound is sharp.

In $\mathbb{R}^2$, the maximal conjecture is a theorem, proved by Córdoba in 1977 using a technique known as the “bush argument.” The proof is a model for higher-dimensional approaches and uses only the tube intersection estimate and the Cauchy–Schwarz inequality:

Theorem 4.4.8: Córdoba's Estimate

In $\mathbb{R}^2$, for every $\delta > 0$ and $f \in L^2(\mathbb{R}^2)$:

$$\|f_\delta^*\|_{L^2(S^1)} \leq C \left(\log \frac{1}{\delta} \right)^{1/2} \|f\|_{L^2(\mathbb{R}^2)}$$

In particular, the Kakeya maximal conjecture holds in $\mathbb{R}^2$ (since $(\log 1/\delta)^{1/2} \leq C_\epsilon \delta^{-\epsilon}$ for any $\epsilon > 0$).

Proof:

By a standard linearization argument, it suffices to prove the dual estimate: if $\{T_j\}_{j=1}^N$ is a collection of δ -tubes with δ -separated directions (so $N \sim 1/\delta$), and $\{a_j\}_{j=1}^N$ are non-negative coefficients, then

$$\left\| \sum_{j=1}^N a_j \chi_{T_j} \right\|_{L^2(\mathbb{R}^2)}^2 \leq C \log \frac{1}{\delta} \sum_{j=1}^N a_j^2 |T_j|$$

Taking $a_j = 1$ for all j gives a bound on the L^2 norm of the sum of characteristic functions, which by duality controls the maximal function.

Step 1: Expanding the L^2 norm. Compute

$$\left\| \sum_j a_j \chi_{T_j} \right\|_{L^2}^2 = \sum_{i,j} a_i a_j |T_i \cap T_j|$$

Split this into diagonal terms ($i = j$) and off-diagonal terms ($i \neq j$).

Step 2: Diagonal terms. The diagonal contributes $\sum_j a_j^2 |T_j| = \omega_1 \delta \sum_j a_j^2$, where $|T_j| = \omega_1 \delta$ is the area of a δ -tube in $\mathbb{R}^2$ (a rectangle of width 2δ and length 1).

Step 3: Off-diagonal terms. For $i \neq j$, the tube intersection estimate (lemma 4.4.4) gives $|T_i \cap T_j| \lesssim \delta^2 / |e_i - e_j|$ in $\mathbb{R}^2$ (since $n = 2$, the bound is $\delta^{n-1}/\alpha = \delta/\alpha$; including the width of the tubes gives δ^2/α after a more careful accounting). Order the directions so that $e_1, e_2, \dots, e_N$ are arranged consecutively around S^1 with $|e_j - e_{j+1}| \sim \delta$. Then $|e_i - e_j| \gtrsim |i - j|\delta$, and by Cauchy-Schwarz ($a_i a_j \leq \frac{1}{2}(a_i^2 + a_j^2)$):

$$\begin{aligned} \sum_{i \neq j} a_i a_j |T_i \cap T_j| &\lesssim \sum_{i \neq j} a_i a_j \frac{\delta^2}{|i - j|\delta} \\ &= \delta \sum_{i \neq j} \frac{a_i a_j}{|i - j|} \\ &\leq \delta \sum_j a_j^2 \sum_{k=1}^N \frac{1}{k} \\ &\lesssim \delta \log N \sum_j a_j^2 \\ &\sim \delta \log \frac{1}{\delta} \sum_j a_j^2 \end{aligned}$$

where the second inequality uses the Cauchy-Schwarz step and the symmetry of the sum. The harmonic sum $\sum_{k=1}^N 1/k \sim \log N \sim \log(1/\delta)$ produces the logarithmic factor.

Step 4: Combining. Adding the diagonal and off-diagonal contributions:

$$\left\| \sum_j a_j \chi_{T_j} \right\|_{L^2}^2 \lesssim \delta \log \frac{1}{\delta} \sum_j a_j^2 = \log \frac{1}{\delta} \sum_j a_j^2 |T_j|$$

The duality between $\sum a_j \chi_{T_j}$ and the maximal function f_δ^* (via testing against f) then yields the claimed bound $\|f_\delta^*\|_{L^2} \leq C(\log 1/\delta)^{1/2} \|f\|_{L^2}$.

The factor $(\log 1/\delta)^{1/2}$ in Córdoba's bound is an artifact of the harmonic sum over tube intersections. It is sub-polynomial in $1/\delta$ (growing slower than any power $\delta^{-\epsilon}$), so the Kakeya maximal conjecture follows immediately. As a corollary, the Kakeya set conjecture in $\mathbb{R}^2$ follows from the maximal conjecture, giving a second proof of theorem 4.3.6.

The structure of the proof reveals why $n = 2$ is tractable. In $\mathbb{R}^2$, the directions live on S^1 and can be linearly ordered. The harmonic sum $\sum_{k=1}^N 1/k \sim \log N$ captures the total pairwise overlap, and this sum grows slowly enough that the Cauchy-Schwarz step produces a useful bound. In $\mathbb{R}^n$ for $n \geq 3$,

the directions live on S^{n-1} and cannot be linearly ordered. The pairwise intersection sum becomes a double sum over an $(n-1)$ -dimensional index set, and the geometric structure of the intersections depends not just on the angle between two tubes but also on their relative positions in space. A direct repetition of Córdoba's approach in $\mathbb{R}^n$ yields the bound $\dim_H(K) \geq (n+1)/2$, which for $n=3$ gives $\dim_H \geq 2$ —weaker than Wolff's $5/2$. The additional gain in Wolff's argument comes from exploiting the *linear* structure of tubes (not just their pointwise intersections):

Theorem 4.4.9: Wolff's Bound

Every Besicovitch set $K \subseteq \mathbb{R}^n$ satisfies $\dim_H(K) \geq (n+2)/2$.

Proof Key ideas:

The proof introduces the “hairbrush” argument, which improves on Córdoba's “bush” argument by exploiting the linear structure of tubes (rather than just their pointwise intersections).

Step 1: Bush versus hairbrush. In Córdoba's argument, the worst case is a “bush”: many tubes passing through a common δ -ball. In a bush, the tubes spread out radially, and their union has volume $\sim N\delta^{n-1}$ (each tube contributes its full volume beyond the common ball). This gives $|\bigcup T_j| \geq c$ when $N \sim \delta^{-(n-1)}$, which is the full conjecture—but only if a bush actually occurs. When tubes are spread out and no bush exists, Córdoba's argument loses efficiency.

Step 2: Wolff's observation. Instead of looking for a point through which many tubes pass, look for a tube T_0 that is intersected by many other tubes. The tubes intersecting T_0 form a “hairbrush” rooted at T_0 . The intersections are spread *along* T_0 (not concentrated at a point), and the tubes emanating from these intersection points spread into the ambient space.

Step 3: Volume estimate. The hairbrush structure produces a volume gain over the naive estimate. By a pigeonhole argument, there exists a tube T_0 intersected by $\gtrsim N^{1/2}$ other tubes (this uses the total intersection count). The $N^{1/2}$ tubes of the hairbrush have δ -separated intersection points along T_0 and δ -separated directions, so their union has volume $\gtrsim N^{1/2} \cdot \delta^{n-1} \sim \delta^{(n-2)/2}$. This leads to the bound $\dim_M(K) \geq (n+2)/2$ (and with more work, $\dim_H(K) \geq (n+2)/2$).

The pigeonhole step is the heart of the argument. The total number of “incidences” (pairs (T_i, T_j) with $T_i \cap T_j \neq \emptyset$ and $i \neq j$) is at least $cN^2\delta$ (from a counting argument using the volume of the union). Distributing these incidences among N tubes, at least one tube T_0 must have $\geq cN\delta \sim N^{1/2} \cdot N^{1/2}\delta$ incidences. The $N^{1/2}\delta$ factor, combined with the geometry of tube intersections along T_0 , gives the improved volume bound.

For $n=3$, Wolff's bound gives $\dim_H(K) \geq 5/2$. The gap between $5/2$ and the conjectured value of 3 resisted all attempts for nearly three decades, until Wang and Zahl closed it in 2025 (see the end of this chapter). For $n \geq 4$, Wolff's bound of $(n+2)/2$ remains the best known general result, though incremental improvements have been obtained using arithmetic combinatorics (Bourgain, Katz–Tao) and polynomial methods (Guth).

The following table summarizes the state of knowledge for the Kakeya set conjecture in low dimensions, illustrating the progression from Wolff's bound to the resolution in $\mathbb{R}^3$:

n	Conjectured $\dim_H$	Best lower bound	Reference
2	2	2 (resolved)	Davies (1971)
3	3	3 (resolved)	Wang–Zahl (2025)
4	4	3	Wolff (1995)
5	5	7/2	Wolff (1995)
n	n	$(n+2)/2$	Wolff (1995)

Exercise 4.4.1

1. Verify the tube intersection estimate (lemma 4.4.4) explicitly in $\mathbb{R}^2$: show that two rectangles of width 2δ and length 1 whose long axes meet at angle α have intersection area $\lesssim \delta^2/\alpha$.
2. Show that the Kakeya maximal conjecture implies the Kakeya set conjecture. Specifically, if K is a Besicovitch set and $f = \chi_{K_\delta}$ (the δ -neighborhood of K), show that the maximal conjecture gives $|K_\delta| \geq c_\epsilon \delta^{n\epsilon}$, and conclude $\dim_M(K) = n$.
3. In $\mathbb{R}^2$, use Córdoba’s estimate directly (not through the maximal function) to show that a union of $N = 1/\delta$ tubes with δ -separated directions has area $\gtrsim 1/\log(1/\delta)$. (*Hint*: let $F = \sum_j \chi_{T_j}$. Compute $\|F\|_{L^1}$ and $\|F\|_{L^2}$, then apply Cauchy–Schwarz: $\|F\|_{L^1}^2 \leq |\text{supp}(F)| \cdot \|F\|_{L^2}^2$.)
4. Compute the volume of the intersection of two δ -tubes in $\mathbb{R}^3$ at angle α , and verify the estimate $\lesssim \delta^2/\alpha$.
5. Show that the “bush” configuration— N tubes all passing through a common δ -ball—achieves the volume lower bound $|\bigcup T_j| \gtrsim 1$ when the directions are δ -separated. This shows the Kakeya conjecture would be trivially true if all Besicovitch sets had a “bush” structure.
6. Explain why the Kakeya maximal conjecture remains open in $\mathbb{R}^3$ even after the Wang–Zahl resolution of the set conjecture. What additional information would the maximal conjecture provide?

4.5 The Finite Field Kakeya Conjecture

In 1999, Wolff proposed studying the Kakeya problem over finite fields as a model for the Euclidean case. The finite field setting strips away all analytic structure—there are no limits, no measures, no covering arguments—and reduces the problem to pure combinatorics. A “line” in $\mathbb{F}_q^n$ is a set of q points $\{x + ty : t \in \mathbb{F}_q\}$ for $x, y \in \mathbb{F}_q^n$ with $y \neq 0$, and a “Kakeya set” is a set containing a line in every direction. The question is: how large must such a set be?

Wolff hoped that techniques for the finite field problem would transfer to $\mathbb{R}^n$. The finite field analogue was resolved in 2008 by Dvir, using an algebraic method—the “polynomial method”—that is entirely different from the combinatorial and analytic approaches used in the Euclidean setting. The proof is short and self-contained, making it a compelling application of algebraic techniques to a combinatorial problem.

Definition 4.5.1: Finite Field Kakeya Set

Let $\mathbb{F}_q$ be a finite field with q elements and $\mathbb{F}_q^n$ the n -dimensional vector space over $\mathbb{F}_q$. A set $K \subseteq \mathbb{F}_q^n$ is called a *Kakeya set* if for every direction $y \in \mathbb{F}_q^n \setminus \{0\}$, there exists $x \in \mathbb{F}_q^n$ such that $\{x + ty : t \in \mathbb{F}_q\} \subseteq K$.

Since $\mathbb{F}_q^n$ is a finite set of cardinality q^n , the question is quantitative: what is the minimum cardinality of a Kakeya set? There are $(q^n - 1)/(q - 1)$ directions (points of projective space $\mathbb{P}^{n-1}(\mathbb{F}_q)$), and each line has q points, so a Kakeya set contains at least $(q^n - 1)/(q - 1)$ lines. If these lines were disjoint, the minimum size would be $q \cdot (q^n - 1)/(q - 1) \sim q^n$. The finite field Kakeya conjecture asserts that even with maximal overlap, the size is still $\sim q^n$.

The analogy with the Euclidean case is as follows. A “Besicovitch set” in $\mathbb{F}_q^n$ would be a Kakeya set of size $o(q^n)$ —much smaller than the ambient space. The finite field conjecture states that no such set exists: every Kakeya set has size $\geq cq^n$, meaning it must occupy a positive fraction of $\mathbb{F}_q^n$. In the Euclidean setting, Besicovitch sets of Lebesgue measure zero *do* exist, so the analogy is not perfect. The finite field conjecture is closer in spirit to the Minkowski dimension statement: the “dimension” of a Kakeya set (measured by the exponent in $|K| \sim q^d$) must equal n .

Before Dvir’s work, partial results on the finite field conjecture were obtained by Wolff (1999), Bourgain–Katz–Tao (2004), and others, using combinatorial methods that mirrored the Euclidean techniques (bush and hairbrush arguments). These gave bounds of the form $|K| \geq cq^{\alpha n}$ for various $\alpha < 1$. Dvir’s proof, by contrast, gives the optimal exponent $\alpha = 1$ in a single stroke, using an entirely algebraic approach.

The key to Dvir’s proof is a vanishing lemma that connects the geometry of lines to the algebra of polynomials:

Lemma 4.5.2: Vanishing on Kakeya Sets

Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set and let $P \in \mathbb{F}_q[x_1, \dots, x_n]$ be a polynomial of degree $\deg P < q$. If P vanishes on K (i.e., $P(x) = 0$ for all $x \in K$), then P is the zero polynomial.

Proof:

Suppose $P \neq 0$ and $\deg P = d < q$. Write $P = P_d + P_{d-1} + \dots + P_0$ where P_j is the homogeneous component of degree j and $P_d \neq 0$.

Let $y \in \mathbb{F}_q^n \setminus \{0\}$ be any direction. Since K is a Kakeya set, there exists $x \in \mathbb{F}_q^n$ such that $\{x + ty : t \in \mathbb{F}_q\} \subseteq K$. Consider the univariate polynomial $Q(t) = P(x + ty)$. Since P vanishes on K and $x + ty \in K$ for all $t \in \mathbb{F}_q$, the polynomial Q vanishes at all q elements of $\mathbb{F}_q$. The degree of Q as a polynomial in t is at most $d < q$. A univariate polynomial of degree $< q$ over $\mathbb{F}_q$ with q roots must be the zero polynomial. Therefore $Q \equiv 0$.

The leading coefficient of $Q(t) = P(x + ty)$ (as a polynomial in t) is $P_d(y)$: expanding $P(x + ty)$, the terms of degree d in t come exclusively from the homogeneous part P_d , giving the leading term $t^d P_d(y)$. Since $Q \equiv 0$, all coefficients vanish, and in particular $P_d(y) = 0$.

This holds for every $y \in \mathbb{F}_q^n \setminus \{0\}$. Since P_d is homogeneous, $P_d(0) = 0$ as well (every monomial in P_d has positive degree). Thus P_d vanishes on all of $\mathbb{F}_q^n$. A nonzero polynomial of degree d in n variables over $\mathbb{F}_q$ can vanish on at most $d \cdot q^{n-1}$ points of $\mathbb{F}_q^n$ (by the Schwartz–Zippel lemma). Since $d < q$, this bound is $< q^n = |\mathbb{F}_q^n|$, so P_d cannot vanish on all of $\mathbb{F}_q^n$ unless $P_d = 0$. This

contradicts $\deg P = d$, completing the proof.

The vanishing lemma has a clean algebraic content: a Kakeya set is “algebraically large” in the sense that no low-degree polynomial can vanish on it. The theorem then follows by a dimension count:

Theorem 4.5.3: Dvir’s Theorem

Let $K \subseteq \mathbb{F}_q^n$ be a Kakeya set. Then $|K| \geq \binom{q+n-1}{n} \geq \frac{q^n}{n!}$.

Proof:

The vector space $V = \{P \in \mathbb{F}_q[x_1, \dots, x_n] \mid \deg P < q\}$ of polynomials of degree $< q$ in n variables has dimension

$$\dim V = \binom{q+n-1}{n}$$

over $\mathbb{F}_q$ (the number of monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 + \cdots + a_n < q$). Consider the evaluation map $\text{ev} : V \rightarrow \mathbb{F}_q^K$ defined by $\text{ev}(P) = (P(x))_{x \in K}$. By lemma 4.5.2, $\ker(\text{ev}) = \{0\}$, so ev is injective. Since the codomain has dimension $|K|$ over $\mathbb{F}_q$:

$$|K| \geq \dim V = \binom{q+n-1}{n} \geq \frac{q^n}{n!}$$

The last inequality follows from $\binom{q+n-1}{n} = \frac{(q+n-1)(q+n-2) \cdots q}{n!} \geq \frac{q^n}{n!}$.

The proof is three lines once the vanishing lemma is established: the polynomial space is too large to be annihilated by evaluation on a small set. The logic is: (1) a Kakeya set cannot support the vanishing of any nonzero low-degree polynomial (by the vanishing lemma), so (2) the evaluation map from the space of low-degree polynomials to $\mathbb{F}_q^K$ must be injective, and (3) injectivity forces $|K| \geq \dim V$. The proof thus converts a geometric property (containing a line in every direction) into an algebraic property (no low-degree polynomial vanishes on K), and then uses linear algebra to extract a size bound.

The bound $|K| \geq q^n/n!$ is sharp up to the constant $1/n!$. The factor $n!$ comes from the binomial coefficient and can be improved: Dvir–Kopparty–Saraf–Sudan (2013) showed $|K| \geq q^n/2^n$ by a refinement involving the multiplicity of vanishing rather than just vanishing itself. The optimal constant remains open, though for large q the bound $|K| \geq (1+o(1))q^n/2^n$ is expected to be close to sharp.

A natural question is whether the polynomial method gives any information about the *structure* of Kakeya sets, not just their size. Over finite fields, the answer is partially yes: the method shows that Kakeya sets must be “algebraically spread” (they cannot be contained in a low-degree variety), and subsequent work has used this to study the geometric distribution of points in Kakeya sets. Over $\mathbb{R}^n$, as the following remark explains, the method requires substantial modification.

Example 4.5.4: Dvir’s Bound in Small Cases

1. Let $q = 3$ and $n = 2$, so $\mathbb{F}_3^2$ has 9 points. The theorem gives $|K| \geq \binom{4}{2} = 6$. A Kakeya set in $\mathbb{F}_3^2$ must contain a line (of 3 points) in each of the 4 directions of $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0), (0,1), (1,1), (1,2)\}$. The 4 lines have 12 points in total, but with overlaps. One can verify that the minimum Kakeya set has $|K| = 7$ (achievable by arranging the 4 lines to

share a common point and one additional overlap), consistent with the lower bound of 6.

2. Let $q = 2$ and $n = 3$, so $\mathbb{F}_2^3$ has 8 points. The theorem gives $|K| \geq \binom{4}{3} = 4$. There are $(2^3 - 1)/(2 - 1) = 7$ directions in $\mathbb{P}^2(\mathbb{F}_2)$. Each “line” has only 2 points ($\{x, x + y\}$ for $t \in \{0, 1\} = \mathbb{F}_2$). With 7 lines of 2 points each, a total of 14 point-memberships must be distributed among at most 8 points. The complement of any single point in $\mathbb{F}_2^3$ has 7 points and can be checked to be a Kakeya set (every direction is represented), so $|K| = 7$ is achievable. The lower bound of 4 is far from tight in this case, reflecting the looseness of the $1/n!$ constant for small q .

4.5.5 Remark: Why the Polynomial Method Does Not Transfer to $\mathbb{R}^n$ The polynomial method is effective over $\mathbb{F}_q$ because a nonzero polynomial of degree d can vanish on at most $d \cdot q^{n-1}$ points of $\mathbb{F}_q^n$ (the Schwartz–Zippel lemma)—a quantitative bound that makes the dimension count work. Over $\mathbb{R}$, no such bound exists: a nonzero polynomial of degree d can vanish on an uncountable set (e.g., the zero set of $x_1^2 + x_2^2 - 1$ is the unit circle, which has the cardinality of $\mathbb{R}$). The discrete counting argument that powers Dvir’s proof has no continuous analogue.

The conceptual barrier is that over finite fields, “small” sets are sets with few elements, and “large” sets have many elements—a distinction captured by cardinality. Over $\mathbb{R}^n$, “small” sets are sets of low Hausdorff dimension, and “large” sets have high dimension—a distinction that cardinality cannot detect ($\mathbb{R}$ and $\mathbb{R}^2$ have the same cardinality but different dimensions). The polynomial method over finite fields exploits cardinality, not dimension, which is why it does not transfer.

A partial descendant of the polynomial method in the Euclidean setting is the *polynomial partitioning* technique introduced by Guth and Katz (2010), which uses a polynomial to partition $\mathbb{R}^n$ into cells of controlled geometry rather than to vanish on a set. The idea is to choose a polynomial P of degree d such that $\mathbb{R}^n \setminus Z(P)$ (the complement of the zero set) splits into $\sim d^n$ cells, each containing a controlled number of tubes. The zero set $Z(P)$ itself captures the tubes that lie on an algebraic variety, which can be handled separately. This technique has led to progress on related problems (such as the Erdős distinct distances problem) but has not yet resolved the Euclidean Kakeya conjecture.

The contrast between the finite field and Euclidean settings encapsulates a broader theme in modern combinatorial geometry: problems that are “purely combinatorial” over finite fields often require analytic and multi-scale techniques over $\mathbb{R}$. The Kakeya problem is a paradigmatic example of this phenomenon, and the interplay between algebraic, combinatorial, and analytic methods remains an active area of research.

Exercise 4.5.1

1. Verify Dvir’s bound for $q = 2$, $n = 3$: compute the number of directions in $\mathbb{P}^2(\mathbb{F}_2)$, construct a Kakeya set in $\mathbb{F}_2^3$ explicitly, and check $|K| \geq \binom{4}{3} = 4$.
2. Show that lemma 4.5.2 fails if the degree condition is relaxed to $\deg P \leq q$ (rather than $\deg P < q$). (*Hint*: consider the polynomial $P(x) = \prod_{a \in \mathbb{F}_q} (x_1 - a)$, which has degree q and vanishes on $\mathbb{F}_q^n$ regardless of whether K is a Kakeya set.)
3. (Schwartz–Zippel lemma) Prove that a nonzero polynomial $P \in \mathbb{F}_q[x_1, \dots, x_n]$ of total degree d has at most $d \cdot q^{n-1}$ zeros in $\mathbb{F}_q^n$. (*Hint*: induct on n . For $n = 1$ this is the fundamental

theorem of algebra over $\mathbb{F}_q$. For $n > 1$, write $P = \sum_{j=0}^d P_j(x_1, \dots, x_{n-1})x_n^j$ and consider $a \in \mathbb{F}_q$ such that $P(\cdot, a)$ is nonzero.)

4. Show that a random subset $K \subseteq \mathbb{F}_q^n$ of size $|K| = q^{n-1}/2$ is, with high probability, *not* a Kakeya set when q is large. (*Hint*: for a fixed direction, the probability that a specific line lies in K is $(|K|/q^n)^q$. Apply a union bound over all directions.)
5. The *joints problem* asks: given L lines in $\mathbb{R}^3$, how many “joints” (points where three non-coplanar lines meet) can there be? Guth and Katz (2010) proved the sharp bound $O(L^{3/2})$ using the polynomial method. Formulate and prove the analogous result over $\mathbb{F}_q^3$: if $\mathcal{L}$ is a set of lines in $\mathbb{F}_q^3$ and J is the set of joints, show $|J| \leq C|\mathcal{L}|^{3/2}$ using a vanishing argument analogous to lemma 4.5.2. (*Hint*: if $|J| > c|\mathcal{L}|^{3/2}$, find a polynomial of degree $< (|\mathcal{L}|/c)^{1/2}$ vanishing on J , then show its gradient vanishes at every joint, contradicting non-coplanarity.)

4.6 The Wang–Zahl Theorem and Open Frontiers

The results presented so far establish the Kakeya set conjecture in $\mathbb{R}^2$ (Davies, 1971) and provide the lower bound $\dim_H(K) \geq (n+2)/2$ in $\mathbb{R}^n$ (Wolff, 1995). In $\mathbb{R}^3$, Wolff’s bound gives $\dim_H(K) \geq 5/2$, leaving a gap of $1/2$ below the conjectured value of 3. This gap was narrowed incrementally over three decades through the introduction of techniques from arithmetic combinatorics, additive number theory, and polynomial methods. The following is a summary of the main advances:

1. Wolff (1995): $\dim_H(K) \geq (n+2)/2$. (Hairbrush argument.)
2. Katz–Łaba–Tao (2000): $\dim_M(K) > 5/2 + \epsilon_0$ in $\mathbb{R}^3$ for an explicit small $\epsilon_0 > 0$. (Additive combinatorics, sticky Kakeya configurations.)
3. Katz–Tao (2002): $\dim_H(K) \geq (2 - \sqrt{2})(n-4) + 3$ for $n \geq 4$. (Sum-product estimates from arithmetic combinatorics.)
4. Katz–Zahl (2019): $\dim_H(K) \geq 5/2 + \epsilon_1$ in $\mathbb{R}^3$ for a small $\epsilon_1 > 0$. (Polynomial partitioning and degree reduction.)
5. Wang–Zahl (2022): Structural classification of near-extremal Kakeya sets in $\mathbb{R}^3$ into “sticky” and “non-sticky” regimes.
6. Wang–Zahl (2025): $\dim_H(K) = 3$ and $\dim_M(K) = 3$ for all Besicovitch sets in $\mathbb{R}^3$.

Each step introduced new ideas that reshaped the landscape of the problem.

The Katz–Łaba–Tao result (2000) was the first to break Wolff’s $5/2$ barrier in $\mathbb{R}^3$. Their approach introduced the notion of “sticky” Kakeya sets: configurations where, at every intermediate scale $\delta < r < 1$, the tubes cluster into thin slabs. They showed that sticky configurations are in some sense the “hardest” case, and handled non-sticky configurations by a refined hairbrush argument. The improvement $5/2 + \epsilon_0$ was small but conceptually significant: it demonstrated that the barrier at $5/2$ was not fundamental.

The Katz–Tao papers (2002) brought arithmetic combinatorics—specifically, sum-product estimates over $\mathbb{R}$ —into the Kakeya problem. The connection is through the “arithmetic Kakeya conjecture”:

if the set of slopes in a Kakeya configuration has additive structure (e.g., is close to an arithmetic progression), then sum-product estimates force the set to be large. This approach established the first bounds that improve over $(n + 2)/2$ in high dimensions.

The Katz–Zahl paper (2019) adapted the polynomial partitioning technique from incidence geometry to the Kakeya setting. By partitioning $\mathbb{R}^3$ using the zero set of a polynomial, they decomposed the Kakeya problem into “cellular” and “algebraic” components, handling each with different methods. This gave a second independent proof that the $5/2$ barrier is not sharp.

The Wang–Zahl papers (2022, 2025) unified and extended these ideas into a comprehensive structural theory of Kakeya sets at multiple scales, culminating in the resolution of the three-dimensional case.

Theorem 4.6.1: Wang–Zahl Theorem

Every Besicovitch set $K \subseteq \mathbb{R}^3$ has Hausdorff dimension 3 and Minkowski dimension 3.

The proof, contained in the preprint “Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions” (arXiv:2502.17655, February 2025), is approximately 130 pages. The core argument is an induction on scales, and the key innovation is a reformulation of the Kakeya problem in terms of a more general “super-Kakeya” condition involving convex sets (called “grains”) rather than tubes. The term “super-Kakeya” refers to the fact that the reformulated problem is *stronger* than the original Kakeya conjecture: it asserts a volume bound for unions of convex sets satisfying a condition that includes the Kakeya condition as a special case. The stronger formulation is paradoxically easier to prove, because it is preserved under the induction on scales (the original Kakeya condition is not, since passing from scale δ to scale $\sqrt{\delta}$ distorts tubes into more general convex sets).

4.6.2 Remark: Key Ideas of the Proof The proof of theorem 4.6.1 has three main components.

1. *Induction on scales.* The problem at scale δ (tubes of radius δ) is reduced to a problem at scale $\sqrt{\delta}$ (tubes of radius $\sqrt{\delta}$). At the coarser scale, the tubes are replaced by “grains”—convex sets of controlled dimensions that capture the large-scale geometry. The inductive step shows that if the volume bound fails at scale δ , it must also fail at scale $\sqrt{\delta}$ with a quantitatively worse failure, which eventually contradicts the trivial bound at scale ~ 1 .

2. *Sticky versus non-sticky configurations.* In their 2022 paper, Wang and Zahl classified near-extremal Kakeya configurations into two regimes. In the “sticky” regime, tubes cluster into thin slabs at every intermediate scale, creating a self-similar structure. In the “non-sticky” regime, tubes spread out at intermediate scales. The two regimes require different arguments: the non-sticky case can be handled by a refined version of Wolff’s hairbrush argument (combined with polynomial methods), while the sticky case requires a new inductive structure using the grain decomposition.

3. *The role of dimension 3.* When a grain in the inductive step degenerates (becomes flat or thin), the problem reduces to a lower-dimensional Kakeya problem: a 2-dimensional problem for flat grains, or a 1-dimensional problem for thin grains. In $\mathbb{R}^3$, these lower-dimensional problems are already solved (by Davies’ theorem and trivial estimates). This “dimension reduction” is the mechanism that closes the induction. In $\mathbb{R}^n$ for $n \geq 4$, the degenerate cases reduce to Kakeya problems in $\mathbb{R}^3, \mathbb{R}^4, \dots$, which remain open, blocking the induction.

The Wang–Zahl theorem settles the Kakeya set conjecture in $\mathbb{R}^3$, but several related problems remain

open. The state of the art and the main open directions are summarized below.

4.6.3 Remark: Open Problems and Directions *The conjecture in higher dimensions.*

The Keakeya set conjecture remains open for all $n \geq 4$. The barrier is the dimension reduction step in the Wang–Zahl argument: degenerate configurations in $\mathbb{R}^n$ reduce to Keakeya problems in $\mathbb{R}^{n-1}$ (and lower), which are not yet solved for $n - 1 \geq 4$. Resolving the conjecture in $\mathbb{R}^4$ would likely require either a different inductive scheme or new ideas to handle the degenerate cases without reducing to lower dimensions.

The Keakeya maximal conjecture. Even in $\mathbb{R}^3$, the Keakeya maximal conjecture (?? 4.4.6) remains open. The Wang–Zahl proof establishes the set conjecture but does not produce the quantitative L^p bounds needed for the maximal statement. Strengthening the result to a maximal function estimate would have immediate consequences for the restriction and Bochner–Riesz conjectures in harmonic analysis.

Furstenberg sets. A *Furstenberg set* is a variant of a Besicovitch set where, instead of requiring a full unit segment in every direction, one requires a subset of Hausdorff dimension $\geq s$ inside a unit segment in every direction. The Furstenberg set conjecture predicts the minimum Hausdorff dimension of such sets as a function of s and n . Wang and Zahl’s techniques yield partial results on Furstenberg sets, but the full conjecture remains open.

The (n, k) -Besicovitch conjecture. An (n, k) -Besicovitch set is a set of measure zero in $\mathbb{R}^n$ containing a translate of every k -dimensional unit disk. For $k > 1$, it is conjectured that no such set exists (in contrast to $k = 1$, where they do exist). The case $k = 1$ is the classical Keakeya problem; the higher- k cases are open for all n .

Connections to number theory. The Keakeya conjecture has implications for problems in analytic number theory, including estimates for exponential sums and bounds on the distribution of primes in arithmetic progressions. Montgomery’s conjecture on Dirichlet polynomials implies the Keakeya conjecture in all dimensions; the resolution of the three-dimensional case does not immediately yield progress on Montgomery’s conjecture, but it strengthens the evidence and provides new techniques that may be applicable. For the number-theoretic connections, see `NateAnalyticNumTheory`.

The resolution of the three-dimensional Keakeya conjecture represents a major advance in geometric measure theory, and the techniques introduced by Wang and Zahl—induction on scales with grain decomposition, the sticky/non-sticky dichotomy, and the interaction with polynomial methods—are expected to have applications beyond the Keakeya problem itself.

The chapter has traced a progression from abstract measure theory (Hausdorff dimension as a refinement of Lebesgue measure) through geometric construction (Besicovitch sets as measure-zero objects with full directional content) to combinatorial analysis (tube intersection estimates and the Keakeya maximal function) and algebra (the polynomial method over finite fields). This progression reflects the interdisciplinary nature of the Keakeya problem: it sits at the intersection of geometric measure theory, combinatorics, algebra, and harmonic analysis. The harmonic-analytic connections—in particular, the relationship between the Keakeya conjecture and the restriction, Bochner–Riesz, and local smoothing conjectures for the Fourier transform—are treated later in this book.

The remaining challenges (higher dimensions, maximal function estimates, Furstenberg sets) provide concrete targets for future work, and the tools developed in this chapter—Hausdorff dimension, Minkowski dimension, tube intersection estimates, and the relationship between directional geometry and volume bounds—are the foundation on which that progress will be built. The reader who wishes to go further is directed to the following references:

1. Wolff, *Lectures on Harmonic Analysis* (AMS, 2003): lecture notes covering the Kakeya problem, restriction estimates, and related topics.
2. Mattila, *Geometry of Sets and Measures in Euclidean Spaces* (Cambridge, 1995): the standard reference for Hausdorff dimension and geometric measure theory.
3. Falconer, *The Geometry of Fractal Sets* (Cambridge, 1985): a more elementary treatment of dimension theory and Besicovitch sets.
4. Wang and Zahl, “Volume estimates for unions of convex sets, and the Kakeya set conjecture in three dimensions” (arXiv:2502.17655, 2025): the proof of the three-dimensional case.

Exercise 4.6.1

1. Verify that Wolff’s bound (theorem 4.4.9) gives $\dim_H(K) \geq 5/2$ for Besicovitch sets in $\mathbb{R}^3$, and compare this to the Wang–Zahl result $\dim_H(K) = 3$.
2. Show that if the Kakeya set conjecture holds in $\mathbb{R}^n$, it automatically holds in $\mathbb{R}^m$ for all $m \leq n$. (*Hint*: embed $\mathbb{R}^m$ into $\mathbb{R}^n$ and use proposition 4.1.8(4).)
3. Explain in your own words why the Wang–Zahl argument does not immediately extend to $\mathbb{R}^4$. Identify the specific step that fails and what additional input would be needed.
4. The Besicovitch set K constructed in theorem 4.3.4 has $m(K) = 0$ and $\dim_H(K) = 2$ (by theorem 4.3.6). Show that K must have $\dim_H(K) \geq 2$ directly from the Kakeya maximal function estimate in $\mathbb{R}^2$ (theorem 4.4.8), without appealing to Davies’ theorem.
5. Show that a Besicovitch set $K \subseteq \mathbb{R}^3$ with $m(K) = 0$ is nowhere dense (has empty interior). Relate this to definition 1.4.12: K is topologically small (meager) yet has full Hausdorff dimension by the Wang–Zahl theorem. Compare with proposition 1.4.13.